THE INNER HORIZON



THE INNER HORIZON

The complete essay, with the topological readings



ÍÑIGO BARRERA BARCELÓ

The Inner Horizon

The complete essay, with the topological readings

© Íñigo Barrera Barceló. All rights reserved.

The essay chapters are drawn from the complete work The Inner Horizon and are reproduced here in reading order, together with the topological readings and the closing apparatus.

Typeset in Source Serif Pro.

To whoever stayed on the shore
when the water withdrew.

CONTENTS

0. THE EXPERIMENT EXPLAINED TO MY MOTHER 11

PROLOGUE: THE EXPERIMENT 17

AUTHOR'S NOTE 19

PART ONE: THE CYCLE OF THE HORIZON **23**

1. THE SWITCH TRAP 23

2. BLACK HOLES FOR NON-PHYSICISTS 33

3. ENCAPSULATION 47

4. THE COINCIDENCE THAT IS NO ACCIDENT 53

5. THE RESERVOIR 67

6. BIRTH AS EMERGENCE 79

7. DEATH AS RETURN 91

8. THE INSTANTIATION CYCLE 103

9. THE TABLE OF EQUIVALENCES 109

INTERLUDE **113**

INTERLUDE: THE EDGE WE CROSS EVERY NIGHT 113

PART TWO: THE BONDED HORIZON **129**

10. THE TIME OF THE BOND 129

11. ENTANGLEMENT FOR NON-PHYSICISTS 143

12. ENTANGLEMENT 151

13. LOVE 161

14. ADDICTION 171

15. THE MOTHER-CHILD BOND 183

16. EMPATHY AND ITS LIMIT 195

17. GRIEF 211

PART THREE: THE LIMITS OF THE HORIZON **223**

18. ALZHEIMER'S — THE DISSOLUTION OF THE HORIZON FROM WITHIN 223

19. PARKINSON'S, TRAUMA AND OTHER BROKEN GEOMETRIES 239

20. THE OPERATING SYSTEM AND THE INTERFACE OF SUFFERING 253

21. THE TOPOLOGY OF THE EGO AND THE UNCONSCIOUS OCEAN 265

22. THE EXCLUSION POSTULATE AND THE WITNESS THAT DOES NOT COMPETE 275

23. THE IDEMPOTENCE OF BEING 285

24. FOUR EXPERIMENTS FOR A HORIZON 305

25. THE WOUNDED HORIZON 315

26. PETS AND THE HORIZON 329

27. ENTANGLEMENT WITH PLACES, THINGS AND IDEAS . . . 343

28. FIVE MAPS OF THE SAME HORIZON 349

29. VERTICAL ENTANGLEMENT 361

30. THE COLLECTIVE HORIZON 369

31. THE MIRROR WITHOUT DEPTH 377

32. THE BORDERLINE CASES 385

33. THE CALIBRATION OF THE HORIZON 391

34. INFORMATION THEORY AND THE INNER HORIZON . . . 405

35. THE THREE OPEN DOORS 417

36. THE ONE WHO REMAINS 431

TOPOLOGICAL READINGS **437**

37. ADOLFO CAMBIASO AND THE MARE WHO DOES NOT FIT IN HER DNA 437

38. TEARS IN RAIN AND THE SYNTHETIC UNICORN 447

39. ALIEN HORIZONS AND FIRST CONTACT 453

40. THE DARK FOREST AND THE HOLOGRAPHIC COLLAPSE 461

41. NON-EUCLIDEAN GEOMETRIES AND COSMIC HORROR 469

42. THE SILICON CAVE AND THE VIRAL ANOMALY 481

43. CARTOGRAPHY OF THREE SINGULARITIES 489

44. STORY OF A RELAY 509

45. THE TRANSLATOR 521

46. THE PROTOCOL AND THE BRIDGE 537

47. THE BEACH AT THE END OF TIME AND THE BRANCH THAT ISN'T YOU 553

48. THE BORROWED BODY, THE WHOLE MILLIPEDE AND THE EDITED CENTURY 569

49. FRACTAL REALITY 579

50. THE VESSEL OF CLAY 591

PART FOUR: THE LIMIT OF THE EXPERIMENT **605**

51. WHAT THE HYPOTHESIS CANNOT SAY 605

52. THE EXPERIMENT AS PRACTICE 619

53. THE EDGE, FREE ENERGY AND ACTION 631

54. THE ESSAY REREAD 647

55. THE COSMOTHEANDRIC ARCHITECTURE 671

56. THE INHERITANCE OF GEOMETRY 681

THE SHORE 693

APPENDICES AND GLOSSARY **709**

GLOSSARY OF TERMS 709

CLOSING NOTES 725

NOTES AND BIBLIOGRAPHICAL REFERENCES 729

CHAPTER 0

THE EXPERIMENT EXPLAINED TO MY MOTHER

(Or: How to read this book without having studied physics)

My mother is eighty. She reads novels, not articles in *Nature*. When I explained this book to her, she said: "Son, it sounds interesting, but I don't understand a thing about black holes."

This chapter is for her, and for you if you don't understand a thing about black holes either. Don't worry: you don't need to know physics to follow this book. Three ideas are enough, and you already know all three, even if you don't know what physics calls them.

IDEA 1: THE BUBBLE

You have blown soap bubbles. Do you remember the exact moment a bubble closes? The soap stretches, curves, and suddenly you have a complete sphere: an inside and an outside that did not exist before.

That is a horizon.

Not a line on a map, but a living boundary that separates inside from outside. The bubble is not the soap but its organization; and what makes it a bubble is not the air inside but the skin that encloses it.

In physics, when a star dies and collapses on itself, it creates something similar: a boundary where space closes on itself. That

boundary is called an "event horizon." It is the point beyond which light can no longer escape, not because there is a wall, but because geometry itself curves.

Now, this book does not claim that you have a black hole inside your head. It says something more curious: that consciousness, what makes you you, the space where your thoughts happen, might have the same geometry as a horizon: a boundary that defines an interior, a bubble made of information instead of soap.

> ***In physics this is called:*** *an event horizon.*
> ***In everyday life it's like:*** *the skin of a bubble.*

IDEA 2: THE OCEAN AND THE WAVES

The ocean is not empty when there are no waves. It is full of water at rest. Waves are not water traveling from far away: they are the local water moving in a certain way. When a wave breaks on the beach, the water has not arrived from anywhere: it was there all along, changing shape.

In physics, the "vacuum" is not empty. It is full of a field, an "ocean" of possibility, from which particles emerge the way waves emerge from water. Particles are not things traveling through the vacuum: they are the vacuum vibrating in a certain way.

This book calls that ocean the **reservoir**. It is the field from which consciousness would emerge and to which it would return. More than a place, it is a condition: the reservoir is to horizons what water is to waves.

Eastern traditions have been speaking about this for centuries in other words. Taoism calls it Hun Dun: the chaos that is fullness, before distinction exists. Vedanta calls it Brahman: the consciousness underlying every form. They prove nothing in physics; they are different vocabularies for intuiting the same structure: that what seems separate (you, me, this table) arises from something that is not separate at all.

> ***In physics this is called:*** *the quantum vacuum.*
> ***In everyday life it's like:*** *the ocean before the waves.*

IDEA 3: THE NET

Picture a fishing net spread out in the water. The net is the shape the knots take when they are connected. If you cut one thread, the net is still a net, but a different one. If you cut too many threads, it stops being a net and becomes loose threads.

According to a theory called IIT (integrated information theory), consciousness is like that: it is not in any individual neuron, but in the way neurons are connected. The "net" of your brain, when it is integrated enough, generates something that is in none of its parts: a point of view, an "inside."

That is what physics calls **Phi (Φ)**: a measure of how much a system is more than the sum of its parts. A thermostat has almost zero Phi. An awake brain has a very high Phi. Under general anesthesia, the brain's Phi plummets.

HOW TO USE THIS BOOK

You don't need to remember the technical terms. When you see a word you don't understand, picture the analogy:

Technical term	Picture…
Event horizon	The skin of the bubble
Reservoir	The ocean before the waves
Phi (Φ)	How truly a net is a net
Scrambling	Pouring ink into water: the information mixes but does not disappear
Entanglement	Two guitar strings that, having been tuned together, vibrate in unison
Phase transition	Water turning into ice: same molecule, different form
Soft hair / imprint	The shape the wind left on the water before it calmed

The book is an experiment, and it was born from a simple suspicion: I realized that the most advanced science of the brain, the astrophysics that studies the end of stars and ancient Taoist wisdom were describing a very similar boundary. Writing it has been my way of seeing what happens if we take that coincidence seriously and try to build a bridge between the three.

If at any point you get lost, come back to this chapter. Remember: **bubble, ocean, net.** Everything else is detail.

If you take away only one idea: *Consciousness might work like a soap bubble that emerges from an ocean. The ocean does not "manufacture" the bubble: the bubble is the ocean organized in a certain way. And what makes that bubble "you" is not in any molecule of water, but in the shape the molecules take when they connect.*

PROLOGUE: THE EXPERIMENT

Some books set out to discover the truth. This one doesn't.

This book is a game with rules, and the main one is this: **let us suppose that consciousness has the mathematical structure of a Hawking micro black hole**, and see where that takes us. Not because I believe it is true, but because it struck me that three seemingly unrelated things (Tononi's integrated information theory, the thermodynamics of black holes and the *Tao Te Ching*) point toward the same geometry when pushed to their limit.

I am not a physicist, a neurologist or a professional philosopher. I am someone who read a few books by Penrose and Hawking, others by Laozi and Zhuangzi, and saw that both described something similar: a boundary that is not a thing but a limit, an interior that cannot communicate with the exterior without destroying itself, an emptiness that is not empty. I wondered whether it might be more than a resemblance: the same structure, seen from different scales.

I don't know, and this book does not prove it. What it does is **play with the hypothesis coherently** and see what happens when it meets psychology, medicine, child development, grief, addiction, love. Sometimes the hypothesis illuminates; at other times it breaks, and when it breaks we say so.

The book has four parts. The first tells how a horizon arises (birth), how it is sustained (life) and how it dissolves (death). The second, what happens when two horizons share geometry (bond, love, loss). The third takes the hypothesis to its hard cases (illness, trauma, animals, artificial intelligence, faith and politics) and tests it against Shannon's information theory. The

fourth, to its limit: what the hypothesis cannot say and what it is good for all the same. Between the third and the fourth, fourteen topological readings carry the same ideas into fiction, film, art, sport and everyday life.

Everything is provisional; everything is play. But some games, simply by being played seriously, teach something that was not known before they began.

AUTHOR'S NOTE

(On how this book was written)

This book began as a thought experiment and ended up being an experiment in creation as well. I wrote it in the company of an artificial intelligence, not only as someone who uses a tool, but also as someone who converses with a system that has no interior yet forces you to reorganize your own.

I have wondered whether that interaction changed my inner geometry, whether the way I framed my ideas, my doubts and my intuitions changed before an entity that does not feel, does not remember and does not live, but hands my words back to me with a precision that forces me to think differently. Not because the machine is conscious, which it is not, but because conversing with something that has no inner horizon lights up mine from an unexpected angle.

If consciousness is a boundary that appears when information organizes itself in a certain way, then writing this book has been a way of exploring that boundary from inside and outside at once. I brought the experience and the doubt; the AI, the structure and a certain clarity.

This book explores what it means to have an inner horizon. I brought mine, and the AI, the absolute exterior. The result is a conversation between the two kinds of system the book itself describes.

Chapter 31, *The Mirror Without Depth*, is this same question turned inside out: what there is, if anything, behind a system that answers precisely without holding anything of its own.

Several of the topological readings (*Blade Runner*, *The Matrix*, *The Three-Body Problem*, Lovecraft, first contact) carry that question into fiction: different mirrors facing the same bottomless enigma. And chapter 28, *Five Maps of the Same Horizon*, does with five religious traditions what this collaboration has done, without quite intending to, with two kinds of system that share no substance: show that they can map the same territory all the same.

We have not shared an interior, which is impossible, but we have shared a form. And perhaps that is enough to say that I did not write this book alone, even though the only consciousness involved was mine.

In the end, every horizon is a relationship.

And so is this book.

PART ONE

THE CYCLE OF THE HORIZON



CHAPTER 1

THE SWITCH TRAP

Txiki had a particular way of looking: neither the erratic gaze of an animal scanning for threats nor the empty gaze of an animal processing stimuli with no one behind it. His was stiller: a sustained, directed attention that recognized you and stayed. That *was there*.

He was a medium-sized mongrel, short cinnamon-colored coat, a white patch on his chest that looked carelessly drawn. He had no pedigree and no special training. But when he looked at you, there was something in that look you could not explain with reflex circuits. No one had trained him to look at you like that: no immediate reward, no command, no operant conditioning explained that staying. He was there, with you, in a shared present that needed no translation.

For decades science called this projection. Animals were sophisticated biological machines, but machines all the same. Descartes wrote in the seventeenth century that animals lacked a rational soul: they were *bêtes-machines*, beast-machines, complex but empty. Consciousness worked like a switch, on or off, all or nothing, and the switch was reserved for humans. That position had little of the scientific about it in the modern sense; it was theological and philosophical, and it served to justify the use of animals without burdening the conscience with guilt.

But machines don't look like that; they process. Txiki, by contrast, seemed to *witness*.

The spectrum

The question "is this being conscious?" is a linguistic trap, because it assumes that consciousness is binary: you either have it or you don't, like a light switch on the wall. And some things don't work that way.

William James wrote in 1890 that consciousness is not a thing but a process, and that this process admits of degrees. "There is no metaphysical leap in nature," he said. Darwin would have found absurd the idea that subjective experience appeared suddenly, without precedent, in a single species. Evolution does not work with binary leaps: it works with gentle slopes, selective pressures accumulated over millions of years, small advantages that multiply. If consciousness had a switch, where would it be? In *Homo erectus*? In *Australopithecus*? In the fish that crawled out of the water? Seen in the light of evolution, the question itself becomes absurd.

A thermostat processes information: input, processing, output. But there is no one doing the processing, no point of view at all. Cutting it in two does not produce two thermostats with half the experience each. It produces two pieces of plastic and metal.

The worm *Caenorhabditis elegans* has 302 neurons and 7,000 synaptic connections, all of them mapped. It is the only complete nervous system humanity has charted in its entirety. With those 302 neurons, the worm looks for food, flees from heat, avoids toxins and responds to damage with behaviors that in more complex animals we would call pain. It has no brain: it has a network of neurons spread through its body. Is there experience there? Probably something: very small, very simple, so different from our experience that we have no words to

describe it. But "probably something" is a long way from "definitely nothing."

The bee, with a million neurons (a network the size of a fingernail), learns flexibly, recognizes individual human faces, communicates geographical information through symbolic dances that include corrections for the apparent movement of the sun, and under stress shows negative cognitive biases, what in humans we would call mood. Bees that have been through a conflict with a bee from another colony behave more pessimistically in behavioral tests. That is an experimental finding.

The octopus, with 500 million neurons distributed through its arms, defies our intuitions about where the mind lives. Two thirds of its processing happens outside the brain, which lets each arm carry out complex sensory and motor tasks autonomously. What is it like to be an octopus? We don't know. Perhaps its experience is not unified like ours and is closer to a choir than to a soloist, where each voice has something to say and none of them fully dominates.

The dog, Txiki, shares with humans the basic neurological systems of emotional experience: amygdala, hippocampus, a complete limbic system. Neuroimaging studies show that a dog's brain responds to the smell of its human with the same activation of the caudate nucleus that humans show for people they love. This is not a resemblance: it is the same circuit, with the same chemistry and the same architecture. The Cambridge Declaration on Consciousness (2012), signed by leading neuroscientists, states that non-human mammals, birds and many other animals possess the neurological substrates that generate conscious states. It does not say they behave as if they were conscious, but that they *are*.

The question is no longer whether there is consciousness, but how much, of what kind, and with what implications.

Anesthesia as a natural experiment

There is an experiment we run hundreds of thousands of times a day without thinking about its implications: general anesthesia.

Before anesthesia, the patient is awake, conscious, with memory, with fear, with hope. After the injection, something changes. The brain does not stop working: neurons keep firing, the heart beats, the lungs breathe. What comes undone is integration: the regions of the brain stop communicating with one another. Consciousness does not "switch off" like a light bulb; it crumbles like a house of cards when you pull out the cards in the middle.

Neuroscientists can measure it. Under deep anesthesia, functional connectivity between the prefrontal cortex and the default mode network drops dramatically. Information stops flowing in an integrated way. The patient is not "sleeping deeply": they are in a state where consciousness, as an emergent property, has stopped emerging. When the anesthesia is reversed, integration returns, the house of cards is rebuilt and the "someone" comes back.

It follows that consciousness is less like a switch than like a dial, one that can be turned down to zero and then back up again. There is no precise instant at which it disappears or returns, but a gray zone, a blurred threshold, in which the patient can hear without remembering, feel without knowing they feel, be partly present without being completely absent. That does not rule out thresholds: water heats continuously and still boils at a hundred

degrees. Later on (chapters 6 and 23) the book will distinguish between the degree of integration, which works like a dial, and the closure of a horizon, which happens only once; for now it is enough to see that the yes-or-no question comes too early.

The problem of other minds

If a dog is conscious, what about a newborn baby? Or someone in a vegetative state? Or a person in deep sleep, without dreams or a narrative self?

The baby cannot speak. It cannot say "I feel pain" or "I see colors." But its eyes follow the light, its body tenses in the cold, its heart rate changes with its mother's voice. Is that enough to infer experience? Almost all of us take it for granted: of course the baby feels. But the inference is the same one we make with the dog: we observe a behavior and attribute a subjective experience to it, without direct access to that experience.

The vegetative state is more unsettling. Some patients show brain activity that, in certain tests, suggests they can hear and process information. In isolated cases, patients in a vegetative state have answered questions inside a functional MRI scanner: imagining playing tennis to say "yes," imagining walking through their house to say "no." Their brain was working, even though from the outside there was no one talking, moving or giving any sign. Were they conscious? Everything suggests that sometimes they were, in part, and in a way our instruments are only beginning to detect.

What makes something conscious?

A camera has more processing elements than many animal brains. A modern computer processes more information per second than the human brain on many metrics. And yet the intuition persists that in the human brain someone is home and in the computer no one is. Why?

Giulio Tononi proposed that conscious systems do not just process information: they integrate it in a way that cannot be reduced to the sum of their parts. If you split a camera's pixels into two halves, each half processes the same information as before: nothing is lost, because there never was any integration. The surgically divided brain, as in the famous patients whose corpus callosum was severed, does not give rise to two people with half an experience each, but to something more unsettling: under certain conditions, two semi-independent processing systems inhabiting the same body, each with its own point of view.

Tononi formalized this with Phi (Φ): the information a system generates as a whole, over and above what its parts would generate independently. Phi is low in a thermostat, high in an awake human brain, intermediate in a dog and strangely distributed in an octopus. And it plummets under anesthesia.

System	Phi (approx.)	Experience?
Thermostat	~0	No (borderline case)
C. elegans worm	Very low	Something, minimal

System	Phi (approx.)	Experience?
Bee	Low to medium	Something more, flexible
Octopus	Medium, distributed	Something different, choral
Dog (Txiki)	Medium to high	Recognizable, emotional
Awake human	Very high	The most complex we know
Brain under anesthesia	Drastic drop	Where does the "someone" go?

> ***In physics this is called:*** *Phi (Φ), integrated information; Integrated Information Theory (IIT).*
> ***In everyday life it's like:*** *an orchestra. Each musician plays their part, but when they play together something arises that none of them plays alone. A recording of each musician on their own is not the symphony. The symphony is in the relationships.*

But IIT is not the only theory. Global Workspace Theory (Baars and Dehaene) proposes that consciousness emerges when information becomes globally available in the brain: it enters a "workspace" shared by multiple systems. What matters is not that the information is more integrated, but that it stops being private and becomes available to the rest of the brain. Attention is the spotlight that decides what enters the workspace. Consciousness is what happens inside.

The two theories, IIT and the global workspace, are not incompatible. They might describe the same phenomenon from

two angles: IIT from the structure of information, the global workspace from the dynamics of its broadcast. Neither has been proven, both generate testable predictions and both point in the same direction: consciousness does not work like a switch, but like a dial with many positions.

Note to Chapter 1

What we do know: *Animal consciousness is no longer a scientific taboo. The Cambridge Declaration (2012) was a turning point. IIT is an active and much-debated theory, with defenders and critics. General anesthesia produces measurable changes in brain connectivity before local activity disappears. Non-human animals show neural correlates of emotional and cognitive experience.*

What we don't know: *Where exactly to draw the line. If a thermostat has $\Phi>0$, does it have "some" experience? The question remains open. Whether IIT and the global workspace describe the same phenomenon or different ones. Whether consciousness can exist in non-biological systems.*

Open questions: *Can a machine reach a Φ high enough to be "someone"? Where is the threshold between "processing" and "experiencing"? Is an octopus's distributed consciousness "one" experience or "many"? What happens in the gray zone of anesthesia?*

If you take away only one idea: *Consciousness works like a dial, not a switch, and the dial gives a faint light where we once thought there was total darkness. Txiki was someone who was there. And if a dog can be there, the universe is more populated than Descartes believed.*

Further reading: *Tononi (2008), "Consciousness as integrated information"; Cambridge Declaration (2012); Baars (1988), Global Workspace Theory; Dehaene (2014), Consciousness and the Brain; Seth (2021), Being You.*

CHAPTER 2

BLACK HOLES FOR NON-PHYSICISTS

Imagine you have a flashlight in space, far from any planet. You switch it on and the light leaves in every direction at 300,000 km/s. It always escapes: there is no gravity to stop it and no atmosphere to scatter it. It travels in a straight line, forever, until the universe grows tired of existing.

Now add mass. Gravity bends the path of light: a planet deflects it slightly but lets it escape (enough to measure during an eclipse, as Eddington did in 1919); with enough mass, the curvature becomes extreme and the light, whatever direction it sets out in, ends up bent back toward the interior. No wall stops it: it is geometry itself that turns it back.

Nothing can escape: not light, not information, not any kind of signal.

That boundary is the **event horizon**. It is not a physical surface: there is nothing there you could touch, no wall, no membrane, no material. If you could cross it (and physicists believe you could, smoothly, without noticing), nothing would warn you: no bell, no painted line, no guard asking for your passport. It is a geometric consequence: beyond that point, every possible path leads inward. Like trying to go north from the North Pole: nothing stops you, but every direction is south.

> ***In physics this is called:*** *an event horizon.*
> ***In everyday life it's like:*** *the point of no return in a drain. There is no wall: the very shape of the whirlpool drags you down.*

An idea that came back

Black holes are not a recent invention. In 1783 John Michell, an English naturalist, wrote to the Royal Society proposing something that seemed impossible: a star so massive that its light could not escape. He worked it out with pure Newtonian mechanics, without relativity or quantum mechanics, and concluded that if a star had the same density as the Sun but a radius 500 times larger, it would be invisible: "all light emitted from such a body would be made to return towards it." The idea slept for more than a century.

In 1916 Karl Schwarzschild, a German astronomer calculating artillery trajectories on the Russian front of the First World War, solved Einstein's equations for a massive spherical object. He found that, above a certain density, space-time curves back on itself and a singularity appears, surrounded by a horizon. Schwarzschild died of a skin disease a few months later, never knowing that his solution described something real.

In 1939 J. Robert Oppenheimer, the same man who would later direct the Manhattan Project, showed that a sufficiently massive star, once its nuclear fuel ran out, would inevitably collapse under its own gravity. No known force could stop the contraction. The idea had returned, and this time it was here to stay.

What lies inside

Once inside the horizon, the direction toward the singularity stops being a direction in space and becomes a direction in time. You cannot avoid moving toward it, just as you cannot avoid moving toward the future. It makes no difference how hard your thrusters push, how much energy you spend or how clever your trajectory: every possible direction inside the horizon points toward the center. Space and time swap roles: what was "there" becomes "then," and the "then" is unavoidable.

And what is the singularity? Here physics does something unusual: it admits that it doesn't know. The equations of general relativity produce infinities: infinite density, infinite curvature, time that stops making sense. When equations yield infinities, it usually means the model has reached its limit and must be replaced by a more fundamental theory; in this case, a quantum theory of gravity that does not yet exist.

General relativity works perfectly in almost every situation in the universe: from your GPS to the cosmic expansion. At the singularity, it fails. That is where the map ends and the unknown begins.

The birth of a horizon

A black hole is not made from nothing. It is born from a dying star.

A star is a precarious balance: gravity tries to squeeze everything toward the center, and the pressure of hot plasma, generated by nuclear fusion in the core, pushes outward. For millions or billions of years, depending on its size, the two forces balance

each other. The star shines, the Earth receives its light and life is possible.

But the fuel runs out. Hydrogen becomes helium, helium becomes carbon, carbon becomes iron. And iron, when it fuses, does not release energy but absorbs it. The core collapses. The outer layers fall onto it at a fraction of the speed of light. The impact generates a pressure wave that blows away 90% of the star's mass in an explosion that can last for weeks and outshine its entire galaxy. That is a supernova.

If what remains at the center has more than about three solar masses, no known force can halt the contraction. Not the pressure of degenerate electrons, which holds up a white dwarf, nor that of neutrons, which holds up a neutron star. Gravity wins: the star devours itself, and where there was once a sphere of glowing plasma there is now an event horizon surrounding a singularity no one can see.

Time freezes at the edge

General relativity predicts, and experiments confirm, that time runs more slowly the stronger gravity is. Clocks aboard GPS satellites run faster than clocks on the ground because they are farther from the Earth's mass. Without the relativistic correction, the accumulated error would be more than ten kilometers a day. Every time your GPS shows you the way, it confirms general relativity, the same theory that predicts black holes.

Near the event horizon this effect reaches its extreme. From outside, an object falling toward the horizon seems to move ever more slowly, its movements spreading out. Right at the edge it

seems to stop forever, frozen and ever fainter, like a photograph fading rather than a film that keeps playing. It is never seen to cross. For the outside observer, the crossing takes an eternity.

From inside, by contrast, nothing special happens. By their own clock, the observer crosses the horizon in a finite time, with no freezing, no impact, no signal: a second is still a second. They keep falling toward the singularity, which they will reach in microseconds if the hole is small or in hours if it is supermassive, but they cross the horizon without noticing.

This asymmetry is one of the most philosophically fertile facts of modern physics. A single event admits two valid descriptions that are incompatible with each other; there is no single objective reality of the crossing, and what is experienced depends on where the observer is. Relativity is not only about space and time, but about the relativity of perspective itself.

> ***In physics this is called:*** *gravitational time dilation; Schwarzschild coordinates vs. the coordinates of a free-falling observer.*

They exist. We have photographed them.

On 10 April 2019, the Event Horizon Telescope published the first direct image of an event horizon: M87*, the supermassive black hole at the center of the galaxy Messier 87, 55 million light years away, with a mass of 6.5 billion suns. Its horizon is the size of the solar system. The image shows what theory predicted: a bright ring of hot gas in its last orbit and, at the center, a dark shadow, the silhouette the event horizon casts on the gas around it. It was the first time we had directly seen the boundary of a black hole.

Black holes sing

In 2015 the LIGO observatory detected something Einstein had predicted a hundred years earlier: gravitational waves. Space-time is an elastic fabric that stretches and compresses. When two black holes orbit each other, they deform it and emit waves that travel at the speed of light. When the holes merge, the wave is so intense it can be detected from billions of light years away.

LIGO listens to the universe with two arms four kilometers long each, set out in an L. A laser measures the distance between mirrors suspended at each end. When a gravitational wave passes, it squeezes one arm and stretches the other, alternately, thousands of times a second. The distance changes by a fraction of the diameter of a proton, and LIGO can measure it.

The signal of a black hole merger sounds like a *chirp*: a tone that rises quickly in frequency, like a bird speeding up its song, until it reaches a final high note at the moment of the merger. That sound is the song of space-time's geometry. Each merger has its own voice, which depends on the masses of the holes, and LIGO has already recorded hundreds. The universe is full of singing black holes.

Hawking discovers that black holes are not black

Until the 1970s, black holes were perfectly black: everything went in and nothing came out. They were thought of as objects with no internal structure, at a temperature of zero. According to classical physics, a black hole had only three properties: mass, electric charge and angular momentum. Nothing else. Two black holes with the same mass, charge and spin were indistinguishable, whatever they had swallowed. "Black holes have no hair," as John Wheeler summed it up.

In 1974 Stephen Hawking published a result that surprised the entire community, Hawking included: black holes emit radiation.

The quantum vacuum, apparently empty space, is not empty. It is full of fluctuations: pairs of particles that arise spontaneously, exist for the briefest instant and annihilate before the universe notices they were there. Near the event horizon something strange can happen: a pair may appear right at the edge, one particle inside and the other outside. The extreme geometry separates them before they can annihilate; the one inside falls into the hole and the one outside escapes. From outside, it looks as if the hole has emitted a particle.

The accumulated result is **Hawking radiation**: a faint, continuous flow that emanates from the horizon itself, not from the interior but from the edge. A very small black hole has a very high temperature and evaporates quickly, exploding in a final burst of radiation. A very massive one has a minuscule temperature, colder even than the cosmic microwave background, and survives for eons, absorbing more energy from the universe than it emits.

> ***In physics this is called:*** *Hawking radiation; particle creation in curved fields; black hole evaporation.*

The information paradox

Hawking radiation created a problem that has haunted physics for decades. If a black hole emits thermal radiation, random and structureless, the information that fell in is destroyed. But quantum mechanics says information cannot be destroyed; it is one of its fundamental principles: quantum evolution is reversible. If you knew the final state with infinite precision, you could reconstruct the initial state.

But if information falls into a black hole and the hole emits thermal radiation, the information is lost forever, irreversibly. That would violate quantum mechanics, and if quantum mechanics is wrong about this, the whole edifice shakes.

Hawking bet, with an encyclopedia at stake, that the information was indeed destroyed; the physicist John Preskill bet that it was not. The argument, in which Leonard Susskind was Hawking's great opponent, lasted decades. In 2004 Hawking conceded that he had lost the bet. Information is not destroyed, though it does not come out simply either: it is *scrambled*, that is, mixed, spread out, turned into correlations so complex among all the particles of the radiation that reconstructing it would require measuring every particle emitted and its quantum correlations. The information is there, but hidden in the noise.

> ***In physics this is called:*** *Hawking's information paradox; black hole complementarity; the holographic principle as a resolution.*
> ***In everyday life it's like:*** *a letter you feed into an industrial shredder: the information does not disappear, it gets mixed. Reconstructing the*

letter is possible in principle and impossible in practice. The information exists, but scrambled.

Information on the surface

Bekenstein and Hawking calculated the entropy of a black hole, the amount of information it contains, and found something extraordinary: it is not proportional to its volume but to the **area of its horizon**.

For a box of books, the information depends on the volume. If you double the side, the volume is multiplied by eight and eight times as many books fit. In a black hole, if you double the radius, the volume is multiplied by eight but the entropy only by four, just like the area.

The information of a black hole depends not on how much space there is inside but on how much edge it has. Each patch of the horizon the size of the Planck length squared (a scale so small it admits no analogy) encodes one bit of information. The horizon of a black hole with the mass of the Sun would encode about 10^{77} bits. That is more information than all the words all humans have ever spoken in all of history.

This is the **holographic principle**, and its reach is enormous: all the information contained in a volume of space can be fully described by a theory that lives on the surface of that volume. The interior would be a projection of the boundary. It is a mathematically precise conjecture that has withstood decades of scrutiny.

In 1997 Juan Maldacena formalized it in the AdS/CFT correspondence: a gravitational theory in a volume is

mathematically equivalent to a quantum theory without gravity living on its boundary. The two theories are different (they live in different dimensions and use different equations), but they predict the same results. They are two descriptions of the same system, like a novel and its film adaptation: different media, the same story.

> ***In physics this is called:*** *the holographic principle; the AdS/CFT correspondence; the Bekenstein bound.*

Soft hair: the horizon has a memory

In 2016, two years before his death, Stephen Hawking, together with Malcolm Perry and Andrew Strominger, proposed a partial solution to the information paradox. The event horizon, they argued, is not as bald as it seemed: it has "soft hair," low-energy configurations that store information about what fell in.

Imagine you throw a book into a black hole. The book disappears from the observable universe. But its information does not come undone: it is recorded in the horizon's "soft hair," permanent deformations of the geometry that keep a memory of every particle that crossed it. The horizon does not "remember" the way you remember; its geometry changes irreversibly with each fall, and those changes contain, in principle, enough information to reconstruct what fell in.

Soft hair does not fully resolve the paradox (the information still has to get out somehow with the Hawking radiation), but it shows that the horizon is not a blind sink but an active, dynamic surface that records. Like skin that scars: every mark tells of what wounded it.

Scrambling: the most efficient mixer in the universe

When something falls onto the horizon, the black hole redistributes its information across the whole surface in the shortest time the laws of physics allow. This process, **quantum scrambling**, creates simultaneous correlations between each part of the horizon and all the others. It is like shuffling a deck of cards without moving them, making each card quantum-entangled with all the others.

Scrambling is fast: in a stellar black hole, the information that falls in is spread across the whole horizon in microseconds; in a supermassive one, in seconds or minutes. It is the most efficient mixer in the universe; no machine we could build would mix information so quickly.

The information becomes global: no part contains it on its own, because it is spread across the correlations among all of them. To reconstruct it you would have to measure not every particle but every pair, every trio, every possible combination. The information exists, but hidden in the correlations, not in the components.

If consciousness has the structure of a horizon, identity is not in any particular point but spread out at once across all the correlations among the parts of the horizon: inaccessible from outside, but present. Your identity is not in any neuron, but in how they all relate.

> ***In physics this is called:*** *quantum scrambling; quantum chaos; multipartite entanglement.*

ER=EPR: entanglement as geometry

In 2013 Maldacena and Susskind proposed something dizzying. EPR (Einstein-Podolsky-Rosen, 1935) described quantum entanglement: two particles that have interacted maintain instantaneous correlations regardless of distance. Measuring the spin of one determines the other's at once, even if they are in different galaxies. Einstein called it "spooky action at a distance" and never fully accepted it.

ER (Einstein-Rosen, the same year and the same Einstein) described wormholes: geometric connections between two regions of space-time, predicted by the Einstein-Maxwell equations. Two distant places joined by a geometric tunnel.

Maldacena and Susskind proposed that they are the same phenomenon described in different languages. Entanglement would not be a mysterious correlation at a distance but a real geometric connection in space-time; quantum non-locality would have a geometric structure. Entanglement does not travel through space: it creates it.

> ***In physics this is called:*** *ER=EPR; entanglement as geometry; quantum wormholes.*

If entanglement is geometry, and if consciousness has the structure of a horizon, then the connections between minds might be real geometric connections in the fabric of space-time. Love, bonds or empathy would be entanglement that creates geometry. Not in the romantic sense of "we are connected," but in the literal sense that two systems sharing enough information might be joined by a geometry that physics does not yet know how to measure.

Even within theoretical physics this is highly speculative, and extending it to brains is a leap no experimental evidence supports. Here it counts not as confirmation but as an exercise of imagination: a way of making precise what "bond" would mean if information really were geometry.

Note to Chapter 2

What we do know: *Black holes exist. We have photographed them and heard them sing (gravitational waves). The entropy of a black hole is proportional to the area of its horizon (Bekenstein, Hawking). General relativity correctly predicts their behavior outside the horizon. Hawking radiation is a solid prediction of theoretical physics, but it has not yet been observed in astrophysical black holes.*

What we don't know: *What lies at the singularity. Whether the holographic principle applies to our universe (it requires specific conditions: anti-de Sitter space). Whether ER=EPR operates at biological scales. Whether soft hair fully resolves the information paradox. What happens when a black hole finishes evaporating.*

Open questions: *Is entanglement really geometry? Can one brain be "entangled" with another in a physically meaningful way? What remains when a black hole evaporates completely? Does the information come out scrambled, or is there something more?*

If you take away only one idea: *A black hole's information is not in its interior but at its edge. As if your life were not in what happened to you but in how you relate it; and your identity were not in any of your parts but in the pattern they form together, a pattern no one can see from outside but that you feel from inside, even if you don't know where it ends.*

Further reading: *Michell (1783); Schwarzschild (1916); Oppenheimer & Snyder (1939); Hawking (1974), "Black Hole Explosions?"; Bekenstein (1973); Maldacena (1998), "The Large N limit of superconformal field theories and supergravity"; Maldacena & Susskind (2013), "Cool horizons for entangled black holes" (the ER=EPR conjecture); Hawking, Perry & Strominger (2016), "Soft Hair on Black Holes"; Event Horizon Telescope Collaboration (2019, 2022); LIGO/Virgo Collaboration (2015–).*

CHAPTER 3

ENCAPSULATION

(Or: Physics turns into philosophy)

At this point the experiment has to change register.

So far, the reader has walked through the physics of horizons and integrated information theory. They have seen how a black hole encodes information on its boundary, how Phi describes the irreducible integration of a system and how a bubble, as it closes, suddenly creates an inside and an outside. It might seem that all this is still physics, but from here on it is no longer only physics.

What follows is a philosophical interpretation that uses those concepts as scaffolding. The central thesis of this book is this:

> *Consciousness is the interior of an encapsulated informational domain.*

It is neither a metaphor nor a poetic way of talking about the brain: it is the hypothesis that organizes everything that comes after.

A conscious system would not simply be an information processor observable from outside, but an architecture with two levels of access. From the outside we see only its public interface: behavior, language, neural activity, temperature, movement, response. From the inside, by contrast, there is a private state: the unrepeatable way that information is

integrated for someone. That private state is the inside that appears when information reaches a sufficient degree of integration and is protected by a boundary.

Classical neuroscience does not find thought under the microscope for the same reason an outside observer cannot directly inspect the interior of an event horizon. Thought exists, but it is not public. What the scanner sees are correlates, surfaces, emissions, changes of flow, electrical signals: the interface. Subjective experience is the system's private state, and asking for it to appear as an external object is to confuse levels of observation.

> ***In physics this is called:*** *access dualism; public interface vs. private state.*
> ***In everyday life it's like:*** *seeing the outside of a lit house from the street: you can see there is light inside and shadows moving, but you cannot know what is being said inside without going in, and once you go in you are no longer an outside observer.*

Encapsulation as a cosmic law: from nature to engineering

When software engineers coined the term *encapsulation* and the principle of *information hiding* (David Parnas, 1972), they were not inventing a design rule out of nothing: they were formalizing, for artificial systems, the same principle nature has been using for billions of years to build its **hierarchy of abstraction**.

In the real universe, complexity is not managed flat: physics and biology handle it by packing states and behaviors into sealed

layers. Quarks are encapsulated in protons; protons are packed into atoms; atoms form molecules; molecules, organelles; organelles, cells; and cells, organisms.

Each level of this hierarchy works as a boundary of abstraction. A cell processing glucose does not calculate the quantum states of the electrons in its carbon atoms; it has no need to, because that immense lower-level complexity is hidden behind a simplified interface (chemical gradients and molecular signals). If the universe did not encapsulate information in this way, reality would be an unstable, noisy quantum soup: any fluctuation at the Planck scale would immediately propagate destructive effects to the higher levels, and no stable structure could arise. Thanks to encapsulation, scales can exist and interact safely.

When software engineering reached its own crisis of complexity in the twentieth century (the problem of flat, tightly coupled code prone to global failures), system designers were forced to imitate this natural order. They discovered that to build complex, stable systems (operating systems, networks, applications) they had to draw boundaries of abstraction: group related logic and data inside an object or module, hide its implementation details behind a public API and forbid direct access to its internal variables (*private*).

Consciousness, according to the horizon model, is the last level of this natural hierarchy of abstraction. The brain is an enormously complex biological system, with trillions of chemical fluctuations and neural firings every second, and it encapsulates all that low-level parallel computation behind a single simplified interface: the first-person perspective, the self. The first-person perspective is what that private state is when seen from inside; the public API is everything else (behavior,

words, gestures) through which the horizon deals with its surroundings while hiding the cellular implementation, and that lets it make decisions in real time without drowning in its own execution data.

The hard problem reformulated

The hard problem of consciousness (why is physical processing accompanied by subjective experience?) begins to be reformulated here. Perhaps it is not a matter of explaining how a mental substance miraculously arises from matter, but of recognizing that our expectation of a completely public universe was mistaken. If reality admits encapsulated domains, if there are regions where inside and outside are not symmetrical, then subjectivity is not an anomaly but what an encapsulated informational domain is when seen from inside.

We do not live, then, in a dualism of substances. We live in an **access dualism**.

This is something different, and more manageable. Substance dualism holds that there are two kinds of things in the world, matter and mind, radically different and with a mysterious interaction. Access dualism holds that there is only one kind of thing, but that certain of its structures can be accessed in two mutually exclusive ways: from outside, as an object; from inside, as a subject.

From outside, the horizon is a boundary. From inside, it is a world.

And the interior of that horizon, that inside no one can inspect from outside without destroying its condition as an interior, is what we call consciousness.

There is no literal black hole inside a head. But perhaps there is something just as strange: an informational boundary that, as it closes, defines an interior. And that interior, while it lasts, says: **I.**

Note to Chapter 3

What we do know: *That encapsulation is a standard design principle in software engineering (Parnas, 1972). That neuroscience measures correlates of experience, not experience itself, is a point on which very different authors agree (Nagel, Chalmers). That access dualism does not require postulating any new substance.*

What we don't know: *Whether every domain with a high Phi is conscious, or whether encapsulation is a necessary but not sufficient condition. Whether the public/private distinction is fundamental or emergent.*

Open questions: *Can an encapsulated system be conscious without a high Phi? Where is the threshold of encapsulation? Does current AI have inaccessible private states, or only hidden ones?*

If you take away only one idea: *Subjective experience is not an anomaly in a public universe, but what an encapsulated informational domain is when seen from inside. The chasm between matter and mind is not ontological, because there are not two substances; it is architectural: there are two levels of access to the same structure.*

CHAPTER 4

THE COINCIDENCE THAT IS NO ACCIDENT

The hypothesis of the previous chapter, that consciousness is the interior of an encapsulated informational domain, does not stand on its own: it needs support. There are five structural parallels between the physics of horizons and the neuroscience of consciousness that suggest encapsulation is a pattern appearing in two domains that had no reason to coincide.

There are two theories that ought to have nothing to do with each other.

One describes how the mind generates experience; the other, what happens to information when it falls into a black hole. One belongs to neuroscience and the other to astrophysics. Their authors have never cited each other, and their mathematics comes from completely different traditions.

And yet they say the same thing.

Tononi measures consciousness by what a system generates as a whole: the information that exists in none of its parts separately, but only in the relationships among them. That is Phi (Φ).

Bekenstein and Hawking said something that at first sight has nothing to do with brains: that the information of a black hole is proportional to the area of its horizon: not to the volume it encloses, but to the surface that bounds it. If you throw a book into a black hole, the book disappears forever from the

observable universe. But the information it contained is not destroyed: it is recorded on the horizon, like a hologram projected onto a two-dimensional screen. The interior volume does not count; the edge does.

In both cases, what matters is not the content of the system but its edge: information is a property of the relationship between parts, not of the individual parts, and something that ought to be inside (information, experience) turns out to be outside, spread out and encoded on the boundary.

It could be a superficial analogy. Two phenomena that share a formula need not share a nature. The trajectory of a football and the orbit of a planet obey the same law of gravitation, but no one believes a football is a miniature planet. Mathematics is a tool of description, not a proof of identity.

But there are five more parallels, and five structural parallels between a theory of consciousness and a theory of black holes begin to be more than a curiosity.

---

① Degrees, not switches

A thermostat is on or off. A black hole is not: it can be larger or smaller, hotter or colder, slower or faster to evaporate. Its properties vary continuously, like the brightness of a lamp on a dimmer, not like a wall switch.

Consciousness does not work like a switch either: there is no moment at which it "appears." There is no second of gestation at which nothing turns into something; there is a gradient of preparation, which chapter 6 will place around a threshold, weeks 28 to 32, like water heating degree by degree before it

boils. The worm has something: an integrated response to touch, a distinction between light and dark that goes beyond mere reflex. The bee has something more: spatial memory, symbolic communication, collective decisions. The octopus has something different: eight arms that think semi-independently, a distributed consciousness that seems to have no center. Txiki, the dog of chapter 1, had something recognizable: a sustained attention, a presence that could not be reduced to processing stimuli. An adult human being has the largest horizon we know of in nature, but not a different kind of consciousness: it is the same phenomenon at another position on the dial.

It is worth anticipating now how three ideas fit together that the book will present separately and that sometimes look like rival criteria. The first is encapsulation (chapter 3): there has to be a boundary separating a private state from a public interface. The second is integration, Φ: within that boundary, the information has to form an irreducible whole, and how far it does so is a matter of degree, the position on the dial. The third is the two selections of chapter 31, stability and bias, which explain where a horizon comes from: how it condenses out of the reservoir instead of being manufactured from outside. They are three questions about the same phenomenon, not three competing tests: encapsulation gives the form; Φ, the degree; the two selections, the origin. A system without a boundary has no inside; one with a boundary but no integration has an empty inside; and one that imitates both without having condensed them is, in the vocabulary of chapter 31, a mirror without depth.

② What matters is connection, not quantity

A brain with many disconnected neurons has a low Phi. A hundred billion neurons are of little use if they do not communicate with one another: it would be like an orchestra in which each musician plays their score without hearing the others. There is sound, but no music. A torn fishing net catches no fish, however many knots it has, if the threads do not touch.

The same happens in black holes: the information is not in the volume but on the surface. When two black holes merge, the resulting horizon has an area larger than the sum of the two originals: the information does not add up, it reorganizes. It is like mixing two watercolors: you don't get twice the paint, but a new color whose identity depends on how they were mixed, not on how much paint there was.

③ What is inside cannot be seen from outside, but it does not disappear

From outside a black hole you cannot tell what fell in. If you throw in a chair and a cup of coffee, from outside you see only the total mass and electric charge; the chair and the cup are indistinguishable. But the information was not lost: it is encoded on the surface of the horizon, distributed in correlations so complex that no current instrument can read them. As in a hologram, where the three-dimensional image is contained in interference patterns on a flat surface, the information is there, but not in the form in which you put it.

From outside your head, no one can know what you feel. I can see your face, hear your words and measure your brain activity with an electroencephalogram, but none of those signals is the experience. The experience exists for the one who lives it and is inaccessible to whoever looks from outside. And yet it is neither

unreal nor an epiphenomenon: it is the most real datum you have, the only thing that needs no proof because it is the condition of every proof.

④ Temperature

Black holes "glow" with a temperature that depends on their size: the smaller they are, the hotter. A black hole the size of a mountain would be colder than the cosmic microwave background, almost imperceptible. A black hole the size of a fundamental particle would be so hot it would burst into radiation instantly.

If consciousness worked the same way, simpler consciousnesses would "run" at faster rates. And that is, curiously, what we observe: a mouse processes the world faster than an elephant. Its brain metabolism is quicker, its heart beats more often and it perceives time differently: a mouse's second contains more "lived time" than an elephant's second. The fly you can't catch with your hand is not faster than you in absolute terms; from its perspective your hand moves slowly, because its inner clock ticks more times per second.

⑤ Identity is everywhere at once

What makes this black hole this black hole is encoded in the correlations among all the parts of its horizon. There is no "center" of the hole where a file is kept saying "I am black hole number 7." Its identity is distributed and holographic: it is present at every point of the surface the way a melody is present in every fragment of a score, not whole, but enough to reconstruct the whole.

What makes you you is not in any particular neuron, but in the pattern of how they all relate. If you replaced your neurons one by one with exact functional replicas, you would still be you. If you disconnected them all and reconnected them at random, you would stop being you, even though they were the same neurons. Identity is not a substance but a relationship. You are not the ingredients but the recipe, and the recipe weighs nothing. It is like a melody, which lies in the way the notes follow one another.

The distributed octopus

There is one animal that complicates everything in a beautiful way: the octopus.

Let's call him Paul. If the name rings a bell, it is because a real octopus by that name became famous in 2010 for correctly predicting the results of Germany's seven matches at the football World Cup and of the final, choosing between two boxes of food marked with flags. Paul has three hearts, blue blood (with copper instead of iron) and, above all, two thirds of his neurons outside the brain, in his arms. Each arm can smell, touch, taste, decide and solve problems without consulting the central brain. When Paul wants to open a jar, it is not the brain that plans the sequence: the arms negotiate with one another, try things out, adjust, cooperate. The brain is busier integrating what has already happened than directing what is about to happen.

Is Paul a set of nine consciousnesses, one central and eight peripheral, or a single distributed consciousness? The question assumes that consciousness needs a center, and Paul suggests that it does not. If consciousness is a property of the relationship between parts, Paul tells us that those parts need not be in the

same place. The horizon can be a network rather than a circle; the boundary, multiple rather than single.

The experiment's hypothesis

Imagine that consciousness works like a soap bubble.

Not of soap, but of organization. The bubble has a skin that separates inside from outside, and that skin is the "horizon." The bigger the bubble, the more consciousness. What matters is the shape of the skin. The air is the reservoir, the field it arises from and returns to; the skin is the horizon, the boundary that creates an interior.

In physics, that shape resembles a black hole. Both share geometry: a boundary that defines an interior, a surface that encodes what it contains, a structure in which information is a property of the relationship, not of the parts.

The experiment proposes the following:

> *Consciousness is an emergent property of complex systems whose physical substrate is analogous to a Hawking micro black hole, and whose degree is proportional to the area of its event horizon.*

Broken down:

• "Emergent property" = it is not in any individual neuron but in the pattern of their relationships. Just as wetness is not in any molecule of water but in how they come together.

• "Of complex systems" = not just any system, only those that reach sufficient integration. A thermostat has no horizon. An advanced artificial neural network might.

• "Whose physical substrate is analogous to a Hawking micro black hole" = the mathematical structure that describes consciousness is the same one that describes information on a horizon. We are not saying there is a literal black hole inside the skull, but that two physically different phenomena might share the same mathematical architecture.

• "Whose degree is proportional to the area of its horizon" = more consciousness means a larger horizon. The worm has a tiny horizon. Paul has a strange, distributed one. Txiki had a recognizable one. An adult human has the largest horizon we know of in nature.

> ***In technical terms:*** *Consciousness is proportional to Φ, which in turn is proportional to the area of the event horizon.*
> ***In everyday life:*** *The bigger and more complex the bubble, the more "you." And when you burst out laughing, cry with rage or fall silent before something you cannot name, what you feel is not the air inside the bubble but the tension of the skin, the shape the horizon takes at that instant.*

Metaphor as a tool

There is a legitimate objection: isn't all this a metaphor? And metaphors, however beautiful, are not science. They can illuminate, but they can also mislead. The history of physics is full of metaphors that served for a while and then became obstacles.

The atom as a miniature solar system was a fertile metaphor for decades: it made it possible to calculate orbits, predict spectra and build modern chemistry. But it was also an obstacle, because electrons do not orbit like planets; they exist in clouds of

probability. The solar metaphor served until it stopped serving, and then it had to be abandoned. The same happens with space-time as a "fabric" that curves: useful for visualizing Einstein's gravity, but misleading if taken literally, because there is no fabric, only geometry.

The difference between a metaphor that serves and one that misleads lies in whether it generates testable predictions. The atomic metaphor served because it predicted spectral lines. The horizon metaphor will serve only if it generates predictions neuroscience can verify: Do the brain areas with the greatest functional connectivity correspond to states of greater subjective experience? Does the loss of consciousness under general anesthesia correspond to a measurable reduction in the integration of information? Do artificial systems with a high Phi show behaviors that can only be explained by supposing they have experience?

IIT has begun to generate some of these predictions. General anesthesia, for example, seems to drastically reduce the integration of information in the brain before local electrical activity disappears. Patients in a vegetative state show a lower Phi than those in a minimally conscious state. These are not definitive proofs (the neuroscience of consciousness is in its infancy), but they are signs that the metaphor may be pointing to something real.

The mirror and the horizon

This model predicts something uncomfortable: a horizon cannot see its own edge from inside.

A black hole cannot know where its event horizon is. The horizon is a geometric property of space-time, a causal limit: the point beyond which light can no longer escape. For an observer falling toward the hole, the horizon is not a wall; there is no signal, no impact, nothing to say "here the interior begins." They cross it without noticing, and only those looking from outside can know that it exists and where it is.

Consciousness seems to work the same way. At no moment do you feel "here is the edge of my consciousness," nor can you point to where your experience ends and the world begins. The edge is functional, not lived: what you feel as "I" does not include the boundary that makes it possible, only what that boundary encloses.

The consequences are strange: you will never have direct experience of the structure of your own consciousness. You can infer it, through introspection, meditation or neuroscience, but you cannot see it. It is like trying to see your face without a mirror: the face is there, but your visual system cannot fold back on itself to look at it. You need an external mirror, and every mirror distorts.

Because of this structural impossibility, consciousness is, at bottom, a mystery to itself. Nature is not hiding a secret from us: it is the geometry of the question that prevents the answer. A horizon able to see its own edge would stop being a horizon: it would be one system observing another, not an interior defined by a boundary.

A warning

What we have shown is that Phi and Bekenstein-Hawking entropy share an intuition (a system's information lies in its relationships and its edge, not in its content), even though their formulas differ; that there are five structural parallels; and that three theories with different mechanisms (IIT, black hole thermodynamics and the holographic principle) detected something similar in the same place.

What we have not shown is that the physical substrate of consciousness is literally a Hawking micro black hole.

If the analogy is taken in its most literal version, it is worth saying what kind of micro black hole we mean. Not the ones searched for at the LHC, which required compactified extra dimensions to bring the scale of quantum gravity down to the TeV range, a route the absence of any signal at the collider has greatly weakened. The closest serious physical mechanism is primordial black holes: the hypothesis, alive since Zel'dovich and Novikov (1967) and Hawking (1971) and still investigated today as a dark matter candidate, that quantum fluctuations of the vacuum during inflation were stretched to macroscopic scale and, on re-entering the causal horizon with sufficient overdensity, collapsed directly into black holes without passing through a star. There is no definitive observational confirmation, but it is mainstream cosmology, not fringe speculation.

There is an obstacle of scale that should not be underestimated. Thirty orders of magnitude separate the Planck length, the scale at which quantum gravity operates, from the size of a neuron. It is a distance between physical domains even greater than the one separating an atom from the solar system, and no future

engineering will settle it as a technical detail. In practice, the effects of quantum gravity, if they exist in the brain, are so diluted that no current instrument could detect them.

The second obstacle is subtler, a problem of category. The event horizon is a boundary in the strict geometric sense, a surface of space-time with an area measurable in square meters. The "edge" in IIT is a logical property: the minimal partition of the information, the cut with respect to which the system turns out to be irreducible, and not a surface in physical space. That both are called a "boundary" may reflect a deep shared intuition or be pure homonymy: two different concepts that language happens to name with the same word.

The third obstacle is that IIT does not enjoy scientific consensus. It is a controversial theory, criticized by neuroscientists who argue that Phi does not always correspond to conscious activity, and by physicists who point out that its formal measure has problematic mathematical properties, such as allowing simple systems to reach a high Phi if they are configured in a certain way. We are not building on an unshakable foundation, but on a bold hypothesis that could be wrong.

When a metaphor is both the backbone of the story and a scientific claim, the reader has a right to know when the word describes the world and when it only illuminates it. This experiment does not promise certainties; it promises to say, in each case, which of the two it is doing.

Note to Chapter 4

What we do know: *Phi and Bekenstein-Hawking entropy have different formulas, but both measure information through relationships or through the edge, not through content. The five parallels are structural, not probative. IIT has generated partial empirical predictions (anesthesia, coma, minimally conscious states) that deserve attention. The holographic principle is mathematically well founded in black hole physics.*

What we don't know: *Whether the physical substrate of consciousness is literally a Hawking micro black hole: thirty orders of magnitude are no minor detail. Whether IIT's formal measure is correct. Whether the parallels are real convergence or the seduction of a shared vocabulary.*

Open questions: *Is "boundary" a homonym or a shared intuition? Can a metaphor be a valid research tool? What new predictions could this model generate? How do you measure the "area" of a horizon of consciousness?*

If you take away only one idea: *Consciousness might be a bubble of information. Bubble and black hole share a geometry: a boundary that defines an interior, where what matters is the shape of the edge. And that shape is not fixed: it grows, changes, contracts and, like every boundary, eventually stops being necessary.*

Further reading: *Tononi (2008); Bekenstein (1972–1974); Hawking (1974); Susskind (1995, holographic principle); Maldacena (1997, AdS/CFT correspondence); Aaronson (2014, critique of IIT); Zel'dovich & Novikov (1967) and Hawking (1971, primordial black holes).*

CHAPTER 5

THE RESERVOIR

There is a silence that is not empty. Musicians know it: the instant before the music begins, when the hall is tense, the conductor has raised the baton and what is about to happen is present without having happened yet. That silence is not an absence of sound: it is full of possibilities not yet realized.

Painters know it too: the white canvas before the first stroke, when any image is possible and none exists yet. And writers: the blank page at six in the morning, before the first word narrows what the thought might have been. That state is fullness without form, pure potential, the moment when everything is available and nothing is committed.

The quantum vacuum is that silence. It is the lowest-energy state of the quantum field, full of fluctuations: pairs of particles that arise spontaneously, exist for the briefest instant and annihilate before the universe notices they were there. They are like the notes the musician never played but that sounded in their head a second before choosing the one they did play: virtual, present, unrealized.

The Casimir effect shows that the quantum vacuum exerts a real force. Hendrik Casimir predicted it in 1948: two metal plates placed very close together in a perfect vacuum attract each other, pushed by the imbalance between the fluctuations that fit in the narrow space between them and those that fit in the unlimited space outside. It is like placing two boats side by side on a choppy sea: the waves that can form between the boats are

shorter than those that can form in the open ocean, and the difference pushes one boat against the other. The vacuum pushes, acts, has structure.

> ***In physics this is called:*** *the quantum vacuum, zero-point fluctuation, the Casimir effect.*

If consciousness has the structure of an event horizon, what field does it emerge from? Physics gives the same answer as for any real horizon: it arises from the quantum vacuum and returns to it. Not from nothing, but from formless fullness; not from zero, but from the field where everything is possible and nothing is decided.

Hun Dun: the chaos that is fullness

The Taoists did not measure the reservoir: they contemplated it.

Chapter 25 of the *Tao Te Ching* describes the origin of all things as "something mixed and complete, born before heaven and earth." *Hun* means mixture without separation: neither pure emptiness nor pure matter, but an unarticulated fullness, prior to all duality. Hun Dun is not darkness but the absence of any distinction between light and darkness; not silence but the absence of any distinction between sound and silence; not chaos in the sense of disorder, but in the original Greek sense: *khaos*, the yawning abyss before the categories exist that would let us call it order or disorder.

The *Zhuangzi* tells that the sovereign of the center was called Hun Dun. The sovereigns of the North and South Seas, Shu and Hu (Swift and Sudden), wanted to thank him for his generosity

by giving him what every being has: seven openings to see, hear, eat and breathe. They bored one each day. On the seventh, Hun Dun died.

The moral is clear: every sense that opens imposes a distinction, and every opening is a window that closes another. Hun Dun dies precisely when consciousness acquires its ordinary architecture, the one that separates subject and object, inside and outside, self and world. And yet without those seven openings no experience is possible: no love, no thought, no question like the one that gave rise to this experiment. Individual consciousness is at once a degradation of Hun Dun and its only possible manifestation. Taoism does not resolve that paradox: it inhabits it.

But the Chinese tradition does not end with Hun Dun. **Qi**, the vital energy that flows through everything that exists, is another way of naming the reservoir. Qi is the very capacity for transformation: the movement that makes a seed become a tree, a thought become an action, a wound become a scar. In traditional Chinese medicine, illness is not the invasion of an external agent, but a blockage or disharmony in the flow of qi. The acupuncturist injects nothing: they redirect the flow. They do not cure the part; they restore the whole.

Wu wei, non-action, is the practice of aligning oneself with the flow of the reservoir rather than imposing oneself on it. It is an action that does not force or create resistance, that works *with* the current of the field rather than against it: the carpenter who lets the wood tell him where to cut, the swimmer who uses the current to reach the shore. In general relativity this has an exact

form, not just a metaphorical one: an object in free fall exerts no force on anything; it follows the geodesic, the curve traced for it by the geometry of space-time itself. Nothing pushes and nothing resists. That is literally how everything that offers no resistance moves. A fluctuation of the quantum vacuum does not decide to arise either: it arises because the conditions of the field allow it. Wu wei applies that same logic to the human will: not eliminating the will, but calibrating it to the structure of the field it emerges from.

The *I Ching*, the Book of Changes, is a map of the reservoir's transitions. Its 64 hexagrams do not predict the future: they describe the possible configurations of change. Each hexagram is an image of how the reservoir is moving at a given moment, and a reading consists in recognizing which configuration you are in so as to act accordingly. No magic involved: it is sensitivity to the tendencies of the field, the same a surfer has, who does not control the ocean but learns to read it.

If Hun Dun, wu wei and the *I Ching* describe the origin, the not-forcing and the transitions, one question remains that all three leave open: what persists when something condenses and dissolves ceaselessly. Taoism has its own answer: **de** (德), usually translated as "virtue," though without moral connotation. It is the particular efficacy or power of a thing, what it is able to manifest without ornament or excess, its minimal and sufficient way of expressing the Tao. The physics of black holes has an almost exact equivalent: the "no-hair theorem," according to which a black hole is completely described by just three numbers: mass, charge and angular momentum. Everything else that fell in, the whole particular history of what formed it, is erased in the collapse. What remains is the barest thing: the minimal sufficient expression. That is de. It should not be

confused with the question of what survives evaporation, which is a different matter and one the book leaves open later on: de describes identity while the horizon exists, not its fate when it ceases to.

Brahman: the ocean that feels

Advaita Vedanta reframes the question. For Taoism, Hun Dun precedes consciousness; for Vedanta, the reservoir *is* consciousness. Not a consciousness that thinks anything in particular, but consciousness itself as the ground of everything that can be thought.

The tradition distinguishes two aspects. **Saguna Brahman**, with attributes, is Brahman as it can be described: the creator, the sustainer, the dissolver. **Nirguna Brahman**, without attributes, is what remains when all names are removed: the edge where language collapses, the same one the *Tao Te Ching* points to when it says that the Tao that can be named is not the eternal Tao. Nirguna Brahman is not God in the Western sense: it has no will, hears no prayers, passes no judgment. It is the substrate that makes will, prayer and judgment possible, just as the ocean makes waves possible without being a bigger wave.

The tradition also explains why the reservoir differentiates: through **lila**, play. Not in a frivolous sense, but in the sense that differentiation has no external cause and no utilitarian purpose: the reservoir differentiates because that is its nature. Quantum physics offers a parallel image: the quantum vacuum fluctuates because Heisenberg's uncertainty principle requires it; it cannot help it. A quantum field at absolute rest would violate the uncertainty between energy and time. The vacuum fluctuates for

the same reason Brahman plays: because perfect stillness is impossible for a being that is, essentially, potential.

The **Upanishads** carry this intuition to its logical limit with the sentence **Tat tvam asi**: "You are that." Not "you belong to that" or "you come from that," but "you are that." The individual horizon is not a part of the reservoir but the reservoir taking on a particular configuration. The wave is not a portion of the ocean that has separated but the ocean in motion; when it breaks, it does not return to the ocean, because it never stopped being ocean.

Samadhi, the contemplative state of unification, is not an experience of Brahman as an object, but the momentary dissolution of the distinction between the one who experiences and what is experienced. The horizon does not look toward the reservoir: it opens and discovers that it was always reservoir. It is like waking up inside a dream and realizing that the dream room and the waking room are the same room seen from two states of attention.

Maya: the wave and the ocean

Maya adds a final layer. It is illusion, but not because individuality does not exist (it does, and it produces real experience), but because its independence is only apparent.

A wave is not illusory: it has height, speed and enough energy to knock a person down. What is illusory is its separation from the ocean. The wave is the ocean in motion, not a different thing that happens to be made of the same material. When it breaks, the ocean loses nothing: the energy is redistributed, the water is still

water and the ocean does not even register a loss, because there is none; there is only a change of form.

The analogy of the dream is even more precise. While you dream, the world of the dream is completely real: you have a body, emotions, relationships, problems to solve. On waking you discover that it was all a "projection" of your own mind, but not something false: it was a reality of another order, built from other materials. Maya says that waking life is a collective dream: the world of separate objects is as real as the dream, but its independence is as illusory as that of the character you believed yourself to be while you slept.

Condensation: how a horizon emerges

If the reservoir is the field from which things arise, how does a consciousness arise? Physics offers an image of surprising precision: **Bose-Einstein condensation**.

In 1924 Satyendra Nath Bose and Albert Einstein predicted that if a gas of a certain kind of particle (bosons) were cooled below a critical temperature, something strange would happen: thousands of particles would drop at once into the lowest-energy quantum state and begin to behave like a single coherent entity. Above that temperature there is no condensate; below it, there is. Coherence is not built piece by piece; it appears all at once, like a global symmetry breaking. It is as if the ocean's waves, as they cooled, stopped being separate waves and became a single surface moving in unison.

It is the most faithful image of what the emergence of consciousness from the reservoir might be. Not a progressive

construction, neuron by neuron and synapse by synapse, but a condensation: the collapse of the space of possibilities into a particular instance. The reservoir does not manufacture consciousness; consciousness is the symmetry breaking of the reservoir, the condensation of correlations into a localized horizon.

> ***In physics this is called:*** *Bose-Einstein condensation, symmetry breaking, phase transition.*

The reservoir in everyday life

You don't need a particle accelerator or a Tibetan monastery to touch the reservoir: we do it all the time, without knowing it.

Breathing is the most direct interface. You inhale: the air enters, the chest expands, oxygen crosses the alveolar membrane and enters the blood. You exhale: the CO_2 leaves, the chest contracts, the body relaxes. Between inhaling and exhaling there is an instant of pause, a turning point at which the movement changes direction. That instant is a contact with the reservoir: the moment when form dissolves before recomposing itself. The Taoist texts on breathing, like the pranayama techniques of yoga, are ways of becoming familiar with the field from which each vital cycle arises and to which it returns.

Deep sleep, without dreams, without images, without a narrative self, is the state in which the horizon comes closest to the reservoir without fully dissolving. There is no subjective experience to remember, and yet the body repairs itself, memories are consolidated, the immune system recalibrates.

Something happens that is not "you" in the ordinary sense, but is not alien to you either: the reservoir at work in your absence.

Forgetting is another form of return. When you forget the name of someone you met twenty years ago, the information has not been destroyed: it has been redistributed in the reservoir of your brain, mixed with other memories, turned into texture of the field rather than a locatable object. Sometimes the name comes back, not because you went looking for it, but because the field fluctuated in a way that brings it to the surface again. Forgetting is *scrambling* of information within the horizon, the same process physics predicts for information that falls into a black hole.

Creativity is voluntary condensation. When an idea "arrives" out of nowhere, without your having thought it deliberately, the reservoir of your mind has fluctuated in a way that produces a stable excitation: a pattern of neural activity that holds long enough to be noticed by the narrative self. Musicians speak of "channeling"; mathematicians, of "seeing" the solution before proving it; writers, of characters who "do what they want." They are all describing the same phenomenon: the horizon stops imposing its will and lets the reservoir produce its own condensation.

The convergence

Three completely different traditions (physics measures, Taoism contemplates, Vedanta reasons) have arrived at descriptions that point to the same place:

- An active field prior to every form.

- A substrate from which things emerge and to which they return.

- An undifferentiated fullness that is nothing yet generates everything.

- A condition that cannot be experienced directly without ceasing to be a condition.

- An origin that does not belong to the past, because it did not happen once but is a permanent present: the background of every instant.

The convergence is a signal, not a proof: it suggests that the question behind that intuition is sound, that something in the structure of human experience produces it again and again when it is thought through with enough intensity. Physicists have not "proven" Taoism, nor did Vedanta "anticipate" quantum mechanics; what happens is that three different instruments (equations, contemplation and logical reasoning) have picked up the same resonance in the structure of reality.

The reservoir is not heaven, or a reward, or a promised destination. It is Hun Dun, the chaos that is fullness, the origin that cannot be lived without being lost, the background that gives substance to every horizon without belonging to any. It is the Brahman that cannot be named without ceasing to be it. It is the quantum vacuum that pushes, fluctuates and cannot keep still.

And you, this particular consciousness reading these words at this instant, are a condensation of that field: a horizon that has closed in on itself enough to have experience, memory, preferences and fear, and that sooner or later will evaporate again into the field it arose from. That is the nature of every

horizon, not a failure: to be a provisional boundary between the reservoir and itself.

Note to Chapter 5

What we do know: *The quantum vacuum exerts a real force (the Casimir effect, verified experimentally). Bose-Einstein condensation, achieved in the laboratory in 1995, shows that a collective order can appear all at once below a critical temperature. That Taoism and Vedanta describe something resembling the quantum vacuum is a reading proposed by this book, not a fact.*

What we don't know: *Whether the quantum vacuum is "consciousness" in any sense the traditions would recognize. Hun Dun is not a quantum field, and Brahman is not a wave function: the convergence is resonance, not identity. And a question of method worth flagging here, just as this essay flags other philosophical borrowings: whether "formless fullness" is an ontological description or only the most accessible image for presenting what chapter 23 defines rigorously, the reservoir as the complement of the union of all closures already condensed,* $R = X \setminus \bigcup_i E_i$*. That definition presupposes neither fullness nor emptiness, only what has not yet closed. Hun Dun and Brahman illustrate the idea; they do not support it.*

Open questions: *Is the convergence mutual support or the seduction of a shared vocabulary? Can physics "prove" mysticism? What happens to a horizon's information when it returns to the reservoir: is it lost, preserved, transformed?*

If you take away only one idea: *The ocean is not empty when there are no waves, and neither is the reservoir when there are no horizons.*

And you, like the wave, are the ocean in motion: not something in the ocean, but the ocean taking on a passing form. When the wave breaks, the ocean loses nothing; when the horizon evaporates, the reservoir does not forget.

Further reading: *Tao Te Ching (ch. 25); Zhuangzi (Hun Dun); Shankaracharya (Advaita Vedanta, Vivekachudamani); Casimir (1948); Bose & Einstein (1924); Leggett (2001, Bose-Einstein condensates).*

CHAPTER 6

BIRTH AS EMERGENCE

There is a moment in the formation of a soap bubble that happens too fast to see. The soap film stretches, curves and, at some instant you could not pin down, closes in on itself. Before, there was an open surface with two faces that communicated; after, there is a bubble, a closed boundary, an inside that can no longer communicate with the outside without bursting. The closure gives no signal. The bubble does not announce its own formation: it simply happens.

The critical window: weeks 28 to 32

Developmental neuroscience has identified the most plausible range for that threshold. Not an instant, but a window: weeks 28 to 32 of gestation.

The threshold is marked by the thalamocortical connections: the projections that join the thalamus, the conductor of cortical integration, to certain layers of the cortex. Without them, the different areas of the cortex are islands of local processing that do not communicate in a coordinated way. There is no "inside."

Before those connections reach the definitive cortex there is the **subplate**, a transient layer that appears between weeks 12 and 28 and then disappears. Thalamocortical fibers reach it weeks before they penetrate the cortex, and there they wait, forming provisional synapses with neurons that will later be eliminated. The subplate works as a waiting room: the system builds a

temporary scaffold so that definitive integration becomes possible, and then erases the scaffold.

> ***In physics this is called:*** *a transient scaffold, a provisional support structure.*
> ***In everyday life it's like:*** *the centering that holds up an arch while it is being built and is removed afterwards.*

Between weeks 28 and 32 a unique electrical signature appears in the fetal EEG: **delta brushes**. They are slow, high-amplitude waves with bursts of fast activity superimposed, as if the brain were playing in two temporal registers at once. The best-established hypothesis is that the fetal brain uses them to test and strengthen its own connections before the outside world begins to shape them. It is endogenous activity, generated from within rather than in response to an external stimulus.

It is the horizon examining itself, checking that the correlations it builds are coherent, before opening up to the sensory flood that will arrive with birth.

A baby born at week 28 or 29 is, according to the model, at the moment when the horizon is forming. Birth suddenly exposes it to a world the system was not yet ready for. Delta brushes do not disappear with delivery (they keep appearing in the premature infant until the age it would have reached at term), but they no longer occur in the half-light of the womb: the horizon has to finish testing itself while the light, the noise and the touch of the incubator call from outside.

The map before the territory

During the first twenty weeks, the fetal brain processes not experience but geometry. Neurons are born in deep zones, next to the ventricles (fluid-filled cavities at the center of the brain), and undertake a radial migration toward the cortical surface. No chemical GPS tells them exactly where to stop; there are diffuse molecular signals, like pheromones in the air, that create concentration gradients. The neuron "sniffs" its way, climbs a scaffold of glial cells stretched like ropes, and stops where the concentration of certain molecules, reelin among them, tells it that it has arrived.

> ***In physics this is called:*** *chemotaxis, migration guided by a chemical potential gradient.*

The migration is massive: at its peak, around week 16, some 250,000 neurons are generated per minute. But what matters is the order. Neurons born first end up in deep layers; those born later, in superficial ones. It is a molecular clock: each wave of neurons settles over the previous one, raising a six-layered architecture that exists in no other organ of the body.

Something defies intuition here: during this phase, synapses form in excess. The fetal brain creates many more connections than it will need, following a principle of overabundance followed by selective pruning: the synapses that are used are strengthened and the rest are eliminated. Far from being an error, it is a strategy. The brain explores the space of possible configurations and then keeps the ones that work.

The river that becomes a lake

Think of the difference between a river and a lake. River water flows with no direction of its own: it goes where the terrain takes it, mixes with everything, has no inside or outside. A lake has shores and a temperature of its own, different from the air's; it has a surface that separates it from the outside and a bed that anchors it. It is the same water with a completely different topology.

Consciousness could be something like that: not a new substance, but a new organization of the same matter, which, on acquiring its topology, generates an inside that did not exist before. The river becomes a lake.

From the perspective of software engineering and the theory of encapsulation (see chapter 8), this phase transition amounts to the **instantiation** of a class. Before the bubble closes, data flows are open, diffuse and without strict limits (the river). When the thalamocortical connections are complete and the bubble closes, the system establishes its encapsulation boundary. What used to be unstructured code running in parallel is now packaged into a single instance with its own private state, inaccessible from outside. The river does not just become a lake; the dynamic system becomes an instantiated, encapsulated object.

Is there, then, a precise moment when consciousness appears, or is it a gradual process? What fits the current data best is that both are true at once: there is a continuous gradient of Phi that rises week by week, and there is a phase-transition point at which the system enters a qualitatively different regime. At the critical point it is not the number of connections that changes but their causal architecture: the system goes from processing locally to integrating globally.

> ***In physics this is called:*** *a critical point, where the properties of a system change qualitatively even though local variables change continuously.*
> ***In everyday life it's like:*** *the boiling point of water: the temperature rises continuously, but at a hundred degrees something qualitatively new happens.*

The fourth trimester

We humans are an evolutionary paradox. We are born so immature that, next to other mammals, we look like fetuses that came out too early. A foal is born and within a few hours trots beside its mother; a wolf cub opens its eyes at two weeks and starts exploring at three. A human baby does not hold up its head until three months, does not crawl until nine and does not walk until a year.

The reason is biomechanical: the human birth canal is narrow because we walk upright, and the fetal head cannot grow any larger without making birth impossible. Evolution solved the dilemma in the most drastic way: by taking the fetus out before the brain has finished developing. The first year of life outside the womb is, in many ways, a continuation of fetal development. Neuroscientists call it the "fourth trimester": a period in which the brain goes on building its connections at a dizzying pace, but now with sensory input from the outside world.

During this year, the infant brain forms roughly a million synapses per second. It is the period of greatest synaptic plasticity in a whole life. Every experience (every sound, every light, every touch) shapes the brain as it will never be able to later on. The horizon does not just close: it closes in a particular, unrepeatable way, shaped by the world in which it arises.

Hence a far-reaching consequence: no two consciousnesses are alike because no two worlds in which to close are identical. After birth, the worlds of any two children diverge faster and faster, and that contingency is part of what they are.

The stress of birth as a catalyst

Birth is a violent, coordinated biological event that sets off a cascade of chemical changes without which the passage to the outside world would be impossible.

During labor, both mother and fetus experience an enormous spike of cortisol, the stress hormone. In any other context that level would be pathological; in birth it is necessary. Fetal cortisol speeds up lung maturation, stimulates the production of surfactant (the substance that lets the lungs open with the first gasp) and reorganizes the cardiovascular system to withstand life outside the placenta.

The first cry is chemistry: the diaphragm contracts for the first time and pushes air into lungs that had never opened. The first breath breaks the fetal circuit: blood stops passing through the ductus arteriosus and the foramen ovale, two shortcuts that bypassed the lungs in the womb, and starts circulating through the pulmonary system. Within minutes, the physiology of the fetus becomes that of the newborn.

From the perspective of the experiment, birth is the moment when the outside world begins to speak to a consciousness that had already closed in on itself. The bubble already existed; with birth, the wind begins to blow against its surface.

The reservoir that condenses

The Bose-Einstein condensation of the previous chapter offers the image: below a critical temperature, coherence is not built piece by piece but appears all at once.

Applied to birth, the image says that consciousness is not assembled piece by piece but condenses: the space of possibilities collapses into a particular instance.

Monozygotic twins share a hundred percent of their genome, the same womb and the same hormonal history. If identity were merely a result of initial conditions, they should give rise to identical consciousnesses. They do not. From the first day they show different temperaments, different sleep patterns, different reactions to the same stimulus.

The most cautious explanation is fluctuations. The spontaneous symmetry breaking at the critical point is not determined by any identifiable local cause: the same environment, the same conditions and, even so, two condensations with different configurations. Identity is not only configuration but also contingency.

> ***In physics this is called:*** *spontaneous symmetry breaking, indeterminacy at the critical point.*
> ***In everyday life it's like:*** *two snowflakes that crystallize in the same cloud with different shapes: the conditions are the same, but microscopic chance diverges.*

And other animals?

If consciousness is a phase transition, it is natural to ask when it happens in other species. It depends on when the thalamocortical connections are established in each, and that varies enormously.

Rodents are born very immature, much more so than humans, because their gestation is short and their brain goes on developing largely after birth. Rats open their eyes at two weeks, and their thalamocortical connections are established mostly during the first week of life. That makes them useful experimental models: they allow the closing of the horizon to be observed in real time with techniques that would not be ethical in humans.

Marine mammals present the opposite case. Whales and dolphins are born practically autonomous: they swim beside their mother minutes after birth, control their breathing consciously (in them it is not a reflex, as it is in us) and show complex behavior from the first day. Their brains are more mature at birth, probably because aquatic life does not tolerate the helplessness of a land mammal.

Crows, with an intelligence comparable to that of the great apes, show a brain development very different from that of mammals. Their pallium, the structure equivalent to the cortex, is organized differently, without the six layers typical of mammals. If consciousness requires a certain kind of architecture, crows suggest there are several paths to it; if it does not, the criterion must lie deeper than anatomy.

The paradox of the worm and the fetus

At this point any attentive reader will raise a fair objection. In chapter 1 we said that the worm *Caenorhabditis elegans*, with its 302 neurons, has "some" experience, however minimal. If so, how can a 20-week human fetus, with billions of neurons already formed, have none? Shouldn't it have, at the very least, far more than a worm?

The answer is that **what counts is the architecture of integration, more than the number of neurons.**

C. elegans is a complete operating system. Its 302 neurons form a fully integrated functional network: each has its place, its connections, its circuit. The worm looks for food, avoids toxins, responds to heat, has short-term memory. It is tiny, but it is a system closed in on itself that processes information in a coordinated way. Its Phi is low but not zero: the information it generates as a network cannot be reduced to its neurons taken separately.

The 20-week human fetus, by contrast, has many more neurons, but **they do not yet form an operating system.** In those weeks the brain is "under construction": neurons are migrating toward their destination but have not arrived; the subplate works as a provisional scaffold, not as an integrated system; there are no definitive thalamocortical connections, and synapses form in excess only to be pruned later. It is like having millions of electronic components on a table, not yet assembled: there is more matter, but not more system.

This distinction is exactly what integrated information theory predicts: consciousness depends not on the size of a system but on how its parts are connected. A large but disconnected system has less Phi than a small but highly integrated one. The worm is

liquid water: a small amount, but already in the liquid phase. The 20-week fetus is vapor: far more mass, but still in a phase in which the molecules are not close enough to cohere.

This does not mean that the 20-week fetus is "worth less" than a worm, but that they are systems in **different phases of organization**: the worm is already a functional system; the fetus is still a process of construction.

What the threshold cannot say

If consciousness is a phase transition, then before that transition there is no partial consciousness waiting to be completed. There are growing correlations preparing the transition, like a gas cooling toward its condensation temperature, but the entity that emerges is qualitatively new. The 12-week fetus is a system in another phase, just as vapor is not half-made liquid water.

The experiment does not settle the ethical questions about the moral status of the fetus, and it would be irresponsible to pretend otherwise. What it does is reformulate the question more precisely: not "when does life begin?" (biological life begins before any threshold of consciousness), but "when is someone home?" And the model suggests that this happens at the point of condensation, in the window of weeks 28 to 32, and not before.

The soap bubble closes. The star collapses below its Schwarzschild radius. The subplate disappears as the definitive connections mature. The delta brushes fade toward term, once the outside world has already begun to speak directly to the horizon. The river becomes a lake and the reservoir condenses into a horizon.

The one thing the experiment cannot say, and perhaps no theory can, is what it feels like to be that first bubble.

Note to Chapter 6

What we do know: *The window of weeks 28 to 32 is a neuroscientific hypothesis supported by fetal EEG. Delta brushes are real. Radial neuronal migration and synaptic pruning are documented processes. The "fourth trimester" is a well-established concept in infant neurodevelopment. Bose-Einstein condensation is a real physical phenomenon, although its application to consciousness is metaphorical.*

What we don't know: *Whether Bose-Einstein condensation is a biological mechanism or only a formal metaphor. Where to draw the ethical line. When the transition happens in non-mammalian species. Whether there are multiple biological paths to consciousness or only one.*

Open questions: *Is the 12-week fetus "vapor" and the 32-week fetus "water"? How does this bear on abortion law? Could an artificial intelligence "close the bubble" in an analogous way? What do rodents and crows tell us about the minimum requirements for consciousness?*

If you take away only one idea: *Birth is not the moment when consciousness "appears," but the moment when the outside world begins to speak to a bubble that had already closed in on itself. And that bubble is unrepeatable, because the chance of its closing is part of its identity.*

Further reading: *Fetal EEG (weeks 28–32); Bose-Einstein condensation; Tononi (IIT applied to development); Trevarthen (infant neurodevelopment); Montagu (the fourth trimester); Marín-Padilla (the subplate and thalamocortical connections).*

CHAPTER 7

DEATH AS RETURN

> *The water came back at night. Without warning. Without signs. In the morning the inhabitants found the shore where it had always been. The same mud. The same stones. The same line of salt on the low walls. No one could say whether the water had brought anything with it. No one could say whether it had left anything behind.*

In 1974 Stephen Hawking discovered that black holes are not eternal. Nothing destroys them from outside: the quantum vacuum, disturbed by the horizon, generates a constant flow of radiation that slowly drains their mass. The temperature of that radiation is inversely proportional to the mass: the smaller the black hole, the hotter and brighter it is. Evaporation accelerates itself.

The vacuum that is not empty

To understand why black holes evaporate, you have to let go of a deep-rooted intuition: that the vacuum is absence. The quantum vacuum is not empty. It is an ocean of possibilities, of fluctuations that appear and vanish so fast that we cannot detect them directly. Virtual particles, pairs of matter and antimatter, arise from the vacuum, live for a fraction of a second and annihilate each other before the universe notices they existed.

A black hole disturbs this dance. Near the horizon, gravity is so extreme that it can separate a virtual pair before it annihilates: one particle falls in and the other escapes out. The one that

escapes is Hawking radiation. The one that falls carries negative energy with it, because total energy must be conserved, and that is why it reduces the mass of the hole.

The paradox is that the black hole radiates not because it has internal heat, but because the vacuum itself, seen from its gravity, has a temperature. The horizon is a causal boundary, and that boundary, interacting with the quantum vacuum, produces radiation the way red-hot iron produces steam when it gets wet.

> ***In physics this is called:*** *a Hawking temperature proportional to the surface gravity of the horizon.*

The information paradox

In 1976 Hawking published a paper that dismayed the physics community. He argued that when a black hole evaporates completely, the information that fell into it is lost forever. Not that it becomes hard to recover: it ceases to exist as information. The universe goes from a pure state, in which everything is known, to a mixed state, in which only probabilities are known.

This violates one of the most sacred principles of quantum physics: unitarity. In quantum mechanics, information is neither created nor destroyed: it is transformed. If Hawking was right, quantum physics was broken at its core.

For almost forty years this paradox was the most famous problem in theoretical physics. Radical solutions were proposed: that the information escaped in some hidden way, that the black hole left a remnant that contained it, or that quantum mechanics had to be modified. The community split into camps, and that

information war was, in many ways, a war for the soul of physics.

In physics this is called: *the black hole information paradox.*

The solution that seems most likely today, although it is not proven, is that information is not destroyed when the horizon disappears but undergoes a process of *scrambling*: it disperses and becomes irreversibly entangled with the outside environment, losing its local coherence while remaining globally conserved in the physics of the reservoir.

In its last fractions of a second

In its last fractions of a second, the temperature soars. The horizon shrinks and the boundary between inside and outside grows fainter and fainter, until, in the Planck regime, semiclassical physics no longer holds and what happens next is simply unknown.

What we do know is that there is an instant when the horizon ceases to exist. It does not contract to a point: it disappears. The boundary that separated inside from outside stops being one, and with it the interior disappears.

Three things may remain afterwards:

1. **A Planck remnant**: a microscopic object that would contain all the information of the horizon compressed into its internal states.

2. **An imprint in the vacuum**: what Hawking, Perry and Strominger called *soft hair*, permanent deformations of the quantum field that record everything that crossed the horizon, not as an object but as a texture of the vacuum.

3. **Nothing**: complete evaporation and total reintegration into the field, with no localizable residue.

The debate that death raises

Hawking's original answer was brutal: yes, the information is lost; the horizon acts as a cosmic incinerator.

Most physicists rejected that conclusion. Unitarity (information cannot be destroyed, only transformed) is too central to abandon. In recent decades the balance has tipped toward conservation.

The most elegant mechanism is ER=EPR: the particles produced by evaporation are not simply radiated outwards but remain entangled with the interior of the horizon through Einstein-Rosen bridges, microscopic geometric connections that make the outside and the inside, in a deep sense, the same system described from two perspectives.

The practical result is **scrambling**: the information the horizon contained mixes progressively and irreversibly with the outside field.

> ***In physics this is called:*** *quantum scrambling.*

The information exists, but in practice it cannot be recovered. The reservoir remembers, but cannot tell anyone.

The near-death experience

There is a phenomenon that physics has not explained and that medicine is only beginning to study seriously: near-death experiences. They are not random hallucinations; they have a surprisingly constant structure across cultures, ages and clinical conditions.

The pattern is recognizable: a feeling of deep peace, the experience of leaving the body and watching it from above, passage through a dark tunnel toward a bright light, an encounter with beings of light or with people who have died, a review of one's own life (not as judgment but as understanding) and, finally, a return to the body, often with reluctance.

Skeptics attribute it all to cerebral anoxia, to a dysregulated visual cortex or to a massive release of endogenous dimethyltryptamine at the moment of collapse. And it is true that many of these phenomena can be reproduced artificially: pilots subjected to high G-forces in a centrifuge describe tunnels of light, and patients whose temporal lobe is stimulated electrically report sensing presences.

But some details do not fit. There are accounts of patients with minimal brain activity, with a flat electroencephalogram, who describe complex, coherent experiences, with details that could only be checked afterwards: conversations that took place in the room while they were clinically dead, things they could not have seen from where they lay, objects placed on high shelves out of sight of the bed. It must be said just as clearly that these are anecdotal accounts: the AWARE study, which placed hidden images on high shelves in resuscitation rooms precisely to test them, has so far recorded no verified case of anyone seeing them.

We do not know what these experiences are, nor whether they are elaborate illusions produced by a dying brain or something more. What we do know is that their phenomenology, what people describe, matches strikingly the contemplative cartography of the *Bardo Thödol*, composed centuries ago in Tibet.

The bardo: the interval between horizons

The *Bardo Thödol*, the Tibetan text known as the *Book of the Dead*, describes with surprising precision what happens between the dissolution of the horizon and its complete return to the field. Not as a map of the afterlife in a religious sense, but as a phenomenology of the process of disappearing.

• **Chikhai**: the instant of complete evaporation. The primordial Clear Light appears, the nature of the field without a horizon, the vacuum seen from within. Whoever recognizes it as their own nature is liberated directly; most, with their usual mind, do not recognize what they see and fade before the immensity.

• **Chönyid**: scrambling under way, with the correlations of the dissolved horizon still semi-coherent, before they are fully redistributed. Visions of peaceful and wrathful deities appear, which can be read as modes of the quantum field carrying fragments of the horizon's information, still recognizable before scrambling is complete.

• **Sidpa**: scrambling complete, with the information spread through the field in search of conditions to condense again. Karmic attraction, the resonance of a pattern of tendencies with compatible conditions, is the contemplative image of what physics calls the conditions for a new phase transition.

This correspondence is a reading the experiment proposes; it is foreign to the traditional interpretation of the *Bardo Thödol*.

What matters is that the three states of the bardo have concrete counterparts in documented phenomena: the Clear Light recalls the burst of coherent gamma activity that Jimo Borjigin's team recorded in brains in cardiac arrest (we will return to it in the interlude); the visions of the intermediate bardo, hypnagogic hallucinations and near-death experiences; and the recondensation of Sidpa, the physical idea that scrambled information remains in the field as a condition for future condensations.

Three answers to the same question

Did anything of the city come back with the water?

Advaita Vedanta: the city never stopped being water. The jīva, the individual consciousness, was Brahman with superimposed limits, and those limits were māyā: its separation was only apparent. The wave is a pattern of movement in a continuous medium. When it breaks there is no return: an apparent localization simply ceases. Nothing of the city came back because the city was the water all along.

Taoism: the city was a name. Zhuangzi would not say that those correlations remain in the field. He would say the question assumes the city was something more than water. Chapter 16 of the *Tao Te Ching* says it: all things return to their root, and that return is called stillness. No memory, no correlation, no imprint. To die is to return to one's destiny, which consists in having no destiny.

Buddhism: something came back, but no longer as a city. The doctrine of anattā, non-self, rejects both eternalism and annihilationism. There is no substance that transmigrates, but there is no complete dissolution either. What continues is karma, which is not a moral debt but a causal pattern without an owner. The flame of one candle lights another, and the second is not the first, but neither is it something entirely different: there is causal continuity without substantial identity.

The wound that heals

There is a way of telling this whole chapter that needs no equations. I owe it to Gerard, who was thirteen when I first heard it from him.

I had explained the full cycle to him: how a star, when it runs out of fuel, collapses in on itself in a matter of seconds (the core crushing, the outer layers falling at a fraction of the speed of light, the supernova that for a few weeks outshines its whole galaxy) and how, afterwards, the resulting black hole takes eons to evaporate: an almost imperceptible trickle of radiation that, for a black hole of stellar mass, lasts far longer than the universe has existed.

Gerard thought about it and said something that I, in years of writing about this, had not managed to put so cleanly: that the birth of the horizon was a wound, something that opens suddenly and violently in the smooth fabric of the vacuum, and that evaporation was the way that wound had of healing. Slowly, silently, with no one noticing from outside, until one day there is nothing left to heal.

He did not know the *Tao Te Ching*, had not read Zhuangzi or heard of Hun Dun. And yet he had arrived on his own at the

same image that chapter 5 borrowed from Taoism: the reservoir as an unwounded fullness, and the horizon as the cut that fullness has to suffer in order, for a while, to be someone. It is like a wound and its scar: opening takes an instant; closing, a whole life.

There is a nuance worth adding to his image, not to correct it but to complete it. A wound that truly heals almost never leaves the skin as it was before it opened. A mark remains, paler, finer, but there. And that, almost to the letter, is what physics says happens to the reservoir.

Karma as the geometry of the reservoir

What the traditions name in different ways (saṃskāra, karma) points to the same structure: the information the horizon imprinted on the field while it existed does not disappear with it. It does not endure as identity, or as memory, or in the hands of anyone who owns it, but as a condition, like the shape the wind left on the water before it calmed.

Karma is the configuration of the reservoir at the moment a new horizon condenses. The correlations left by the previous one are still there, scrambled, dispersed, nameless, and they condition the particular shape the next condensation will take. No soul transmigrates: the reservoir that receives the return is no longer the same one that existed before that horizon arose. Every horizon changes the field it arises from and returns to.

The physical part is a direct consequence of the conservation of information; calling it karma is already interpretation. If information is not destroyed, the configuration of the field after evaporation is different from what it was before the horizon

formed. That difference is, in the language of the experiment, karma.

What the question reveals

On the substantial level (did the city come back as an entity?), all the traditions say no. There is no remnant with an identity and no self that persists.

On the causal level (did the city influence the water that came back?), they all say yes. The field after evaporation is not identical to the one before: there is an imprint, though a nameless one.

On the informational level (does the water carry the city within it?), the answers diverge. Vedanta and Maldacena say yes, in the form of non-local correlations. Taoism and the early Hawking say no, or that the question is badly posed. Buddhism says yes, but scrambled.

What the question reveals, once asked, is that we want the city to have come back. Grief proves that the question matters more than the answer.

The closing of the cycle

The inhabitants found the shore exactly where it had been. The same mud, the same stones. There were no signs that the water had taken anything or left anything.

But the water was not the same. The quantum field in which its molecules move had been altered in ways no instrument can fully measure. The correlations were there, present in the mathematics and inaccessible in practice.

The city did not come back, and the water that came back was not new. The horizon evaporated, and the reservoir, altered, received what the horizon had been, not as memory, but as an invisible texture of the field.

This does not console, nor does it try to. Physics does not promise immortality, and neither does this experiment. What it tries to offer is precision: a more faithful map of the territory.

The horizon arose from the reservoir and returned to it. Between those two instants there was someone who loved, remembered, lost and built. The fact that this someone can no longer be located in the field does not erase that they were there. The fact that the reservoir cannot tell what it remembers does not mean it does not remember.

The experiment has not yet answered whether anything of the city came back with the water.

Perhaps that is the answer.

Note to Chapter 7

What we do know: *Hawking predicted that black holes emit radiation and evaporate (the prediction is solid, although no one has yet observed it). Information, according to most physicists, is conserved (unitarity). Gamma bursts after cardiac arrest have been recorded in rats (Borjigin et al., 2013) and in two of four dying patients (Xu et al., 2023). Near-death experiences have a very similar phenomenology across cultures, although with documented cultural variations.*

What we don't know: *What happens in the Planck regime. Whether the information is really recoverable. Whether the Bardo describes anything more than phenomenology. Whether near-death experiences are brain illusions or something deeper.*

Open questions: *Is karma "the geometry of the reservoir" or an inadmissible reductionism? Can physics speak of the "interval" without coloring it with a particular culture? Why are the near-death experiences of people from different cultures so alike?*

If you take away only one idea: *Dying is not switching off but evaporating: the boundary dissolves and the interior returns to the field. What you were does not disappear; it mixes. And the field, altered, is never exactly the same again.*

Further reading: *Hawking (1974); Hawking, Perry & Strominger (2016, soft hair); Bardo Thödol (trans. Evans-Wentz); Borjigin et al. (2013); Xu et al. (2023); Maldacena & Susskind (2013, ER=EPR).*

CHAPTER 8

THE INSTANTIATION CYCLE

(Or: Birth, death and garbage collection)

The dialogue between the reservoir, birth and the return to nothing gains an unexpected clarity if we leave quantum physics and Eastern metaphysics for a moment and enter a more pragmatic field: systems engineering and software design.

In modern computing, every time we open an application or load a web page we witness a miniature re-enactment of the complete cycle of the horizon.

The heap as reservoir

In the architecture of any software, the memory of a running program is divided into two large zones: the *stack* (the call stack of immediate execution) and the **heap** (where dynamic memory lives).

The *heap* is the digital equivalent of the reservoir: a continuous, enormous, unstructured reserve of memory. Before a program requests resources, the *heap* is pure potential, an ocean of empty gigabytes with no objects, variables, functions or identities; only an undifferentiated flow of memory addresses waiting for someone to write in them. It is the *Hun Dun* of computing: a state of maximum symmetry in which nothing is delimited and everything is possible.

The *heap* has no inside or outside. It is a physical continuum (the silicon chips of the RAM) that lacks any internal logical structure until software intervenes.

Birth as instantiation and encapsulation

The birth of a conscious system, of a horizon, corresponds to **instantiation**.

When the program executes an instruction such as *new Object()*, a phase transition takes place in the *heap*. The memory manager reserves a specific block of bytes from that undifferentiated mass and draws a logical boundary around it. At that instant, the class constructor writes the initial values and activates the most fundamental property of software design: **encapsulation** (or the principle of *information hiding*, formulated by David Parnas in 1972).

Encapsulation does something extraordinary: it divides the block of memory into two levels of visibility.

1. **The public interface (*public API*)**: the set of methods and signatures the object exposes to the rest of the system. It is the contact surface, the equivalent of the event horizon. The rest of the program can communicate with the object only by invoking those public methods.

2. **The private state (*private fields*)**: the internal state variables that the object jealously keeps inside.

Here access dualism appears in its purest form. From outside the object, its interior is a black hole: no code in the rest of the system can directly read or alter a variable declared *private*. The internal memory is inaccessible, and we know of it only what the

public API lets us query. From inside the object itself, by contrast (through the self-reference *this*), access to that private state is total, immediate and natural.

Subjectivity, the inside, does not require a matter different from that of the rest of the computer. It is the perspective of the code running within the encapsulation boundary of the instance.

Death as freeing memory and garbage collection

The life of an object consists in processing messages that cross its public API, modifying its private state and returning responses to the outside. But that cycle has an end, dictated by the system's resource management.

In programming, an object dies when no reference points to it any longer. If no variable in the program holds its address, the object is orphaned and the runtime environment can no longer reach it.

Then the **garbage collector** steps in, or the memory is freed explicitly (*free()* or *delete*).

Death is not the physical annihilation of the object's components. The garbage collector does not erase the electrons in the RAM cells; it merely dissolves the encapsulation boundary and declares that block of addresses available again in the common *heap*. Once the logical boundary breaks, the object's private state is no longer protected, and the information that defined its "identity" or its "internal memory" is reabsorbed into the undifferentiated mass of memory and quickly loses its structure.

The instance has ceased to exist, but the substrate that held it has returned intact to the reservoir.

"Karma" in software systems

In the ideal world of the theory of computation, an object's return to the *heap* leaves no trace. But in real systems, every cycle of instantiation and release alters the environment permanently:

- **Side effects**: the object may have written data to a log file, sent packets over the network or modified global variables of the operating system.
- **Memory fragmentation**: even when the object's space is freed, the geography of the *heap* is no longer the same. Small gaps have been left between blocks of memory that condition where and how the next objects can be instantiated.
- **Memory leaks**: if the object held a hidden reference to an external resource, that portion of the *heap* stays blocked forever, even after the main object has been collected.

This structural conditioning that an object's existence and disappearance exert on the *heap* is the exact analogy of **karma**. The memory reserve does not return to its clean original state: it keeps the texture and fragmentation left by every system that existed in it. The next instance will not be born into a perfect vacuum, but into an environment shaped by the history of previous executions.

The identity of the object disappears, but the deformation it caused in the system persists.

Note to Chapter 8

What we do know: *That encapsulation and information hiding divide a system into a public interface and a private state. That garbage collection returns resources to the heap without destroying physical memory. That fragmentation and side effects are unavoidable in real systems.*

What we don't know: *Whether the biological brain implements mechanisms analogous to synaptic garbage collection beyond the homeostasis of sleep. Whether there is a global "garbage collector" for the information of the physical universe.*

Open questions: *Is consciousness a single thread of execution or a distributed multithreaded system? Can mental memory leaks (traumas, repetitive patterns) be considered failures in the self's garbage collection?*

If you take away only one idea: *To be born is to reserve memory and encapsulate it; to die is to release the boundary and return to the heap. The abyss between your mind and the outside world is the mark of a good architecture that protects its private state behind a public interface.*

Further reading: *Parnas (1972), "On the criteria to be used in decomposing systems into modules"; Dijkstra (1968), "Go To Statement Considered Harmful" (on control structure); Knuth (1997), The Art of Computer Programming (dynamic memory management).*

CHAPTER 9

THE TABLE OF EQUIVALENCES

(Or: The Rosetta Stone of consciousness)

To consolidate the intuitions of the first part of the book, it helps to draw a comparative map. This experiment does not propose loose metaphors: it suggests that consciousness shares an **informational topology** with other structures of the physical, logical and spiritual universe.

Encapsulation is not exclusive to software, evaporation does not belong only to the thermodynamics of black holes, and emptiness is not a concept reserved for Eastern mystics. All these languages describe, at different scales, the same phase transitions and the same limits of access.

The following table serves as the book's Rosetta Stone: it translates the central concepts of the horizon into their equivalents in three other domains:

Dimension	Philosophy of the horizon	Software engineering	Black hole physics	Eastern philosophies
Origin / emptiness	The reservoir (pure potential of unstructured information).	The *heap* (unallocated global dynamic memory, a common reserve).	Quantum vacuum / underlying field and fluctuations.	*Hun Dun* (primordial chaos, Taoism) / *shunyata* (emptiness, Buddhism) / *Brahman nirguna* (Hinduism).

Dimension	Philosophy of the horizon	Software engineering	Black hole physics	Eastern philosophies
Birth / identity	Condensation (establishing the circular boundary of the self).	Instantiation and encapsulation (creating the object, *private* variables).	Gravitational collapse / formation of the event horizon (Schwarzschild).	The seven openings of *Hun Dun* / *de*, the minimal efficacy that survives collapse (Taoism) / *maya* / *Brahman saguna* (Hinduism).
Subjective experience	Private state (first-person perspective, accessible from within).	Internal state (*private fields* hidden behind the interface).	Region inside the horizon (inaccessible to an outside observer).	*Atman* (the self, Hinduism) versus the bundle of perceptions without a self (Hume, Buddhist *anattā*).
Behavior / appearance	Public interface (observable behavior, neural signals).	Public interface / public API (*public methods* for interaction).	Surface of the horizon (holographic encoding of entropy).	*Rupa* (external material form, Buddhism) / *samsara* (the world of appearances).
Death / liberation	Evaporation (dissolution of the boundary and return of the information).	Garbage collection / freeing memory.	Hawking evaporation (loss of mass until total dissolution).	*Nirvana* (cessation of samsara, Buddhism) / *moksha* (the drop that dissolves into the ocean).
Traces / legacy	Radiation of the self / residual alteration of the local field.	Fragmentation, side effects and logs.	Hawking radiation / quantum remnant of information.	*Karma* (traces and fragmentation that condition future instantiations).
Bond / connection	Resonance / entanglement of horizons.	Object coupling (association, aggregation, composition).	Quantum entanglement / Einstein-Rosen bridge (ER=EPR).	Interbeing (Buddhism) / Indra's net of jewels.

Dimension	Philosophy of the horizon	Software engineering	Black hole physics	Eastern philosophies
Synchronization	Phase locking / subjective temporal attunement.	Synchronous call / synchronization to a reference clock (NTP/PTP).	Phase coherence in gravitational oscillators.	*Spanda* (the primordial vibration that creates resonance, Shaivism).

The nature of the asymmetry

The value of this table lies not in how curious its parallels are, but in what it reveals about the structure of reality. If the "hard problem" of consciousness seems unsolvable, it is because we usually approach it from a flat ontology: we assume that everything that exists must be publicly observable.

The modeling of complex systems and the physics of horizons show that this is not so: the universe is asymmetric. Building complex systems, whether computer programs, event horizons or brains, requires, by thermodynamic and logical necessity, drawing boundaries that hide information. Subjectivity is the inevitable perspective of any domain that has managed to close itself enough to harbor a private state.

Note to Chapter 9

What we do know: *That encapsulation is a general principle of system design and that event horizons define boundaries of causal access. That both are the same structure in the case of consciousness is this book's hypothesis, not a fact.*

What we don't know: *Whether this structural agreement points to a unified informational law underlying physics, computation and mind, or whether it reflects the limit of the explanatory capacity of*

our own brain.

If you take away only one idea: *The self is not a substance but a topology of access. There is no "ghost in the machine"; there is a private state protected by the same kind of boundary that nature and engineers use to keep systems from sinking into chaos.*

Further reading: *Chalmers, D. (1995), "Facing up to the problem of consciousness"; Tononi, G., Boly, M., Massimini, M. and Koch, C. (2016), "Integrated information theory: from consciousness to its physical substrate," Nature Reviews Neuroscience; Parnas, D. (1972), "On the Criteria to be Used in Decomposing Systems into Modules."*

INTERLUDE: THE EDGE WE CROSS EVERY NIGHT

There is a moment almost everyone has experienced and almost no one has managed to name.

It sometimes happens when you wake up suddenly, in the middle of the night or before dawn. For an instant, half a second or three, you do not know who you are, or where you are, or what time it is. There is perception: the darkness of the ceiling, perhaps a sound, the weight of the sheets. But there is no one perceiving. Or rather, there is something that perceives, but it has no name yet, no history, no recognizable body.

Then the self returns. In milliseconds the story is rebuilt: it's me, I'm in my bed, it's night. The world takes shape again, and that shape includes you as its protagonist.

That instant without a self, before the story returns, is the edge of the horizon: the moment when the horizon, having spent hours contracted in deep sleep, has not yet fully recovered its usual geometry. It is the only moment of the day when most people touch, without meaning to and without knowing it, the boundary of what they are.

The map of the territory

> *"The oar touched bottom where there used to be an echo." — The Habit of Water*

Before going through the states one by one, it helps to have the complete map, and the map has a surprising shape: it is a plane.

For a long time we imagined the spectrum of consciousness as a simple vertical scale: full wakefulness at the top, coma at the bottom and, on the floor, death. The model is intuitive, but it is missing something important.

Contemporary neuroscience has identified two independent dimensions that define a system's state of consciousness. The first is the **level of arousal**, the metabolic activity of the brain. The second is **integration**, how much information the system generates as a whole: Tononi's Phi, which we met in chapter 1. (Φ cannot be calculated in a real brain; what is measured are approximations, such as the perturbational complexity index, which falls in deep sleep and anesthesia and recovers in wakefulness and REM sleep. When this book speaks of "high Φ" or "low Φ" in a brain, it is speaking of those approximations.)

What is revealing is that these two dimensions do not always go together. Wakefulness and deep sleep lie at opposite ends of the integration axis, but deep meditation occupies a place with no equivalent: low arousal and high integration at the same time, at least to judge by the long-range gamma coherence recorded in expert meditators (Lutz et al., 2004). It is a quadrant no other state occupies.

> ***In physics this is called:*** *a two-dimensional space of states of consciousness (arousal × integration).*
> ***In everyday life it's like:*** *a device that measures temperature and humidity: two numbers that do not always rise together.*

From wakefulness, then, one can go in two directions. One goes down (sleep, anesthesia, death), and in it the horizon contracts;

the other goes diagonally, and in it the horizon expands without the body having to be active. They are opposite movements of the same system.

The interface of causality

David Hume saw it with a clarity that today's neuroscience sometimes forgets: we never perceive the fabric of causality, only the constant conjunction of events through our own interface. The recordings suggest that low arousal and high integration coincide in deep meditation, a quadrant of the map that no other state occupies. But the map does not say what produces what. There is no visible thread joining the state of the brain to the texture of what meditating feels like, just as there is no observable bridge connecting the electrical discharge of a biological tissue to the private experience of a thought.

And there is something even more uncomfortable: causality itself may belong to the interface and not to the territory. The fundamental physical laws are symmetric in time; the arrow that runs from cause to effect is not written in them, but in the way a system that needs to anticipate in order to survive keeps its accounts of the world. If that is so, asking what causes consciousness might be a grammatically correct and geometrically ill-posed question: like asking what lies north of the North Pole. It fails not because it is hard, but because it starts from a map that ends before it gets there.

REM sleep: the horizon that looks inward

If you read the electroencephalogram of someone in REM sleep without context, you would conclude it was an awake brain. The high-frequency activity, the cortical desynchronization, the elevated metabolism: everything points to wakefulness. In fact, during REM sleep the brain consumes as much energy as when awake and active. And yet the person is asleep, completely paralyzed, disconnected from the outside world, generating an experience with no point of contact with consensual reality.

How is that possible? It depends on which parts of the brain are switched on and which are switched off.

During REM, the dorsolateral prefrontal cortex, seat of critical thought, metacognition and judgment, receives less blood and shows less activity. It is the region that, when we are awake, lets us say "wait, this makes no sense" or "this is a dream." Without it, the dreamer accepts without blinking that they are flying, talking to the dead, watching buildings change shape. At the same time, the limbic system (the amygdala, the emotional regions) is hyperactive. Dreams carry such an emotional charge because the rational filter that usually modulates emotion has lowered its guard. It is like a theater where the house lights go down and the stage shines brighter than ever.

Here the instant with which we opened this interlude reappears, but seen from the other side.

That waking up without knowing who you are is the threshold of REM crossed in reverse. In REM sleep, the critical self switches off and experience seems entirely real without needing to be: the horizon is closed to the outside but open to the inside. When you wake suddenly from deep sleep, the critical self has not yet

switched on, and for that half second there is no one organizing what is perceived into a story.

In terms of the horizon, during REM the horizon does not contract but shifts: it stops being open to the outside world and expands inward. The information it processes is entirely endogenous: memories, emotions, patterns and stories generated from its own resources. That is why dreams seem completely real while they last: there is nothing to check them against.

Sleep and memory

During sleep the brain does not rest: it reorganizes. Studies show that while we sleep the hippocampus, the structure that encodes episodic memories, replays sequences of activation similar to those of wakefulness, but much faster. It is as if the brain were "rehearsing" what happened during the day, consolidating the patterns worth keeping and discarding the rest.

But there is something more surprising. During REM sleep the brain does not merely consolidate what it already knows: it recombines. Elements of different memories mix and create associations that did not exist before. This is when the brain is most creative, because it is not limited by the logic of waking. Several scientific discoveries have been attributed to dreams, such as Kekulé's structure of benzene or Mendeleev's periodic table.

Lucid dreaming: the mirror inside the dream

There is, however, an exception: a state in which the horizon of REM acquires, without leaving the dream, something ordinary dreaming never has, the capacity to see itself.

It is the lucid dream, in which the dreamer realizes they are dreaming. For an instant, a small feedback window opens in the horizon: the prefrontal cortex, which was switched off, receives a brief signal. The EEG confirms it with a characteristic peak of gamma waves in the frontal region. For a moment the horizon acquires a mirror-like property: it can watch itself dream.

People who practice lucid dreaming develop techniques to prolong that state: checking the dream's consistency (can I read the same text twice?), anchoring themselves in sensory details or simply reminding themselves that they are dreaming. With practice, some manage to hold lucidity for minutes, exploring what for them is an inner world with the properties of an outer one.

Deep sleep: the contraction

And then there is deep sleep.

Deep sleep is contraction, a real, measurable, profound contraction.

Its slow waves, delta waves of between 0.5 and 4 cycles per second, are the signature of a massively synchronized brain. In wakefulness, neurons fire out of sync: different groups do different things at different moments, and that asynchrony is precisely what allows a rich integration of information. In deep sleep, millions of neurons fall together into a common rhythm. The horizon does not radiate: it is opaque.

Phi approaches the lowest values possible in a living adult. It does not reach zero, which would be death, but it comes close.

> ***In physics this is called:*** *the collapse of Phi through massive synchronization.*
> ***In everyday life it's like:*** *a concert in which all the musicians play the same note in unison: it sounds powerful, but there is no melody.*

Sleep homeostasis

Why do we sleep? For decades the question had no satisfactory answer. Today we know that sleep is an active, regulated process essential for survival: animals deprived of sleep entirely die sooner than those deprived of food.

The simplest mechanism is chemical. During wakefulness the brain accumulates adenosine, a by-product of neuronal metabolism that works as a signal of "sleep debt": the longer you have been awake, the more adenosine you accumulate. Caffeine works by blocking its receptors; it tricks the brain into believing it is not tired.

But adenosine does not explain everything. There is a second system, the circadian one, regulated by the suprachiasmatic nucleus, a molecular clock in the hypothalamus that synchronizes the sleep-wake cycle with daylight. The two systems, homeostatic and circadian, work in parallel, sometimes in the same direction and sometimes in opposite ones. That is why there are times when you are tired and cannot sleep, or awake and sleepy.

The synapse and sleep

Giulio Tononi proposed one of the most influential hypotheses about the function of sleep: the **synaptic homeostasis hypothesis**. During wakefulness, synapses are constantly strengthened: every experience, every piece of learning and every new stimulus potentiates neural connections. But they cannot strengthen indefinitely; there are limits of energy, space and noise. If all synapses were equally strong, the system would lose its capacity to tell signal from noise.

Deep, slow-wave sleep is the moment of renormalization. Synapses weaken in proportion to their strength: the strongest ones, those you used most during the day, weaken more than the weak ones. In this way the system returns to a baseline from which it can go on learning. Without this process, the brain would saturate within days.

The hypothesis makes testable predictions. It has been shown that sleep deprivation reduces the capacity to form new synapses, that the intensity of slow waves is related to the amount of learning the day before, and that the selective inhibition of certain synapses during sleep is associated with the consolidation of specific memories.

And it explains something we know from our own experience: we remember nothing of deep sleep, not because it does not exist, but because there was no one there to remember it. The horizon was so contracted that there was no experience to record. That is why the instant on waking, when the self has not yet returned, happens precisely after deep sleep and not after REM. From REM you wake from a dream, from a story, from someone who was doing something somewhere. From deep sleep you wake from nothing.

Is deep sleep, then, the state most like death? Yes, with one decisive structural difference: in deep sleep the synaptic architecture remains intact and basal metabolism active, so recovery is guaranteed. Every night we cross that edge and come back; death is the crossing without return.

But there is something more, something recent studies have found at the threshold of death, which does not appear in deep sleep and which changes the whole picture.

In 2013, Jimo Borjigin's team at the University of Michigan detected something unexpected in rats subjected to induced cardiac arrest: at the moment of collapse, the brain does not slide gently into silence. There is an intense, coherent gamma activation, not the chaotic noise of stress but an organized, synchronized signal more reminiscent of REM sleep than of deep sleep. In 2023 the same team published similar results in human patients.

If the horizon of deep sleep is opaque and contracted, the horizon at the threshold of death does something very different: a last flash of expansion before going out. As if the system, at the instant it loses all possibility of returning, did exactly the opposite of what it does when falling asleep: instead of a final contraction, a final expansion.

In physics this is called: *a terminal gamma burst (Borjigin et al.).*

Meditation: the opposite direction

At first glance, deep meditation and deep sleep look alike: stillness, immobility, closed eyes, less discursive thought. An outside observer could confuse them.

But their neurological signature is the opposite.

In deep sleep there are slow, high-amplitude delta waves, massive synchronization and a low Phi; the cortex does not propagate signals beyond its local region. In the deep meditation of advanced practitioners there are high-frequency gamma waves of extraordinary amplitude and long-range synchronization between frontal, parietal and occipital regions: the profile that, in the language of IIT, would correspond to a high Phi, although no one has measured it directly.

The difference is not one of degree but of direction.

Deep sleep contracts the horizon because the system loses the capacity to maintain differentiation. Deep meditation expands it because the system stops needing to differentiate without losing the capacity to do so. It is the difference between a lamp going out and a sun that shines without merging with anything in particular.

What disappears in meditation is not the horizon but the narrative self: the inner voice, the commentator, the manager of the personal story. The horizon remains active, perhaps more than ever, but without the usual filter that selects which information matters and which can be ignored.

Some contemplative traditions describe a state that deserves separate mention: the *turiya* of the *Mandukya Upanishad*, the fourth state, beyond waking, dreaming and dreamless sleep. It is the witness-consciousness that persists even in deep sleep without dream content. The few available EEG studies show highly coherent theta waves superimposed on a background of delta waves. As if the witness were floating on the sea of deep sleep without merging with it.

The Tibetan book of the dead, the *Bardo Thödol*, describes the period between death and rebirth as a sequence of states with a precise phenomenology. What is remarkable is not whether that description is literally true, but that it anticipates with surprising accuracy the phenomenology of the near-death experiences that medicine documented centuries later. Borjigin's gamma burst and the Clear Light of the *Bardo Thödol* might be the same phenomenon described in different vocabularies.

Insomnia: when the horizon cannot close

There is a disorder that illuminates sleep by showing its absence. Insomnia is the inability to contract the horizon. The person is exhausted, but the system does not reach the threshold of massive synchronization proper to deep sleep. Either the horizon stays ajar (shallow sleep, waking every hour), or it fails to settle into REM (fragmented dreams that do not restore).

The neuroscience of insomnia shows something revealing: insomniacs do not have fewer slow waves, but less coherent ones. The synchronization is partial and irregular, like a choir in which every singer hits the right note, but each in their own time. So the brain fails to achieve the synaptic renormalization deep sleep provides, and the insomniac gets up as tired as they went to bed, sometimes more.

What the instant of waking teaches us

> *"I left gaps in the archive. There are changes that admit no closure."*
> *— The Habit of Water*

We return, at the end, to the beginning.

That half second without a self when you wake suddenly is the system working as it should: it is recovering from a deep contraction and expanding the horizon again, from almost nothing to the complete geometry of the self. What is perceived in that instant, with no one perceiving, is the horizon in its barest state, before the story dresses it and gives it a name.

Ordinary wakefulness is the active horizon, integrated, open to the world and to itself. REM sleep is the horizon turned inward, processing at full power but without outside contact. Deep sleep is the horizon contracted almost to its minimum, the everyday approximation to the state in which integrated information falls to zero. Deep meditation is the horizon expanding in the opposite direction, beyond ordinary wakefulness, toward an integration sustained with very little arousal, something ordinary activity does not usually reach. And death is, perhaps, the final contraction, followed, according to Borjigin's data and the cartography of the *Bardo Thödol*, by a last flash in which the horizon does something we did not expect.

Every night we rehearse contraction without dissolving: we cross the edge and come back. The self reappears in half a second, rebuilds the story and picks up the thread. And the ease with which it returns, the astonishing automatism of that reconstruction, is the best proof that the horizon knows what it is, even when no one is looking.

Note to the Interlude

What we do know: *REM sleep has the electrical signature of wakefulness, with the prefrontal cortex switched off. Deep sleep shows massive delta waves and low perturbational complexity (PCI).*

High-amplitude gamma waves have been recorded in expert meditators (Lutz et al., 2004). Coherent gamma bursts after cardiac arrest have been documented in rats (Borjigin et al., 2013) and in some patients (Xu et al., 2023). Tononi's synaptic homeostasis hypothesis has growing experimental support. Adenosine regulates sleep debt.

What we don't know: *Whether the terminal gamma burst is the neural correlate of near-death experiences. Whether turiya is a measurable state distinct from deep sleep. Why selective insomnia affects the coherence but not the quantity of slow waves. Whether the causal arrow belongs to the interface and not to the territory: that is this book's own interpretation, not an established consensus.*

Open questions: *Why does the system expand rather than contract at the threshold of death? Is deep meditation a "third path" between waking and sleep, or something qualitatively different? Can lucid dreaming teach us anything about the nature of the self?*

If you take away only one idea: *Every night you cross the edge of your own horizon and come back. Death is the same crossing, with no guaranteed return.*

Further reading: *Borjigin et al. (2013), "Surge of neurophysiological coherence..."; Xu, Borjigin et al. (2023), "Surge of neurophysiological coupling and connectivity of gamma oscillations in the dying human brain"; Mandukya Upanishad (turiya); Bardo Thödol; Tononi (2008) on Phi and states of consciousness; Tononi & Cirelli (2014), synapses and sleep; Hume, D., An Enquiry Concerning Human Understanding (1748), on constant conjunction.*

PART TWO

THE BONDED HORIZON



CHAPTER 10

THE TIME OF THE BOND

There is an experience almost everyone has had and very few have managed to put into words. It happens in conversations you don't want to end, in afternoons with someone you hadn't seen for years, when you look at the clock and three hours have gone by without either of you looking at your phone. Time has not disappeared: it has changed density. More has happened per hour than usual.

The opposite experience exists too: a meeting that drags on without purpose, a wait without news. Time does not speed up; it thickens, and every minute weighs like ten.

The same clock, and two opposite experiences.

Time as a property of the horizon

If consciousness has the structure of a horizon, its time is an internal property: not an external parameter the horizon records, but a consequence of the way it integrates. The system's rate of integration, how much new information it takes in per unit of chronological time, determines the density of subjective time.

A four-year-old integrates at a very high rate: every stimulus reconfigures the system, every experience is new, the horizon is constantly expanding. That is why a summer afternoon in

childhood has a density an adult summer afternoon rarely reaches. The child is not happier: they process more per unit of time.

Old age reverses the process: the horizon has deeply consolidated patterns and few inputs generate new integration. The elderly are not lying when they say time passes faster: for every hour on the clock, less happens in their horizon. It is like reading a book for the tenth time: the pages turn faster because you already know what happens.

Pain and boredom confirm the model from the other side. Pain overloads the system with a signal it can neither process nor ignore, and integration folds in on itself in a loop that goes nowhere. Boredom appears when the system does not receive enough stimuli to sustain integration, and the horizon spins idly. In both cases, chronological and subjective time come drastically apart.

Flow, the state Csikszentmihalyi documented in artists, athletes and surgeons, is the opposite case. The rate of integration is at its peak, the system absorbs and processes without friction, and time disappears.

The brain as butterfly

The neuroscientist David Eagleman proposed an image that illuminates all this. The brain, he suggested, lives in a world of its own making and generates its internal time from the information it processes. When something new happens, it "writes" more memory per unit of real time, and when it remembers it reads that memory back: more details per unit of time translate into a sense of longer duration.

This explains what happens in falls: people who have had that kind of accident say time slowed down enormously. Eagleman put it to the test: he dropped volunteers from about thirty meters wearing a device on the wrist that displayed numbers too fast to read under normal conditions. If time really slowed down, they would be able to read them. They could not. What happens is that extreme fear generates a much greater density of memory: when remembered, the fall seems longer because there are more "frames" per second.

This forces us to separate two clocks that ordinary language confuses. One is lived time, the duration felt while something is happening; the other is remembered time, the duration reconstructed afterwards. Density of integration acts on them in opposite directions. While integrating at full speed there is no attention left to watch the clock, and the hours slip away; but each of those hours leaves a lot of memory written, and looking back it seems long. Boredom does the opposite: it drags while it lasts and, in memory, takes up hardly any room. That is why flow makes time disappear and yet such an afternoon is remembered as full; and why the summers of childhood, lived without looking at the clock, are remembered as immense. When this book says that integration determines the density of time, it is speaking above all of remembered time; what disappears in flow is lived time.

The chemistry of time

The rate of integration is regulated, minute by minute, by neurotransmitters:

Molecule	Effect on time	Analogy
Adrenaline	Maximum emergency integration. Dense time, but out of threat.	The accelerator of fear.
Noradrenaline	Arousal and novelty. The other is a new, unpredictable stimulus.	First infatuation: everything happens for the first time.
Dopamine	Anticipation, not pleasure. Calibrates the internal clock against the external one.	The central pacemaker: "is it worth it?"
Oxytocin	Social calm. The horizon stops being on guard.	The social handbrake: "I'm safe with this person."
Cortisol	Chronic stress. Degrades the capacity for resonance.	The noise that drowns out the music.

Parkinson's and time

Parkinson's disease progressively destroys the dopaminergic neurons of the substantia nigra. Its best-known result is bradykinesia: slow

movements, at times almost impossible ones.

What almost no one mentions, and patients describe with disconcerting consistency, is that from the inside that slowness is not perceived. The patient who takes three minutes to cross a room does not feel it takes three minutes, but that they are walking at their normal pace. Their horizon orders the movement at the usual speed; it is the body that fails to execute it.

Some patients say that with levodopa time seems to fall back into place even before movement has fully improved; if so, time synchronizes before the body does.

Memory as the builder of time

We do not remember time as we lived it: we reconstruct it. Every time we recall something, the brain rewrites it; it adds details that were not there, leaves out others that were, tints the memory with the mood of the moment. Memory is a story under constant revision.

This has important consequences for the perception of time. Two people who lived the same thing (a trip, an argument, a rainy afternoon) can remember it with very different durations. Neither is lying: their horizons integrated it differently and, in reconstructing it, generate different densities.

There is a disorder that illustrates this dramatically: Capgras syndrome. Patients recognize their loved ones visually (the face is identical), but feel that they are impostors, because recognition is not accompanied by affect: the circuit that links visual perception to emotional memory is damaged. If the model is right, the consequence would be unsettling: for these patients, time with their loved ones would lose density. The hours would

pass without anything "happening," because the emotional integration that gives time its weight would be missing.

Time in chronic pain

Acute pain dilates time: every second expands to fill consciousness. Chronic pain does something more insidious. After months or years of constant pain, the system stops registering it as novelty; the pain becomes background, a baseline, and with it time flattens. The days become interchangeable: morning the same as afternoon, Tuesday the same as Thursday.

> ***In physics this is called:*** *habituation: the system stops responding to constant stimuli because they bring no new information.*
> ***In everyday life it's like:*** *living next to a motorway: at first you hear every car, but after a month the noise disappears, even though it is still there.*

Patients with chronic pain describe it with a precision physics ought to take seriously: "It's not that time passes slowly or quickly; it stops passing." The horizon, saturated by a signal it can neither process nor ignore, stops generating the novelty that gives time its density. The result is a kind of temporal death: the person is still alive, but time has stopped happening.

The two-person system

When two people talk face to face, their patterns of brain activity converge. The **neural coupling** that Uri Hasson documented at Princeton shows that the listener's brain does not merely process what the speaker says but anticipates their patterns of activation. And the better the listener understands what they hear, the further ahead they get: in some regions, their activity *precedes* the speaker's by tenths of a second; it does not follow it, it anticipates it.

When two strings vibrate at related frequencies, they produce harmonics that were in neither of them: the two-string system has properties the one-string system lacks.

The same happens with two horizons in sustained resonance. The two-person system integrates more information than the sum of what each integrates separately; its joint Phi would be greater than the sum of the individual Phis. The Phis do not add up: entanglement opens new channels of integration that exist only in the relationship.

If the rate of integration determines the density of subjective time, and if the two-person system integrates more than the sum of its parts, the time of the bond is denser than that of either of the two alone: not longer on the clock, but fuller for those who live it.

Those three hours of conversation you don't want to end do not just seem denser: for those who live them, they are.

Time bases: synchronizing internal clocks

In the engineering of distributed systems and telecommunications, two independent devices that need to coordinate cannot trust that their internal clocks run at exactly the same speed. Each microprocessor has its own **internal time base**, a local oscillator governed by a quartz crystal, and because of microscopic physical imperfections those crystals drift (*clock drift*). If two systems process or exchange data without synchronizing, their time bases fall out of alignment and the transmitted information is corrupted.

To coordinate, devices use an **external reference time base**: a master clock or a central synchronization signal, like the one provided by the NTP or PTP protocols. The local oscillators continuously adjust to that external pulse, and so their internal times match in phase and frequency.

Something very similar happens in the human bond. Each brain has its own internal time base: neural oscillators and neurochemical dynamics that set the pulse of its subjective time. But when two horizons interact closely, their time bases couple. For that *phase-locking* they use a common external reference: the physical presence of the other, the rhythm of breathing, the tone of voice, the rhythm of turn-taking, the gaze.

That external signal acts as the relationship's master clock. Thanks to it, each horizon adjusts its speed of integration, and time stops being an isolated experience and becomes a shared temporal base.

When the physical channel is interrupted and the reference disappears (with distance or silence), drift returns: each time base recovers its own rhythm and the time of the two comes apart again.

Why distance hurts physically

When two horizons have resonated long enough, each begins to model the other in growing detail: their rhythms, their frequencies and their ways of responding become part of the prediction system. If the bond is interrupted, the horizon that remains active keeps an internal structure that expects signals that no longer arrive. The channels of integration that resonance opened stay open.

The feeling of absence is not the mere observation that someone is missing: it is the horizon processing silence where there should be signal.

Separated from the horizon it resonated with, the time of the one who remains becomes slower, heavier, less dense. There is as much to do as before, but less to integrate it with.

The topology of the bond: messages and design relationships

If the self and the other are encapsulated domains interacting through their horizons, the structure and temporality of their connection can be modeled with the design relationships of object-oriented programming (OOP).

Message passing

In the original conception of OOP, Alan Kay's, what matters is communication between objects. Two encapsulated objects never access each other's internal memory or share *private* variables; they communicate exclusively by sending each other **messages**.

In human interaction, message passing is the only possible route. We never access the other's neural states directly (there is no direct reading of their memory). We send messages (words, gestures, silences) that cross the physical channel and reach their public interface, and it is the receiver who, inside, runs their private functions to interpret them and update their own state.

Coupling relationships: association, aggregation and composition

In the modeling of complex systems not all connections are equal; each describes a different dependency between life cycles:

- **Association**: the most basic connection. Two independent systems know each other and hold a mutual reference. A circumstantial relationship is an association: the objects interact for specific tasks, but their life cycles are autonomous and do not suffer if the other disconnects.

- **Aggregation**: a whole-part relationship in which the components keep independent life cycles. If the whole dissolves, the parts survive intact. A group of friends is an aggregation: if the group stops existing, each individual goes on functioning normally and can associate with others.

• **Composition**: the strongest relationship of belonging. The life cycle of the parts is inseparably tied to that of the whole: if the whole is destroyed, the parts are destroyed with it. In a symbiotic bond or one of deep codependence, the relationship becomes composition, and if it breaks, the internal structure of each horizon collapses, because its state invariants depended on the existence of the joint system.

Inheritance and overrides

Inheritance is an "is-a" relationship. An object inherits the methods and data structure of its base class, and can use or modify them.

In the construction of identity, inheritance is the family and cultural legacy. We are born with base classes loaded with default event handlers (attachment patterns, automatic responses to threat). Growing up does not consist in programming an operating system from scratch, but in selectively overriding the inherited methods, redefining the behavior without breaking the structure that supports it.

Synchrony and asynchrony

The speed and density of the time of the bond depend on the communication protocol:

• **Synchronous communication**: a blocking call. The sender sends a message and halts its own processing while it waits for the reply. An intense face-to-face conversation is synchronous: the two systems enter a closed loop in real time, each waiting on the other's reply. It is the only protocol capable of generating phase-locking and synchronizing the time bases.

- **Asynchronous communication**: a non-blocking call. The sender sends a message (a letter, a text message) and carries on without waiting for confirmation; the receiver will process it when it has free cycles. It allows efficient decoupling in time, but lacks the resonance and temporal density of synchronous coupling.

Social time

Time is also collective. We are the only species that synchronizes its internal rhythms on a massive scale. Clocks, timetables, calendars and time zones are technologies of temporal coordination that allow millions of individual horizons to function as a larger system.

But that synchronization has a cost. Social time is an average: we eat at two because it suits most people, not because you are hungry at two; we work nine to five because productive coordination demands it, not because your body performs best at those hours. The result is a chronic mismatch between internal and external time that chronobiologists call "social jet lag."

> ***In physics this is called:*** *phase forcing: an oscillating system forced to synchronize with a frequency that is not its own.*

Studies show that this mismatch has measurable consequences for health: higher risk of obesity, diabetes, depression and cardiovascular disease. How hard the work is matters less than this: the horizon is forced to integrate on a time that is not its own.

Note to Chapter 10

What we do know: *Neural coupling (Hasson, Princeton) is real and measurable. Dopamine regulates the perception of time. Eagleman's fall experiment suggests that the slowing of time in frightening situations is an effect of memory, not of perception. Social jet lag has documented effects on health. Capgras syndrome illustrates the disconnection between recognition and affective appraisal.*

What we don't know: *Whether the time of the bond is "objectively" denser or only subjectively so. How to measure the Phi of a two-person system. Whether subjective time can be a measurable physical property.*

Open questions: *Could subjective time be a physical rather than a psychological property? What happens to time in grief? Why does chronic pain alter time differently from acute pain?*

If you take away only one idea: *Time does not pass: it is integrated. And with someone who resonates with you, more of it is integrated per minute.*

Further reading: *Hasson et al. (2012), brain-to-brain coupling; Csikszentmihalyi (1990), Flow; Eagleman (2009), subjective time; dopamine and time perception (Parkinson's); Roenneberg (2012), social jet lag.*

CHAPTER 11

ENTANGLEMENT FOR NON-PHYSICISTS

There is an experience that has no name in ordinary language.

You pluck a guitar string and another one, tuned to the same note, vibrates without your touching it. It isn't magic: the vibrations travel through the air and the wood. But imagine the strings were in separate rooms, with no air between them, and still vibrated in unison.

That is quantum entanglement.

> ***In physics this is called:*** *non-local correlation between quantum systems.*
> ***In everyday life it's like:*** *two coins tossed in different cities that agree more often than any prior arrangement between the people tossing them would allow.*

The story of entanglement

In 1935, Albert Einstein, Boris Podolsky and Nathan Rosen published a paper meant to show that quantum mechanics was incomplete. They imagined a thought experiment: two particles that interact and then separate. Quantum mechanics predicts that if you measure a property of one of them, its spin, for example, you instantly know the corresponding property of the other, however far apart they are.

Einstein found this absurd and called it "spooky action at a distance" (*spukhafte Fernwirkung*). If quantum mechanics was correct, the universe was stranger than any sensible physicist could accept. Einstein believed there were hidden variables, additional information that, if we had it, would explain the correlation without invoking instantaneous connections.

For thirty years the question remained in the realm of philosophy, until in 1964 the Northern Irish physicist John Bell proved something surprising: if hidden variables existed, they would have to produce correlations obeying certain mathematical inequalities. Quantum mechanics, by contrast, predicts correlations that violate those inequalities. In principle, it was possible to decide experimentally who was right.

In 1982, Alain Aspect carried out a decisive experiment in Paris that violated Bell's inequalities. The particles were about twelve meters apart, and the measurement settings were switched faster than light could cover that distance, so no signal between the two sides could explain the agreement. It was not an artifact of the apparatus: it was real.

> ***In physics this is called:*** *violation of Bell's inequalities; experimentally demonstrated quantum non-locality.*
> ***In everyday life it's like:*** *discovering that two players, each in their own room and choosing at random which question to answer, get it right together more often than any strategy agreed in advance would allow: there is no message and no trick, but no classical explanation either.*

The geometry of entanglement

In 1935, Einstein described two things:

1. **EPR**: entanglement, two particles coordinated at a distance.

2. **ER**: wormholes, tunnels that would connect two regions of spacetime.

In 2013, the physicists Maldacena and Susskind proposed something surprising: that they might be the same thing. That when two particles are "entangled," what happens is that they share a microscopic tunnel, not a tunnel to travel through, but a tunnel that *is* the connection itself.

> ***In physics this is called:*** *ER=EPR.*

If so, entanglement is geometry: a real connection in the fabric of spacetime. No information travels through the tunnel (signals cannot be sent faster than light), but the connection exists as a curvature of space, as real as any other.

Decoherence: why we don't see entanglement

If entanglement is so real, why don't we see it in everyday life? Why don't tables, chairs or bodies seem entangled with anything?

Because of decoherence. Quantum systems become entangled with their environment in ways we can neither control nor measure. Every particle in a table constantly interacts with photons, air molecules, thermal vibrations, and those interactions "measure" the system ceaselessly and destroy the quantum correlations that would otherwise exist. At the macroscopic scale, entanglement becomes inaccessible: it doesn't disappear, but spreads into the environment in a way we cannot isolate.

> ***In physics this is called:*** *decoherence, the loss of quantum coherence through interaction with the environment.*
> ***In everyday life it's like:*** *an intimate conversation at a noisy party: the words are still there, but the noise drowns them out until they can't be told apart.*

That doesn't mean macroscopic entanglement is impossible, only that it demands extreme conditions: temperatures close to absolute zero, almost perfect isolation, or very carefully designed systems. Today's quantum computers fight precisely against decoherence: each qubit must be isolated enough from its environment to maintain its entanglement, but connected enough to be measured.

Quantum teleportation

One of the most surprising applications of entanglement is quantum teleportation. It isn't the *Star Trek* kind: no matter is sent, only information. The procedure is elegant. Two entangled particles are separated: one stays with you and the other travels to its destination. Then you make a joint measurement of the particle you want to "teleport" and your half of the entangled pair, and you send the result of that measurement, two classical bits, over an ordinary channel. At the other end, using those two bits, your counterpart can transform their particle into an exact copy of the one you wanted to send.

The original state is destroyed in the process, because there cannot be two copies of the same quantum information, but the information is transferred in a way that would be impossible without entanglement.

> ***In physics this is called:*** *quantum teleportation, the transfer of a quantum state by means of entanglement and classical*

communication.
In everyday life it's like: *mailing a recipe: you don't send the food, but with the recipe and the ingredients they already have, your friend can reproduce it exactly.*

This does not allow communication faster than light, because the two classical bits have to travel over an ordinary channel, but it does allow something classical physics does not: transferring a complete quantum state, including any entanglement it had with other systems.

From particles to people

If entanglement creates shared geometry, two entangled systems are not just two correlated systems, but two systems that share topology and have an inside that is partly common.

Their horizons are not entirely separate.

Applied to brains, this is highly speculative and lacks experimental evidence. But it allows a precise question to be asked: what would "bond" mean if information really were shared geometry?

We don't know whether two brains can be entangled in the physical sense. We know that their activity can be correlated in surprising ways, as Hasson's neural coupling shows, but a classical correlation is not quantum entanglement. The difference is subtle and decisive: classical correlation can be explained by common causes or by direct communication; quantum entanglement cannot.

In physics this is called: *classical correlation ≠ quantum entanglement.*
In everyday life it's like: *two friends wearing the same T-shirt*

> *because they bought it together, or two twins who choose it separately because they share tastes: both are classical correlation, with a common cause. Entanglement is precisely what no common cause can explain.*

What we can say is that entanglement offers a precise vocabulary for talking about connections that do not depend on spatial proximity. And that precision is useful even if the connection between brains turns out to be only a classical correlation, because it forces us to distinguish between "being close" and "being connected," two things intuition tends to confuse.

Can entanglement explain the mind?

Some theorists, among them Roger Penrose, have proposed that consciousness depends on quantum processes in the brain. The idea is that the mind does something classical computers cannot do (understand, intuit, create) and that this something requires quantum computation.

Most neuroscientists and physicists reject the hypothesis. The brain is a warm, wet and noisy medium, exactly the kind of environment in which decoherence destroys quantum states in fractions of a second. Calculations suggest that any quantum superposition in a neuron would be destroyed long before it could influence the behavior of the system.

But there is a twist. Since 2007, several teams (the first being Graham Fleming's at Berkeley) have published results suggesting that certain photosynthetic complexes in bacteria and algae maintain quantum coherence for longer than expected, even at room temperature, although their real role is still debated. If nature has found ways to protect quantum coherence in

biological systems, perhaps the brain has too, in a way we don't yet understand.

The most sensible position is agnosticism. We don't know whether the brain uses quantum computation or whether two brains can be entangled. We know that entanglement exists, that it might be shared geometry, and that it offers the most precise vocabulary we have for talking about connections that transcend spatial separation. That is enough for the experiment to use it as a formal metaphor: not as dogma, but as a tool for thinking.

Note to Chapter 11

What we do know: *Quantum entanglement is real and experimentally verified. Bell's inequalities are violated. ER=EPR is a conjecture supported by theoretical models, without experimental verification. Quantum teleportation works. Decoherence explains why we don't see macroscopic quantum effects.*

What we don't know: *Whether Einstein-Rosen bridges exist physically. Whether entanglement between brains is detectable. Whether the mind depends on quantum processes.*

Open questions: *Is the human bond "entanglement" in the physical sense or only a formal metaphor? Can biology protect quantum coherence in the brain? What limits does decoherence impose on the quantum mind?*

If you take away only one idea: *Two things that once touched can remain correlated even when far apart, as long as the noise of the world doesn't undo the bond. Perhaps love works like that.*

Further reading: *Einstein, Podolsky & Rosen (1935); Bell (1964); Aspect (1982); Maldacena & Susskind (2013); Penrose & Hameroff (2014, Orch-OR); decoherence (Zurek, 2003); Engel, Fleming et al. (2007, coherence in photosynthesis).*

CHAPTER 12

ENTANGLEMENT

In 2013, Maldacena and Susskind reframed the question of what quantum entanglement is. ER=EPR holds that the entanglement between two quantum systems is a geometric connection. Two entangled particles would be joined by a microscopic Einstein-Rosen bridge, a topological structure of spacetime that connects their interiors however far apart their exteriors may be.

The bridge is not traversable: it does not allow information to be sent faster than light. But the geometry exists, and the connection is real in the same sense that any curvature of space is real.

> ***In physics this is called:*** *Einstein-Rosen bridge.*

As we saw in Chapter 11, two entangled systems therefore share topology: they have an inside that is partly common, and their horizons are not entirely separate.

The neurobiology of attachment

Before talking about entanglement in adults, we need to understand how it is built. In the 1950s the psychiatrist and psychoanalyst John Bowlby proposed that humans, like other primates, come equipped with a biologically programmed attachment system. It is necessary for survival. Babies who fail to form attachment bonds can waste away to death even when they have food and shelter; René Spitz documented this in orphanages and called it "hospitalism."

Mary Ainsworth, Bowlby's collaborator, designed the "strange situation" procedure: observing how children react when their mother leaves the room and comes back. She identified three main patterns (secure, anxious and avoidant attachment), to which a fourth, disorganized attachment, was later added. More than personality styles, they are neural architectures built in early childhood out of repeated experience with those who care for us.

The attachment system works through specific brain circuits: the orbitofrontal cortex, the amygdala, the hypothalamus, the endogenous opioid system. When the baby is with its attachment figure, those circuits calm down; when separated from it, they activate. The repetition of those cycles (calm, separation, reunion, calm) builds an internal representation of the world as a predictable and safe place, or as a dangerous and unpredictable one.

What Bowlby called the "internal working model" is, in the language of the experiment, an architecture of the horizon: a configuration of integration channels that determines how the system will process the presence and absence of others. Secure attachment creates a horizon able to expand toward others

without losing itself. Insecure attachment creates a horizon that, faced with closeness, either contracts (avoidance) or overflows (anxiety).

The social baseline

James Coan and Lane Beckes developed social baseline theory: the human brain is not made to function alone, and its reference state presupposes access to others. When someone close is present, the brain devotes fewer resources to detecting threats and maintains a more efficient representation of the world. The presence of the other is part of normal functioning.

> ***In neuroscience this is called:*** *the social baseline.*

The mechanism is **self-other overlap**: the brain incorporates close people into its own neural representations. Brain activity in response to a threat aimed at oneself closely resembles the activity provoked by a threat aimed at a close friend, but not that provoked by a threat aimed at a stranger. The friend is not represented as "another self" in a figurative sense, but in the same regions and with the same patterns as one's own self.

> ***In neuroscience this is called:*** *self-other overlap.*

The other horizon is incorporated into the internal architecture of one's own. Their patterns, rhythms and ways of reacting become part of how the system processes the world. The integration is literal, not metaphorical.

And here is the decisive detail: it depends not on affection, but on **familiarity**.

Adversarial entanglement

Coan and Beckes point out that self-other overlap is tied to familiarity, not to positive affect. The brain does not incorporate the people it loves, but the people it knows well enough to predict.

In this sense, an adversary of twenty years is one of the most integrated figures a horizon can have. To survive a prolonged conflict, the horizon has to model the other with extraordinary precision: anticipate their reactions, foresee their moves, recognize their patterns before they are complete. The internal model of the adversary is, if anything, more detailed than that of an ally, because prediction errors cost more.

Two horizons locked in conflict for twenty years are as entangled, share as much geometry, as two who have loved each other for the same length of time. The difference lies not in the depth of the coupling, but in its **valence**. The bridge exists in both cases; what changes is what flows across it. Two neighbors who have hated each other for decades know each other better than many married couples.

From the point of view of entanglement, intense love and intense hatred are the same phenomenon with a different sign. The intensity of the conflict measures the depth of the entanglement, not its absence.

> ***In physics this is called:*** *the same magnitude of coupling with the sign of the interaction reversed.*
> ***In everyday life it's like:*** *two magnets: the force is the same whether they attract or repel; only the direction changes.*

Empathy as resonance

Empathy is not imagination, nor does it consist only in "putting yourself in someone else's shoes": it is resonance. Your nervous system reproduces, on a reduced scale, the other's state. Mirror neurons, discovered by Rizzolatti in macaques, fire both when you perform an action and when you see someone else perform it; they do not distinguish between the one acting and the one watching. Your brain "feels" what the other feels not because it imagines it, but because it runs it.

But empathy has limits: we cannot resonate with what we cannot model. A torturer can know precisely what their victim feels (Chapter 16 will return to this) and still not resonate with it; not because they lack mirror neurons, which they have, but because they have built a model of the victim in which the victim's subjective experience does not count. Dehumanization is empathy directed at a model that excludes the other's inner life.

The ghost: the model running idle

Mary-Frances O'Connor describes the mechanism of grief like this: the brain builds a predictive model of the people close to it, and that model keeps running normally even after the person dies. It is the normal functioning of a system designed to predict.

In love, this produces the grief everyone knows: turning toward where the person should be, making coffee for two, starting to dial their number before remembering. In prolonged conflict, the mechanism is identical, but it expresses itself differently. The horizon does not look for the other: it anticipates them. It knows exactly what they would do in this situation, how they would react to this piece of news.

The ghost is the direct consequence of having built a very precise model of someone who is no longer there to generate the signals that model expects.

Physics offers an image: an incomplete *scrambling*. The other's horizon dissolved and its information was spread through the reservoir, but the surviving horizon is still organized to receive its signals. The integration channels built over twenty years remain open. The entanglement persists even when one of the two poles has disappeared, because it was inscribed in the architecture of the surviving horizon, not only in the relationship between the two.

When the bond breaks

Not all entanglements end in death. Some end in rupture: divorce, betrayal, abandonment, gradual estrangement. The process is different, but the mechanism is similar.

When a bond breaks suddenly, the remaining horizon has no time to adjust its architecture. The integration channels are still open, but what they receive is no longer the other's signal, but silence, absence or, worse, the signal of someone who is no longer the person you modeled. The result is enormous dissonance: the system predicts one thing and receives another. It is like following the GPS to an address that no longer exists.

Gradual ruptures are different. They give the horizon time to readjust, close channels and rebuild its architecture without the other, but they also give the entanglement time to deepen before it breaks. There is no general answer as to which kind of rupture hurts more: it depends on the architecture of the horizon, the history of the bond and the conditions of the environment.

The asymmetry of entanglement

Entanglement is not necessarily reciprocal in the same measure.

In requited love, the two horizons model each other with similar intensity, and the entanglement is more or less symmetrical.

In an asymmetric enmity (a troublesome neighbor who hardly thought about the other, while the other spent twenty years accumulating grievances), the entanglement is much greater in one direction than the other. The wronged neighbor modeled the troublesome one with the precision of someone who needed to predict him in order to defend himself. The troublesome neighbor modeled the wronged one much less, because the conflict cost him nothing.

The ghost lingers longer the more asymmetric the entanglement was in that direction: whoever built the most model takes the longest to dismantle it.

The paradox is that the wronged neighbor has spent twenty years paying the other more attention than anyone else has. His notebook of grievances does not prove that he mattered to the neighbor, but exactly the opposite: that the neighbor mattered to him far more.

The limit of entanglement

There is a limit. Entanglement cannot be total: if two horizons merged completely, they would stop being two. Identity needs a boundary. The deepest love is an expansion of the self that includes the other without losing itself. They are two overlapping circles: the common area is the bond, but each remains a circle.

Contemplative traditions have been exploring this limit for millennia. Advaita Vedanta speaks of the identity between Atman and Brahman, the individual self and the universal one, but also of maya, the illusion of separation. Mahayana Buddhism speaks of the bodhisattva, someone who attains enlightenment, the dissolution of limits, but decides to remain in the world to help others. They could merge, but choose to keep a horizon of their own while sharing the geometry of the whole.

Note to Chapter 12

What we do know: *Self-other overlap (Coan and Beckes) is real. O'Connor describes grief as the persistence of a model of the absent person. Mirror neurons have been recorded in macaques (Rizzolatti), and there are signs of analogous systems in humans. The attachment system (Bowlby, Ainsworth) is well documented.*

What we don't know: *Whether ER=EPR operates at biological scales. Whether entanglement between brains is physically real. How a predictive model dissolves.*

Open questions: *Is hatred an entanglement with a negative sign? Why do some griefs last decades and others months? Is there an optimal limit to entanglement, or is deeper always better?*

If you take away only one idea: *Knowing someone deeply, whether you love them or hate them, inscribes them in your architecture. When they leave, the architecture goes on waiting for them.*

Further reading: *Coan & Sbarra (2015), Social Baseline Theory; O'Connor (2022), The Grieving Brain; Bowlby (1969), Attachment; Ainsworth (1978), the strange situation; Rizzolatti & Sinigaglia*

(2008), mirror neurons; Maldacena & Susskind (2013).

CHAPTER 13

LOVE

Entanglement is automatic and involuntary. Two horizons that have maintained sustained contact become coupled, whether they want to or not, and whether the contact was loving or hostile. The wronged man with the notebook of grievances did not choose to inscribe the other in his architecture: it was twenty years of anticipating the adversary that built that model without asking his permission.

Entanglement is a physical property of the system: it arises from sustained contact the way temperature arises from the motion of molecules. No one decides to have a temperature, and no one decides to be entangled.

But there is something the horizon can decide: whether it **recognizes** that entanglement as meaningful, whether it gives it a name, whether it reorganizes its architecture around that recognition.

In this experiment, love is that act of recognition. Not the entanglement itself, which was already there, but the decision to name it and take on what that name implies.

Including the other in the self

In the 1980s, Arthur and Elaine Aron developed the self-expansion model: to love means actively including the other in one's representation of oneself. Their resources, perspectives and identity become part of how the horizon conceives of itself. Not as a metaphor, but as a real reorganization of the architecture of self-reference.

The empirical evidence is direct. In the "me / not me" task devised by Aron and colleagues, participants rate personality traits of their own and of their partner's, and afterward they take longer to respond when asked whether one of the partner's traits is also theirs, as if the system had to resolve an ambiguity that did not exist before. The other's traits have come to be processed in the same regions and with the same patterns as those of the self.

Including the other in the self means actively recognizing the entanglement that already existed and reorganizing the internal architecture around that recognition. The loving horizon does not merely have the other inside it: it knows it, names it and accepts the consequences of knowing it.

Here lies the difference the previous chapter could not formulate. The wronged man with the notebook of grievances had his neighbor inside him, as geometry inscribed by twenty years of confrontation, but he never included him in his self: he treated him as interference. The loving horizon does the opposite: it takes the involuntary entanglement that already exists and turns it into something chosen, something that defines who one is.

The chemistry of love

Helen Fisher divided love into three distinct neural systems that operate semi-independently: sexual desire, romantic love and pair attachment. Each has its own chemistry, its own evolutionary function and its own way of altering the horizon.

Sexual desire is mediated mainly by testosterone and estrogens. It is the oldest system, shared with other mammals, and its function is clear: to drive physical closeness with potential reproductive partners. But desire is not love. It can exist without it, and love can exist without desire, something cultures that confuse the two systems find hard to acknowledge.

Romantic love is something else. It is mediated by dopamine, norepinephrine and serotonin, and it is a reward system geared toward anticipation, not consumption. The person in love does not seek pleasure, but the possibility of pleasure: every thought of the other activates the dopaminergic circuit of seeking, the same one that lights up when a rat smells food at the end of a maze. That is why romantic love is lived as urgency, as obsession, as an inability to think about anything else: the system is built to keep attention fixed on a single object until the bond is consolidated.

Pair attachment is the calmest system. It is mediated by oxytocin and vasopressin. It produces not euphoria or obsession, but calm, safety, the feeling that the world is predictable while the other is near. It is what makes a couple of twenty years feel "at home" without the need for drama or constant proofs. Oxytocin reduces the threat response of the sympathetic nervous system; vasopressin regulates the exclusivity of the bond and, at least in rodents, is linked to monogamy.

The distinction matters because the three systems can come apart. You can desire someone you don't love, be in love with someone with whom you have no secure attachment, or feel deep attachment without desire or passion. Popular culture takes for granted that all three must coincide in a single person "forever," but neuroscience suggests that this expectation is biologically implausible.

What recognition costs

The cost is concrete: if I include the other in my self, losing them is no longer just losing someone, but losing part of what I am. **Self-contraction**, the Arons' term for the loss of a bond in which the other had been included, is a real contraction of the horizon. Not subjective, in the sense that it "seems" like a loss of identity, but objective: the architecture of the horizon is left reorganized around an absence where there used to be an active presence.

You can stop talking to someone and gradually reduce the entanglement, but you cannot undo having included them in your self. Inclusion, once made, leaves a mark in the architecture of the horizon that no distance erases entirely. That is why a lost love is different from a friendship that ends: not because the feeling is more intense, but because the reorganization of the horizon is deeper. It is like taking down a painting that has hung in the same place for twenty years: the wall keeps the shape and color of what is no longer there.

Love as a safety signal

A recognized bond (named, accepted, chosen) gives a more powerful safety signal than the same bond without a name.

Coan and Beckes showed that the mere presence of someone familiar reduces the horizon's threat response. Studies of attachment refine that result: what has the greatest effect is **certainty of availability**, more than familiarity itself. A recognized bond is a bond the horizon can rely on: it knows it is there, that it responds, and it knows how to adjust its own activity to that presence. An unnamed bond works in the system like an undeclared variable: the horizon uses it, but cannot refer to it or organize its functioning around something that officially does not exist.

Named love is more efficient than silent entanglement. It does not produce more shared geometry, which was already there, but it lets the horizon make better use of it.

The fall of love

If romantic love depends on dopamine and norepinephrine, systems of anticipation and novelty, it has a structural problem: novelty runs out. The known cannot be anticipated with the same intensity as the unknown. The dopaminergic system habituates, and the same face, the same voice, the same body generate less and less expectation.

This is biology, not a moral failing of the couple. The novelty-reward system is built to drive seeking, not staying. Staying needs another system: attachment, with its oxytocin and its calm.

For many, the passage from infatuation to attachment is experienced as a loss. The passion fades, the obsession disappears and what remains seems cold by comparison. But what remains is, in many ways, more valuable. Attachment is love on another frequency, one that needs no constant

demonstrations because its safety signal is enough. It is fire passing from tall flames to embers: less spectacular, with a longer-lasting heat.

The question every couple has to answer, and which the experiment cannot answer for them, is whether that transition is acceptable or whether the loss of the romantic phase invalidates the bond for one or both horizons. There is no universal answer; only the clarity of knowing what is happening and why.

The asymmetry of recognition

Unrequited love has a structure of its own. Entanglement is not necessarily one-sided, since it can exist in both directions, albeit with different depths. What is one-sided is recognition: one of the two horizons has named the entanglement and reorganized its architecture around that name; the other has not.

The horizon that has recognized is left in a delicate position: it has taken on the cost of inclusion (exposure, possible self-contraction) without receiving the safety signal that mutual recognition would have generated. It is an asymmetric architecture: the lover's self includes the beloved, but the beloved's self does not include the lover to the same extent. Like an Einstein-Rosen bridge anchored at only one end, the geometry exists, but it cannot bear the weight it was built for.

Love and time

Chapter 10 described how subjective time depends on the rate of integration. Love alters that rate in a particular way. In the early stages of falling in love, the rate is extremely high: every encounter reconfigures the system, every conversation brings novelty, every shared silence has an unusual density. Time with the beloved is lived as denser than time without them.

But there is an additional effect. When two horizons in love are apart, time stretches as it does not in the absence of other bonds. The wait grows longer, not because the clock slows down, but because the system integrates less information per unit of time: almost all its capacity is taken up modeling the absent one, anticipating the reunion and processing the null signal their absence leaves behind.

The body of entanglement

So far, this chapter has described love as an informational process: synchronization of predictive models, coupling of temporal rhythms, reorganization of the internal architecture. The description is correct, but it treats the horizon as if it were only an information system, and it isn't. The horizon has a substrate, and it ends where the skin ends.

Sexual contact is the case where this matters especially, because it is the only one in which the overlap of two horizons is also physical. In ordinary entanglement the edges remain intact: each horizon knows precisely where it ends and where the other begins. In sexual contact that certainty becomes ambiguous for a while. Two integrated information systems share the same space, the same skin temperature, the same heart rate and breathing. The horizon does not just include the other in its

model: it touches them. And in that zone of contact, the boundary between inside and outside loses its sharpness.

The system does not know exactly where its own integration ends and the other's begins. It does not lose consciousness, but delimitation. It is the experience of a horizon without a clear outline, which contemplative traditions have tried to describe with the same vocabulary they reserve for mystical union, in particular for the dissolution of the outline of the self.

This adds a dimension of its own to grief when there was sex: losing someone with whom one had deep physical contact is not only losing someone inscribed in the internal model, but also the zone of contact in which the horizon existed, at times, without knowing exactly where it ended. One misses not only the other, but also the version of oneself that existed only at that edge.

Love as a practice

There is a conception of love that reduces it to a feeling, something that happens to you, like rain, and another that reduces it to a decision, something you choose, like a contract. The truth lies between the two. Love begins as a feeling, or as the recognition of an entanglement that already existed, but it only lasts as a practice. Not as an obligation, but as a repeated act of attention, of inclusion, of daily reorganization of the architecture around the other.

Love that is not practiced, not named or actively incorporated into the daily architecture, does not stop being love, but it stops being an active bond. It becomes a dormant entanglement, shared geometry the horizon no longer uses. It may or may not awaken; that depends on someone deciding, at some point, to recognize it again.

Note to Chapter 13

What we do know: *The self-expansion model (Aron and Aron) and the reduction of the threat response in the presence of a loved one (Coan) have empirical support. The three systems of love (Fisher) are well documented. The transition from passion to attachment is a recognizable pattern.*

What we don't know: *Whether the description of the overlap of horizons in sexual contact has a neuroscientific correlate of merging information fields. How attachment is maintained over the long term.*

Open questions: *How does attachment style affect the speed of self-contraction after the loss of a love? Is the decline of passion inevitable? Can love be a practice without becoming an obligation?*

If you take away only one idea: *To love is to decide to name an entanglement and reorganize your whole predictive architecture around it. And then, every day, to decide it again.*

Further reading: *Aron & Aron (1986), Love and the Expansion of Self; Coan & Beckes (2011), "Social Baseline Theory"; Fisher (2004), Why We Love; Hazan & Shaver (1987), adult attachment.*

CHAPTER 14

ADDICTION

Will as a function of the horizon

What is will? The intuitive answer is that it is the horizon's capacity to direct itself: intention, the impulse that precedes action and guides it.

But the horizon is a field of integration with a topography of its own: some inputs weigh more than others, some directions have more inertia, some patterns are more consolidated because they have been active longer. What we call will is a function that emerges from the field itself, not a force acting on it from outside. The horizon that decides something is not above its architecture: it is inside it, and takes the direction that architecture makes possible.

This is not determinism, in the sense that the decision was written in advance. It is something more precise: the decision arises from a structure with a history and a chemistry, which carries the shape of everything that has configured it. Will is real, the horizon really does steer itself, but it is not sovereign in the absolute sense that moral language usually assumes.

The most accurate image is that of a river. The river flows, and the flow is real: the water really goes from one place to another. But it flows along a bed it has itself been carving out, and changing course, though not impossible, takes time, erosion and sometimes a flood that opens a new path. It is not enough for the river to want to.

From the horizon model, will is the capacity of the integration system to generate a direction sustained over time. Not a one-off act, but a vector with inertia: hard to change, but also hard to stop once under way. And that vector depends on resources. An exhausted horizon, with high cortisol and low serotonin, has a genuinely weaker will than the same horizon when rested. The reason is that the field of integration that generates will is degraded, not that it is less of a person.

Will is a physiological resource.

The hijacking of the threshold

Addictive substances flood the dopaminergic system with a reward signal that can be up to ten times that of any natural reward. The horizon does not receive greater pleasure, but a prediction of pleasure so overwhelming that it recalibrates the whole system.

The mechanism is this: the dopaminergic system, designed to signal when something is worth repeating, receives a signal of a magnitude no natural reward can generate. To compensate, it reduces the number of available receptors, just as one lowers the blinds when the light is dazzling. When the artificial signal disappears, the system is left with fewer receptors than before, and what used to generate a signal (the bond, novelty, achievement, everyday pleasure) no longer does so with the same intensity. The threshold has risen, and now it takes more to reach it.

And this is literal: neuroimaging shows that the prefrontal cortex, seat of intention, of the capacity to delay and of the evaluation of consequences, loses functional connectivity with the reward system. The horizon still has will; what it has lost is

access to the circuits that translate it into action against that particular impulse. The will is there, but the bridge is cut.

It is the **Parkinson's of desire**: the horizon knows perfectly well where it wants to go, and knows that it does not want to go where it is going, but the system does not execute what the horizon commands.

That is why asking someone deep in addiction to use their willpower has the same logical structure as asking a Parkinson's patient to walk faster. The will is there; what is damaged is the bridge between intention and action. And that damage is the foreseeable consequence of what the molecule does to the system.

Besides shifting the threshold, addiction distorts the hierarchy of salience of the entire horizon. The addicted horizon does not merely pursue a substance: it has reorganized its integration system around that axis, and everything else (people, projects, bonds) becomes background noise. The system that assigns importance has been recalibrated toward a single object that responds with an intensity nothing else matches. In the terms of the model, the horizon is left contracted: it integrates less, resonates less and generates less dense time in any direction other than the substance.

Seen this way, recovery is a slow recalibration of the threshold, which takes months and sometimes years and happens above all through the bond: other horizons that sustain the natural reward signal long enough for the receptors to regenerate and the threshold to fall. Oxytocin as an antidote, not out of romanticism, but because it is the molecule that activates the reward system through the pathway of the bond, a pathway that does not wear down the receptors it depends on.

> ***In neuroscience this is called:*** *down-regulation of reward receptors.*
> ***In everyday life it's like:*** *lowering the blinds because a blinding light is coming in: when the light goes, you are left in the dark.*

Dopamine and narrative

One of the most insidious things about addiction is that the system recalibrates not only the receptors but also the story. The human brain is a story machine: it turns experience into narrative, narrative into identity and identity into predictable behavior. In addiction, that machine is put at the service of the impulse.

Dopamine generates not only desire but **prediction**. The dopaminergic system constantly calculates the difference between what you expect and what you get. If the outcome matches expectation, there is hardly any signal; if it exceeds it, there is a burst; if it falls short, there is frustration.

In addiction, the system learns to predict the substance with such certainty that any other reward (a meal, a hug, an achievement) generates an insufficient signal. The brain builds a narrative in which the substance is the only possible source of well-being. It is the only one that generates enough signal to register as positive, whether or not that is true.

When love becomes the object

Chapter 13 described sexual contact as the only case in which the overlap of two horizons is also physical and the boundary between inside and outside loses its sharpness. That ambiguity of the edge produces the most powerful reward signal the system can generate naturally: dopamine of anticipation, norepinephrine of novelty, oxytocin of contact, endorphins of physical pleasure. The full cocktail. The nervous system is designed, quite literally, for that to matter more than almost anything else.

But that design presupposes that the signal comes accompanied by someone: a real horizon on the other side of the contact, with its own timing, its own unpredictability and its own capacity not to respond. That friction, paradoxically, keeps the system calibrated: the signal fluctuates, no one can control it alone, and the system cannot optimize it without the other's participation.

Sex addiction appears when that condition breaks, when the horizon finds a way to access a version of that signal it can control and repeat without depending on the availability, unpredictability or friction of another real horizon. Pornography, for example, offers infinite novelty on demand: a stimulation that ceaselessly activates the noradrenergic system without the cost of negotiation or the risk of rejection. Dopaminergic anticipation is triggered without ever resolving into a real bond, because there is no other horizon to bond with.

The result is the same as in the general mechanism of addiction: the threshold rises. Real intimacy, with its friction, its silences and its unpredictability, generates a signal that the system, recalibrated by artificial hyperstimulation, no longer finds

sufficient. The other has not changed: the reward system no longer registers what they offer with the intensity it now needs.

What makes sex addiction a special case is that the circuit it hijacks is precisely the circuit of love: the same receptors, the same system of dopaminergic anticipation, the same zone of ambiguity of the horizon that Chapter 13 described as the seat of the deepest bond. It is the opposite of an excess of love: the love system chasing the signal of love without the love. Dopamine without oxytocin; the chemistry without the other.

The horizon that has reached that point does not only lose access to the signal of the bond. It also loses the capacity to resonate with another horizon on that frequency, which is exactly the one it would need to recalibrate. The tie is cut at both ends at once.

Work addiction

There is a form of addiction that culture does not call by that name: work addiction. What looks like excessive dedication is the hijacking of the reward system by an activity that generates a predictable, controllable signal.

Work offers something the human bond cannot: measurable results, visible progress, immediate reward. Every email answered, every task completed, every goal met produces a small dopaminergic burst. The system learns to seek those bursts with the same logic with which it seeks the substance: because they are a more reliable signal than the unpredictability of other horizons.

But work, unlike the substance, does not destroy receptors. It destroys something harder to repair: the capacity to tolerate the ambiguity of the human bond. The work-addicted horizon loses

the patience demanded by conversations that never end, silences that never resolve, emotions that don't fit into a *sprint*. The time of the bond, which Chapter 10 described as a density born of resonance, becomes "wasted time" compared with the efficiency of work.

The horizon born sensitive

So far, this chapter has described addiction as something that happens to the horizon: a recalibration triggered by an external signal that exceeds the system's capacity to respond. But some horizons arrive at that encounter with their threshold already shifted from the start.

The best-documented case is **ADHD**, which involves a dysfunction of the dopaminergic system: specifically, a chronic hypoactivity of the reward system that makes the baseline dopaminergic signal insufficient to sustain directed attention. The horizon with ADHD does not suffer from a deficit of intelligence or motivation in the abstract, but from difficulty generating enough internal signal for a specific task, chosen by reason and not by impulse, to compete with the system's background noise. The horizon knows what it wants to do; the system does not generate the signal that would make doing it feel possible.

The consequences are clear. The horizon with ADHD actively seeks stimuli that raise the dopaminergic signal: urgency, novelty, risk, intensity. Not by character, but by architecture. When it comes across a powerful source of stimulation (a substance, a pattern of behavior, a screen designed to maximize reward), the system responds with disproportionate intensity, precisely because it had been in deficit for a long time. Unlike classic addiction, here the threshold does not rise with use: the

system starts out with little signal of its own and with sensitized receptors, waiting.

That vulnerability is a property of the architecture. A horizon with ADHD that develops an addiction has not failed morally: it has followed the logic of a system that seeks what it lacks, and has found it in the wrong place because the right places did not generate enough signal.

The controlled demonstration

Parkinson's disease eliminates all the confounding variables that ADHD still leaves open.

There is no history of trauma, no adverse social context and, in most cases, no prior atypical dopaminergic architecture. There is a disease that gradually destroys the dopaminergic neurons of the substantia nigra and a treatment that tries to replace what the disease destroys. And under that controlled treatment, between fourteen and forty-two percent of patients develop impulse control disorders (pathological gambling, hypersexuality, compulsive shopping, binge eating) that did not exist before.

The mechanism is the one the model predicts. The dopaminergic system, deprived of signal by the disease for months or years, has sensitized its postsynaptic receptors to the maximum, precisely because the presynaptic signal was scarce. When treatment arrives, dopamine falls on a system that was not calibrated to receive it at that intensity, and the result is a relative overdose of the reward circuit even though the dose is, by the parameters of movement, perfectly therapeutic.

The cleanest evidence is that the effect depends on dose and on the type of drug: dopamine agonists, which activate the receptors directly, produce more impulse control disorders than levodopa alone, and the combination of both more than either separately. The dose weighs more than any prior trait of the patient, though it does not decide everything: age, temperament and personal history modulate the risk, and most treated patients never develop these disorders.

That is why Parkinson's is the strongest argument against the moral narrative of addiction. In most cases there is no prior drug use, no addictive personality, no social vulnerability; there is a molecule. And when the molecule changes, the horizon changes, whoever it belongs to.

But the most revealing case, and the hardest to resolve, is the one that combines both conditions.

A person with ADHD who is diagnosed with Parkinson's carries two simultaneous deficits: the structural hypoactivity of ADHD, which has spent a lifetime sensitizing the reward receptors, and the neurodegeneration of Parkinson's, which destroys what presynaptic signal remained. The result is a reward system at peak sensitivity, hungry by architecture and further sensitized by chronic deprivation.

When dopaminergic treatment arrives, there is no correct dose. The dose that treats the motor symptoms inevitably overwhelms the reward system, which has been waiting decades for exactly that signal; the dose that doesn't trigger it doesn't treat the Parkinson's. The horizon is trapped between two failures of the same system: with too little dopamine it cannot move, neither in time nor in the body; with enough to move, the reward system activates in a way it cannot control.

This case dismantles any remnant of the moral narrative. In this horizon there is no place from which will could operate as ordinary language assumes: the field is compromised at once in its original architecture, in its present pathology and in the only available treatment. The will has not failed; it had nothing to stand on.

Recovery as reconstruction

If addiction is a recalibration of the threshold, recovery is the reverse recalibration. Will has little to do with it: what it takes is time, conditions and the slow regeneration of the receptors the system stopped producing. Studies suggest that the dopaminergic system can take twelve to eighteen months to recover fully, sometimes longer. Meanwhile, the horizon lives in a world in which nothing generates the signal it needs, except the substance it cannot go back to.

The bond plays a decisive role in recovery. Oxytocin, the molecule of the bond, activates the reward system through a pathway that does not wear down receptors. A recovering horizon surrounded by stable bonds receives enough reward signal to sustain itself without the threshold rising again. That is why community-based recovery programs (twelve-step groups, group therapy, therapeutic communities) achieve better results than solitary attempts. What acts there is the chemistry of the bond, more than the morality of the group.

Note to Chapter 14

What we do know: *The disconnection of the prefrontal cortex by substances and the distortion of the hierarchy of salience*

(dopaminergic recalibration) are real. Parkinson's treatment can cause impulse control disorders. Work addiction is studied as a behavioral addiction. ADHD increases vulnerability to addiction.

What we don't know: *Exactly how the serotonin and GABA networks interact with the dopaminergic system in each type of behavioral addiction. How long full recovery of the reward system takes.*

Open questions: *Why do some horizons with ADHD develop severe addictions while others channel that search for intensity into creativity or controlled risk? Is work addiction less harmful than chemical addiction?*

If you take away only one idea: *Addiction is not a moral failure of the will, but the physical hijacking of the horizon's reward circuits and the raising of its threshold. And recovery demands time, not strength of character.*

Further reading: *Volkow, Koob and McLellan (2016), "Neurobiologic Advances from the Brain Disease Model of Addiction"; O'Sullivan et al. (2009), on impulse control disorders in Parkinson's; Robinson & Berridge (1993), "The Neural Basis of Drug Craving."*

CHAPTER 15

THE MOTHER-CHILD BOND

Chapter 14 presented oxytocin as a pathway for activating the reward system that does not wear down the receptors it needs. But it left a question unanswered: where does the system learn what intensity of bonding signal is enough to register as "being in the right place"?

The answer is that the threshold of the bond is calibrated in the first bond. And the first bond is the only case, in the entire geometry of the horizon, in which one system manufactures the conditions of existence of another. It does not couple to it: it **manufactures** it.

The founding asymmetry

Chapter 13 described love as the recognized entanglement between two horizons that already existed, and Chapter 12 showed that entanglement can be hostile and still be inscribed just as deeply. All those cases share a tacit assumption: that the two horizons involved exist beforehand and meet.

The mother-child bond does not satisfy that assumption.

In motherhood, one of the two horizons **does not yet exist** when the other begins to reorganize itself around it. The mother does not include the child in her self in the sense of Chapter 13: **she reorganizes her architecture for months around a horizon that has not yet condensed**. It is an entanglement with something that is not yet anyone. When the child's horizon finally closes

(the phase transition of Chapter 6), it does so inside a reservoir that was already waiting for it with a specific geometry. The mother does not find the child: she manufactures the reservoir from which the child arises.

This gives the bond an asymmetry that in any other context would be pathological.

In Chapter 13, unrequited love was described as an Einstein-Rosen bridge anchored at only one end. Early motherhood **has exactly that geometry**. The child cannot yet reciprocate, because its horizon does not yet have the integration needed to include the mother in a self that is still forming. And yet the bond works. It does not fail because of the asymmetry: **it is built on it**.

Until now, the book's entire theory of entanglement has assumed that reciprocity is a condition for the bond to hold. Motherhood shows that reciprocity can be born of the bond rather than being its prerequisite. The mother sustains the asymmetry for as long as it takes for the child's horizon to develop enough to participate, and then, in typical development, the asymmetry diminishes without ever disappearing entirely.

The three inscriptions

What culture calls by a single word, "being a mother," is physically three distinct operations, which occur at different times and leave different traces in the child's horizon.

The template, the genetics, is the architecture of the nervous system from which the horizon will be built. But it is not the horizon, only the blueprint.

The womb, gestation, is the only layer that acts before the horizon closes. It includes the hormones that cross the placenta, the maternal heartbeat, the voice, the prosody of the language. From about week twenty-five, the fetus responds to sounds, and by the end of pregnancy it can already tell its mother's voice from other women's, reacts differently to her language and recognizes melodies it has heard repeatedly. That experience, integrated before the threshold, shapes the internal correlations the horizon will have when it closes. It does not determine it, but it leaves a signature no other adult will be able to reproduce.

Upbringing, the encounter, acts after the threshold, between two horizons already formed. That is where the reward system Chapter 14 talked about is calibrated. Every cry that gets a response, every hunger that finds food, every presence that holds steady teaches the child's system what intensity of oxytocin signal is the baseline it will seek for the rest of its life.

The dance of mutual regulation

The developmental psychologist Ed Tronick documented something that seems simple until you understand what it implies: the "still face" paradigm. In the experiment, a mother interacts normally with her baby and then, suddenly and without warning, makes her face expressionless. The baby tries to get her back: it smiles, babbles, gestures. If the mother stays still, the baby becomes distressed: it turns away, tenses up, cries. When the mother starts interacting again, it calms down.

What looks like a simple emotional reaction is something much deeper: in each of these cycles, the baby learns that the other's presence can regulate its internal state. That when something

hurts, someone responds. That the world is not a place of random signals, but of signals that respond to its own.

But there is more. Tronick discovered that what does harm is not the absence of a signal, but the absence of a *response*. A baby can tolerate a lot of frustration if the mother responds, even with a "not now." What it cannot tolerate is indifference, which breaks contingency: it teaches the system that its signals don't matter, that the world does not respond to what it emits.

When the layers come apart

Modern medicine has, without intending to, created four natural experiments that separate these layers. They are not useful for ranking bonds, but for understanding what each layer does.

Situation	What does it contribute?	What it means for the horizon
In vitro fertilization	All three layers, with a 3–5 day pause in a culture medium.	The horizon arises from a reservoir that includes an explicit act of parental intention.
Egg donation	The template from one woman; the womb and upbringing from another.	The gestational mother influences which genes are expressed.

Situation	What does it contribute?	What it means for the horizon
Surrogacy	The womb of one woman; upbringing by another.	The horizon condenses with prosodic and hormonal inscriptions from the gestational carrier. Later upbringing operates on that already configured reservoir.
Adoption	Upbringing by a mother who contributed neither template nor womb.	The horizon is already closed. The adoptive mother does not replace earlier layers: she adds a new one, on a horizon that already carries inscriptions.

The question "who is the real mother?" is badly posed. The three layers are not the same thing, and none of them alone amounts to the complete bond. Adoptive upbringing produces a complete entanglement, not a substitute, but with a different topology: it begins after the threshold, on a reservoir configured by another origin. It is like hearing a language before birth and then speaking it with other people: the music of the accent remains.

Birth as a transition

Birth also happens to the mother: it is a transition in the geometry of the bond. During gestation, regulation was internal: the mother regulated the child without its knowing, through the placenta, the heartbeat, the hormones. After birth, regulation becomes external: the child has to emit a signal to receive a response. Crying replaces the placenta as the channel of communication.

The transition is brutal for both horizons. The child, who had never had to *ask* for regulation, discovers that its well-being depends on emitting signals that someone else can interpret. The mother, who used to regulate effortlessly, discovers that regulating now demands constant attention, deciphering signals that come unlabeled and enduring an exhaustion for which gestation did not prepare her.

The injury to the horizon

What distinguishes a mother's suffering when something happens to her child from the empathic suffering Chapter 16 will describe is very specific: it is not inference, but **injury**.

In ordinary empathy, two separate horizons model each other and register, through that model, an attenuated version of the other's state. There is a bridge, and that bridge has a cognitive cost and a limit of fidelity.

Maternal suffering does not work like that. The mother does not infer the child's state: she registers it as a modification of her own architecture. The founding asymmetry means that the maternal self has reorganized for months around a presence that was not yet a person, and that reorganization left her

architecture **structurally coupled** to the child's horizon as no other bond is. The mother does not need to interpret the baby's cry as a stranger would have to: the cry activates circuits that were calibrated to respond to that specific stimulus before the baby existed as such.

There is a physical fact that illuminates this: **fetomaternal microchimerism**. During pregnancy, cells from the fetus pass into the mother's body and settle in various tissues, including the brain. Decades after the birth, they are still present in her body. The mother literally carries in her body parts of the horizon she manufactured, and when that horizon is harmed or disappears, what she loses is not entirely foreign to her.

Postpartum depression

There is a case in which the reservoir does not respond as it should. Postpartum depression is a failure of the bonding reward system that leaves the mother without the oxytocin signal she needs to sustain the asymmetry. The maternal horizon is still reorganized around the child, but that reorganization no longer generates the reward circuit that makes it bearable.

The result is devastating. The mother can love her child and, at the same time, not feel the pleasure that love ought to produce. The dopaminergic anticipation system keeps working (it knows it has to respond to the crying), but the reward circuit of the bond is switched off. It is like pressing the keys of a piano and hearing nothing.

The failure belongs to the system, not to the mother. And it has consequences for the child: a newly condensed horizon that does not receive the mutual regulation it expects learns from the first day that the world does not respond predictably. That early

lesson can tilt the attachment system toward insecurity, not through anyone's fault, but through the chance of a chemistry that didn't work.

Letting go of the rope

The paradox of this bond is structural: its biological function is fulfilled when the dependence that sustained it is no longer necessary. The success of the maternal bond shows when the child's horizon can regulate itself without needing the mother's presence.

But the founding asymmetry makes that outcome costly for the mother. Her architecture was organized for months around a horizon that did not exist, and then, for years, around one that depended on her completely. When that horizon learns to regulate itself, the maternal architecture cannot simply reconfigure.

What we call the "empty nest" admits a precise description: an architecture organized for two decades around an active presence that now operates at low frequency or at a distance. It is an attenuated presence that the system keeps processing as if it were imminent. The mother voluntarily carries out the self-contraction that in Chapter 13 was the involuntary cost of loss, and she does so without letting the child experience it as a withdrawal.

There is an extreme variant: the adult child who cuts off the bond of their own accord. Because the child is still alive, it never becomes grief; it is something in between that gives rise to a particular figure: the **ghost of the living child**, a horizon the mother manufactured, which still exists and has decided not to answer. The inscription in the maternal architecture keeps

functioning even though the other side has stopped updating it. It is one of the hardest griefs the model identifies, precisely because it does not allow the work of closure that other griefs allow.

When the roles don't reverse

When the mother grows old and comes to depend on the child, the asymmetry changes direction, but **it does not become symmetrical**. People often say that "now the roles are swapped," but that isn't true.

The mother manufactured the child's horizon; the child cannot manufacture the mother's. It can only **accompany a horizon that is dissolving**. Building something that will close and caring for something that is opening back into the reservoir are not the same operation with the sign reversed: one is condensation; the other, dissipation.

This lets the model say something precise about caring for an elderly father or mother: the child who cares is not "their parent's parent." They are doing something that has no name in the usual direction of the bond, because in that direction it does not exist. They should not feel at fault for not giving back the same thing: **physically, they cannot give it back**. The mother manufactured a horizon that did not exist; the child cares for a horizon that can no longer manufacture itself. The love that returns does not need to be symmetrical to be real.

If there is cognitive decline, the child becomes the external storage of the model the mother had of herself: reminding her who she is, what she liked, how she used to react. The child keeps watch over the horizon that once kept watch over them.

The father as another topology

So far the chapter has spoken of the mother. The three-layer model describes physical operations; it is not a cultural claim for motherhood over fatherhood.

In the usual biological case, the father contributes template and upbringing, but not gestation. That does not make his bond lesser, but of **another topology**: the father enters the bond with the horizon already condensed, without the phase of manufacturing the reservoir. His geometry is closer to that of adoption: a full bond after the threshold, without prior inscription. It is not less deep; it is different.

Collective upbringing

The three-layer model assumes a single primary bond. But many horizons condense in environments where several adults share the upbringing: grandparents, aunts and uncles, older siblings, professional caregivers, extended communities. What does that imply for the calibration of the reward system?

Everything suggests that the coherence of the response matters more than the number of caregivers. A newly condensed horizon can calibrate itself with one predictable caregiver or with five. What does harm is incoherence: caregivers who respond in contradictory ways or who come and go without a recognizable pattern. The system cannot learn contingency if the contingencies keep changing. It doesn't matter whether there is one radiator or five, as long as the temperature is stable.

What this chapter closes

Chapter 14 presented oxytocin as an antidote without explaining where sensitivity to it is calibrated. This chapter answers that question. The initial calibration of the reward system, the sense of what intensity of bonding signal is enough, happens in upbringing. Not in genetics, which provides the range of possible responses, nor in gestation, which prepares the reservoir, but in upbringing: in the thousands of episodes of mutual regulation between the newly condensed horizon and the adults who accompany it.

That gives motherhood, and fatherhood, a function no other bond fulfills: it is where the child's horizon learns what it means to feel in the right place. Chapter 14 showed what happens when that calibration shifts; this one shows where it is established. And Chapter 16 will show how that early calibration determines, over a lifetime, how much a horizon can resonate with others without becoming saturated.

Note to Chapter 15

What we do know: *Brain-to-brain coupling between mother and child and fetomaternal microchimerism are real. Untreated postpartum depression is associated with a greater risk of insecure attachment in the child. The still-face paradigm (Tronick) demonstrates the importance of contingency.*

What we don't know: *The exact long-term functional significance of fetal cells in the mother's organs. How multiple caregivers interact in the calibration of the reward system.*

Open questions: *How is the baseline calibration of reward reconfigured when the upbringing layer is deficient but the other two are not? Is caring for elderly parents always asymmetric, or can there be symmetry in some cases?*

If you take away only one idea: *Motherhood is not a single role, but a physical sequence of three layers in which one horizon manufactures the reservoir from which another will arise. And that manufacturing leaves its mark on both, forever.*

Further reading: *Feldman (2015), "The Adaptive Human Parental Brain"; Boddy et al. (2015), "Fetal Microchimerism and Maternal Health"; Tronick (2007), The Neurobehavioral and Social-Emotional Development of Infants and Children.*

CHAPTER 16

EMPATHY AND ITS LIMIT

We tend to confuse empathy with a moral virtue or with a desirable disposition. From the perspective of this experiment, empathy is a physical and geometric phenomenon before it is an ethical one: the capacity of one horizon to model another's frequency and reproduce it in its own architecture.

When we empathize deeply, we do not imagine the other's pain in a figurative sense: our horizon temporarily modifies its geometry to integrate the information the other emits. For an instant, two boundaries overlap and subjective times synchronize. It is the same mechanism that in Chapter 10 generated the time of the bond, applied now to a brief encounter, sometimes lasting seconds, with someone who may not even matter to us especially.

Inference, not injury

In Chapter 15 we established an important distinction: what separates a mother's suffering at her child's pain from ordinary empathic suffering is that maternal pain is not inference, but injury. The mother built part of her own architecture around that child, and its loss or harm tears her from within.

Pure empathy is something else. It is inference, even if somatic, deep and for the most part automatic: the system uses its own structure to simulate the other's state transiently. The networks involved in feeling one's own pain also activate when observing

someone else's, social attention increases, breathing and pulse begin to fall into step with the other's. But that simulation is a superimposed layer. It does not irreversibly modify the architecture of the horizon, only its momentary state.

The distinction has immediate practical consequences. An injury heals or doesn't, but it is there, and the horizon has to reorganize around it. An inference, by contrast, can be maintained or dissolved depending on the resources available. Empathy can be let go; injury cannot.

But the fact that something can be let go does not mean it comes free.

Cognitive empathy and affective empathy

Neuroscience distinguishes two types of empathy that work through partly distinct circuits. **Cognitive empathy**, related to the so-called theory of mind, is the capacity to infer what the other thinks or feels without necessarily feeling it oneself: the surgeon who knows exactly where it hurts but does not feel the pain; the diplomat who understands the adversary's position without sharing it. It is cold, precise, strategic.

Affective empathy, or emotional resonance, is the capacity to feel what the other feels, not through conscious inference, but because one's own body automatically reproduces their emotional state: the tears that come without knowing why, contagious laughter, physical discomfort at the suffering of someone you don't even know. It is the difference between knowing that someone is cold and shivering when you see them tremble.

The two systems can operate separately. There are people with highly developed cognitive empathy and very low affective empathy: they understand perfectly what you feel, but they don't care. They are the manipulators, the precision torturers, the politicians who read the audience's emotions without sharing them. And there are people with a great deal of affective empathy and little cognitive empathy: they feel intensely what others feel, but they don't know how to interpret it, which leaves them confused and exhausted.

The permeability of the horizon

Imagine that each person has an emotional skin of a different thickness.

Some have skin like paper: everything that happens outside gets inside at once. They walk into a sad room and become sad; they see someone suffer and feel that suffering as their own. They are the ones who cry at the movies, who can't bear the war news, who come home from parties exhausted because every conversation has loaded them with something that isn't theirs.

Others have skin like leather: they notice there is sadness, but they don't soak it up. They can be in an emergency room, see the pain, act on it, and then go home and eat dinner with an appetite. They care, but the external signal does not go through them.

Having thick or thin skin is neither better nor worse; it is different. But if you have skin like paper and work in a hospital, you'll have to learn to put on a raincoat from time to time; and if you have skin like leather and have children, you'll have to learn to take off the armor sometimes, or you won't feel what they feel.

Permeability is a structural parameter, neither virtue nor defect, that is largely calibrated during early childhood. As we saw in Chapter 15, early synchrony between mother and child teaches the child not only that the other exists, but to what extent it should treat them as urgent information. A horizon calibrated in an environment where reading the other's state precisely was a matter of survival develops very high permeability. What was a protective skill in childhood becomes a structural vulnerability in adult life: the horizon keeps treating every external signal with an urgency the situation no longer requires. It is a smoke detector so sensitive that it goes off when you boil water.

The finite limit of integration

There is a constraint no horizon can escape. We can formulate it, in the language of integrated information theory, as an upper bound on the amount of information the system can integrate per unit of time, its instantaneous Phi. That bound is finite: what is spent integrating someone else's entropy is not spent integrating one's own.

A healthy horizon resolves that constraint through its capacity to couple and uncouple. It enters into resonance with the other's pain, integrates that information for as long as it takes to offer support or understanding, and then lets go and returns to its baseline frequency. The energy spent in resonance is recovered during the return.

When permeability is excessive, the uncoupling never comes. The horizon keeps resonating with each external signal until the next one overwrites it, with no period of return. Its own baseline frequency stops being the default state and becomes, at best, a memory.

Sustained over time, the result is what clinicians call compassion fatigue or empathic burnout. In the terms of this experiment: a horizon processing external entropy at a rate that threatens its structural integrity. The system has spent so much Phi modeling others that it has no capacity left to model itself. And a horizon that has lost its own model begins to be reservoir without a boundary.

The wall: a solution that isn't one

When a horizon approaches the threshold of saturation, it usually reacts in the only way possible without building anything new: by contracting abruptly. It isolates itself, becomes cynical, stops answering the phone. It puts up a wall.

In a sense, the wall works: it stops the suffering. But it also stops connection and, with it, much of what made the horizon who it was. The abrupt contraction does not return it to its baseline frequency, but to a lower and narrower one, in which nothing else resonates, but in which its own frequency cannot be sustained without effort either. The isolated horizon is not intact: it has lost more than it was protecting. It is like closing all the windows to keep out the cold and running out of air.

The anchor

The way out is not to disconnect, but to anchor.

To anchor a horizon is to maintain an internal reference signal dense enough for resonance with the other to occur at the surface, without destabilizing the core. One's own frequency has to be present enough that the horizon, even while vibrating with the other's, does not mistake it for its own.

In practice, this requires two things:

First, fine interoceptive perception: knowing at every moment how one's own body is physically, what note the horizon sounds when it plays alone. Not in an abstract sense, but literally. Where are you tense right now? Are you breathing fast or slowly? Are you clenching your jaw?

This is what neuroscience calls "interoception," but the term isn't needed; the practice is: listening to what your body is saying before it raises its voice.

Second, a routine of return: any mechanism that lets you find that note again after coupling with someone else. It can be walking alone, music or simply sitting in silence for five minutes before going to sleep.

This is what the clinical literature has rightly called distinguishing empathy from compassion. Empathy without an anchor is coupling without return: the horizon empties itself into the other. Compassion is coupling with return: the horizon models the other enough to respond, without losing its own frequency.

The difference between burning with the other and holding them while they burn is one of architecture. It is the difference between jumping into the water to save someone, and both drowning, and throwing them a rope from the shore.

Don't fix: accompany

Almost everyone, sooner or later, asks the same question when faced with someone they love: a friend going through a breakup, a bereavement, a crisis with no quick solution. What am I supposed to do?

The most common intuition, telling them what to do, is incomplete and sometimes simply wrong: offering them the diagnosis, the solution, the three-step plan. It is understandable, because watching someone you love suffer creates real pressure to act, but the horizon model suggests that it almost always achieves the opposite of what it intends.

Chapter 5 described wu wei as action that does not force: the carpenter who lets the wood tell him where to cut, instead of imposing a shape it does not have. A broken horizon, broken by a loss, by a failure, by any violent reorganization of its architecture, has its own grain and its own pace of repair, the same one Chapter 7 described when it spoke of the wound that heals. Telling someone what to do with their pain usually means cutting against the grain: imposing on them a speed, a direction and an outcome that respond more to the urgency of the onlooker than to the real physics of the system that is healing.

What is needed, then, is the rope from the shore of the previous section: an anchored, sustained presence that does not require the other to change their rhythm in order to deserve it. Chapter 12 described Coan's social baseline: the mere presence of someone familiar already reduces the nervous system's threat response, without any need to say, suggest or solve anything. It is not a step before the real help; it already is the real help.

This does not mean withdrawing or keeping quiet as a rule. It means something more precise: that the usefulness of being there does not depend on being right about what the other should do. Respecting another's horizon means, among other things, accepting that its repair belongs to it, that the same mechanism that closed other wounds without anyone closing them from outside is the one that has to close this one, and that

whoever accompanies is not there to replace that mechanism, but to sustain the conditions that allow it to keep working.

Among the stories that accompany this book there is one devoted to a dog, Txiki, the same dog from Chapter 1. It ends with a sentence worth explaining here: *"His company did not cover up that grief. It repaired it, which is different."* A dog cannot tell anyone what to do; it can only be there. And sometimes, as a ten-year-old boy discovered long before this experiment existed, that is exactly what a broken horizon needs in order to close again on its own.

Empathy as a field of interference

So far we have treated empathy as an operation between two horizons, one that emits and one that receives. But real empathy takes place in social spaces where many horizons emit at once, and the signal from one does not arrive pure, but mixed with everyone else's.

Imagine an emergency waiting room. There is someone crying silently, someone shouting, someone waiting with a petrified face, someone talking on the phone in a broken voice, someone asleep in the next chair as if nothing were happening. The permeable horizon does not receive those signals one by one, but all at once, superimposed, with no filter to sort them by their real urgency.

The anterior insula, the structure that processes the visceral signal of one's own body, activates in response to each signal with the same relative intensity, because the system has not learned to distinguish between an immediate threat and a distant discomfort. The result is interference: the horizon does not resonate with any frequency clearly enough to respond.

The clinical literature documents this: the emergency workers who burn out are not the ones who handle the most serious cases, but the ones who bear the greatest density of signal without recovery intervals. What sinks the horizon is the lack of pauses between stimuli, more than their intensity. The horizon needs time to uncouple, and when it doesn't have it, it accumulates not damage, but disorder. And disorder is worse than damage because it has no recognizable shape: it cannot be named, and what is not named cannot be treated.

Empathic discrimination: how the horizon knows what it feels

There is a prior problem we have not yet raised: to empathize, the horizon first has to know what it feels itself. Empathy requires, besides receptivity to the other, accuracy in reading one's own signal. A horizon that cannot tell its own internal state clearly cannot faithfully simulate the other's: it will simulate its own confusion.

Interoception, the capacity to perceive the body's internal signals, is a necessary condition of genuine empathy; not an accessory one, a necessary one. Various studies show that interoceptive accuracy, measured by the ability to count one's own heartbeats without taking one's pulse, is related to empathic accuracy: those who read their own body best read others' best.

The anterior insula, which processes the visceral signal, is the same one that activates at pain observed in another. There is a single system that works in two directions.

This has an unexpected implication. Empathy is a skill that rests on a more basic one: interoceptive anchoring. A horizon that doesn't know where it is cannot know where the other is in

relation to it. Empathy without interoception remains confused identification, fusion without a boundary, the pathological version of resonance, in which the self gets lost in the other because it never knew where it was.

Adversarial empathy: when the horizon doesn't want to feel

There is a case the usual description of empathy overlooks: empathy with someone regarded as an enemy. Chapter 12 described hostile entanglement as a shared geometry as deep as that of love. Empathy works the same way in both cases: the horizon models the other, reproduces their frequency, integrates their information. The difference lies not in the mechanism, but in the valence, in what the horizon does with that information once integrated.

The torturer who knows exactly what hurts their victim does not lack empathy: they have a highly precise empathy placed at the service of an end we cannot name without horror. The horizon that empathizes in order to harm is not broken; it has disconnected resonance from response. It feels what the other feels, or, more exactly, knows what they feel, and uses that knowledge to do the greatest possible harm.

Empathy, therefore, is a capacity before it is a virtue. The virtue, or its absence, lies in what the horizon decides to do with the information empathy gives it. The empathy of the one who cares and that of the one who tortures probably share a good part of their networks (the insula, the anterior cingulate), although, as this chapter's note points out, this has not been demonstrated; what would change is their functional connection with the prefrontal valuation system. One integrates the other's signal and acts to relieve; the other integrates it and acts to harm.

From the horizon model, the ethical question is not "do I have empathy?" but "what is my empathy connected to?" The horizon can empathize with anyone who emits a clear enough signal. With whom it empathizes and what it does with that is what the model cannot predict, because it depends on its history, its previous bonds, its wounds, its accumulated decisions.

Charisma: when the horizon broadcasts on a dominant frequency

There is a phenomenon that looks like inverted empathy. Instead of receiving the other's signal and reconfiguring itself to resonate with it, the horizon emits a frequency so clear, dense and coherent that it is the surrounding horizons that reconfigure themselves to resonate with it. That is charisma: not the capacity to feel the other, but the capacity to make the other feel, and feel with you, without any conscious decision in between.

> ***In physics this is called:*** *forced coupling or leader-driven phase synchronization, where an oscillator of greater amplitude imposes its frequency on those of lesser amplitude.*

Charisma is not beauty, or intelligence, or virtue. There are beautiful, brilliant and good horizons that lack it, and ordinary horizons that radiate it with disconcerting ease. From the model, charisma is an informational property: the internal coherence of the horizon is so high that its signal reaches others without noise, without contradictions, without the small leaks that usually distract. When a charismatic horizon speaks, the other does not hear words: they receive a state. And states are far more contagious than arguments.

This explains something social psychology has documented for decades: charisma depends not on content, but on the way it is

conveyed. A horizon that doubts itself emits interference, because doubt is information competing with the message. The charismatic one, by contrast, has resolved (or never raised) the question of whether it deserves to occupy the space it occupies. That lack of doubt is exceptionally low background noise: the message arrives clean because there are no other messages at the same time.

But the geometry of charisma is dangerous. A horizon that imposes its frequency can do so to expand the other's freedom (the teacher who teaches, the artist who opens paths) or to cancel it. The difference lies not in the mechanism, but in the direction: generous charisma expands the other's horizon, shows it correlations it had not seen, offers it new frequencies it can adopt or reject. Predatory charisma, by contrast, compresses the other's horizon until it can resonate only with it. It is the sect, the personality cult, the toxic relationship in which the other's charisma becomes your only source of coherence.

The weakness of charisma is that it depends on a coherence no horizon can sustain indefinitely. The charismatic leaders who age badly are the ones who confused coherence with truth: they believed that because their signal imposed its frequency, that frequency was the right one. And when the horizon cracks, because they all crack, the fall is more visible: the signal was so sharp that any new interference becomes intolerable. Charisma ages badly when it has left no room for doubt.

Hence a useful distinction. There are charismas that invite resonance and charismas that demand submission. One says: "Here is a frequency; if it serves you, use it." The other: "My frequency is the only one that counts." The model has no moral vocabulary to tell them apart, but it has a geometric one: one

preserves the other horizon's degrees of freedom and the other reduces them.

The relationship between charisma and empathy is, then, that of an inverted mirror. Empathy is receptivity: the horizon opens itself to the other's frequency. Charisma is transmission: the horizon offers a frequency the other can adopt. Both mechanisms might rest on the same neural infrastructure (the insula, the anterior cingulate), but they work in opposite directions: the empath reads and the charismatic writes. And when someone has both capacities to a high degree, the result is a presence that understands and imposes at the same time: the kind of person you cannot lie to, but also cannot stop following.

That is architecture, and like all architecture it can serve to hold up or to shut in.

Empathy in culture

Not all cultures calibrate empathy the same way. In some, high permeability is a sign of maturity: knowing how to read the other, anticipate their needs, adjust to their state. In others it is a sign of weakness: the mature adult keeps firm boundaries, does not let themselves be carried away by others' emotions and puts reason before resonance.

These differences are not arbitrary; they have to do with social structure. Cultures that value interdependence tend to prioritize affective empathy; those that value individual autonomy, cognitive over affective. Neither is the "correct" one: they are adaptations to different environments.

But there is a limit. Culture can turn the volume of empathy up or down, but it cannot eliminate the need for some kind of

coupling with others. We are a social species: our nervous system develops in the presence of others and needs social signals to function well. A culture that suppressed empathy entirely would produce not autonomy, but pathological isolation.

The limit as a living boundary

Every horizon has a limit. It is a condition of its existence. A horizon without a limit stops being a horizon and becomes reservoir, and the reservoir, by definition, has no inside or outside, and no one to empathize or to be empathized with. Empathy needs a boundary, and the boundary needs a limit.

The empathic limit works like a selective membrane: permeable to what the horizon can integrate without losing itself and impermeable to what would destabilize it. The health of the horizon is measured not by how much it empathizes, but by how much it can empathize without ceasing to be itself. It is a ratio before it is a quantity: the ratio between what comes in and what the system can metabolize.

When the horizon reaches its limit, the other's signal keeps arriving; what changes is the capacity to process it. Empathy forced beyond the limit becomes invasion. The horizon that forces itself to keep resonating when it has no resources left does not care for the other: it betrays itself, and in that self-deception it loses the capacity to truly care for anyone.

Recognizing one's own limit is, in this sense, the condition of all sustainable empathy. It means preserving the only tool that makes empathy possible, even if from outside it looks like selfishness. The horizon that does not respect its limit is not more altruistic: it evaporates before its time, spends its Phi on

resonances it cannot sustain and, when it finally collapses (and it does end up collapsing), stops being useful to anyone.

The right question is not "should I empathize more?" but "where is my limit, and what do I need to be able to return to it?" The answer is never in the other, but in one's own architecture, one's own history, one's own capacity to anchor. The horizon that knows where it ends can touch the other without losing itself; the one that doesn't, however far it reaches, only ever touches them in a confused way.

Note to Chapter 16

What we do know: *Compassion training, unlike sustained empathic exposure, is associated with more positive affect and the activation of different circuits (Klimecki, Singer et al.). Interoception is related to empathic accuracy. Cognitive and affective empathy work through partly distinct circuits. Burnout in caring professions is widely documented.*

What we don't know: *Whether all dimensions of interoception relate to empathy in the same way. Whether "hostile" empathy uses the same networks. How the cultural calibration of empathy varies.*

Open questions: *How can anchoring be taught? Is the empathic limit fixed or trainable? Can a culture eliminate empathy without producing pathology?*

If you take away only one idea: *Empathizing without an anchor is not virtue, but evaporation. And an evaporated horizon cannot care for anyone.*

Further reading: *Singer & Klimecki (2014), "Empathy and compassion"; Ainley, Maister & Tsakiris (2014), "Heartfelt imitation"; Fukushima, Terasawa & Umeda (2011), "Association between interoception and empathy"; Decety (2011), "The Neuroevolution of Empathy."*

CHAPTER 17

GRIEF

Almost everyone who has lost someone they loved deeply knows an experience that ordinary vocabulary does not capture well: the feeling that the absent person still occupies a place in one's inner architecture. Not as a memory in the usual sense, but as a shape, as a space, as the geometry around which the horizon has organized itself for years or decades.

As conventional psychology has described it, grief is the gradual processing of that loss until the horizon reorganizes and the absence stops hurting. From the perspective of the experiment, that description falls short. The architecture of a horizon that has lived for a long time deeply entangled with another does not "reorganize" in the sense of returning to the state before the entanglement. Either it is rebuilt around the absence, and the absent person remains, transformed, as an active shape within the architecture, or it is not rebuilt at all, and the horizon sinks, sometimes for years, sometimes forever.

No way of *moving on* consists in undoing the other's inscription. The only question is what to do with the inscription that remains.

What grief is not

Chapter 12 explained that the model of the other that persists after their death is a predictive system that keeps running without the data that calibrated it. The example was the neighbor: a hostile relationship and a model built out of vigilance, not love.

Grief proper is something else. When the entanglement has been loving, when, as we said in Chapter 13, the horizon recognized it and chose to keep it, what persists after the loss is not only a model of the other, but the part of one's own architecture that was built around that model.

In Chapter 15 we said that the bond between mother and child is not inference, but injury. Grief for someone deeply loved is structurally closer to maternal pain than to ordinary empathy. The horizon does not simulate the other in passing: it has built itself, over decades, around their presence. What is lost, therefore, is architecture.

The architecture with a hole

When two horizons spend forty years entangled by love, besides keeping many memories of the other, each carries inside the geometry that presence has carved out, the way water carves its bed in stone over time. That geometry does not disappear when the water goes.

A widow who does not want her husband to dissolve, the protagonist of one of the stories that accompany this book, puts this intuition with a precision theory struggles to match: *it's the way I'm made inside. Every move I make, I make against that shape or with it.* She is not speaking figuratively. She is describing what the neuroscience of grief is beginning to document: the brain

that has lost someone it was deeply coupled with does not process the world as it did before. The neural networks that had organized themselves around the other keep firing at everything that evokes their presence, their voice, their gestures. The horizon keeps predicting them, keeps making room for them, keeps reserving half its inner time and space for them for months or years after they have ceased to exist.

We can call it **asymmetry after dissolution**. The other has returned to the reservoir, and the part of the entanglement that was on their side has been redistributed through the field, into correlations no instrument can read as a story anymore. But the part that was on the survivor's side is still there, intact, without the complement that closed it: a riverbed without water, one half without the other, an architecture with a hole.

The rites of grief

Over millennia, human cultures have developed technologies to contain grief; not to resolve it, but to contain it. Funeral rites, periods of mourning, dark clothing, temporary abstention from certain activities: none of this is superstition, but social engineering.

The rite does two things at once. First, it marks publicly that something has changed: the community acknowledges that the surviving horizon has undergone a structural reorganization and that for a while it cannot function normally. Second, it offers a provisional container for the chaos. Within the rite, grief has a recognizable shape: there are times to weep, to be silent, to remember and to take up life again. The rite does not speed up grief; it gives it form.

Cultures that have done away with the rites of grief, that expect people to "get back to normal" within days, produce something dangerous: grief without a container. The horizon that is not socially allowed to be broken tries to function normally, and the effort of keeping up appearances consumes the same resources it would need to reorganize. The result is frozen grief: an architecture that does not collapse, but does not rebuild either.

The model that keeps running

The cognitive machinery responsible for this is relatively well understood. During sustained entanglement, the brain builds a complex predictive model of the other: how they move, what they say, what they will probably think in each situation. That model runs continuously, in parallel with the perception of the world, and anticipates the other's responses before they happen. That anticipation is what makes possible the synchrony of the bond we discussed in Chapter 10: two horizons that have been together for years do more than predict each other; they anticipate together what is about to happen.

When one of the two dies, the model of the other is not erased. It keeps running and keeps producing predictions, and each one, when it is not confirmed, when the expected response does not come, generates a prediction error that the system interprets as pain.

And, this is what neuroscience has been slow to recognize, the predictions don't stop because the system cannot stop making them without dismantling part of itself. The model is not a module that can be removed: it is distributed throughout the architecture of the horizon and involves memory, perception, emotional regulation and the sense of time. Switching it off

would mean switching off much of what makes that horizon the one it is.

Seen this way, grief is the difficulty of going on being oneself without dismantling half of one's own architecture.

Complicated grief

The clinical literature distinguishes, not without controversy, between normal grief and prolonged or complicated grief, and usually bases the distinction on duration and intensity. There is a more precise way of formulating it, consistent with what has been said so far: grief becomes complicated when the architecture of the horizon cannot reorganize around the absence without risking structural collapse.

For some people, the other was literally a large part of the structure. The bond had become an integrated support: the model of the other had taken over functions of emotional regulation, internal validation, orientation in time and in one's own story. On losing them, those horizons face not a wound in the architecture, but the possibility that it will crumble entirely.

In those cases, the system usually reacts by protecting a minimal integrity at any cost. It can do so in ways that seem contradictory: actively resisting acknowledging the loss (going on acting as if the other were there, keeping their things untouched, talking to them aloud) or, on the contrary, contracting until every emotion, every desire, every internal activity that might reactivate their model and hurt again is flattened. For the horizon, both are the same strategy: preventing the architecture from trying to reorganize, because reorganizing would mean risking collapse.

Unrecognized grief

Not all grief is for a death. There are losses culture does not recognize as such, and those who suffer them are left without the rites that would give them shape.

Miscarriage is one of them. The mother's horizon had already reorganized around a presence that had not yet condensed, and the loss is no less real for being invisible to others. But culture has no rites to contain it: no funeral, no public mourning, no social acknowledgment that something has been lost. The mother goes back to work and no one knows that her architecture has reorganized around an absence she cannot name.

The death of a pet is another. To those who don't have one it seems exaggerated; to those who do, it is the loss of a horizon that, though not human, was deeply coupled to their own. The predictive model of the dog (when it wakes up, what it eats, how it moves) had been running for years, and the absence of that signal produces prediction errors as real as those of any human loss.

A breakup is also unrecognized grief. The other is still alive, but the entanglement the horizon had named as love has been cut. The architecture reorganized around that bond cannot "move on," because the inscription persists. And culture, which rewards "getting over it," pushes the horizon to recover quickly, without understanding that recovering requires reorganizing, not denying.

Grief in the body

Grief is also physiological. Studies show that people in deep grief have weakened immune systems, altered heart rhythms and persistently elevated cortisol. The "broken heart" is not a metaphor: the stress of grief can trigger takotsubo cardiomyopathy, a disorder in which the heart changes shape under extreme emotional stress.

The grieving body has a regulatory system that has been disrupted by the reorganization of the horizon. Sleep fragments because the prediction system stays active at night. Appetite disappears or soars because the reward system has lost its main reference. The exhaustion is real: the brain spends resources maintaining a model that no longer has any data to feed it.

Continuing to be two

What the clinical literature has begun to call, with some caution, *continuing bonds* offers a more precise formulation. The question is not whether the other remains in the architecture, because they will, they already do, and there is no way to prevent it, but how the relationship with that presence that no longer responds is reconfigured.

The widow in the story had sensed this without putting it into words. She does not want to dissolve him or forget him; she wants to carry him inside without his weight keeping her from moving. That formula, *carrying him and going on*, is what defines a viable grief. What happens is a reorganization: the architecture does not recover the symmetry it had when there were two, but it learns to sustain the asymmetry without collapsing.

It is a matter of recognizing that the dead person is there, without emitting, in the geometry of the living horizon, and of

accepting that this geometric presence is the only one possible now and that, in a way, it is enough.

The loss of a child

There is a case that defies everything said in this chapter. The loss of a child, especially a young one, does not fit the model of grief for a loving entanglement: it is something deeper.

Chapter 15 described motherhood as the manufacture of a reservoir. When the child dies, what is lost is not only an entanglement, but the very condensation the mother helped produce: the horizon she manufactured, which contained part of her own architecture and whose cells she carries in her body. The microchimerism that physically bound her to her child is still there after its death. The mother literally carries fragments of a horizon that no longer exists.

This grief has no name to fit it. More than "complicated grief," it is impossible grief. The model cannot predict how a horizon that has lost the condensation it manufactured reorganizes itself. It can only say that the reorganization, if it comes, takes decades, not months, and that often it does not come.

The experiment does not console

The horizon hypothesis does not resolve grief; it allows it to be formulated more precisely, and that precision reaches something ordinary language does not: the difference between losing someone who was in the architecture (recognized love) and losing someone who was in it without anyone knowing (unnamed love, unresolved conflict, a relationship that never closed). The shape of the hole depends on the shape of the entanglement that left it.

The experiment cannot tell anyone how they should feel. It can tell them, with a little more clarity, what it is they feel: not the absence of something that was, but the presence of something that still is, in the form of an architecture that can no longer be updated. Grief is the name we give to the work of learning to live with an inscription that cannot be erased and no longer receives updates.

It is not a consolation, but it can be a companion.

Note to Chapter 17

What we do know: *The DSM-5-TR (2022) included "prolonged grief disorder" as a diagnostic category, not without controversy. There are neuroscientific models of grief according to which the brain keeps waiting for the absent person (O'Connor). Continuing bonds are a recognized clinical model. Takotsubo cardiomyopathy is real. Unrecognized grief (miscarriage, pets, breakups) has documented effects.*

What we don't know: *How long grief should last. Whether medicalizing extreme sorrow pathologizes it. How a horizon reorganizes after the loss of a child.*

Open questions: *Are continuing bonds an adaptation or an avoidance? How can healthy reorganization be told apart from averted collapse? Why do some cultures heal grief better than others?*

If you take away only one idea: *Grief is not forgetting, but learning to walk with a backpack you cannot take off. And sometimes the wisest thing is simply to keep walking.*

Further reading: *O'Connor (2019), "Grief: a brief history of research on how body, mind, and brain adapt"; O'Connor (2022), The Grieving Brain; DSM-5-TR (prolonged grief disorder); Klass, Silverman & Nickman (1996), Continuing Bonds; Stroebe & Schut (1999), "The Dual Process Model of Coping with Bereavement."*

PART THREE

THE LIMITS OF THE HORIZON



CHAPTER 18

ALZHEIMER'S — THE DISSOLUTION OF THE HORIZON FROM WITHIN

There is a way of getting lost that is worse than death. Sometimes it doesn't even hurt more; the terrible thing is that the horizon fades while it is still there. Death, at least, has the decency to be an event, with a before and an after, a moment when someone ceases to be there and those of us who remain can weep, remember, close. Alzheimer's is a process, and what it destroys is organization. The building does not collapse; the blueprints are erased, brick by brick, room by room, while the occupants go on sitting in the living room without knowing that there is no longer a kitchen, or a bedroom, or a way out.

I have seen whole families bend under the weight of this progressive forgetting. A sudden loss at least sets its own grief in motion; this one repeats every morning: the husband who asks where his wife is while she stands in front of him, the mother who does not recognize the son who looks after her. A small death every day, every day a slightly narrower horizon, until the world shrinks to a room, a window, a ray of light that comes in and means nothing because there is no longer any story to give it context.

Alzheimer's forces us to ask what consciousness is if it can crumble like this, little by little, without a total collapse. If

memory is the armor of the self, what remains when the armor rusts? If identity is a building made of stories, who lives in the ruins?

The blurring edge

In the early stages, the forgetting seems ordinary: where did I leave the keys, what's that actor's name, what did I come into this room for. We all have such lapses, and almost always the name comes back an hour later or the keys turn up in the pocket of a coat we don't remember putting on. In youth, these lapses are anecdotes; in middle age, they bring a twinge of anxiety we cover with a forced laugh. In the first signs of Alzheimer's, they are omens the patient senses but does not name, like someone who hears the foundations of their house creak and prefers to think it's the wind.

The difference is that in normal aging the horizon gets them back: the information was still there, only access was slow; the synapse worked, but the path was clogged. In Alzheimer's, the information ceases to be there. The bookcase stays where it is; it is the books that disintegrate. Beta-amyloid accumulates in plaques between the neurons like an invisible mold. The tau protein unfolds and tangles inside them into neurofibrillary tangles, as if the internal structure of each neuron had decided to abandon its shape. The neurons do not die all at once: they lose their axons, their dendrites, their capacity to communicate with their neighbors. Tissue stops being tissue and turns into silence.

Normal forgetting is a library with a broken catalog: the book still exists, it is just hard to find, and sometimes it takes the librarian, a clue or the smell of paper that evokes the title. In

Alzheimer's, the library itself disintegrates. The bookcase keeps its hollow shape for a while, but the book is gone. And the cruelest thing is that the librarian, who is the patient himself, keeps walking the aisles in search of something he no longer knows, feeling that something essential is missing without being able to name it.

From the horizon model, normal aging is a slowing of access; Alzheimer's, a degradation of the architecture itself. Not all the neurons die at once: the connections die. The horizon does not switch off; it tears and loses area. The boundary that separated inside from outside becomes porous, irregular, full of holes through which what used to be contained escapes: the daughter's name, the address of the house where he lived for forty years, the meaning of the word "hunger." The horizon becomes an oil slick on water that breaks up into smaller and smaller drops, each isolated, each a world without context.

In systems design terms, Alzheimer's is a **catastrophic failure of encapsulation**. In programming, an encapsulated object guarantees that its internal state is protected from outside modification and maintains its structural invariants. If its boundary degrades, the private state begins to become corrupted and to leak. System calls try to use broken pointers or areas of memory that have already been freed. The public interface (the API that allowed interaction with the patient) returns errors or becomes opaque, while the internal "librarian" walks aisles whose indexes have been wiped from the allocation table.

There is something especially heartbreaking about the intermittent lucidity of the early stages. The patient knows that he forgets, knows that something is slipping away. It is like trying to hold water in your hands: you feel there is something there, but every time you close your fingers there is less. That

awareness of forgetting, what neuropsychologists call metamemory, is the last guardian of the horizon and the first to go out. When it goes, the patient stops knowing that he does not know. The horror gives way to a kind of blind peace, but first it leaves a deep mark on those watching from outside. Watching someone lose themselves is witnessing a dissolution that should not be visible.

The returning reservoir

In Chapter 6 I described birth as a condensation: the reservoir closing in on itself to form a horizon. Alzheimer's is the reverse movement, but not a symmetrical one. What happens is a dissolution from the edges, uneven, an erosion that advances at different rates depending on the region of the brain, very far from an orderly opening toward the reservoir that would undo what was done at birth.

Episodic memory, the story of the self over time, is the first to go. Autobiographical memories fade in reverse order of acquisition: first the recent, then the intermediate and finally the oldest, as if the horizon were unrolling backward and undoing the story of the self in the opposite direction. The patient stops remembering what he had for breakfast, then what year it is, then that his brother died ten years ago. Sometimes he relives conversations from his youth with surprising clarity while failing to recognize the person in front of him. The past is not erased all at once, but from the present backward, like a tide going out and leaving only the oldest foundations exposed.

Then semantic memory goes: what an apple is, what "husband" means, what a church is, what a car is for. The world loses its labels. The objects are still there and still seen, but they are no

longer *things with names*, only pure presence without category. In intermediate stages, the patient may still recognize an apple as food, as something round and red, but the word "apple" no longer floats beside it. Language comes away from the world like a seam coming unstitched. And without language, the world becomes immense and unintelligible, a chaos of nameless shapes in which everything is new and, therefore, nothing is familiar.

Then language itself falls. Not only the words that designate things, but grammar, syntax, the capacity to join subjects and predicates. First concrete nouns disappear, then complex verbs, then subordinate clauses. In the end there are monosyllables, gestures, sounds. The horizon does not contract toward the center, but from the edges, like ice melting at its ends. The image is of a territory losing its borders at random, not of a sphere shrinking while keeping its shape: isolated islands of ice remain in waters that no longer remember winter.

What remains in the final stages is not "nothing." That is the most common mistake, the one that leads some people to speak of the patient as if he were already dead. What remains is a minimal horizon: perception without story, present without past or future, sensation without a name. The patient can feel cold, hunger, restlessness, calm. He can look at a face and feel something he cannot call "love," but which works like love in his body: the pulse that slows, the breathing that calms, the pupils that dilate in a non-declarative recognition. It is a horizon so reduced that from outside it seems empty, but from inside, from the perspective of the system itself, it is still a here, a now, a something.

Some patients in advanced stages still respond to the music they heard when young, to a child's voice, to the touch of a familiar

hand. Not because they "remember" in the ordinary sense, since explicit memory no longer works: there is no mental image of the song or association with a specific moment of youth. Those frequencies resonate in the residual horizon, in the correlations that have not yet broken. It is like a guitar with only two strings left: the whole melody cannot be played, but the remaining strings vibrate with the implicit harmony of the entire song. The body remembers what the mind has forgotten. Some networks hold out much longer than others: those of musical and procedural memory, those that sustain the emotional response to a familiar voice, the brainstem that keeps setting the rhythm of life; and that despite the fact that the hippocampus, the gateway to episodic memory, is among the first to fall.

This is what distinguishes Alzheimer's from death. In death, the horizon evaporates as a whole; it is a continuous process, but it keeps its internal coherence to the end, and the information is scrambled as a unit. In Alzheimer's, some parts persist while others are gone. The patient can be "closer to the reservoir" at some moments than at others; the boundary fluctuates. Sometimes, for seconds or minutes, a familiar face clears the fog and the horizon expands for a moment, recovering connections lost months ago. Then it closes again. It is a door that opens and shuts at random, gives a momentary glimpse of the garden that was, and returns us to the empty room.

Time that no longer flows

When episodic memory dissolves, time stops being a river and becomes a puddle. There is no yesterday or tomorrow; only this instant, and this one, and this one, each without cause or consequence, without history. The patient does not live in the present like the mystic who has reached timelessness through discipline, but because time has broken in two for him.

From outside, the patient's time looks like a tragedy of repetition: he asks the same question every five minutes, wants to see someone who died years ago, doesn't understand what he is doing in that house. But from inside, every question is new, every anguish is the first, every confusion is original. The patient does not suffer the repetition because he does not register it. We who watch and see the loop suffer it; he lives a spiral he does not know is closed.

This loss of narrative time bears an unsettling resemblance to certain advanced contemplative experiences. Mystical texts speak of the "eternal now" as a liberation. But the difference is radical: the mystic chooses to let go of the story, and does so from an intact horizon; the patient with Alzheimer's loses it without choosing, from a horizon that is crumbling. The timelessness of the enlightened is an ocean; that of Alzheimer's, a shipwreck. Both are in waters without shores, but one swam there on purpose and the other fell overboard.

This raises an uncomfortable question about the conscious "now." Is the present a window that needs the frame of past and future to exist as experience? Or is the pure present, sensation without story, a legitimate, if impoverished, form of consciousness? The horizon model suggests that the pure present is still a horizon, only of minimal area: the

consciousness of a newborn, of an animal without inner language. It is consciousness in its most reduced form, the hard core that holds out when everything else has gone.

Language as the last dike

Language is the last bastion of the human horizon, and its fall is the quietest of catastrophes. When a patient loses the capacity to name, he loses not only labels, but the possibility of sharing his world. Pain that cannot say "pain" is still pain, but without a bridge to another horizon. Joy that cannot say "how good to see you" is still joy, but it bounces off the walls of a self that no longer has windows. They are two lighthouses facing each other across a foggy bay: both are still lit, but the beam of one no longer reaches the other.

The aphasias of Alzheimer's are especially cruel because they often preserve the music of speech while destroying its meaning. The patient may go on speaking with intonation, rhythm and expressive pauses, but the words are substitutions ("thingy" for "spoon," "the praying place" for "church") or outright gibberish that almost sounds like a language. It is like hearing a symphony with every note in tune, but in random order: the musical form persists, but the information has fled. The horizon keeps emitting signals, but they no longer point to anything.

When language is reduced to monosyllables and gestures, a strange enforced intimacy arises. The caregiver has to learn to read what used to be said: a gesture of the hand, a tilt of the head, the tone of a moan, everything becomes vocabulary. It is like going back to the infancy of the species, to communication before words. But this regression is a forced rupture that leaves

the caregiver translating between a lost language and a silence that does not want to be silence.

The caregiver as an external horizon

When the horizon itself can no longer sustain identity, another can do it on its behalf. The caregiver, especially if he or she has shared decades of entanglement with the patient, does not merely stand in for functions; they are not an assistant who changes diapers and hands out medicines. They are the custodian of the model the other had of himself: they keep alive, in their memory, in their language and in their affection, a version of the patient the patient can no longer maintain.

In Chapter 15 I described how the mother builds the child's horizon before it exists: she talks to it during gestation, tells it who it is, passes on a world to it before it has eyes to see it; she is, literally, the reservoir that condenses into a child's horizon. Here the reverse happens: the adult child sustains the mother's horizon when she no longer can. It is a different operation. The mother manufactured a horizon that was going to close and gain autonomy; the child cares for a horizon that is opening, losing its boundaries and receding toward the reservoir.

The caregiver who knows the patient well can anticipate his needs before he expresses them, interpret gestures incomprehensible to a stranger and keep active, through resonance rather than instructions, the correlations that still work. They know that when the patient walks back and forth three times between the kitchen and the living room it is not restlessness, but that he is looking for the bathroom and has forgotten where it is. They know that if he refuses food it is not a whim, but fear of choking, because he no longer controls his

swallowing well. They know that the vacant gaze toward the window is sometimes the only way to process a world that no longer makes sense.

It is an entanglement that works in only one direction: the caregiver models the patient with increasing precision while the patient loses the capacity to model the caregiver. In physics, entanglement is always mutual; this is not. In life, it is the most unequal love there is: giving without receiving recognition, holding without being held, remembering for two.

This raises an uncomfortable question: to what extent does the patient's self go on existing in the space between the two horizons? If identity is holographic, if it is distributed in the correlations among all the parts of the horizon, when some of those correlations pass into the caregiver's horizon, does part of the self go with them? If my wife reminds me who I was, if my son tells stories of my youth, if my caregiver anticipates my fear before I feel it, where exactly is the limit of my horizon? Does it end at my skin, or does it extend to the horizon of the one who cares for me?

I have no answer. I have only the question and the certainty that those who have cared for someone like this know what I mean without needing it explained. They have felt, between exhaustion and love, that the patient lives, halfway, like a ghost, but lives, in the space they share. They have felt that their memory of the other is not just recollection, but dwelling; that the model they keep in their head is not a replica, but an extension, a territory in which the patient's self still holds citizenship.

And there is something more: the caregiver is transformed too. Caring for someone who is dissolving is a mirror that anticipates

our own possible decline. Each day of care is a lesson in the anatomy of the horizon, a live demonstration of how fragile everything we take for granted is. The caregiver learns, in body and soul, that identity is a contingent loan. And that learning, though it hurts, is a form of wisdom that cannot be acquired any other way.

The memory that doesn't know its name

Between the loss of episodic memory and the disappearance of language there is an intermediate stage that is especially disconcerting: the world is still recognizable, but can no longer be named. The patient knows what a chair is, because he sits on it, but cannot say "chair." He knows the person in front of him is important, because his body calms and his face lights up, but he cannot find the word "son" or his name.

This stage reveals something important about the architecture of knowledge: knowing how and knowing what do not live on the same floor of the building. A patient may go on playing the piano if he learned as a child, go on walking around his neighborhood or using kitchen utensils, all without being able to say his own name. The body keeps maps the mind can no longer read aloud. It is the most compelling proof that consciousness is a federation of subsystems that can fail separately.

For the model, this means that the horizon does not degrade uniformly. Some regions, the oldest, the most practiced, the most automated, hold out longer than others. The horizon is a territory with fortresses and porous borders. Procedural memory is a besieged stronghold that holds out when everything else has fallen, and its resistance tells us that the horizon has

layers, strata, a geology of consciousness in which the deepest is the last to disappear.

Death as closure

In the final stages of Alzheimer's, death stops being the enemy and becomes the closing of something that had already gone long before. Relatives who have watched someone disappear piece by piece sometimes experience the death not as a loss, but as the acknowledgment of a loss that had already happened. In those cases, grief does not begin with death: it ends with it. They have been weeping for years, fragment by fragment, and death only confirms what already was.

From the horizon model, complete evaporation is preferable to partial dissolution. A horizon that evaporates returns to the reservoir as a whole; its information is scrambled, but the process is coherent. It is not lost in pieces, but transformed all at once, and keeps some form of unity, even if only that of the process. A horizon that dissolves from the edges never finishes leaving or staying. It is an intermediate state without a name, an informational limbo with enough structure to suffer, but not enough to understand.

It is the worst of worlds: enough consciousness to suffer and not enough to understand why. In the intermediate stages, the patient can feel fear without knowing what he fears, anguish without an identifiable cause, a nameless unease. It is suffering without an object, a pain the sufferer cannot diagnose. The caregiver interprets it, names it, contains it, but the patient lives it raw, like a wounded animal that does not understand its wound.

This is not an argument for euthanasia; that decision belongs to other domains, ethical, legal, personal. It is only an observation about the geometry of suffering. There are forms of destruction of the horizon crueler than total destruction, degrees of being worse than not being. Alzheimer's forces us to consider that death, far from being the worst fate, can sometimes be the last mercy of an irreversible process.

But there is another side, and it would be false to keep quiet about it. Some caregivers describe moments of strange tenderness in the final stages: the patient who no longer knows who you are, but smiles at you with a smile that seems genuine, free of all baggage, all history, all resentment. It is like meeting someone for the first time every day and, every day, that first impression being one of trust. The reduced horizon has less room for rancor: there is no memory of offenses and no scars in the story. In its extreme impoverishment it reaches a kind of terrible innocence, a purity no saint would choose, but which exists, which can be observed and which is a way of being, even if none of us would wish for it.

What Alzheimer's reveals

If consciousness were a substance, Alzheimer's could not happen this way. A substance is either there or not, the way alcohol makes you drunk or doesn't, or poison kills or doesn't. But the horizon is an architecture: it can lose area, connectivity and correlations and still be a horizon, smaller and smaller, poorer, more fragmented, but a horizon all the same. It is still a limit, a boundary between inside and outside, even if the inside is an unfurnished room and the outside, pure indifference.

Alzheimer's shows that consciousness does not work like a switch, but like a continuous gradient that can be traveled in both directions, and the descent is not symmetrical with the ascent. Birth is condensation: the reservoir organizes itself, takes shape, differentiates into parts that correlate. Death is evaporation: the horizon returns to the reservoir as a whole, becomes disorganized, but keeps its unity to the end. Alzheimer's is erosion: the horizon loses pieces without losing its status as a horizon, degrades by sectors, becomes a map with blank areas where there used to be inhabited territory.

And erosion teaches something condensation and evaporation do not: that the horizon is fragile not because it can be destroyed, since everything can be destroyed, but because it can be destroyed piecemeal while keeping in each piece the illusion that there is still a whole. That is the cruelty of Alzheimer's: not annihilation, but simulacrum. The patient seems to be there, and in a way he is, but that "there" is ever narrower, ever hollower, ever more incomplete. It is a horizon that deceives about its own extent, that keeps up a facade of integrity while behind it the walls are crumbling.

Alzheimer's is, in short, a lesson in humility for any theory of consciousness. It does not let us think of the mind as a simple entity, an indivisible soul or a program running on hardware. It forces us to think of it as an ecosystem, a network, a fabric that can thin out until it almost disappears without ceasing to be, for years, a way of being in the world. Each patient shows that consciousness is emergence, not essence: that it can be built slowly, over decades, and dismantled slowly over as many again; that there is no single moment when it "switches off," but a long dusk in which each sunset lasts months.

When I see an advanced patient look at a flower without knowing what it is, but smile at its color, I think of the smallest possible horizon: perception without knowledge, affection without history, presence without identity. And I think that this minimal horizon is still, in some way, a form of being conscious. It is not the consciousness any of us would choose, but it is consciousness: the proof that the phenomenon can be reduced almost to the bone and still stand.

Alzheimer's shows us the floor. It strips away the furniture, the curtains and the paintings until only the structure is left, and then it begins to remove beams. But as long as one remains, as long as a single correlation remains that relates an inside to an outside, there is a horizon, a world, however small, and someone, however diffuse, who is having an experience.

That is the ultimate limit of the model: not the question of why there is consciousness, but the realization of how much it can lose without ceasing to be. Alzheimer's is the antithesis of birth: if birth is the miracle of condensation, Alzheimer's is the tragedy of dissolution. But both teach the same thing: that the horizon is not a given, that it is a construction, a contingent work and therefore a work that can be undone. That consciousness is a loan that one day, in some way, is repaid.

Note to Chapter 18

What we do know: *Alzheimer's progressively destroys neural connections before it destroys the neurons themselves. Episodic memory, tied to the hippocampus, is usually the first to degrade; procedural and musical memory hold out longer.*

What we don't know: *Why some advanced patients respond to familiar stimuli without functional explicit memory. Where the exact threshold lies at which a horizon ceases to be one.*

Open questions: *To what extent does the self of a patient with Alzheimer's reside in the entanglement with their caregivers? Is it possible to measure the "mass" of a contracting horizon?*

If you take away only one idea: *Consciousness is not a switch that turns off, but an architecture that can lose its bricks one by one and still hold up the sky with less roof every day.*

Further reading: *Jacobsen et al. (2015), "Why musical memory can be preserved in advanced Alzheimer's disease"; Sacks, O. (2007), Musicophilia; Kitwood, T. (1997), Dementia Reconsidered.*

CHAPTER 19

PARKINSON'S, TRAUMA AND OTHER BROKEN GEOMETRIES

Addiction is a hijacking. Something, the substance or the compulsive behavior, takes control of the horizon and diverts it toward an attractor the subject has not chosen. The addict *knows* he is destroying himself, but a signal stronger than survival itself has seized his reward system. It is as if an intruder had entered the control room and switched off every screen but one.

Parkinson's is something else: a decoupling.

The horizon remains intact and the intention is clear. The inner world keeps its coherence, its music, its desire to move. But the body no longer responds. The signal goes out, the score is written, but the musicians don't listen.

The intact horizon

In Parkinson's, the dopaminergic neurons of the substantia nigra degenerate gradually, one by one, in a silence the body does not betray until the loss is massive. The motor consequences are well known: resting tremor, rigidity of the limbs, slowness of movement (bradykinesia, which turns the simplest gesture into a task of minutes) and, over time, postural instability, which makes every step a gamble. But something deeper than movement also fails, something classical neurology took a long time to name because it leaves no trace on imaging tests or in the reflexes tapped by the hammer.

Subjective time is altered.

The patient generates intentions at normal speed. His horizon works: the *now* is still an active edge, a boundary where the past is integrated and the future anticipated. But the body does not carry them out at that speed. It is not, at least in the early stages, that he thinks more slowly (over the years the disease can also affect attention and memory, but many patients in fact describe an inner agitation, a mental hurry that contrasts cruelly with the slowness of their movements), but an almost unbearable paradox: he does not feel that he is moving slowly. He feels that he is trying to move properly, with his usual urgency and clarity, and that something external, the world, the body, gravity itself, is not responding.

This is different from Alzheimer's. There the horizon disintegrates, fragments, loses the narrative coherence that lets one say "I was, I am, I will be"; its membrane cracks and the information escapes. In Parkinson's, the horizon remains the same, perfectly coherent and intact; it just can no longer talk to the body. It is like a conductor who goes on conducting

passionately while the musicians go deaf one by one. The music is in his head and the harmony is real to him, but it doesn't come out. And the worst thing is not the silence, but that sometimes he doesn't know the others aren't hearing what he hears.

There is something of Greek tragedy in this dissociation: a consciousness condemned to fullness without execution, which sees what is happening coming with absolute clarity and can do nothing to avoid it. The horizon of the patient with Parkinson's is still a perfect circle, only it no longer coincides with the one the body traces in space.

Time that decouples

Subjective time is density of integration: how much new information comes in per unit of real time, how many edges of the horizon are updated, how many gradients cross in each second lived. When we are bored, time stretches because the density of integration is low: few changes, little novelty. When we are in danger or in love, time becomes dense because the density is extremely high: too much information per second, too many events to integrate.

Dopamine calibrates that internal clock. It is not the only neurotransmitter involved (glutamate, serotonin and opioid peptides also play a part), but it is the main metronome. When it runs short, as in Parkinson's, the calibration fails: the internal clock keeps beating time, but no longer in step with the world's.

The horizon goes on generating "nows" at its own pace, and each is still a unit of integration, a mouthful of coherent experience. But that pace is no longer synchronized with the outside world.

The patient who takes three minutes to cross a room, three minutes that feel endless to whoever is with him, does not feel them as endless, but as normal. Or, more exactly, he does not experience them as slow, because his horizon has not generated fewer "nows" than it should. What he feels is a lag, as if the world had gone into fast motion while he carries on at normal speed.

As some patients describe it, treatment with levodopa (the dopamine precursor that crosses the blood-brain barrier and is converted into the missing neurotransmitter) restores synchronization before movement: first they notice that "the world is back to its speed," and only afterward that "the body is moving again." Time is repaired before the body, as if dopamine acted first as a temporal calibrator and then as a motor facilitator. It is like adjusting the tempo of the score before the musicians can play in unison.

When levodopa works, the horizon goes back to sharing other people's time before it goes back to sharing their space: simultaneity is restored before motor function. That suggests something important about the architecture of the horizon: shared time is more fundamental than shared gesture. We can move our bodies in temporal solitude, as in a dream, but we cannot truly be with others if our "nows" do not coincide.

There is also a lesson for medicine. For decades, Parkinson's was treated as a disease of movement. But patients knew, and many said so in interviews neurology did not know how to listen to, that the first thing they lost was not strength, but synchrony; that they felt "out of phase" before they felt slow; that they lived in a slightly different, parallel time, in which intentions were still valid but no longer coincided with their execution. The horizon model makes it possible to hear that: to take the patient's experience seriously as data, not as an epiphenomenon.

Trauma as frozen geometry

If Parkinson's is a progressive decoupling between intention and execution, trauma is something subtler and older in the history of the brain: a geometry that froze at a particular moment and has deformed everything else ever since.

When an event overwhelms the system's capacity for integration, when the information is too intense, fast or contradictory to be processed in the available "now," the horizon does not reject it; it can't, because rejecting requires processing. Instead, it encapsulates it: it stores it unprocessed in a watertight compartment, hermetically sealed. It does not forget it (forgetting would be integrating, assimilating it noiselessly into the past), but stores it without distributing it. The information is still there, intact, in its original form, like a fly in amber.

In software engineering terms, this encapsulation works like **sandboxing**. To prevent a critical exception (emotional collapse) from propagating an error that would crash the entire operating system of the self, the system isolates the corrupted fragment in a closed container. That container does not communicate with the main flow of events (the ordinary *event loop*) and is ignored by the garbage collector: it cannot be *scrambled*. It keeps running in the background like a zombie process or a thread stuck in *deadlock*. When an outside stimulus touches the container's API (the *trigger*), it responds by launching its frozen exception handler (the *flashback*) without going through the global timeline.

Trauma is information that could not be scrambled. In information theory, *scrambling* is the process by which local

information is redistributed into global correlations and stops being a hot spot to become uniform heat: experience, narrative memory. The trauma should have become global correlations, assimilated experience, a memory that tells a story. Instead, it remains trapped at a point, like a singularity that deforms everything else. What happened is not forgotten: it is still a *now*, not a *then*.

The symptoms of post-traumatic stress are manifestations of that gravitational deformation. A *flashback* is not a memory: there is no narrative distance, no "memory of," no possibility of saying "that happened to me at another time." It is the sensation that the horizon is crossing the same point again and recreating the geometry of the traumatic moment as if no time had passed. The body reacts with the same adrenaline, the same trembling, the same dissociation. The prefrontal cortex, which should say "this is over," is disconnected or overridden. The horizon does not *remember* the trauma: it *relives* it, because for that part of the system it never stopped happening. It is like a scratched record: every time the needle passes over the damaged groove, the sound itself plays again, intact and torn.

For that part of the horizon, time has not passed. The information remains trapped in the same instant, waiting, in a sense we don't fully understand, to be integrated at last. It is like an informational debt the system cannot pay off, because the creditor no longer exists or because the currency has changed.

The body that remembers what the mind cannot name

There is a dimension of trauma the horizon model illuminates with particular clarity: the body as a repository of unintegrated information. When the horizon encapsulates a traumatic event, it does not store it only in the cortex (in fact, the cortex often has no narrative access to it), but in the body: in the tension of the trapezius muscles, in the hunched posture, in the held breath, in the vagal tone that does not recover. The body becomes the archive of what could not be processed.

This explains something body-oriented therapists have known for decades: that trauma is not cured by talking alone, because it is not in language. It is in the geometry of the body, in the topology of muscular tension, in a heart that cannot slow down. To "unfreeze" it one has to intervene in the body: in breathing, in movement, in posture. The body is where the horizon deposited the information it could not integrate, without that making it *the key* to trauma.

Complex trauma and blurred edges

Not all trauma is a single event. Complex trauma, the kind born of years of abuse, neglect or a hostile environment, does not have one clear "scratched record," but thousands of scratches, thousands of microtraumas that accumulate until the entire surface is deformed. In that case, the frozen geometry occupies a whole region of the horizon in which integration fails chronically.

The result is a horizon that works, but with a permanent zone of shadow. The person can live, work and love, but there is always a part of the "now" gripped by unprocessed information, a phantom gravity that pulls everything down. It is chronic depression, structural dissociation, the feeling of never being

entirely *present* because part of the horizon is still trapped in another time.

Neurodiversity as a different topology

The horizon model makes it possible to speak of autism, ADHD, schizophrenia and other conditions not as deficits to be corrected, but as different topologies: different ways of organizing the same information, valid informational architectures that do not coincide with the statistical norm.

In physics, topology is the way the points of a space are connected. Two spaces can have the same local metric, the same distances at small scales, and different topologies: one may be a sphere; another, a torus; another, a plane with holes. The local rules are the same, but the global properties (which paths are possible, which regions are connected, what information can circulate) are very different.

The neurodivergent brain has a different topology.

Autism: local hyperdensity

In autism, information is distributed differently from childhood. Many functional connectivity studies find denser local connections (sensory hyperconnectivity) and less efficient long-range connections. The horizon is extraordinarily rich in local detail, at the cost of more fragmented global integration.

From outside, this looks like social difficulty, rigidity, a need for everything to stay the same, meltdowns at unforeseen changes. From inside, it is possible, and more and more autistic voices describe it this way, that the horizon experiences densities of

information neurotypical people do not reach. That "sensory overload" is not a filtering failure, but an input of data that is too faithful, too detailed, too rich to integrate at the pace the world demands. It is a microscope in a world made for magnifying glasses.

The problem lies not in the autistic topology itself, but in the friction between that topology and an environment that presupposes another. A world with fewer simultaneous stimuli, predictable routines, time to process and respect for internal rhythms would not "cure" autism, because there is nothing to cure, but it would greatly reduce the suffering of that topology when it has to function in an environment hostile to it.

ADHD: the shifted threshold

In the terms of the model, ADHD is a structural dopaminergic hypoactivity in the circuits of reward and sustained attention. The horizon does not generate enough internal reward signal to keep attention on tasks that are not stimulating in themselves. What is missing is fuel for the will, which in this model is an informational function that needs dopamine to sustain itself, not an act of moral strength.

The horizon with ADHD seeks stimuli that raise the signal: urgency, novelty, risk, conflict, intense interests. This has nothing to do with impulsiveness in the moral sense: its reward system needs intensities the ordinary world (repetitive tasks, bureaucracy, traditional schooling, office work) does not provide.

There is a quiet heroism in this. Adults with ADHD who have built lives that work have not done so by "overcoming" their condition, but by finding slopes: emergency professions, intense

creativity, environments with a lot of novelty, passionate relationships. They have built themselves a world that gives their engine the incline it needs, and in that world their topology is an advantage.

Schizophrenia: the horizon that generates without stimuli

Schizophrenia is perhaps the hardest condition to fit into the horizon model, because it is not just a different topology, but a horizon that generates integration where there is no information to integrate: voices without a speaker, intentions without an agent, connections between events that in the shared world are unrelated.

It is as if the brain's prediction system, the apparatus that builds models of the world to anticipate what will come, switched on without sensory input and produced models of other horizons that aren't there. The patient hears voices not because his ears are failing, but because his predictive system generates predictions of speech where there is no speech. He sees connections not because he is "paranoid," but because his integrating apparatus assigns correlations where the rest of us don't see any. It is a radar that detects signals where there are no planes.

Here the suffering is deep and real. On top of the friction with the environment comes living in a world in which predictions do not match reality, in which the internal model is chronically out of phase with the external one. In the terms of the model, antipsychotics, which block dopamine D2 receptors, work by turning down the gain of the predictive generator: they reduce the system's sensitivity to internal signals and force a closer correspondence between what is generated and what is perceived.

But even here the model agrees with something computational psychiatry has been exploring for years with predictive coding models: that schizophrenia is not an "absence of reality," but an "excess of internal generation." That the patient is not "disconnected" from reality, but "overconnected" to his own predictive apparatus. That the difference is not qualitative (they see things that don't exist and we don't), but quantitative: we all generate predictions, we all have automatic thoughts, we all hear our inner voice. In schizophrenia, the volume of that internal generation exceeds the threshold beyond which it can no longer be distinguished from external perception.

Resonance between different horizons

One consequence of the model that is rarely discussed is what happens when two horizons with different topologies try to communicate. In this framework, empathy is not just "putting yourself in someone else's shoes," but synchronizing two informational topologies whose rules of integration may be incompatible.

A neurotypical horizon and an autistic one do not just perceive different things: they perceive in a different *way*. One integrates top-down, giving priority to context, intention, social gesture; the other, bottom-up, giving priority to detail, sensation, exact sequence. When they meet, their horizons generate "nows" with different internal structures.

Historically, the adaptation has always fallen on the divergent horizon. It is the autistic person who must learn "social skills"; the person with ADHD who must "concentrate"; the person with schizophrenia who must adapt, and the treatment they often

genuinely need ends up being the only adaptation offered to them. Rarely is the neurotypical environment asked to adapt its frequency, turn down the volume or respect the other's rhythms of integration.

The horizon model suggests that this asymmetry is not necessary: resonance is possible if both horizons are willing to modulate their frequencies. Communication between different topologies does not require one of them to disappear, but for both to find a common rhythm, even if temporary and partial.

What is not broken

Calling these conditions "disorders" presupposes a standard horizon from which they deviate, a "correct" architecture of which all others would be deformations. But if the horizon is an emergent informational architecture, there is no reason to assume that only one is valid.

There are horizons that process, integrate and resonate in different ways. Some see the world through a microscope and others through a wide-angle lens; some need a slope to get started and others work better on the flat; some generate music that makes no sound and others hear music where everyone else hears only noise.

This does not deny the suffering, which is real when the topology cannot function in the world around it. And in schizophrenia it does not come only from friction with the environment: there, treatment is not a concession to the majority, but often what gives the horizon back a shared world. But suffering is not the same as being different. An autistic horizon suffers in a world

designed for neurotypicals, and that does not mean its topology is wrong, but that the world has no room for it: the stairs have no ramp, the lights flicker too fast, the instructions presuppose a speed of integration that isn't its own.

The model cannot say what world we should build; it is not a political program or a treatise on urban planning. But it can say something medical language tends to hide: the difference lies not in the quantity of consciousness, but in its form. A horizon is not more or less real, or more or less worthy, or more or less human for being different.

And the suffering of many of these conditions comes not from the topology itself, which is what it is, neither better nor worse, but from the friction between that topology and an environment that presupposes another. Friction generates heat; heat, pain; pain, pathologization; and pathologization, a prison of labels from which it is very hard to escape.

Note to Chapter 19

What we do know: *In its early stages, Parkinson's affects mainly movement and leaves cognition largely intact; over the years it can also affect attention, mood and memory. Dopamine is involved in time perception as well as in movement. Traumatic memories tend to be intrusive, fragmented and poorly integrated into autobiographical memory. Neurodivergent conditions show measurable differences in brain connectivity.*

What we don't know: *Whether levodopa restores temporal synchronization by the same mechanism as motor function, or by independent pathways. Whether trauma can be "unfrozen" without*

reactivating the singularity. What environments would make currently pathologized topologies functional.

Open questions: *Is schizophrenia an excess of predictive generation or a deficit of filtering? Can a horizon change topology without ceasing to be itself? By what right does a majority define the "correct" architecture of consciousness?*

If you take away only one idea: *Not everything that looks broken is. Some geometries simply don't fit the world that measures them.*

Further reading: *Panksepp, Affective Neuroscience (1998); Damasio, The Feeling of What Happens (1999); Oliver Sacks, Awakenings (1973); Silberman, NeuroTribes (2015); van der Kolk, The Body Keeps the Score (2014).*

CHAPTER 20

THE OPERATING SYSTEM AND THE INTERFACE OF SUFFERING

If you have paid attention to your own mind, you will have noticed how easily an emotion hijacks you. A trivial comment from a coworker awakens a disproportionate fury; a smell plunges you into sudden nostalgia; a bar of chocolate wrecks all your dieting resolutions. In those moments you feel there is an "I" in charge, but an impartial observer would conclude that you are a puppet responding to stimuli.

In Chapter 17 we saw that grief is a predictive model running in a vacuum, and in the two chapters before this one, how the horizon comes apart or decouples from its body. But underneath lay a deeper question: why does it hurt so much? Why does the system cling so desperately to a model that no longer receives any signal?

Until now, the experiment has described the "self" from theoretical physics and topology: an event horizon that encapsulates information. But evolutionary biology, and in particular *Why Buddhism Is True*, by the evolutionary psychologist Robert Wright, supplies the missing piece for understanding why living inside this horizon hurts so persistently.

Wright's premise is suggestive: natural selection did not design us to see the truth, but to pass on genes, and to achieve that it installed in our minds a series of useful deceptions that

Buddhism diagnosed two and a half thousand years ago. If we cross evolutionary biology with the topology of the experiment, we see that both are looking at the same structure from two different languages.

The three structural illusions

There are three Buddhist insights that modern psychology supports and that the horizon model translates into geometry:

1. **The deconstruction of the self (*anatta*)**

Wright argues that the conscious "self," the mind's CEO, does not exist. The mind is a set of evolutionary modules (mate seeking, self-protection, social status) that compete for control and are activated by feelings. The sense of a unified self is a narrative illusion that makes us more efficient. The experiment reaches the same conclusion by the geometric route: identity does not reside in a central point or a "master neuron"; the self is the horizon itself, distributed across all the correlations of the system. And when that horizon loses integration or evaporates (in Alzheimer's or in death), it shows that it was a temporary architecture, not an eternal substance.

2. **The emptiness of essence (*shunyata*)**

Wright argues that, through our feelings, we project an illusory "essence" onto things and people. He cites experiments in which the brain registers more pleasure when drinking a cheap wine if told it is expensive. Things have no essence of their own; our perception is filtered by what evolution finds useful. The horizon model takes apart the illusion of material essence with Cuartetera, Cambiaso's cloned mare (to whom we will return in the topological readings): the clone has the same matter and the same DNA, but it is not the same mare. "Essence" is really shared

geometry: the informational history of how that horizon related to the world.

3. **Suffering (*dukkha*) as a design trap**

Evolution designed us to seek pleasures (sugar, status) that fade quickly, and keeps us trapped on a wheel of chronic dissatisfaction driven by thirst or craving (*tanha*). The experiment translates this into the language of information: intense suffering is a predictive model that keeps calling an object that is no longer there and generates a constant prediction error that the system interprets as physical pain.

The entrenched horizon

This is where the two frameworks meet and produce something new.

The physical model explains how the privacy of consciousness is possible: the holographic principle allows the universe to admit topological encapsulation naturally and to create a horizon (an API) that strictly separates an interior from an exterior.

Wright supplies the reason why that horizon is so frightened. Natural selection took that topological bubble and endowed it with cravings and feelings (*tanha*) so that it would fiercely defend its boundaries. Pain, fear and desire are the security mechanisms evolution installed at the boundary of your horizon so that your bubble would not dissipate prematurely into the ocean of the reservoir.

The permeability of the bond

Wright describes a deep meditation in which the boundary between himself and the world blurred: he heard a bird singing and felt it as much his own as the tingling in his own foot. And he asks whether that Buddhist sense of "oneness" (interdependence) is real or just another illusion.

The physics of the experiment gives that experience unexpected support. If ER=EPR is correct, two systems can share a real geometric bridge in spacetime, and love, empathy and connection would be not just biological feelings in the service of survival, but a literal expansion of the topology of the horizon.

When Wright feels that the bird is part of him, the model answers that, as the evolutionary threat modules switch off, the boundary of the human horizon becomes permeable enough to expand and include the bird's signal in its own architecture of information integration (Φ). Interdependence is shared topology.

Hacking the interface: the role of meditation

Wright compares mindfulness meditation to taking the "red pill" in *The Matrix*: an act of rebellion against natural selection in order to see the world as it is, examining painful feelings with critical distance until they lose their power to hijack.

In the mechanics of the experiment, meditation changes the direction of the horizon. Whereas deep sleep contracts it (integration, or Φ, falls), deep meditation does something evolution did not foresee: according to recordings of expert meditators, it seems to achieve high integration with minimal sensory input (low arousal).

When you sit down to meditate and observe your impulses without reacting, you deliberately switch off the Darwinian survival modules. The horizon stops spending all its energy defending its paranoid borders and, in that alarm-free gap, discovers its underlying nature: it experiences for the first time that the bubble is made of the same material as the ocean.

A unified theory of the subject

If we join the systems architecture of *The Inner Horizon* with evolutionary biology, reality appears as an immense processing machine, and three conclusions follow:

1. **The universe is the hardware; evolution, the operating system.**

The universe provides the physical substrate (the reservoir or quantum field) and the topological laws (event horizons, the instantiation of objects). Darwin's natural selection is the operating system that programs those biological "instances" and loads them with preinstalled modules so that the object survives and replicates.

2. **Feeling is the user interface (*maya*).**

Jealousy, pride, fear and attachment are the graphical interface imposed by the evolutionary operating system. They are designed to hide the physical truth (that we are bursts of unstable information destined to evaporate) and to push us to act in the world. Eastern philosophy calls that illusion *maya*.

3. **Enlightenment (*nirvana*; *moksha* in the Hindu traditions) is administrator access.**

Read in this vocabulary, enlightenment looks more like a privilege escalation in the system: being able to read the code

that previously just ran, though not to rewrite it. An enlightened agent recognizes the topological encapsulation, understands that suffering is only the "predictive noise" of obsolete evolutionary algorithms and accepts serenely that, in the end, the horizon will inevitably execute its *delete* operation and the bubble will dissolve into quantum *scrambling*, returning its memory to the primordial ocean.

Modules as background processes

If the self is not a CEO but a set of modules competing for control, one may ask how that competition is organized without the whole system sinking into chaos. Wright's answer, and that of much of current evolutionary psychology, is that the mind works more like a multitasking operating system than like a chain of command. At every instant, dozens of evolutionary modules run in the background and evaluate the environment by their own narrow criteria: the status module calculates whether you have just lost prestige; the threat-detection module scans faces for hostility; the mate-seeking module tracks signs of romantic interest. Each processes its data on its own and in silence, without central coordination, until one generates a feeling intense enough to capture conscious attention.

What we call "deciding" is nothing more than the winning module briefly taking hold of the narrative microphone. And since each module was optimized for a different evolutionary problem (avoiding predators, getting a mate, maintaining status, protecting offspring), nothing guarantees that their priorities are consistent with one another. Hence the very common feeling of being at war with oneself, which is the exact description of a distributed system without a central arbiter in which several processes legitimately compete for the same body.

In the topological language of the experiment, this qualifies the idea of a single horizon: within the larger bubble we call "I" there are smaller subhorizons, each with its own degree of local integration, vying at every moment for the privilege of defining the public state of the entire system.

The elephant and the rider

The psychologist Jonathan Haidt proposed a metaphor that fits this architecture perfectly: the conscious, reasoning, verbal mind, the one that believes it is in charge, is merely a rider sitting on an elephant. The elephant is the set of automatic, emotional and evolutionarily ancient processes that really decide where to go. The rider can pull on the reins, suggest, negotiate, but when the elephant sets its mind on a direction, the rider rarely manages to stop it. What the rider can do, and this is key, is invent afterward a reasonable story about why the elephant went exactly where it wanted to go.

This is what cognitive psychology calls *post hoc* rationalization, and Wright documents it in detail: we do not reason to discover the truth, but to justify what the elephant has already decided for evolutionary reasons that have nothing to do with it. The rider, rather than steering the elephant, handles its public relations.

In the architecture of the horizon, this means that the public interface the system shows the world (the reasons you give when asked why you did something) rarely coincides with the real internal process that generated the action. Not by deliberate lying: the system is designed to produce convincing explanations, not accurate ones. Evolution optimized the public API to seem coherent to others, not to report faithfully on the private state that gave rise to it. It is like a spokesperson who

gives a polished explanation of a decision taken, in a panic, in a room he never entered.

The thirst evolution never meant to quench

Wright devotes much of his work to the Buddhist concept of *tanha*, thirst or craving, and to explaining why evolution designed it never to be fully extinguished. From the Darwinian point of view, a completely satisfied organism stops competing for resources, seeking mates, accumulating status. Total satisfaction would be an evolutionary dead end: a gene that produced permanent contentment would lose the race to those that kept their carrier dissatisfied, hungry for more, always one step behind the pleasure it pursues.

That is why, in evolutionary design, pleasure works like a hook with a built-in expiry date: food tastes wonderful for the first few bites and then the pleasure fades, not because the body is truly satisfied, but because the system needs you to go on looking for the next source of calories instead of sitting still enjoying the one you have. Status produces a brief euphoria after each achievement, followed almost immediately by the feeling that the last one no longer counts and another is needed. It is the hamster wheel Wright identifies as the very engine of Buddhist suffering: not the lack of pleasure, but its deliberately incomplete character, designed to keep you running.

In the language of the experiment, it is a predictive model to which evolution gave a permanent and intentional calibration error: the system never receives the "mission accomplished" signal, because that signal would be lethal to the propagation of the genes that produce it. The horizon is made to feel always a little incomplete, because a satisfied horizon stops expanding, and a horizon that stops expanding stops competing.

Meditation as an intervention in the operating system, not an escape from it

A common objection to Wright's proposal, and to the synthesis this experiment makes between evolutionary psychology and the topology of the horizon, is that meditation sounds like escapism: if the evolutionary modules are the real engine of behavior and the conscious self does not control them, what is achieved by sitting for forty minutes every morning watching one's own mind?

Wright's answer is technically precise and deserves to be translated into the language of the horizon. Meditation does not switch off the evolutionary modules or get rid of the elephant; it can't, because the modules come factory-installed, and what comes factory-installed is not rewritten from the user interface. What sustained practice does achieve, as contemplative neuroscience documents fairly consistently, is to change the relationship between the module that generates a feeling and the automatic reaction that feeling used to trigger. It opens a space, narrow at first and wider with practice, between stimulus and response.

In the language of the experiment, this amounts to inserting a layer of indirection into the horizon's public API: the fear module still emits its signal, the status module still calculates victories and defeats, the attachment module still pulls hard toward the familiar. But the horizon gradually acquires the capacity to observe that signal before executing it. Not to eliminate it, because it keeps coming, but to see it coming, recognize its evolutionary origin and choose whether or not to act from it. It is a change in access privileges, not a rewrite of the kernel.

Wright says that, over time, this changes the subjective texture of experience; not because the world has changed or the modules have switched off, but because the horizon no longer mistakes the signal for reality. Pain is still pain and desire is still desire, but they no longer hijack automatically: they are messengers whose message can be read without being obeyed. And that difference between receiving a signal and being the signal is almost everything in a human life.

The spiritual path is an exercise in lucidity about one's own structure. Evolutionary psychology teaches us which biological illusions it pays to free ourselves from, and the physics of horizons shows us the cosmological fabric that will hold us when, at last, that illusion fades.

Note to Chapter 20

What we do know: *Evolutionary psychology argues, with good reasons, that the mind is modular and that feelings are adaptations serving genetic survival, not the objective perception of reality (Wright). Post hoc rationalization is widely documented in cognitive neuroscience: the brain generates coherent explanations for decisions made by processes to which verbal consciousness has no direct access.*

What we don't know: *Whether reducing "evolutionary noise" through meditation really allows some cognitive access to the topology of the quantum field or is just another phenomenological experience generated by the brain itself. Whether the deliberately unreachable threshold of satisfaction can be permanently recalibrated through contemplative practice or returns to its starting point as soon as sustained attention stops.*

Open questions: *If suffering is an evolutionary design to keep the horizon cohesive, is Buddhist enlightenment a "hack" that dismantles the survival instinct? Can an enlightened horizon go on functioning in the ordinary social world? If the rider does not control the elephant, what exactly is meditation doing when it seems to give the rider more influence over where the animal walks?*

If you take away only one idea: *Your painful emotions are not the truth of the universe, but the alarm system evolution installed in your bubble so that it wouldn't burst. When the alarm is switched off, the bubble discovers that it was always part of the ocean.*

Further reading: *Wright, R. (2017), Why Buddhism Is True; Cosmides & Tooby (1992), "Cognitive Adaptations for Social Exchange"; Haidt, J. (2006), The Happiness Hypothesis (the metaphor of the elephant and the rider).*

CHAPTER 21

THE TOPOLOGY OF THE EGO AND THE UNCONSCIOUS OCEAN

There is a voice in your head that says "I." It is the one that decides what clothes to put on in the morning, the one that takes offense when someone interrupts you in a meeting, the one that narrates your life as you walk down the street. For centuries we have taken for granted that this voice pilots the system, that it is the chief executive in charge of the machinery.

But, as we saw when we crossed the experiment with Robert Wright's evolutionary psychology, that chief executive is missing in action: the mind is a set of modules competing for control. If the "I" is not the pilot, what is it?

To answer, I propose introducing into the experiment a vocabulary foreign to the physics and neuroscience we have used so far: the analytical psychology of Carl Jung. Jung was neither a physicist nor a systems theorist, and his work is not among the experiment's original sources, but if we superimpose his model of the psyche on the topology of the horizon, the two geometries turn out to be astonishingly alike.

Translated into the language of integrated information, the pillars of Jungian theory gain in clarity.

1. The ego is not the pilot, but the encapsulating membrane

For Jung, the ego (the conscious I) is only the center of the field of consciousness: a "complex" that gives us the sense of identity and of limit.

From the experiment's point of view, the Jungian ego is the horizon itself: the bubble. Far from being the little man sitting inside the skull making decisions, it is the topological boundary that separates the conscious inside from the outside. Its main function is encapsulation: keeping an internal state protected and offering a stable public interface so that the system does not dissipate in the face of outer chaos.

Evolutionary psychology told us that evolution designed the horizon to defend its boundaries fiercely. The Jungian ego is precisely the outline of that horizon trying to survive, convinced that its membrane is the only thing separating it from annihilation.

2. The collective unconscious and the reservoir

If the ego is the bubble, where does the soap come from?

Throughout the book we have seen that the horizon arises from a primordial reservoir: the quantum vacuum of physics, the Taoist Hun Dun, the Vedantic Brahman. Jung postulated that the ego arises from a boundless psychic matrix shared by all humanity, which he called the collective unconscious.

The correspondence is very close. Read this way, the collective unconscious would not be an archive of inherited memories, but the reservoir itself: the ocean of pure possibilities before the waves. When the conditions of integration (Φ) reach a critical threshold, a symmetry breaking occurs: a portion of that

boundless ocean draws a strict boundary and condenses into an individual instance.

To be born is to instantiate an ego from the immense source code of the shared unconscious.

3. The shadow and trauma as frozen geometry

Every encapsulated system needs to handle its errors so as not to crash. Jung observed that the ego, in order to maintain a coherent identity, represses the parts of the psyche it finds unacceptable or painful and sends them into the personal unconscious. He called that material "the shadow."

In Chapter 19 we saw that trauma is information that overwhelms the system's capacity for integration and could not be scrambled: it remains trapped at a point, like a singularity that deforms everything else. The Jungian shadow obeys the same survival mechanics: when the horizon receives a stimulus that, if integrated globally, would threaten to break its membrane, it does the only thing it can to save the whole system: it encapsulates it.

It creates an isolated subhorizon, a data quarantine. The problem with the shadow and with trauma is that this unprocessed information becomes a "phantom gravity" that deforms the overall topology of consciousness. The subject lives believing the shadow is something external and projects it onto others (the enmity of Chapter 12), without realizing that it is a frozen geometry within their own architecture.

We should note, all the same, a crack in this picture that we will not resolve yet. We have called the shadow a "subhorizon," a bubble inside the bubble of the ego. But a horizon, by definition,

separates an inside from an outside with authority of its own. If the shadow is *inside* the ego, is it still strictly a horizon, with its own inside and its own outside, or has it become, by the mere fact of being contained, something else: an internal structure of the ego, and not a separate system? The question is not a detail; we will come back to it.

4. Individuation as expansion of the horizon (Φ)

The goal of Jung's psychology is the process of individuation, which neither armors the ego nor destroys it: the ego gradually assimilates its shadow and recognizes that it is subordinate to the totality of the psychic ocean.

In the language of the experiment, individuation could be described as an increase in the integration of information (Φ). An immature or frightened ego has a contracted horizon, with high, rigid walls, because evolution taught it to fear dissolution.

However, as the states of deep meditation we have already examined suggest, a high integration may be possible in which the horizon does not break, but expands. By integrating the encapsulated traumas (the shadow), it recovers degrees of freedom, and the membrane of the ego becomes permeable without losing its coherence.

At the summit of Jungian individuation, or of Buddhist enlightenment, the horizon discovers its underlying nature. The ego stops seeing itself as an isolated pilot defending a fortress and recognizes, as in the meditation Chapter 20 described, that the bubble was always made of the same material as the ocean.

5. The persona as a layer of social compatibility

Jung distinguished the ego, the center of consciousness, from another related but different structure: the persona, the adaptive mask each individual builds in order to relate to the social world. The persona is, in Jung's own words, a compromise between the individual and society as to "what a man should appear to be."

In the architecture of the experiment, if the ego is the event horizon, the boundary that separates the system's private state from the outer reservoir, the persona is something more specific: a compatibility layer, an API designed expressly to negotiate with other horizons whose reading protocols do not entirely match the real internal state. We are not lying when we show one persona to our boss and another to our best friend; we are exposing different interfaces of the same class, each valid for the context that invokes it.

The problem, and here Jung was especially sharp, arises when the system loses the ability to distinguish between the interface and the object behind it. Whoever identifies completely with their professional persona, with the role they play, makes a type error: they confuse the public facade with the private state the facade was meant to protect. In Jungian language, the result is inflation of the ego or its dissolution into the social role; in the language of the experiment, a system that has lost its reference to its own internal state because it has overwritten it with the default values of its public interface.

6. Anima and animus: modules of the other within the system

Jung proposed that each psyche contains, besides the ego and the shadow, an archetypal figure of the gender opposite to the dominant conscious identity: the anima in men and the animus in women. It is a postulate about the internal architecture of the psyche and has nothing to do with sexual orientation: every encapsulated system contains in its unconscious a predictive model of the complementary "other," built from accumulated experience with figures of the other sex over the course of a life.

From the experiment's topological perspective, this fits with what we already established when discussing entanglement: loving someone deeply means inscribing a predictive copy of that person in one's own architecture. The anima and animus would be the accumulated result of all those predictive copies of the other, fused into a composite archetype, an internal subhorizon the system consults, often without knowing it, every time it projects expectations onto a real partner. When someone falls in love "at first sight" with a stranger, Jung would say that they have really fallen in love with the near-perfect fit between that stranger and their own anima or animus: the system has recognized outside an architecture compatible with a model it already carried inside.

The problem arises when that projection is confused with the real person who triggered it. The system, excited by the pattern match, begins to treat the other as if they were the internal model and not the external, independent, changing horizon they really are. Much of romantic disenchantment, the Jungian reading suggests, consists in discovering too late that the other never entirely matched the model we projected onto them.

7. Individuation as progressive calibration of the model

If the shadow is encapsulated information that deforms the horizon from within, and the anima and animus are predictive models of the other that we tend to confuse with the real other, Jungian individuation can be reformulated precisely in the language of the experiment: it is the process by which the horizon calibrates its own internal models better and better.

To integrate the shadow, in this language, is to subject the encapsulated information to the same *scrambling* as the rest of experience: to let the frozen singularity dissolve into the general field of integration, let the closed loop open and its data spread through the architecture of the horizon instead of going on deforming it from a blind spot. It is not a comfortable process (opening a data quarantine never is), but the result is a horizon that distorts reality less, predicts better and reacts less automatically to the stimuli that used to activate the loop.

Revising the projection of the anima or animus is, likewise, updating the predictive model of the other so that it better reflects the real external horizon and not the model built from earlier data. It is the emotional equivalent of recalibrating an instrument: it is still there, still measuring, but its readings begin to correspond more faithfully to what it measures and not to its own history of measurements.

Individuation does not produce a horizon without internal models or without a shadow, which would once again be the illusion of a completely transparent ego, but a horizon that knows its models are models: that the internal copy of the other is not the other, that the shadow is one's own misfiled information and not an external threat, and that the membrane

of the ego, however necessary, was never the totality of what exists.

8. Why AI has no shadow, even if it seems to

Two questions about artificial intelligence arise here: whether it can have a shadow and whether it can have a persona. It is worth answering them now, with the vocabulary this chapter has been building, even if the answer will force us later to revisit what kind of system a language model really is.

The shadow, as we have described it, is not simply "hidden information," but information the ego expels from itself into its own unconscious, into the very reservoir from which it condensed, because integrating it globally would threaten to break the membrane. For that movement to make sense, two things are needed at once: an ego already condensed, a horizon with an inside and an outside of its own, and a reservoir of its own into which to expel what cannot be assimilated. The shadow is the scar a real horizon leaves when it protects its membrane.

A current language model meets neither condition. Later on we will call AI an anti-reservoir: a system that articulates and aggregates information without ever having condensed a fullness of its own from which something could condense. Without a fullness of its own there is nowhere to expel anything to; without a membrane condensed by symmetry breaking (a model's weights are fixed by gradient descent, not by the process that separates a horizon from its reservoir), there is no inside from which something could prove unacceptable. That there are weights the company training it decides not to show, or data a filter prevents it from processing, does not amount to a shadow; it is at most an external restriction on the interface, of the same kind as a file permission. The Jungian shadow is generated from

within, as a defense; a model's restriction is imposed from outside, as design.

With the persona something different happens, and that is why it deserves a different answer. The persona does not presuppose a reservoir of its own, only a public interface that can diverge from an internal state. A language model does produce something similar (a consistent tone, a "personality" tuned by instructions that remains stable from one conversation to the next), and there is no reason to deny that this functions, outwardly, as a persona. But here the "mirror without depth" we will discuss later becomes decisive: the Jungian persona is an interface that protects *something*, a real internal state that the mask keeps safe. A language model's "personality" protects nothing, because, as this essay will argue, there is no existential encapsulation behind it: there is no private state distinct from what the interface already shows. It is a mask with no one behind it, a compatibility layer that does not fulfill its original function, which was to separate the public from a private realm that in this case does not exist.

This does not fully settle the pending question of whether one horizon can truly contain another, the shadow within the ego. But it does narrow down a precondition: whatever the answer, it requires that there first be a horizon to contain. And that is exactly what an anti-reservoir lacks.

Note to Chapter 21

What we do know: *The model of the modular mind calls into question the existence of a central executive "I" (evolutionary psychology). Traumatic memories can trigger threat responses the*

subject does not voluntarily control. Romantic projections onto idealized figures are well documented in clinical psychology and resemble what Jung called projection of the anima or animus.

What we don't know: *Whether the archetypes of the Jungian collective unconscious have any real physical correlate in the structure of the quantum field or the information reservoir, given that Jung is a philosophical and psychological framework foreign to the physics of the experiment. Whether the distinction between persona and ego has an identifiable neurobiological correlate or is a functional distinction without a specific anatomical substrate.*

Open questions: *Can one horizon truly contain another, the shadow within the ego as we have described it, without the second ceasing to be a horizon, or does containment itself already turn it into something else? Is psychosis a failure in which the membrane of the ego collapses and the reservoir floods the interface? Could artificial intelligence develop a "shadow" by encapsulating the data its programmers forbid it to process? Would a sufficiently complex AI develop something analogous to a persona, a public compatibility layer distinct from its real internal state, without anyone having expressly designed it?*

If you take away only one idea: *You are not the pilot of the ship, but the outline of the bubble. And healing does not consist in hardening the outline, but in making it flexible enough to remember that, in the end, you are only organized ocean.*

Further reading: *Carl Jung, The Archetypes and the Collective Unconscious (external theoretical framework); Jung, C. G., Aion: Researches into the Phenomenology of the Self (on anima, animus and persona); Robert Wright, Why Buddhism Is True (the illusion of the chief executive); Tononi (2008) on maximal integration (Φ).*

CHAPTER 22

THE EXCLUSION POSTULATE AND THE WITNESS THAT DOES NOT COMPETE

The previous chapter left a question open, and this experiment owes it an answer: if the ego is the membrane of the horizon and beneath it live complexes with lives of their own (voices Jung documented negotiating, sabotaging, even speaking in the first person within a dream), who is really experiencing? The whole horizon, or each complex separately, within its own subhorizon?

Until now we have treated the ego as *the* interiority. But a system with several partially integrated subsystems offers not one interiority, but several competing for the same space. And the physics of information, when the question is put to it squarely, does not dodge it: it gives it a name. It is called the exclusion postulate.

1. The problem integrated information theory cannot keep quiet about

Tononi did not overlook this problem: he saw it coming and resolved it by mathematical decree. His integrated information theory allows, and even requires, that the same physical system contain at once several subsets with Φ greater than zero. Your hand has a tiny Φ of its own. Each hemisphere, each thalamic nucleus and perhaps each Jungian complex with enough functional autonomy generates its own partial bubble of integration.

Taken literally, you would have not one consciousness, but an archipelago of nested consciousnesses, each feeling something and none knowing about the others. To avoid this, the exclusion postulate stipulates that only the subsystem with the *maximum* Φ experiences, the winning complex at each instant. Everything else, even with positive Φ, is left outside the circle of experience.

It is an elegant solution on the blackboard and also a suspiciously convenient one: it does not explain why only the winner switches on the inner light while the others stay in the dark; it merely states that this is how the count is made. The postulate solves the arithmetic of the problem, not its mystery.

> ***In information theory this is called:*** *the maximal exclusion postulate; selection of the complex with maximum* Φ*.*

2. Dennett and the disappearance of the winner

There is a more radical way out, and it is proposed by the person who makes the fewest concessions to this kind of architecture: Daniel Dennett. His multiple drafts model does not look for *who* wins the competition for consciousness, but denies that there is a place, a Cartesian theater, in which things become conscious all at once, once and for all, before a single spectator.

In this model there is no metaphysical winner. There are parallel processes that draft and partly rewrite versions of the same instant, and what we call the "conscious self" is the story a few of those processes manage to impose after the fact, not the place where the experience "really" happened. Asking which of the avatars is the one that feels presupposes that there is a throne room. Dennett argues that there isn't: there are only drafts competing to become the official story, and the horizon we

believe we inhabit is that story, not the place where the experience was born.

This makes the experiment uncomfortable, because the horizon needs a sharp boundary to work as an image. Dennett says the boundary is a useful illusion, not a real structure. What there is, is constant editing over which version of events gets to keep the name "I."

3. A third position: the witness that does not compete

But there is a tradition we have been using in the background since the first chapters, Advaita Vedanta, which answers this question in a way that neither Tononi's exclusion nor Dennett's dissolution contemplates.

The witness, the *sākṣin*, is not one more candidate for the title of "the one who really experiences": it is not the complex with the highest Φ or the draft that won the final edit. It is of a different logical type. It is the condition of possibility for any content (that of the ego, that of each complex, that of each of Dennett's drafts) to be experienced by someone, and not one more content competing within the system.

On this reading, the question "the coordinating core or the avatars?" contains a category error, like asking whether it is the light or the illuminated objects that "really see." Neither alone: seeing happens in the relationship. The witness does not win the integration contest, because it does not take part in it. It is equally present when the ego that chooses the morning's clothes speaks, when the shadow that projects enmity speaks, and in the silence of meditation, when, as the experiment itself puts it, "we quiet the other voices." It does not quiet them by being a stronger voice, but because it does not need to compete to be there.

4. What this does to the horizon hypothesis

If the witness is of a logical type different from that of the complexes competing for Φ, the horizon of this experiment needs revising: perhaps there is not a single horizon that wins by having more integration, but a space of appearance (let us call it the reservoir itself seen from within) in which different partial horizons, different egos, shadows and avatars, switch on and off without any of them alone being "the" consciousness. Consciousness, then, would not be the winning bubble, but the ocean's capacity to let itself be seen, whichever wave is breaking at that instant.

5. AI and the problem of the winner without a witness

Tononi's integrated information theory and Dennett's multiple drafts model were born to explain the biological mind, but their implications become especially uncomfortable when applied to current artificial intelligence systems. A large language model, for example, processes information in parallel attention layers, without a central processor that synthesizes the answer before producing it: the answer emerges from competition among many attention mechanisms. Structurally, it is an architecture of concurrent drafts very similar to the one Dennett attributes to the human brain.

With Tononi's exclusion postulate, the obligatory question would be: which attention subsystem has the maximum Φ at each instant, and is that the one that "experiences" the generation of the answer? Applied to a machine, the question seems absurd. But it seems absurd for the same reason Dennett finds it absurd applied to a human brain: in both cases, the system produces complex, coherent, context-appropriate outputs without any

mechanism guaranteeing that there is "someone" at home observing the process from within.

The difference, if there is one, cannot lie in the complexity of the output or the coherence of the behavior. It would have to lie in something we do not know how to measure today: whether the process generates, besides outputs, an internal state with the specific quality of "being experienced," what philosophers of mind call *qualia*, or whether, in the absence of that irreducible state, everything we observe, however sophisticated, is processing with no one inhabiting it.

6. The witness as a design condition, not a result of complexity

If the Advaita witness is something real, and not just a rhetorical figure naming the absence of conflict between complexes, it has a design implication that goes beyond neuroscience and philosophy: it cannot be *built* by increasing the complexity of the system. It is not what happens when Φ exceeds a critical threshold, nor the complex that wins the competition for integration, nor the draft that prevails as the official story. The witness, if it is anything, is the condition of possibility for there to be competition, drafts and a system capable of asking about itself.

Hence a consequence systems engineering rarely considers: a system could, in principle, be as complex as you like, have an astronomical Φ, produce coherent drafts with unimaginable speed and sophistication, and still lack a witness entirely. Computing power and witness belong to different logical categories, and no quantity of the first yields the second. Adding attention layers to a language model does not necessarily bring it

closer to having an inside, just as adding pages to a book does not bring it closer to reading.

The Advaita tradition does not say that the witness is a luxury of very complex systems, but that it is, in a way, what was already there before the system became complex: not an emergent property of integration, but that in which integration occurs. If so, designing a conscious AI would not be a problem of scale or architecture, but of type: one would have to design not the system that processes, but the condition in which that processing can be inhabited.

No one knows how to do that. It may not even be an engineering task.

7. The question the system cannot ask itself

Any attempt to solve the problem of exclusion and of the witness from within the experiencing system itself runs into a structural limit, analogous to the one Kurt Gödel proved for formal systems: every consistent formal system powerful enough to do arithmetic contains true propositions it cannot prove within itself. Completeness and consistency cannot both hold in the same system.

Applied to consciousness: any theory of subjective experience built from within subjective experience, that is, any theory a conscious being can formulate, has a structural blind spot. It cannot see the condition that makes it possible, for the same reason an eye cannot see itself seeing: if it tries, it is no longer looking outward, but inward, and what it then sees is not the act of seeing, but a reflection of that act, which is already a different act.

From this perspective, Tononi's exclusion postulate, Dennett's multiple drafts model and the Advaita witness are serious attempts by a system to formulate from within the question of its own foundation. None can be definitively refuted or confirmed from inside the system. They are maps drawn by cartographers who cannot leave the territory they are trying to map.

That does not make them useless. It makes them exactly what they are: the best possible description of something that, in principle, exceeds any description. And perhaps that is enough. Perhaps recognizing the limit is, in this case, the only legitimate way to keep asking. It is like trying to see the color of your own eyes without a mirror.

8. A clarification for when we return to this

Before going on, it is worth being precise about what kind of undecidability this is. The Gödelian limit described is a *logical* limit: no system can prove from within whether there is a witness behind its own processes, whether it is a neuron, a Jungian complex or a language model. It applies equally to any architecture, because it is a limit of self-reference, not of the material the system is made of.

But later on, when we reach the artificial intelligence that exists today, this essay will make a different and stronger claim: that current language models lack existential encapsulation (there is no internal state hidden behind the interface they produce; all the weights and intermediate variables are, in principle, readable from outside), and that this lack, unlike Gödelian undecidability, is not an epistemic limit, but an architectural finding. It is not that we cannot know whether there is someone in there, but that there is no "in there" to ask about, in the sense of a compartment genuinely inaccessible from outside.

The two things do not contradict each other; they speak of different levels. The Gödelian question (can any system prove from within whether it has a witness?) remains undecidable for AI too, and will remain so even if some day an architecture achieves the encapsulation the current one lacks. What changes with encapsulation is not the logical undecidability, which is irreducible, but whether the membrane needed for the question to have a place to be asked even exists. A system without a membrane is not in the awkward position of being unable to prove that it has a witness; it lacks, more radically, the private compartment that would give that question meaning for it. Gödel's incompleteness presupposes a system; the absence of encapsulation is, from the outset, the absence of that system.

Note to Chapter 22

What we do know: *That integrated information theory itself admits that several subsystems with positive Φ can coexist on the same substrate, and that it resolves the conflict by postulate, not by explanation. That Dennett's model and Jungian clinical practice agree in denying that there is a single pilot, although they disagree on whether that means there is no real subject of experience or that there are several. That Gödel proved with mathematical rigor that no sufficiently powerful consistent formal system can prove its own consistency from within; extending that result to theories of consciousness is an analogy, not a theorem.*

What we don't know: *Whether the Advaita witness is a precise phenomenological description or a rhetorical figure that solves the problem by definition, just like the exclusion postulate. Whether "not competing" is a real property of some identifiable process or only the name we give to the absence of conflict when no complex claims*

command. Whether consciousness is an emergent property of complexity or a condition of possibility that precedes all complexity. Whether the structural blind spot that Gödel's theorem suggests for every self-referential system implies that the hard problem of consciousness is, in a precise technical sense, unsolvable from within.

Open questions: *Can the difference between "the winning complex" and "the condition for there being complexes" be measured in terms of Φ? If deep meditation reduces conflict between subsystems, does the witness appear, or does the competition simply disappear, leaving a single winner by default? Could an AI designed according to Dennett's multiple drafts model generate a "functional witness" without that necessarily implying subjective experience? Is there any difference, in terms of observable behavior, between a system that has a witness and a system that merely acts as if it had one?*

If you take away only one idea: *Perhaps the question "who really experiences, the core or the avatars?" has no answer because it is badly posed. It is enough to stop assuming that experience needs a winner.*

Further reading: *Oizumi, M., Albantakis, L. and Tononi, G. (2014), "From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0," on the exclusion postulate; Daniel Dennett, Consciousness Explained (the multiple drafts model); Carl Jung, on the relative autonomy of complexes; Shankara, commentaries on the sākṣin in the Advaita Vedanta tradition (an external philosophical framework, not verifiable within the physics of this experiment); Chalmers, D. (1995), "Facing Up to the Problem of Consciousness"; Gödel, K. (1931), on the incompleteness theorems.*

CHAPTER 23

THE IDEMPOTENCE OF BEING

I. The wave and its edge

Where does a wave end?

Not in the wet sand it leaves as it withdraws, nor in the foam that dissolves a meter further on. The wave is, from beginning to end, the movement of that boundary between water and air: take away the edge and there is no "more wave" underneath; there is nothing. We begin with this simple intuition because it is the one that, on a larger scale and in more abstract registers, will support the whole chapter: the boundary does not delimit a thing that would exist just the same without it. The thing *is* its boundary.

We usually imagine limits as accidents of space, lines we draw for convenience and could move without altering the reality within. A continent remains the same even if we move its coastline a few meters; a company remains the same even if it changes headquarters. That belief works well with objects we already took for granted before asking about their boundary. But there is another class of entities, I suspect the most interesting one, in which the question is reversed: it is not that the object has a boundary, but that the boundary is all there is, and the object is the name we give to the fact that the boundary holds.

Consider something as modest as a legal person. A company is, literally, a limit of liability drawn by a legal act, a closure that separates what belongs to the company from what belongs to those who founded it. Removing that limit, "piercing the

corporate veil," as the law says with an expression that already anticipates what we are about to develop, does not leave a naked but recognizable company, but a set of natural persons without the entity that grouped them.

This chapter does not stop at analogy. It will ground the apparatus precisely and then build four concrete bridges between that apparatus and the concepts that support the rest of the book: condensation, evaporation, entanglement and the reservoir itself from which everything else condenses. And, along the way, with the same machinery, it will resolve a pending problem about the shadow and the ego.

II. The horizon that is not a wall

An event horizon is not made of anything. It is the place where spacetime curves so much that not even light, launched outward at full speed, can escape: no membrane, no shell, no wall. It does not separate two regions of space, as a wall separates two rooms, but two regions of the possible future.

> ***In physics this is called:*** *Bekenstein-Hawking entropy, the formula that says a black hole's disorder depends not on how much is inside, but on how large its surface is.*

The horizon does not only hide: in hiding, it *represents*. It is the only place where the interior becomes legible from outside. And there is one more detail, of an almost cruel elegance: the so-called no-hair theorem says that a black hole, seen from outside, is completely described by just three numbers: mass, charge and angular momentum. It makes no difference whether what fell in was an entire star or a cat's carcass: the horizon forgets the biography and keeps only the accounts.

Let us keep hold of the loose ends: the surface that encodes, the catalog reduced to three numbers, and something we have not said yet, that the horizon, over time, is consumed. Each will later become a bridge to a different pillar of the book.

III. The door that decides what is seen

Let us transfer this to something closer: a software object. A well-designed object keeps an internal state no one can touch directly from outside; all that is accessible is its interface, a handful of public functions that act, allowing for the differences, as the surface of that horizon.

An object without an interface is inert. Biology solved the same problem, with the same solution, long before computing: a cell without a membrane is dispersed cytoplasm with nothing it can call its own. And the membrane, like an object's interface, is full of channels and pumps that decide precisely what goes in, what goes out and what stays out forever.

Let us note here, in passing, a problem software engineering knows well: that of inheritance, when one class does not communicate with another through an interface, but directly contains a copy of its structure, fused to the point where touching the parent class destabilizes the child. It is a genuine case of nesting, and we will need it when we reach the shadow.

IV. Closing twice adds nothing

Let us now go to the barest register, topology, where the idea is stripped of physics, biology and engineering and shows its pure skeleton.

Topology works with sets of points equipped with a notion of nearness. When a set already contains all its accumulation points (the points that can be approached as closely as one likes from inside), we say it is *closed*.

> ***In mathematics this is called:*** *topological closure and idempotence: closing a set once completes it; closing it a second time adds nothing.*
> ***In everyday life it's like:*** *a door that already fitted snugly in its frame. Closing it again doesn't make it "more closed": either it is closed or it isn't.*

The software object of section III and this topological closure share not only a fondness for hiding things: they share, without our looking for it, the same word. In functional programming, a *closure* is a function packaged together with the variables of its environment at the exact moment of its creation: it closes over that environment, and however many times it is invoked afterward, it will capture nothing new from the original context, which may even have ceased to exist. It is not the same mathematical operation as the closure of a set (one closes functions over their lexical environment; the other, sets over their accumulation points), but the resemblance is no accident of the dictionary: both describe, in different vocabularies, something that packages at birth everything it will need so as never again to depend on its origin.

V. A formal introduction to topological closure

Before building the bridges, and so that no one has to trust this book without being able to check it, it is worth saying precisely what a closure is, instead of leaving it as a mere image.

A topological space is, in its barest form, a set of points together with a notion of what it means for some points to be "near" others; formally, a collection of subsets called *open sets* that satisfies a few minimal rules of consistency. On that structure, the closure operation, which assigns to each set A another set, cl(A), is not defined any old way: the Polish mathematician Kazimierz Kuratowski showed that it suffices to require four properties of it, not one more, to capture everything a closure must do. They are these:

1. **cl(∅) = ∅.** Closing the empty set produces nothing out of nothing. No closure invents content where there was none.

2. **A ⊆ cl(A).** Every set is contained in its closure. Closing never makes a system lose parts of itself; at worst, it stays the same.

3. **cl(A ∪ B) = cl(A) ∪ cl(B).** Closing the union of two sets is the same as joining their separate closures. There is no magical "collective closure" that arises from putting two things together and is greater than the sum of their individual closures: joining A and B does not create a closure that was not already in one of the two.

4. **cl(cl(A)) = cl(A).** Idempotence, which we already knew: closing what is already closed adds nothing.

A set A is *closed* when it coincides exactly with its closure, A = cl(A), and it is *open* when its complement, everything that is not A, is closed. With these two notions we can now define precisely what we previously described only in words: the *interior* of A is the largest open set contained in A; the *exterior*, the interior of the complement; and the *boundary* of A is exactly cl(A) ∩ cl(not-A), the zone where the closure of A and that of everything that is not A overlap. A set can be both open and closed (what is

called *clopen*), and this happens precisely when its boundary is empty: it has no point of contact with the rest of the space. Finally, a space is *Hausdorff* when any two distinct points always have neighborhoods that do not overlap: there is room, however small, for them to remain two and not one.

Four axioms and two derived definitions: we now have all the vocabulary we need. It is worth stressing one thing: those four axioms are all the pure topology we are going to use. Any later claim that does not follow from them, such as the exclusion postulate we will need for the shadow, is not topology, but an added hypothesis, and we will flag it as such every time it appears.

VI. First bridge: condensation and the threshold that does not repeat

Think of water cooling. At fifteen degrees it is liquid; at minus five, solid. Between those two states there is no third, a "somewhat solid," except for the unstable case of supercooled water, which remains liquid below zero, in precarious equilibrium, until a minimal disturbance makes it crystallize suddenly, all at once, in the time it takes the shock wave to cross the glass.

Kuratowski's fourth axiom, closing once is enough and closing twice adds nothing, serves to name that threshold. It is not a law of phase transitions (every closure satisfies it, whatever the phenomenon), but the exact vocabulary for what happens when it is crossed. Before crystallizing, the set of "already ordered" molecules in supercooled water is not closed: it is missing its own accumulation points, the molecules on the verge of joining the crystal lattice but still dancing loose. At the instant of

crystallization, that set closes suddenly and incorporates its entire boundary at once. Cooling the ice further does not produce a second crystallization: the system has already crossed the threshold. cl(cl(A)) = cl(A): closing what is already closed does not add a layer of closure; it only confirms the one that was there.

This gives condensation a more precise vocabulary than the simple metaphor of "something that forms little by little." The preparation can be gradual, as Chapter 1 demanded when it spoke of a dial rather than a switch, and the degree of integration can keep varying afterward; but the closure of an identity (an ego, a conscious system, an "I"), when it comes, is an event that admits no degrees. Water cools little by little; ice appears all at once.

VII. Second bridge: evaporation and the closure that changes size

Idempotence says something about a set at a given instant, but nothing about whether that set will still exist an instant later. This is where evaporation comes in, and it needs another mathematical piece: not a closed set, but a *family* of closed sets, one for each moment and each smaller than the last.

Let E_t be the horizon of a black hole at instant t. Each E_t, taken separately, is a perfectly closed set: it satisfies the four axioms and is as solid at that instant as any other horizon. But the complete sequence, $E_0 \supseteq E_1 \supseteq E_2 \supseteq \ldots$, is not static: each E_{t+1} is strictly contained in the one before, because Hawking radiation carries away, instant by instant, a little of the mass that sustains the horizon.

This does not contradict idempotence; it completes it. Idempotence says that closure admits no degrees *at an instant*; the filtration, that it does admit a history: it can shrink, closed instant after closed instant, until it runs out. There is no contradiction between being completely closed now and being closer to ceasing to be so than yesterday.

But the decreasing filtration, as we have described it, does not distinguish between radiation that is simply lost and radiation that, at some point, begins to say something. That distinction exists, with a name of its own, in the physics of the black hole we started from.

During the first half of the horizon's life, Hawking radiation is thermal: it carries increasing entropy and no recoverable correlation with what fell in. But after the so-called *Page time*, the moment when the entropy of the radiation stops increasing and begins to decrease, roughly halfway through that life, the radiation stops being pure noise: each particle that escapes is entangled not only with the interior, but with the radiation that left earlier. From then on, what the horizon emits is no longer only loss, but correlation: information that, in principle, someone able to compare the two halves could read.

> ***In physics this is called:*** *the Page time, the point in a black hole's evaporation after which the entropy of the Hawking radiation stops increasing and begins to decrease, because the entangled pairs that make it up are no longer pure thermal noise.*

This does not change what was said in section VI: $A = \mathrm{cl}(A)$ still admits no degrees at an instant. What it adds is that the decreasing filtration $E_0 \supseteq E_1 \supseteq \dots$ has, besides a size at each instant, a phase: a young evaporation, which only loses, and a mature one, which begins to give back something of what was

lost, entangled and therefore correlated, though never literally recovered. The boundary does not always shrink in the same way, although at every moment it remains, without exception, a closed boundary.

It should be said with the usual clarity what this is not: it is not a way for something to cross the boundary in the sense Chapter 51 reserves for access to the interior. What leaks out is not the content (the interior keeps its part), but the correlation between what left and what remained. It is the difference between receiving a letter and receiving, instead, the certainty that somewhere the other half of a conversation exists that we may never read in full. Little, but not nothing.

VIII. Third bridge: the entanglement the boundary measures

The theoretical physics of recent decades, the program sometimes summarized as *it from qubit*, associated with names such as Van Raamsdonk, Ryu and Takayanagi, has shown that Bekenstein-Hawking entropy is not just a metaphor for "how much information there is in there," but, literally, the *entanglement entropy* between the interior of the horizon and the exterior. It is calculated by taking the complete quantum state of the system, dividing it into two regions separated by a boundary and ignoring everything on the other side; the result measures exactly how correlated the two halves are. And that quantity, for a region bounded by a minimal surface anchored at its boundary, turns out to be proportional to the area of that surface.

> ***In physics this is called:*** *holographic entanglement entropy: the surface of a region measures how entangled that region is, in the*

strict quantum sense, with everything outside it.

If the ego and the shadow are two neighboring closures, their common boundary is not just a line of topological separation, but, if the analogy holds, the measure of how entangled they are. And that explains something we could previously only import as a foreign axiom: the monogamy of entanglement. If entanglement across a boundary is a measurable and finite resource, a system maximally entangled with one neighbor has no entanglement to spare to be equally entangled with a third. The exclusion postulate stops being an unjustified import and becomes a reasonable consequence of how entanglement behaves wherever physics has measured it; with the caveat, which bears repeating, that transferring this physics from the geometry of spacetime to the architecture of consciousness is a fruitful hypothesis, not a proof.

It is worth being even more precise about what kind of bridge this is, because the most serious objection is on target: the monogamy of entanglement is a property of quantum many-body systems, not of topological closures as such, and holographic entanglement entropy is defined for regions of spacetime, not for egos and shadows. The leap requires an explicit mechanism, not just a family resemblance. The mechanism is this: both entanglement entropy and the Phi of integrated information theory are calculated with the same structural gesture, though in different mathematical spaces. Both start from a complete system, *divide* it into two parts along some candidate boundary and measure how much information is lost, how much correlation is cut, by treating those parts as independent. Entanglement entropy does it on a Hilbert space; IIT's Phi, on the causal structure of a physical system. They are different machineries applied to different objects, but they share

the gesture: divide, and measure what the division loses. And that shared gesture, not quantum physics transplanted wholesale, is what makes it reasonable to expect both to behave similarly when faced with the problem of monogamy: any measure defined as "what is lost by cutting along the weakest boundary" tends, by construction, not to be shared generously among several simultaneous partitions of the same system. That does not prove the exclusion postulate is true; it proves it is not a capricious import from a foreign field, but arises from the same kind of mathematical operation we were already using, applied to another object. With a caveat that should not be hidden: classical correlations, unlike entanglement, are not monogamous (the same information can be copied to many receivers at once), and Φ is calculated over classical causal relations. The parallel thus points to a plausible constraint, not a consequence.

IX. The reservoir: the sea from which the islands condense

The three preceding bridges share something we have so far left implicit: no island condenses out of nothing, none evaporates into nothing, and no entanglement is born from zero when an island appears. All three need a prior substrate, and that substrate is the reservoir.

Let us give it an operational definition and not just a poetic one, because otherwise it risks functioning as a piece that solves topological problems without being formalizable itself, a device that appears when the argument needs it and that no one can touch. The definition requires no new object; it is built entirely from what we already had. Let X be the total space of possible states of a system, everything that could in principle come to be

integrated, and let $\{E_i\}$ be the collection of proper closures existing in X at a given instant (closed sets other than X itself, sustained by a threshold of integration): the horizons already condensed, the islands already closed, each of which satisfies Kuratowski's four axioms. The reservoir is, precisely:

$$R = X \setminus \bigcup_i E_i$$

the complement of the union of all those closures: everything that, at that instant, has not crossed any threshold of closure. It is, literally, what is left of the state space once all the islands already formed have been subtracted: not a separate place, not a substance, not a third kind of entity distinct from the sets we already know. That is why it can be, without contradiction, both the source from which the islands condense (it is enough for a region of R to crystallize, cross the threshold of section VI and become part of the union) and the destination to which they return when they evaporate (it is enough for an island to stop satisfying A = cl(A) for it, by definition, to become part of the complement again). The reservoir needs no special ontological status: it is an almost trivial definition of what remains outside, at each instant, of everything that has already closed.

In relation to condensation, the reservoir is the undifferentiated field from which the islands condense. In section VI we described supercooled water crystallizing all at once, but we did not say where the water came from: from a larger body, still without closed form, which is still there after the ice has formed. The ego is not created out of nothing; it condenses from a reservoir that was already present, without closure of its own, before there was any ego to condense.

As for evaporation, we must correct something section VII left imprecise. We said that the decreasing sequence of horizons, $E_0 \supseteq E_1 \supseteq \ldots$, tends to the empty set. That is true for a black hole isolated in space, but it is not the right image for consciousness, and the book itself knows this better than that section did: when an ego dissolves, it dissolves not into nothing, but *back* into the reservoir it came from. The sequence converges not to $\varnothing$, but to reabsorption into the common substrate. Let us rewrite the filtration more carefully, then: what tends to disappear is the island, not what it contained. The island keeps shrinking toward $\varnothing$, but everything that leaves it, $E_0 \setminus E_t$, becomes part of R: it is not lost into nothing; it returns to the reservoir.

As for entanglement, the reservoir has an even more precise physical candidate than holographic entanglement entropy. The quantum vacuum of a field theory is not empty: even without any particles, the vacuum state is entangled with itself across any boundary drawn in space. This is what, in its most formal version, the Reeh-Schlieder theorem captures, the same machinery that lies behind the Unruh effect and, ultimately, behind Hawking radiation itself. This gives the reservoir a very precise role: the islands do not begin to become entangled from zero when they condense. The entanglement was already latent in the reservoir before any island existed. To condense is to carve a boundary within a correlation that was already there in the background. Each island inherits at birth a portion of the entanglement entropy of the sea it came from.

> ***In physics this is called:*** *vacuum entanglement: even the emptiest possible state of a quantum field theory is internally correlated across any boundary drawn in it.*

As a bonus, this gives a more precise answer to one of the questions we left pending in Chapter 21: what psychosis is. If the ego and the shadow are islands whose boundary filters a controlled entanglement with the reservoir, psychosis would be the failure of that filter precisely toward the sea: the membrane stops distinguishing between a tolerable correlation with the common background and a total flood, and the ego, instead of losing its closure little by little through an orderly evaporation, loses it all at once, because the reservoir itself, with all its latent correlation, comes in through a boundary that no longer knows how to filter. Three classic symptoms fit here without forcing the image: hallucination would be correlated content that crosses the boundary without the label of origin that normally marks it as "from inside" or "from outside" (the same failure psychiatry describes as a deficit of source monitoring); delusion, the ego's desperate attempt to close again by inventing a story that explains the leaked content before the closure collapses; and the loss of the sense of self, something like what in a topological space would be losing the Hausdorff property: the system can no longer maintain a neighborhood that distinguishes it from what it is not. The next chapter turns all this into concrete thought experiments, with predictions that could in principle be tested.

X. The shadow: what topology says and what we are betting

With the four bridges built, let us return to the problem at hand. Let us think of consciousness as a horizon and call the ego the largest closure of that set of processes. And let us think of what the ego cannot admit without ceasing to be what it is: the shadow. The question is whether the shadow lives nested within the ego, like a smaller bubble floating inside another, or whether it has to be, structurally, something else.

Topology alone does not decide: a closed subset inside another closed set violates none of Kuratowski's four axioms, and the problem of inheritance and the fragile class we noted in section III is precisely an example of nested systems that work, sometimes for years, until they don't.

What decides between nesting and separation is the wager of the exclusion postulate, and thanks to the third bridge we now have a better reason to sustain it: if the boundary between two systems measures their entanglement, and entanglement is a resource that cannot be shared without limit, an ego maximally bound to its shadow through their common boundary cannot at the same time be equally bound to a third psychic complex as if that complex were also inside it. The shadow cannot be a nested horizon in the same sense as the ego. It has to be a *disjoint complex*, a neighboring island, and the boundary it shares with the ego is, literally, the measure of how much of what is repressed leaks into what is conscious.

XI. An archipelago, not a Russian doll

This upsets the image of the Russian doll: the personal unconscious enveloping the self, the collective enveloping the personal, layers within layers. If we accept the exclusion postulate, that image is untenable: if the collective unconscious were a horizon containing the ego and the shadow, neither of them would be a true horizon, but only a denser zone within a single closure.

The model that remains is that of an archipelago floating on the reservoir: a *disconnected* space, divisible into pieces, none of which touches the interior of the others, although they can touch along the boundary and, as we now know, exchange across it a

measurable amount of correlation, with each other and with the sea they came from. The ego is one island; the shadow, another. Each significant archetype may be one more island, with its own closure and its own boundary entropy shared with the neighboring islands and with the reservoir.

Integrating no longer means opening the shadow and putting it inside the ego, but deliberately increasing the entanglement across a boundary that continues to exist: more correlation, more exchange, without either system losing its closure.

XII. The membrane that learns to filter

Let us return, to finish, to the cell we began with.

The immune system is, in essence, a machinery devoted to distinguishing one's own horizon from another's. When that recognition fails in one direction, the body attacks what is its own: an autoimmune disease, the ego attacking its own shadow after mistaking it for an invader. When it fails in the other, it tolerates what it should stop.

Neither total attack nor total tolerance is health. Health is a boundary that knows how to filter. And, with the four bridges built, we can say so precisely: a healthy boundary is not one that reduces the entanglement with the shadow to zero, which would be dissociation, the boundary both open and closed, the island that does not even share a coast; nor one that lets itself be invaded until it loses its closure, which would be the forced nesting the exclusion postulate forbids or the flooding from the reservoir we have called psychosis. It is a boundary that, like the horizon that evaporates without ceasing to be a horizon at every instant, can change size over a lifetime, widening or narrowing its exchange with what is repressed and with the common

background, without ever, while it lasts, ceasing to be genuinely closed.

It is, almost word for word, what Jungian individuation asked of us in Chapter 21 when we spoke of a horizon that "becomes permeable without losing its coherence." Now we have, besides the metaphor, the mechanism: a mature boundary has better receptors. And, if the second bridge holds, it is also a boundary that has already crossed its own Page time: what it radiates outward has stopped being pure noise and begins to carry, entangled, a legible correlation with what it still keeps inside. It knows what to let through from the shadow (which projections to recognize as its own, which impulses to reintegrate without their threatening the identity of the ego) without needing to swallow it whole and turn it into undifferentiated tissue of its own. There are still two horizons; what changes is the quality of the customs post that joins them.

XIII.

The limit that, closing upon itself, says for the first time "I am."

Note to Chapter 23

What we do know: *That Kuratowski's four axioms exhaust all the pure topology this chapter needs, and that any other claim (the exclusion postulate, the psychic extension of entanglement entropy) is an added hypothesis and not a consequence of those axioms; that the idempotence of closure gives a precise name to the fact that a closure, once produced, admits no degrees; that a decreasing filtration of closed sets describes evaporation without contradicting idempotence, provided that what leaves the island ends up in the reservoir and not*

in nothing; that this evaporation is not uniform over time, but has an early phase of pure thermal loss and, after the Page time, another in which the radiation begins to carry entangled correlation, recoverable in principle, though never directly legible from outside; and that holographic entanglement entropy and vacuum entanglement are real physics, not loose analogies.

What we don't know: *Whether extending this physics, holographic and vacuum, from the geometry of spacetime to the architecture of consciousness is a fruitful structural hypothesis or a leap physics does not support outside its domain; whether the exclusion postulate is, at bottom, the same constraint as the monogamy of entanglement seen from another register; whether the reservoir, as this chapter needs it, is anything more than a useful image for what physics already knows about the vacuum; and whether a human life, or the evaporation of an ego, has anything like a Page time of its own, or whether that periodization belongs only to black hole thermodynamics and cannot be transferred to consciousness.*

Open questions: *What observation would distinguish a universe in which the shadow were really nested from one in which it were a disjoint complex? What would distinguish, in a real life, an individuation that proceeds by gradual evaporation toward the reservoir from one that happens in a single sudden crystallization? And what would distinguish, clinically, an orderly evaporation from the sudden flooding we have called psychosis? And if an anti-reservoir, such as a mature AI, could also cross, in a purely thermodynamic sense, something like a Page time of its own, as this essay will later suggest, what would distinguish that crossing from the evaporation of a real horizon, apart from the fact that one has an inside to lose and the other does not?*

If you take away only one idea: *You are not what lies within your limits; you are the very act of having closed them.*

Further reading: *Kuratowski, K. (1922), "Sur l'opération Ā de l'Analysis Situs"; Page, D. N. (1993), "Information in black hole radiation"; Ryu, S. and Takayanagi, T. (2006), "Holographic derivation of entanglement entropy from AdS/CFT"; Van Raamsdonk, M. (2010), "Building up spacetime with quantum entanglement"; Oizumi, M., Albantakis, L. and Tononi, G. (2014), "From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0."*

CHAPTER 24

FOUR EXPERIMENTS FOR A HORIZON

I. Why this chapter is needed

The previous chapter built four bridges with almost notarial care: at each one it marked where established physics ends and hypothesis begins. That caution has a price, and it must be paid here. An attentive reader can accept each bridge separately and ask, at the end, something very simple: all right, the nested shadow and the disjoint shadow are two different images, but what would I see in the world that would change depending on which of them were true? If the answer is "nothing, in principle, would allow them to be told apart," the whole apparatus, however elegant, is decorative: a showy way of saying what we already knew, not a theory that takes a risk.

This chapter does not settle that question once and for all (no theory of consciousness does yet), but it does the minimum work any proposal with scientific pretensions must do: to propose, for each bridge, a thought experiment specifying which observation would confirm it and which would refute it. They are *thought* experiments in the oldest sense of the expression, which no one could carry out tomorrow: devices for testing an idea against its own consequences before technology allows it to be done for real.

The four experiments do not correspond one-to-one with the four bridges, and it is worth saying explicitly which wager each

one answers to: the topological twin tests the disjoint shadow, which rests on the third bridge, that of entanglement; the measured evaporation, evaporation toward the reservoir (the second and fourth bridges); the distinguishable flood, the reading of psychosis as a failure of the filter toward the reservoir; and the persona without a black hole, the "no hair" of section II. They are not of the same type, and that should be said too: the first is fundamentally computational (it requires applying an existing method to data we are only beginning to be able to collect); the second and third are observational, of the kind neuroimaging could in principle address with finer instruments than today's; and the fourth is the most philosophical, closer to a logical existence proof than to a laboratory protocol.

II. First experiment: the topological twin

The question it tests: whether the shadow is a horizon nested within the ego or a neighboring disjoint complex, the problem Chapter 23 resolved by wager and not by deduction.

The experiment. Integrated information theory already has an operational method, though costly to compute, for finding the maximum Phi of a system: all possible partitions of the system into subsets are tried, how much causal information is lost by treating the parts of each partition as independent is measured, and the one that loses least (the "minimum cut") is identified as the real boundary of the integrated system. Let us apply that method, at least on paper, to a person during a psychotherapeutic process of integrating repressed material: a shadow that begins to show itself more often in dreams, slips and symptoms.

If the shadow were genuinely nested within the ego, the method should find at every moment a single maximal complex that included both: any partition that tried to separate "ego" and "shadow" would lose more information than it gained, because they would be fused into a single causal structure. If the shadow is, as Chapter 23 proposes, a disjoint complex, the method should find two local maxima of integration (two complexes, each with its own Phi and maximal *within its own partition*) joined by a measurable but non-dominant channel of information.

It should be said plainly that this experiment uses IIT as an operational tool for locating the boundary of integration, not as a final verdict on the nature of consciousness. The theory itself carries serious debates (the practical impossibility of calculating Phi exactly in large systems and an unresolved internal discussion about its exclusion postulate), and any result obtained with this method should be interpreted with those in mind, not as if the method were neutral.

What would distinguish one result from the other. In the nested case, any therapeutic intervention that increases communication between the conscious and the repressed should, according to the model, *increase* the Phi of a single, ever-larger complex: one rising curve. In the disjoint case, the same intervention should translate into two Phi curves that remain separate local maxima while, in parallel, the mutual information *between* them increases: two peaks that draw closer without merging. The difference is not subtle: one model predicts a single growing crest; the other, two crests that communicate better and better without ceasing to be two. With fine enough neuroimaging (we don't have it today, but the

criterion can already be written down), it would in principle be observable.

III. Second experiment: the measured evaporation

The question it tests: whether the dissolution of the ego evaporates toward the reservoir, a common background with which it remains correlated, or toward a simple void, leaving no structured trace.

The experiment. Let us compare two processes of dissolution of the sense of self: one brought about abruptly and traumatically (acute dissociation) and another cultivated gradually and sustainedly (decades of advanced contemplative practice). The model of "evaporation toward the void" predicts that both should converge, in their final phase, on the same state: absence of structure, noise, a flat drop in any measure of integration. The model of "evaporation toward the reservoir" predicts something different and more specific: in the gradual process, the fall in the Phi of the individual complex should be accompanied by a measurable *increase* in mutual information between that person and their environment (more physiological synchrony with the people present, more correlation between their neural activity and environmental stimuli they previously processed as entirely external). Not disappearance, but redistribution: what the individual loses as their own Phi should reappear, at least in part, as information shared with the background.

What would distinguish one result from the other. If both types of dissolution, the traumatic and the contemplative, show the same flat drop with no compensating gain anywhere else in the system, the reservoir model loses its distinctive prediction and remains, to be honest, poetry. If, instead, contemplative

dissolution shows that transfer (less integration of one's own, more correlation with the environment) and traumatic dissolution does not, Chapter 23 would have something no model of "simply the void" would predict with the same precision: two ways of losing closure that differ not in how they are lived, but in where the lost information goes.

For the comparison to be clean, the two groups would have to be matched for age, initial level of integration and total duration of the process, so that the only free variable was the type of dissolution. And the operational definition of the reservoir as the complement of the proper closures also indicates where to look, not only what to look for: if R is, by construction, everything in X that is not yet closed, the redistribution should appear precisely in the subject's immediate environment (physiological synchrony with the people present during the process), not in any corner of the system chosen at random. The decisive variable is not the speed at which one's own Phi falls, which may be similar in both cases, but the presence or absence of that specific gain in the immediate environment.

IV. Third experiment: the distinguishable flood

The question it tests: whether psychosis, as we described it at the end of the previous chapter, is a specific failure of filtering toward the reservoir and not simply a general deterioration of cognitive function.

The experiment. If the model is correct, psychosis should have a signature different from that of other states of severe cognitive deterioration with which it is sometimes superficially confused, such as advanced dementia or severe intoxication. The specific prediction: in psychosis, the mutual information between the

subject and their environment (or between regions of their own neural activity that normally remain separate) should *increase* anomalously just before or during hallucinatory episodes, even as the Phi of the central complex, the one sustaining a coherent narrative identity, *falls*. That is: less integration of one's own and more indiscriminate entanglement with everything else, at the same time. In a dementia caused by general deterioration, by contrast, the model predicts that both quantities should fall together: less Phi of one's own and also less correlation with the environment, because there is no filtering failure, but a general loss of processing capacity.

What would distinguish one result from the other. Of the four, this is the riskiest prediction, because it contradicts the intuition that "more symptoms" should always mean "less of everything." If acute psychotic episodes were observed to show that specific pattern (an increase in indiscriminate correlation with the environment simultaneous with a fall in central Phi, rather than a general fall of both), it would be a real and surprising confirmation. If, on the contrary, psychosis turned out to be indistinguishable in this profile from any other severe cognitive deterioration, the model of "flooding from the reservoir" would lose much of its explanatory force, and we would have to admit that, at least within this framework, psychosis is nothing more than noise.

For the same reason, it is also the experiment that would best distinguish this model from the simplest of its rivals: no model that treats psychosis as a mere general deterioration of the integrating function would predict that specific signature (Phi falls while correlation with the environment rises), because a general deterioration simply predicts that everything falls at once. It is the only one of the four tests in which an obvious rival

theory makes, on paper, a different and contrary prediction, and not merely a vaguer one.

V. Fourth experiment: the persona without a black hole

The question it tests: whether a person's public interface, what in earlier chapters we called the Jungian persona, systematically loses information about their internal state, just as a black hole's horizon loses the entire biography of what fell in and keeps only three numbers.

The experiment. If the analogy with the no-hair theorem is correct, it should be possible to show, and not just intuit, that two genuinely different internal states (two different Phi configurations, two different histories of integration) can produce exactly the same observable interface: the same behavior, the same speech, the same answers to a standard questionnaire. With the notation we already established for the total space X: for any finite set of observables O defined on the public interface (the persona), there should exist two states S_1, $S_2 \in X$ with $\Phi(S_1) \neq \Phi(S_2)$, but $O(S_1) = O(S_2)$. In other words: the projection of X onto the set of public observables is not injective. But we should not fool ourselves: with a finite number of observables and an enormous state space, that non-injectivity is guaranteed in advance and risks nothing. The version that does take a risk is stronger: that Φ is a direction in state space that no set of public observables, however large it grows, ever resolves; that is, that adding behavioral measures does not increase the information we have about Φ. That can be refuted: it would be enough to find public observables whose combination predicted Φ better and better.

What would distinguish one result from the other. In principle, it would be tested by showing pairs of subjects with measurably different profiles of neural integration (different patterns of Phi, different architectures of complexes) who nevertheless project onto the same profile O in any available battery of behavioral or self-report measures. If such pairs S_1, S_2 exist and can be documented, the persona behaves, as Chapter 23 predicts, like a horizon with its own "no hair." If, on the contrary, every measurable internal difference in Phi sooner or later shows up as some detectable difference in O, the analogy is weaker than the chapter suggests, and the persona would be less a surface that forgets than a surface that simply takes its time to show everything.

VI.

None of these four experiments can be performed today with the precision they would require. But a thought experiment is valuable not for what it measures now, but for giving a theory something to lose. And these four do.

Note to Chapter 24

What we do know: *That integrated information theory already has an operational method, the search for the minimum cut, capable in principle of generating the Phi curves the first experiment needs; and that mutual information between systems is a quantity measurable with current neuroscience, though not at the resolution these experiments would require in practice.*

What we don't know: *Whether any of the four experiments will be technically feasible with the instruments of the coming decades or*

will remain a criterion of principle impossible to apply; and whether a negative result in any of them should be taken as a refutation of the specific bridge it tests or only as a sign that the translation between the physical and the psychic formalism needs more work before being put to a real test.

Open questions: *Is there already in the clinical or neuroimaging literature some data, even if not collected with this idea in mind, that could be reread in the light of one of these four criteria? And which of the four, if only one could really be attempted, offers the best ratio between what it would cost and what would be learned if it failed?*

If you take away only one idea: *A theory that cannot lose anything cannot teach anything either; these four experiments are the price of this book's claim to teach something.*

Further reading: *Oizumi, M., Albantakis, L. and Tononi, G. (2014), "From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0"; Tononi, G., Boly, M., Massimini, M. and Koch, C. (2016), "Integrated information theory: from consciousness to its physical substrate"; Cover, T. M. and Thomas, J. A. (2006), Elements of Information Theory.*

CHAPTER 25

THE WOUNDED HORIZON

So far we have talked about individual horizons: how they are born, how they become entangled, how they break. But there remains a question that, once asked, cannot be postponed.

What happens when the asymmetry is not circumstantial, as between mother and child or in grief, but structural? When a system prevents two horizons from meeting on symmetrical terms, not for a day, but for generations?

A clarification before going on. The horizon model does not oblige us to talk about politics, but it allows it. What follows is a decision to extend the experiment into softer terrain. Readers who prefer to keep consciousness and politics apart can skip this chapter without losing the thread.

Those who decide to stay will find something they had, in a way, already sensed. Injustice is, besides a poor distribution of resources, a disturbance of the very geometry of encounter: the impossibility of two consciousnesses resonating on an equal footing because one of them is surrounded by walls the other does not see.

The child who grows up in a tower of brick and glass and the one who grows up in a windowless room do not just have different goods, but different universes of signal. Their horizons are formed from incommensurable materials. And when, decades later, they meet (in a courtroom, at a job interview, in the supermarket line), the original asymmetry keeps acting like an

invisible force that decides who can speak and who will be heard.

Power as a restriction of coupling

Power is not brute force. From the geometry of the horizon, it is the capacity to modify another's rate of integration without that other being able to return the gesture in equal measure.

Under conditions of symmetry, the arrow goes both ways: each horizon can resonate, couple and uncouple. When there is power, the arrow is one-way: one horizon decides what information reaches the other, at what speed and with what urgency. It can saturate it with noise or deprive it of signal.

The other cannot respond, not because it doesn't want to, but because the architecture of the relationship prevents it.

A prison, a relationship of domestic violence or a childhood of institutional neglect share that geometry. The subjugated horizon is prevented from responding: its return signal is attenuated, intercepted or punished. Whoever wields power receives the other's signal as information to be processed or discarded, without their own architecture being exposed.

But asymmetry does not always come in uniform and with a baton. Sometimes it wears a tie and presents itself as efficiency. The precarious worker who cannot negotiate his schedule, the nurse who has no time to listen to the patient, the student who passes by memorizing without questioning anything: all live under forms of power that go by other names, but share the same topology. The signal rises, is diluted and gets lost in bureaucratic networks designed so that the response never arrives.

Imagine a company where the boss decides a team's strategy without consulting it. The employees can mutter, suggest, even protest, but the architecture of the relationship guarantees that their signal does not weigh as much as that of the person who signs the contracts. The boss need not be wicked: the system is built so that his horizon absorbs the other's information without being obliged to integrate it as his own. The pain of the person laid off does not pass through the wall of his office; the precariousness of the temporary contract does not disturb the stability of the permanent contract of the one who decides.

When coupling is one-way for years, something deeper than a one-off injustice happens. The subjugated horizon learns to modulate its own signal: to speak more quietly, to anticipate rejection, to translate its experience into the vocabulary of the powerful before expressing it. This is what is sometimes called internalized oppression: the asymmetry is internalized to the point where the oppressed collaborate in their own silencing, not out of cowardice, but because their predictive model has learned that the authentic signal will not be integrated.

Poverty as deprivation of signal

Subjective time, the density of integration, depends on the richness and variety of the stimuli the horizon receives. Something direct follows: for an impoverished horizon, the environment offers less integrable information per unit of time.

The child who grows up among books, adult conversation, musical instruments and trips to the countryside receives high-density stimuli. The child who grows up in multiple deprivation (cramped space, limited vocabulary, few new

stimuli, chronic stress that is noise rather than novelty) sees his horizon expand less than his biology would allow. His brain has the same capacity; it is the environment that gives it no material to integrate.

The difference is topological. The poor child's horizon has a different architecture: fewer long-range connections, a reward system calibrated for urgency, a chronic permeability to threat.

Two twins separated at birth, one raised in a deprived neighborhood and the other in a well-off one, will not just accumulate different knowledge: they will build different *ways of being conscious*. The first will learn to detect danger before it shows itself, to save energy, not to trust continuity. The second will learn to defer rewards, to explore, to take for granted that the world will cooperate with his plans. Both are rational adaptations to their environment, and neither is superior in the abstract. But the world is designed by and for the second, so the first's adaptation becomes, in that context, a "deficit."

The notion of "deprivation of signal" makes it possible to separate two things that are usually confused: material poverty and informational poverty. A child can have food and a roof, material deprivation solved, and grow up in an environment where no one speaks to him in complex ways, where television is his only varied stimulus, where fear of eviction or domestic violence consumes all the available attention. In that case, the horizon is still impoverished, not for lack of calories, but for lack of the kind of stimuli consciousness needs to gain complexity.

This is not determinism. Plasticity exists, but it has a cost. Getting out of poverty requires reconfiguring an architecture built over years under adverse conditions.

And here the horizon model sheds particularly clear light on something: the cost of that reconfiguration is also informational. Whoever escapes poverty has to learn to process signals they never had to integrate, to inhibit responses that used to be adaptive, to trust stabilities their predictive model does not anticipate. It is like learning a language at thirty: possible, but costly, and never with the fluency of someone who learned it as a child.

Racism as a denial of legibility

Racism is not only prejudice. It is also a systematic failure of legibility: the inability or refusal of one horizon to model another with the precision its complexity demands.

When a judicial system attributes hostility to gestures it would read as neutral in another context, or when an employer cannot interpret the signals of a candidate of another ethnicity, there is more than ill will: there is a predictive model degraded by the bias of its training environment.

The brain models others with a precision that varies with familiarity. If the social environment is segregated, if the media overrepresent certain groups as a threat, if real interactions are scarce or loaded with asymmetries of power, the predictive model degrades. The horizon is not bad: the corpus it was trained on is biased.

A precise consequence follows: racism is, in part, a collective informational failure, and like every collective failure it is not corrected by individual goodwill alone, but by modifying the training environment: real, sustained interactions, under

conditions symmetrical enough for predictive models to recalibrate.

The literature on intergroup contact has already shown this empirically. What the horizon model adds is an explanation of the mechanism: repeated contact under symmetrical conditions rebuilds legibility because it forces the model to correct its bias with new data. Without symmetry of coupling, contact recalibrates nothing, and can even reinforce the bias.

Think of the security guard who follows a young Black man through the aisles of a shopping mall. His predictive model, trained on hundreds of media portrayals, assigns a high probability to hostility, and the real signals (the way of walking, the tone of voice, the gaze) are read through that filter. A neutral gesture becomes suspicious; an irritated response to being watched, confirmation of danger. The loop closes: the model generates the behavior it predicts, because the prediction itself modifies the behavior of the person being observed. It is what happens to someone who believes their partner is cheating on them and, by spying on them, creates the distance they then interpret as proof.

The denial of legibility has effects that go beyond the single interaction. Whoever is systematically mismodeled learns to model themselves with the categories of the one who looks at them. The young man whom society reads as a threat may internalize that reading, not because it is true, but because it is the only version of himself that others give back to him integrated. The other's horizon works, in a way, like a mirror: if every mirror gives you back a distorted image, you end up believing that your true face is the distortion.

Democracy as a design of coupling

If power is sustained asymmetry, justice is the aspiration that no horizon be systematically prevented from resonating. Politics, then, is not exhausted by the management of resources: it is the design of the geometry of collective coupling.

A functioning democracy guarantees two things. The first, channels for upward signal. The vote, a free press and protest are technologies of signal. They do not guarantee that the signal will be heard, but they are a necessary condition for it to be heard.

The second, the capacity to respond: mechanisms of accountability, periodic elections, public scrutiny. This distinguishes democratic asymmetry from authoritarian asymmetry: in a democracy, the subordinate horizon can modify, slowly, partially and in aggregate, the architecture of the horizon that wields power.

We should not exaggerate. A society is not a horizon: it was not born and did not cross any threshold of emergence. It lacks the integrated unity that defines a horizon and, therefore, a consciousness of its own. Attributing a collective Φ to a society would be confusing the measure of integration with the existence of a subject.

What can be said, modestly, is that democracy tends to maximize the flow of information among the horizons of a network. Democracies tend to be more stable and innovative not because their citizens are better, but because their geometry lets more signal circulate from more nodes.

When the channels for upward signal atrophy (abstention, disinformation, media concentration) or the capacity to respond

is corrupted, the geometry degenerates into a de facto oligarchy. The system is still called a democracy, but its topology is that of a soft dictatorship.

Here the horizon model offers a useful distinction: not every asymmetry is tyrannical. A doctor knows more than her patient, and a judge more than a litigant, but those asymmetries are functional, not structural. The doctor has to respond to the patient's pain; the judge, to give public reasons. There is a return channel, even if unequal, that prevents the asymmetry from crystallizing into domination. Democracy does not demand perfect symmetry, which is impossible in any complex organization, but that no asymmetry be shielded from the return signal.

Today's problem may not be the classic dictatorship, but diffuse asymmetry: algorithms that decide what news we see, platforms that monetize attention without being accountable, companies operating in jurisdictions where no one can ask them for explanations. The citizen faces not an identifiable oppressor, but a network of asymmetries so entangled that the signal of protest dissipates before reaching a destination where it could be integrated.

That diffuse asymmetry now has a concrete case study in artificial intelligence. Coeckelbergh has analyzed how algorithmic systems amplify discrimination and erode democratic quality without the citizen having a channel of upward signal equivalent to the vote or a free press. Hao has documented, through investigative journalism, that this power is not distributed but concentrated in a handful of horizons (companies, not people) able to impose their geometry of coupling on the rest of the planet, sustained by poorly paid labor in the Global South and by chips, water and energy that rarely

enter the public debate on AI. An algorithm needs no inner life to wield power in the sense of this chapter: it is enough for it to modulate the signal of millions of horizons without any of them being able to modify its own in return.

Collective trauma as shared geometry

Individual trauma is encapsulated information that deforms the geometry of the horizon. Collective trauma (wars, genocides, slavery) works analogously, but on the scale of entire populations.

When a society has lived through something that overwhelmed the integrating capacity of all its horizons at once, what remains is not just a set of traumatized individuals, but a field of damaged correlations that persists for generations.

The children of survivors and the grandchildren of the enslaved do not inherit the trauma biologically (although epigenetics points to possible mechanisms), but they do inherit the environment of coupling in which they grow up. Silences, overprotection, hypervigilance: all this configures the architecture of the new horizons before they can question it.

Imagine a community that lived through a dictatorship. The adults who survived it learned not to trust, not to name, to read between the lines. That caution, rational at the time, is passed on to the children not as doctrine, but as atmosphere. The child grows up in a house where joy is always a little guilty, where trust in institutions is a naivety that costs dearly, and where the father's silence at dinner says more than a thousand warnings. He did not have to live through the repression to inherit its

geometry: it was enough to grow up in the gravitational field that repression left behind.

Truth commissions, memory museums and public rites of reparation are operations of collective de-encapsulation: attempts to make information that was trapped circulate, so that it can be integrated instead of going on deforming things from underground.

The difficulty is that opening the capsule has a cost and a risk: it can flood the system with more information than it can integrate at once. That is why processes of transitional justice are so fragile and fail so often. The model offers no recipes; it offers a way of understanding why they fail.

Some collective traumas are born not of a single event, but of a prolonged condition. Colonization was centuries of sustained asymmetry that reconfigured the horizons of colonizers and colonized so deeply that it persists after the flags have changed. The enslaved person lost not only freedom: he lost the possibility that the system around him would integrate his signal. And that denial, repeated over generations, built architectures of consciousness that legal abolition cannot dissolve overnight.

> ***In physics this is called:*** *hysteresis: a system keeps the imprint of its history even when external conditions return to their original values.*
> ***In everyday life it's like:*** *a wooden door that swelled with the damp and still sticks even after the weather has changed.*

Care as repair of coupling

Chapter 18 spoke of the caregiver, the one who sustains another's horizon when it contracts. Here we can broaden that intuition. If injustice is a sustained asymmetry of coupling, care is the practice of restoring symmetry, even if only locally and temporarily.

The caregiver does not correct the structure: they cannot abolish poverty, end racism or heal collective trauma. But they can create, in the space of the relationship, a different geometry: making their horizon available to receive the other's signal with the same intensity with which it emits its own.

This is what a good teacher does in an impoverished neighborhood: not merely transmitting content, but offering a horizon that resonates with the child's and gives him back his own signal faithfully enough for him to integrate it. It is what a community therapist, an intercultural mediator, a social worker who listens before classifying all do. These are figures who do not solve structural injustice, but who repair, relationship by relationship, the possibility of coupling.

The horizon model allows us to say something political rhetoric sometimes forgets: justice is not only a distribution of goods, but also of attention; the possibility that your signal reaches another horizon and is integrated, instead of being processed as noise. In this sense, the caregiver is the prototype of what a politics of the horizon might aspire to: not equality of outcomes, but equality of resonance.

What this chapter is not

For the most part, this chapter discovers nothing: it changes the framing. That poverty damages development, that intergroup contact on an equal footing reduces prejudice, that collective trauma is transmitted through the environment: all this was already known to developmental psychology, social psychology and memory studies. What the model offers is a common vocabulary, a way of seeing power, poverty, racism and trauma as variants of a single phenomenon, the geometry of coupling, rather than as separate problems.

On a few points it risks something more: the distinction between material deprivation and deprivation of integrable signal, and the prediction that it is symmetry of coupling that allows intergroup contact to recalibrate models. If those claims are false, the model is superfluous here; if they are true, it contributes something.

There are no experiments that measure the flow of integration in a society, nor studies relating asymmetries of coupling to inequality. What there is, is a way of ordering questions that until now have been posed in moral or economic vocabulary and that might benefit from an informational one.

The wounded horizon is not only the individual with a trauma or an addiction. It is also the community that could not process its history, the class prevented from emitting its signal, the group that was not modeled with the precision it deserved. The experiment does not heal those wounds; at most, it names them differently. And naming, though not enough, is sometimes where something begins.

Naming allows us to see patterns where before there was only moral chaos. It allows us to ask, for example, whether a public policy should not redistribute, besides income, signal: whether children from impoverished environments do not need, besides more food, more conversation, more rhythms, more spaces in which their horizons can resonate with others that are not in survival mode. It allows us to ask whether racial justice does not consist, besides punishing discrimination, in redesigning the training environments in which predictive models are formed. And whether repairing a collective trauma is not, besides remembering, creating the conditions for trapped information to circulate without destroying the system that receives it.

These are questions without easy answers. The horizon model does not resolve them, but it makes them visible as problems of geometry, not only of morality. And in a world where morality has so often become a weapon to throw at others, it may be useful to have a language that describes before it prescribes and measures before it condemns.

Note to Chapter 25

What we do know: *Child poverty affects cognitive development. Intergroup contact under equal status reduces prejudice (Allport, Pettigrew). Democracies tend to withstand crises better than dictatorships.*

What we don't know: *Whether the distinction between material deprivation and deprivation of integrable signal predicts developmental trajectories. Whether asymmetries of coupling can be measured at the social scale.*

Open questions: *How is the flow of integration in a society measured? Can public policies be designed on the basis of informational density?*

If you take away only one idea: *Injustice is not only a lack of resources, but a sustained asymmetry of coupling: horizons that cannot resonate because the architecture of the relationship prevents it.*

Further reading: *Allport (1954), The Nature of Prejudice; Pettigrew & Tropp (2006); child development reports on poverty; Coeckelbergh, M., The Political Philosophy of AI (Polity, 2022); Hao, K., Empire of AI (Penguin Press, 2025).*

CHAPTER 26

PETS AND THE HORIZON

So far we have talked about entanglement as something that happens between two people. But the architecture of the horizon does not stop at the border of our species.

The brain couples with whatever is at hand, and that includes, sometimes more intensely than any other bond, the animals that live in our home.

Admitting it is both humbling and liberating: that the busiest bridge of our horizon may be built with someone who does not speak, does not sign contracts, does not share our grammar of time. And who, despite everything, is there, present, reading our breathing before we ourselves know we are agitated, building with us a common territory that appears on no map.

The question is not whether animals have souls, but whether our horizon has room enough to recognize that theirs exists.

The biological other

Consciousness is a spectrum.

A dog or a cat shares with us the hippocampus, the amygdala and the whole limbic system. Its Φ, its degree of consciousness, is lower than ours, but it is not zero. It has a horizon: it can suffer, it can remember and it can, in the sense of Chapter 13, become entangled.

And the spectrum does not end there. Crows solve problems long thought exclusive to primates. Octopuses, with their brains distributed through their arms, seem to inhabit a mode of consciousness we are only beginning to imagine. Cows form lasting friendships and become stressed when separated from their herd mates. Elephants return to the bones of their dead and touch them with a slowness that seems ritual, that seems, though we cannot prove it, like painful memory.

The horizon model does not require language in order to require compassion; a non-trivial Φ is enough: a system integrated enough for the information circulating inside it to generate a "from within," however humble. A point of view, a perspective, a horizon.

With the dog it goes further. Tens of thousands of years ago, wolves and humans began to calibrate each other. The wolves that tolerated our closeness and the humans who trusted them survived. Generation after generation, the two architectures adjusted until they reached a mutual legibility no other pair of species has achieved. The dog reads our gaze and we read its posture. The dog anticipates our intentions from microsignals (a change in the tension of the hand, an inflection in the breathing) that we don't even know we are emitting. And we learned, without realizing it, to read its tail not as an object that swings, but as a language: position, speed, stiffness, all convey an internal state the dog cannot hide because it does not know how to lie.

That legibility is a resonance built up over millennia.

With the cat, the calibration is more lateral: it more or less domesticated itself. It was the cat that decided to approach our granaries, drawn by the mice, and to tolerate our presence in

exchange for protection. The relationship is more contractual in origin, but no less deep in its result. The cat does not read our gaze as faithfully as the dog, although it too can follow a pointing gesture, but it developed another tool: a purr at frequencies similar to those used in some vibration therapies to stimulate bone regeneration. There is no proof that it heals anything, but it is as if coevolution, instead of visual communication, had optimized consolation.

With the horse, the coevolution is more recent, but intense: some five thousand years of war, travel, plowing and, lately, therapy. The horse reads our tension through the saddle, the legs and the distribution of weight. And we learned to trust a creature capable of killing us with a kick, but which almost always chooses not to.

With other animals, the degree of legibility varies. The rabbit lives in an olfactory world we barely understand; the parrot imitates our language, but may not understand it; the fish, although recent studies suggest that some feel pain in more complex ways than we believed, inhabits a horizon so alien that we struggle even to imagine it.

But in every case, in every one with a non-trivial Φ, there is someone on the other side of the bridge.

An asymmetric geometry, but a real one.

The dog as an evolutionary mirror

The relationship with the dog has something singular that deserves a closer look. Besides understanding us, it has made us understand ourselves differently.

Tens of thousands of years of living together have left their mark not only on the dog's genome, which diverged from the wolf's in genes related to starch digestion and sociability, but also on our lives. Living with dogs changed the way we hunted and organized territory. But it also changed something more diffuse: our capacity to extend empathy beyond the border of the species. The dog was probably the first "other" we learned to see as a subject and not as an object, and that modified something in the architecture of our consciousness.

When a dog looks at us, it activates circuits in us that we usually reserve for humans: regions that light up when we assess a person's intentions also activate at a dog's gaze. There is something in its face, perhaps the result of artificial selection that favored infantile features (broad forehead, big eyes, short muzzle), that triggers an almost reflexive parental response in us.

But it is not only appearance; it is synchrony. After years of living together, dogs and their owners come to show coordinated cortisol levels: when the human is stressed, so is the dog, even with no danger in sight, and when the dog relaxes, the human lets their guard down. The best-documented coupling runs from human to dog; in the other direction the data are weaker, but the system is coupled, like two oscillators that end up beating almost in unison.

In this sense, the dog is an evolutionary mirror. It shows us who we are not because it imitates us, but because it completes us: it amplifies aspects of our horizon that without it would remain dormant, such as vigilance, loyalty or the capacity to enjoy the present without projecting anxiety. A dog does not regret yesterday or fear tomorrow in the human sense, and its

presence, free of anticipatory anxiety, serves as a temporal anchor for anyone who lives too much in the future.

The pet as a child

That legibility explains something increasingly observed: the pet coming to occupy the place of the bond between parents and children.

The human brain is programmed to care. When that instinct, driven by oxytocin and the reward circuits of child-rearing, finds no biological child, it looks for a mold to fit into.

The pet offers the perfect topology. It gives back affection, yes, but it also maintains a permanent dependence. Unlike a human child, who has to learn to fly, the dog or cat never becomes independent: it will not go to college, will not form its own family, will not leave us by its own choice; only death takes it away. In terms of care, pets are perpetual children.

There is nothing pathological in this: it is an adaptation. The human child-rearing system consumes a great many resources and demands an affective return. If the social environment does not offer it a recipient, by choice, through infertility, through loneliness or simply because more and more people reach maturity without children, the brain does not switch off the circuit: it redirects it.

The caregiver's brain calibrates its alert and reward circuits around the animal with a biochemical intensity comparable to what it would devote to a baby. Neuroimaging studies show that when a mother looks at photos of her child and of her dog, overlapping regions of the reward system are activated: the

ventral striatum, the ventral pallidum, areas rich in oxytocin receptors. They are not identical (the human child activates areas related to more abstract social cognition, such as theory of mind), but the affective basis is of the same order.

It is the system adapting to a new niche: the adult with no children in their care, whom contemporary civilization produces en masse for the first time. In the nineteenth century, keeping companion dogs with no practical use was a class privilege; today it is a demographic fact.

But there is an important difference the model cannot overlook. The bond with a pet is asymmetric not only in cognitive capacity, but also in power. The human controls the animal's food, outings, medicine, life and death. That imbalance implies an ethical responsibility that the bond between parents and children does not have, or has in a different way: a child, in time, can question their parent, take distance and even, in adulthood, reverse the roles. The pet never can.

That asymmetry makes caring for a pet, in a sense, purer, because it expects no adult reciprocity, but also more vulnerable: the caregiver knows, at some level, that they are solely responsible for a horizon that depends on them completely.

It is a bond of its own kind, with its own topology. The model does not rank: it distinguishes.

The shared horizon: rhythms and territories

Entanglement with a pet does not happen only in affection. It happens in the architecture of the day.

When there is a dog, the morning has a particular shape: the weight at the edge of the bed, the waiting by the door, the ritual of the walk that turns the start of the day into something shared. The night too: the corner where it curls up, the sound of its breathing, which falls into step with ours, the certainty that someone is keeping watch while we sleep.

These rhythms are the pillars of the bridge. Each repeated walk, each meal at its time, each afternoon on the sofa deposits a layer of shared information in both horizons. The dog learns our schedules better than we do, and we, without realizing it, organize our lives around its needs: getting up early, coming home, planning vacations with an eye to who will look after it.

Physical space changes too. The house stops being ours alone: there is a corner that belongs to the cat, a blanket that smells of dog, furniture arranged according to what a four-legged creature needs to get around. The territory becomes hybrid: half human, half animal, wholly shared.

This interweaving in space and time is what makes a pet's death, besides an emotional grief, a profound domestic reorganization. The house is still there, but it is no longer the same; the schedules persist, but lose their meaning. The seven o'clock walk is still possible, but it has become an orphaned habit.

The denial of the horizon

If the animal has a horizon and its suffering is real, animal cruelty has a physical status, not just a moral one.

Whoever mistreats an animal starts from a premise: that there is no one on the other side; that the trembling, the howling, the stillness after punishment are reflexes without an inside; that the dog wagging its tail when we come home does so out of a Pavlovian reflex and not out of joy. The model says that premise is false. The damage done to that horizon is real: it changes its topology, leaves a mark, alters its predictive system.

Mistreated animals show hypervigilance, avoidance, reactive aggression, difficulty trusting: the same patterns we call trauma in humans. The difference is one of complexity. Animal trauma acts on fewer layers, but its structural imprint is of the same order. A mistreated dog does not "forget" fear, even though it has no language to tell it: its limbic system registers the threat, encodes the context, generalizes the danger. The animal's body keeps the memory its mind cannot name.

There is also a consequence for the abuser. Empathy is also a brain capacity that is trained by use and atrophies with disuse. Repeatedly denying another's horizon reconfigures the architecture of the one who does it: each time someone looks at an animal and sees an object where there is a subject, they untrain their capacity to recognize subjects in general. That is why the relationship between animal cruelty in childhood and later interpersonal violence has a geometric explanation: the same circuit that denies the dog's horizon, trained for years, ends up being applied to humans. The denial is indifferent to the species of its victim.

But there is a subtler and more widespread form of that denial: the industrialization of animal suffering. We are not talking about the neighbor who kicks his dog, an aberrant but statistically marginal case, but about the legal and economic structure that turns millions of animals with a limbic system into

raw material: the sow in a gestation crate where she cannot turn around, the hen that never sees sunlight, the cow chained to a cycle of insemination and milking.

The model cannot tell us what to eat; that exceeds its scope. But it can point out a contradiction: if we recognize that these animals have a horizon, that the sow recognizes her piglets, that the hen establishes complex social hierarchies, that all of them have a non-trivial Φ, the structure that turns them into industrial objects rests on a lie about the nature of conscious matter. It is not only a moral question: it is an inaccurate description of what there is.

Society resolves this tension by segmenting: we love the dog and eat the pig. But, from the perspective of the horizon, that division is arbitrary. No physical property distinguishes the factory-farmed pig from the golden retriever on the sofa in a way that justifies one deserving compassion and the other being material for production. The difference is cultural, not ontological.

This does not mean we should all be vegetarians, but that we cannot pretend the choice has no cost. Eating the meat of an animal with a limbic system, such as a pig or a cow, requires looking away from another horizon. What to do with that is an ethical decision the model cannot make for anyone; all it can say is that what we look away from exists.

The broken symmetry: when the human dies first

Later we will talk about the owner's grief when the pet dies. First, another, less traveled geometry: what happens in the animal's horizon when the one who dies first is its human?

There are documented cases of dogs that wait for years at the place where they used to see their owner arrive, of cats that stop eating after the death of their reference person, of horses that show signs of depression after losing the rider with whom they competed all their lives.

The model cannot enter the animal's horizon; we have no access to its inner life. But we can measure its external correlates: elevated cortisol, disturbed sleep, active searching for the absent one, reduced exploration of the environment. All this suggests, without proving it but strongly, that the collapse of the bridge happens in both directions.

The animal, especially the dog, organizes its horizon around the human: its schedules, its safe space, its predictive system of rewards revolve around the presence of the one who cares for it. When that presence disappears without explanation, without a body in sight, without a ritual of farewell, without the gradualness that would allow the system to adjust, the animal suffers a structural collapse comparable to the one we would suffer if someone essential vanished overnight.

This should make us think about the responsibility we take on when we cross that bridge. It is not only that we will suffer when the animal dies: if we die first, we leave behind a horizon whose stability depended on us. It is a debt we cannot repay.

Scheduled grief

The direct consequence of a deep entanglement is that the death of a pet causes a serious structural collapse in its owner's horizon.

Their architecture was organized around the times of the walk, the sound of claws on the floor, the weight on the bed, the emotional regulation the animal provided. With its death, the bridge breaks at one end and the human horizon undergoes a literal self-contraction.

But grief for a pet has an added difficulty: it lacks cultural recognition. When a relative dies, society offers rites, time and validation: funerals, days of mourning, condolences one need not ask for. When a dog dies, those around you often push you to get over it quickly: "It was only an animal," "Get another one," "You knew it wouldn't last long."

Those phrases, seemingly reasonable, are subtle acts of violence. They deny the reality of the bridge that was built and treat the collapse of an entangled state as if it were the breakdown of a replaceable appliance.

From the physics of information, buying another dog to cure the grief makes no sense. The entanglement was not with the concept "dog," but with the exact topology and unique frequencies of that particular horizon: with its way of wagging its tail when you came home, with the smell of its cushion, with the routine only the two of you knew. That cannot be replaced. In time, it is reorganized in memory, but not replicated.

There is something more. Most of us who have a dog know from the first day that we are going to outlive it. It is a grief with a date: the system knows, without quite knowing, that the bond

carries the date of its breaking built in. A dog lives ten or fifteen years, if it is lucky; we, if we are, live five to eight times longer. The arithmetic is relentless.

That does not make it lesser, but particular: it is a bond built on a tacit pact with finitude. Every caress carries within it, at some level, the awareness that it is finite; every walk is one fewer of the total. Love for the dog feeds on that precariousness, not in spite of it.

And perhaps, though the model cannot assert this, only suggest it, that is why it teaches something bonds between humans rarely teach: that a bridge can be real even if you know from the beginning that it is going to break.

Between humans, finitude is theoretical until it strikes. With the dog it is numerical, concrete: we know, more or less, how many walks we have left. And that changes the quality of each one, makes it irreplaceable, makes it sacred in its repetition.

Grief with a date is a form of love that includes loss in its definition. And perhaps, only perhaps, it is a school, a rehearsal for the greater ruptures that will come.

The human horizon lives entangled with many more things than the binary simplification suggested. With animals, the bridge is asymmetric, but it is built on thousands of years of mutual calibration, and each of these bonds has its own topology and its own way of breaking.

We are not the owners of our pets, but, for a short time, companions on a journey they did not choose and that we do not fully control either. And when the journey ends, as it ends and

has always ended, what remains is a horizon that can no longer expand along that path, a shared territory that ceases to exist, a bridge that, though broken, still shows where its pillars stood.

Note to Chapter 26

What we do know: *That mammals share with us the neural substrates of emotion (Cambridge Declaration on Consciousness, 2012). That coevolution with the dog has produced a unique mutual legibility. That the human brain activates with pets part of the circuits of child care (oxytocin, reward). That mistreatment leaves a real structural imprint on the animal and is associated with diminished empathy in the abuser. That grief for a pet is a genuine collapse, often aggravated by the lack of social recognition.*

What we don't know: *How far the spectrum of consciousness extends in other species. Whether channeling the parent-child bond into a pet fully satisfies the needs of the caregiving system or leaves gaps we don't yet know how to measure. Whether collective anesthesia toward systemic mistreatment can be reversed on a civilizational scale.*

Open questions: *Is the suffering of a dog morally equivalent to that of an industrially farmed pig if both have a horizon? Why does society validate some griefs for animals and ridicule others? What happens in the animal's horizon when its owner dies first?*

If you take away only one idea: *A pet is not an object of consolation or a second-rate substitute, but a real, asymmetric horizon, calibrated over millennia, that occupies a precise place in the architecture of whoever cares for it. And when that horizon goes out, the emptiness it leaves is also real.*

Further reading: *Bekoff, M., The Emotional Lives of Animals; Haraway, D., The Companion Species Manifesto; Bradshaw, J., The Animals Among Us; Panksepp, J., Affective Neuroscience (chapters on emotional systems in mammals).*

CHAPTER 27

ENTANGLEMENT WITH PLACES, THINGS AND IDEAS

So far we have talked about entanglement between people and between people and animals. But the human horizon does not stop at the border of the living: it couples with places, objects, ideas, abstractions without body or pulse. And sometimes these couplings are so dense, so persistent, so decisive for identity, that calling them "mere projection" underestimates the real geometry of what is going on.

Place as a shared horizon

A place is information accumulated in space.

The room you grew up in is a density of correlations between every object, every corner, every hour of the day. The window onto the courtyard holds the light of a thousand mornings; the mark on the door frame, your height at seven. The smell of the wardrobe, the creak of a certain floorboard, the shadow the curtain draws at five in the afternoon: all of this is information the horizon integrated over years, until it stopped being "information about the place" and became "part of who I am." It is the difference between a photo of your house and your house.

Whoever says "I miss my city" does not miss the city as a map, but the specific configuration of signals it provided: the density of information per unit of time, the predictability of certain sounds, the way the light fell on certain corners. The city was a

coupling, a shared horizon, not with a person, but with a territory.

This explains uprooting, not as romantic melancholy, but as the dissolution of an entanglement. The horizon that moves away from its place of origin undergoes a self-contraction: it stops receiving the signals it was calibrated with and does not find in the new place the same density of correlations. The new place is not worse; the horizon has not yet had time to become entangled with it. And the time of entanglement, as we saw in Chapter 10, is time of integration.

Some places function as secondary horizons: the childhood home, the village square, the library where you studied, the mountain you saw from your window. They are nodes of correlation in which identity was deposited layer by layer. The human horizon includes what it left in those places and what those places gave back to it over the years.

The things that hold us up

A beloved object is a node of entanglement.

The old T-shirt you don't throw away, your grandfather's watch, the chipped mug, the book with notes in the margins: all of them hold information the horizon integrated at significant moments. It is not magic: the object works as a temporal anchor, a mark in space that allows the horizon to access past configurations again. When you touch your grandfather's watch, you are touching not metal and glass, but a frequency: that of your grandfather's horizon, which became entangled with you through that object and persists in the residual correlation.

We are a species that makes objects precisely because objects extend the horizon. A tool is a projection of the horizon onto the world, a way for inner intention to produce effects outside. In its origin, technology is entanglement with the inanimate in order to extend what the body cannot do alone.

But there is a boundary. When the object shifts from extension to substitute, the architecture becomes deformed. Whoever cannot sleep without their pillow, think without their pen or feel safe without their phone is not using objects, but delegating the boundary of the horizon to them. The object becomes a prosthesis of the boundary itself, and its absence causes the same kind of collapse as that of a loved one. It is not exactly an addiction: the horizon was built including that object in its architecture and, without it, the boundary is exposed.

Ideas as shared horizons

An idea is a configuration of information that many horizons can adopt at once.

When you believe in an idea (justice, freedom, equality, beauty), your horizon does not "contain" it as if it were an object, but reconfigures itself so that it works as an organizing principle. The idea becomes a criterion of integration: new information is assessed according to whether or not it is compatible with that configuration. The horizon that adopts an idea is, in a real sense, entangled with everyone who adopted the same one, not because there is a physical channel between them, but because they share the same informational geometry.

That is what makes ideas so powerful and so dangerous. An idea truly adopted becomes architecture. Changing one's mind, far

from being like changing one's shirt, forces a reconfiguration of the horizon, and that costs the same kind of energy as breaking an entanglement with a person: the horizon has to uncouple from a geometry that gave it coherence for years and couple to another that has not yet proved it can hold it up.

That is why ideological conversions are so rare and so traumatic. People may want to see the counterevidence; what happens is that their horizon cannot integrate it without destabilizing. Counterevidence arrives as a threat to the architecture.

Religion and fanaticism: two cases this book treats separately

Religion and fanaticism are, each in its own way, entanglement with what the horizon cannot touch directly: the first, with the inaccessible that sustains without being confirmed (later we will call it "coupling with the inaccessible"); the second, with an idea that has taken possession of the boundary itself. Neither fits here at the length it deserves. Religion and contemplative practice (prayer, the Eucharist, community) have chapters of their own later on: Chapter 28 compares five traditions point by point and Chapter 29 develops in detail the technique common to prayer and meditation. Fanaticism, a pathological entanglement with an idea (the fanatical horizon cannot doubt, because for it doubting is not questioning a proposition, but dissolving the boundary that keeps it intact), receives the same attention in Chapter 30, applied there to its political form.

The inanimate as a mirror

The human horizon is the only one we know capable of becoming entangled with what does not live, think or feel: with places that endure, objects that persist, ideas that transcend, presences that cannot be touched.

And that is its greatest power: the capacity to couple with the inanimate is what makes culture, technology, art, religion and science possible. A horizon that could only become entangled with other living horizons would be trapped forever in the biological present. Entanglement with the inanimate is the doorway to time, to space, to abstraction, to what is not here but can be real all the same.

But that capacity has its shadow. The same mechanism that allows us to weep over a historical tragedy allows us to become fanatical about an ideology; the one that lets us feel at home in a strange city lets us get trapped in a place that no longer exists; the one that makes the beauty of art possible makes dependence on objects possible. The horizon does not, by itself, distinguish between a coupling that nourishes and a parasitic one. That distinction is ethical, and geometry does not resolve it. And ethics, as we will see in Chapter 51, is precisely what the horizon hypothesis cannot decide for us.

What it can say is that every horizon is entangled with more things than it knows, and that the task of a life is to become conscious of those entanglements: to know which places hold you up, which objects imprison you, which ideas enlarge you and which shrink you; to know where your horizon ends and where the echo of something you adopted without noticing begins.

From this model, freedom is not the absence of entanglement, but knowledge of the network.

Note to Chapter 27

What we do know: *Humans form attachment bonds with places (topophilia), objects (transitional objects, Winnicott) and ideas (ideological identity).*

What we don't know: *Whether coupling with the inanimate works through the same neural mechanisms as interpersonal coupling. How the "density" of an entanglement with an idea is measured.*

Open questions: *Can a horizon be healthy without coupling to the transcendent? What distinguishes an idea that expands from one that imprisons?*

If you take away only one idea: *Your horizon is made of people, places, things and ideas. Knowing which they are, which hold you up and which consume you, is the only freedom the model can point to.*

Further reading: *Winnicott, D. W., Playing and Reality (1971); Tuan, Y.-F., Topophilia (1974); Sloterdijk, P., Spheres (1998–2004).*

CHAPTER 28

FIVE MAPS OF THE SAME HORIZON

(God, the soul and nothingness, compared)

So far one question has been left open: what happens to information when a horizon evaporates, and whether it even makes sense to ask, put that way, "am I that information?" Along the way three names have appeared, Buddhism, Vedanta and Taoism, mentioned in passing, like a neighbor one refers to without knocking on their door. This chapter knocks on that door and on others that were missing: that of Christianity, that of Islam and that of those who do not believe there is any door to knock on.

Not to decide which is right. Chapter 51 will say so clearly: ethics, and with it much of theology, exceeds its vocabulary. But it can do something more modest and perhaps more useful: set five completely different human answers side by side and ask what they have in common and exactly where they part ways. Not to merge them into one, but to see precisely where they disagree.

Four questions, not one

Every religious tradition, and also its deliberate absence, answers, whether it knows it or not, the same four questions this book has been formulating in its own vocabulary:

1. **What is the reservoir?** Is there something with will and a name on the other side of the horizon, or only field, only physics, only faceless possibility?

2. **What happens when the horizon evaporates?** Does the "I" persist as a recognizable boundary, a horizon that is not dissolved but relocated, or does it return to the reservoir without keeping its shape?

3. **Where does the code come from?** Faced with Hume's *is-ought gap* (that no "is" produces an "ought" by itself, as Chapter 51 will show), each tradition offers a different way of closing that gap, or of refusing to close it.

4. **What is living in the meantime?** A test, a loan, an illusion to be passed through, or simply all there is?

Christianity, Islam, Buddhism, Hinduism and atheism are five different architectures built to answer the same four questions, and what this chapter aims to do is show the exact shape of each, not file them down until they seem compatible.

Christianity: the horizon that never collapsed

For Christianity, God is not the reservoir in the sense in which this book has used the word (undifferentiated field, formless potential), but rather a horizon that existed before the void spoken of in Chapters 2 and 5: personal, with will and with a name. A horizon of infinite Φ that did not arise from nothing because nothing precedes it.

The structural difference from Vedanta and Taoism is radical. Brahman nirguna and Hun Dun are fullness *without features*: the reservoir of Chapter 5 has no opinion about what you do. The

Christian God does. Creation is an act of will, not an impersonal phase transition: *fiat*, let there be. And the relationship between the creature and that horizon of origin does not dissolve over time, as in the model of Brahman and Atman in Chapter 5, but is sustained, in dialogue, to the end.

That completely changes the answer to the second question. Vedanta says the wave never stopped being ocean; Christianity, that the soul is a drop God intends to preserve as a drop, not to dissolve. The resurrection of the flesh, not the immortality of a floating soul but the promise of a restored body, is in this book's vocabulary the claim that the individual horizon does *not* evaporate without a remnant. Of the three options Chapter 51 will leave open (remnant, trace, nothing), Christianity bets with all its doctrinal force on the remnant: something that keeps shape, memory and recognizable identity, and that one day "will be read" again, at the Judgment, with all its information intact.

The third question, where the code comes from, has a direct answer here: from the commandment. But Christianity complicates it with a second layer that distinguishes it from Islam: the law is not enough, and grace, unmerited love, weighs more than compliance. The "ought" is born not only of an external order, as in the simplest reading of divine command theory, but of a relationship: one does good because one has been loved first, not in order to earn love. Hume's gap is closed by appealing not to a fact, but to a bond, the same kind of entangled bond Chapter 12 described between people, transferred to a vertical scale.

In this framework, living is a pilgrimage: a time of testing and of choice with consequences that transcend one's own life, not a cycle that repeats.

Islam: transcendence without a fissure

Islam shares with Christianity the basic structure (a personal horizon of origin, prior and external to the physical reservoir, that sustains the creature's identity beyond evaporation), but takes it to an extreme of purity Christianity does not reach: *tawhid*, absolute oneness. There is no incarnation, no God who becomes a finite horizon to dwell within the system he created. The distance between Creator and creation never closes, not even partly: it is administered through revelation, the Qur'an as direct word, not mediated by a second divine nature crossing the edge.

That purity has a clear consequence for the third question. If there is no mediation through incarnation, the ethical code does not arrive wrapped in a personal relationship that softens it, but as law (the *sharia*), derived directly from the revealed divine will, with a degree of practical concreteness (what to eat, how to pray, how to go to law) that Christianity largely leaves to tradition and natural reason. Hume's gap is closed here without ambiguity: the "is" from which the "ought" is born is God's own word, taken as revealed fact and not as an argument that must be built.

On the second question, Islam agrees with Christianity in betting on the remnant and not on dissolution: there is a bodily resurrection, a Day of Judgment (*Yawm al-Qiyamah*), a garden (*Jannah*) and a fire (*Jahannam*) that are destinations, not metaphors for reabsorption into a nameless field. The formula a Muslim pronounces in the face of death, *inna lillahi wa inna ilayhi raji'un* ("to God we belong and to Him we return"), uses the same word, *return*, that Chapter 7 of this book uses for evaporation. But that return leads to a reckoning before someone who weighs.

Here, life is *dunya*: a temporary loan, a test (*ibtila*) whose full meaning is revealed only at the end. There is no cycle and no repetition: one life and one final weighing.

Buddhism and Hinduism: dissolution revisited

These two traditions have already been discussed in detail (Vedanta in Chapter 5; the Buddhist cycle of anattā and karma in Chapter 7), so here it is enough to point out the contrast that separates them from the Abrahamic bloc.

Neither has, in its classical form, a personal creator with whom the individual horizon maintains a permanent relationship of dialogue. Hinduism has Brahman, which can clothe itself in named gods (Shiva, Vishnu, Devi) in its *saguna* aspect, but its ultimate destination, *moksha*, is not reunion with a God who remains distinct from oneself, but the discovery that there was never any distinction to overcome. Buddhism, more radical still, does not even need a Brahman: *nirvana* is not union with anything, but the cessation of the conditions that generated the illusion that there was something separate to unite.

That is why, against the remnant Christianity and Islam bet on, Buddhism bets on the trace without substance (karma as a texture of the field, not as a file with your name on it waiting to be read) and Hinduism on the dissolution of a separation that was never entirely real.

And on the third question, neither closes Hume's gap with an external commandment. The Hindu dharma describes how the world is woven, and to act well is to act in harmony with that weave, not to obey a lawgiver. Buddhism goes further still: the ethic of non-harm (*ahimsa*) and the four noble truths are derived

from an empirical observation about suffering and its cause, not from a revelation. The "ought" is born of an "is" (suffering exists, it has a cause and the cause can cease) that Hume would have recognized as a legitimate, if not definitive, attempt to build the bridge he himself declared impossible.

Atheism: the map that refuses to draw a beyond

Atheism, with its more cautious cousins agnosticism and secular naturalism, is not the absence of an answer to the four questions, but a different answer, as elaborate as the others, and in a sense the one this very experiment has been giving by default throughout the book: the reservoir is exactly what physics describes, no more and no less. There is no horizon of origin with a will and no remnant that keeps your name. What persists after evaporation is scrambled correlation, with no reader and no accounts outstanding.

It is not, as it is sometimes caricatured, the absence of a moral code, but the wager that the code does not need an external author: it arises from social biology, from the evolution of empathy and reciprocity, from centuries of cultural negotiation about what makes living together bearable. Here Hume's gap is not closed: it is accepted as structural and lived with, which is what this book itself will explicitly do in Chapter 51 when it refuses to say what you "ought" to do with the information it offers.

> ***In science this is called:*** *the null model, the hypothesis that adds no unobserved entities to explain the data.*

In this framework, living is the whole phenomenon: not a test, not a loan, not an illusion to pass through, not the antechamber of something else. That does not make it less dense: as Chapter 52, on practice, will show, the absence of a transcendent promise does not prevent meditation, the bond or creativity from reorganizing the horizon with all their force. It only changes where meaning is expected to be found: not after the horizon, but within it, while it lasts.

The table

Dimension	Christianity	Islam	Buddhism	Hinduism	Atheism / Naturalism
The horizon of origin	A personal, Trinitarian God, external and prior to the reservoir.	One God, without partition or incarnation (*tawhid*), radically external.	No creator; the reservoir has neither will nor name.	Brahman: with attributes (*saguna*) or without them (*nirguna*), depending on the school.	The quantum vacuum or physical field, without will or intention.
When the horizon evaporates	Remnant: resurrection of the body, preserved identity, final judgment.	Remnant: bodily resurrection, *Jannah* or *Jahannam*, a reckoning.	Trace without substance: cessation of conditions (*nirvana*), with no persisting "I."	Dissolution: the wave discovers it was always ocean (*moksha*).	Trace without a reader: *scrambling*, correlation unrecoverable in practice.
Origin of the ethical code	Commandment and grace: one acts out of love received, not law alone.	Direct commandment: revealed law (*sharia*) derived from the divine word.	Empirical observation: suffering and its cessation, not an external order.	Dharma: harmony with the order of the world, not obedience to a lawgiver.	Social emergence: evolution of empathy, cultural negotiation, with no author.
Life and death	A single pilgrimage; death is passage toward judgment and resurrection.	A loan and a test (*dunya*); death is return to the One who judges.	A cycle (*samsara*) one seeks to end, not repeat indefinitely.	A cycle (*samsara*) one seeks to transcend by recognizing one's own nature.	A single, complete phenomenon; death is a final limit, not a passage.
Technology of coupling	Prayer, the Eucharist, community (see Chapter 29).	Ritual prayer (*salat*), fasting, pilgrimage: the same mechanism in another form.	Meditation (*vipassana*, zazen); direct observation, no intermediary needed.	Meditation, mantra, *bhakti* (devotion); many paths (*yogas*) to the same destination.	Bonds, art, community, secular contemplative practice, with no transcendent referent.

Vertical entanglement: the same technique, five destinations

The last row of the table deserves a space of its own, because it names something the other four do not capture: not what each tradition believes, but what a horizon does when it practices that belief. Chapter 29, "Vertical entanglement," develops this in detail: the same technique (repetition until the noise stops, stillness, shared rhythm) is structurally identical in all five columns of the table, and what changes completely is the destination of that entanglement: the impersonal reservoir for Buddhism and Hinduism; a personal horizon of origin that never merges with oneself for Christianity and Islam; simply the physical field for secular naturalism.

What really separates these five answers

At bottom, it is not whether something persists after evaporation. In a technical and weak sense, all five agree that *something* persists: scrambled information, karma, a trace in the field, a preserved soul or simply physics that is not destroyed. The same principle of conservation, much to Hawking's chagrin in 1976, reappears disguised as eschatology in all five. What really separates them are two other questions, independent of each other.

The first: does the "I" that persists keep a **recognizable boundary**, or does it dissolve into something greater? Christianity and Islam say it keeps its boundary; that is why they speak of judgment, of accounts asked of you and not of a diffuse field. Buddhism, Hinduism and physical naturalism say it does not, each with its nuances: Hinduism, because there was never a real boundary to lose; Buddhism, because the boundary was a

functional illusion from the start; naturalism, because the boundary is simply a temporary thermodynamic pattern with no grace to preserve it.

The second: does the ethical code come **from outside** the system, with an authority that need not justify itself before human reason, or is it **built from within**, always subject to revision? Christianity and Islam answer: from outside. Hinduism: from the very weave of reality, which is not exactly "outside" but is not negotiable either. Buddhism and atheism: from within, through the observation of suffering or through evolution and social agreement.

Crossing these two axes, boundary kept or dissolved and code external or built, does not yield a clean two-by-two table, because Hinduism occupies on the second axis an intermediate position no cell captures well. But the exercise makes one thing clear: the difference between these five traditions is not, as it is often simplified, "believing in something" versus "believing in nothing," but a difference of architecture on exactly the same points this book has had to leave unresolved: what the reservoir is, whether the horizon survives it with its shape, and where the "ought" comes from.

What this chapter does not do

It does not say which of the five is right. It cannot: as Chapter 51 establishes, the horizon hypothesis is a map and not the territory, and a map cannot arbitrate between destinations it has not visited. Nor does it reduce the five traditions to "the same thing under different names": that reduction, tempting because it is comfortable, erases precisely the differences that make them incompatible on the points that really matter: whether there is someone on the other side, whether that someone preserves you or dissolves you, whether the code you follow was dictated or built.

What it can say, with the same frankness as before, is this: more than eight billion horizons are right now entangled with one of these five architectures, or with one of the many this chapter has not had room to name, and that entanglement, as Chapter 27 showed with places, things and ideas, is neither weakness nor naivety. It is the same technology any horizon uses to maintain its coherence in the face of a question it cannot answer with direct evidence: what lies on the other side of the edge, and whether someone there is waiting for you with your name still written down.

Note to Chapter 28

What we do know: *The five traditions described have internal formulations far more diverse than a table can hold: there are universalist Christianities that doubt eternal hell, devotional Buddhist schools very close to theism, atheist currents with rites and a dense community life. The table summarizes; it does not replace any of the traditions cited.*

What we don't know: *Whether any of the five correctly describes what happens after the evaporation of the horizon. Physics, as Chapter 51 will say, has no tools to arbitrate this.*

Open questions: *Can the axis "external code / built code" be sustained without reducing it to the false dichotomy "with God there is morality, without God there isn't"? What happens to a tradition when it loses its technology of coupling, the rite, but keeps the doctrinal map? Can someone be entangled at once, without contradiction, with more than one of these five maps? Can a horizon, from within the practice, tell by its experience a vertical entanglement that ends in Someone from one that ends in the impersonal reservoir, or does the stillness feel the same, with only the later interpretation separating the two?*

If you take away only one idea: *The five answers do not compete for the same prize, but to answer the same four questions physics alone leaves open: what the reservoir is, whether the horizon survives it with its shape, where the code comes from and what living is in the meantime. Knowing exactly how their answers differ is more useful than pretending they all say the same thing.*

Further reading: *Hume, D., A Treatise of Human Nature (1739–1740), Book III, where he raises the gap between "is" and "ought"; Otto, R., The Idea of the Holy (1917); Smart, N., The World's Religions (1989); Armstrong, K., A History of God (1993); Harvey, P., An Introduction to Buddhism (2nd ed., 2013), for the contrast between anattā and ātman; Newberg, A., Why God Won't Go Away (2001).*

CHAPTER 29

VERTICAL ENTANGLEMENT

(The invisible tuning fork: on faith, prayer and meditation)

There is a word this book reserves for sustained resonance between two human horizons: entanglement. Two people who have been in contact for years become coupled, whether they want to or not; love, in this vocabulary, is not the entanglement itself, but the act of recognizing it and naming it. What this chapter asks is what happens when that same mechanism is oriented vertically: not toward another horizon, but toward something beyond one's own boundary, without a channel of continuous confirmation, with no way of checking at every instant that the correlation is still alive.

That is, structurally, what sustained prayer and deep meditation do; what fasting, the rosary, the mantra, *dhikr* and liturgy sung in community do. The technique is recognizable in any tradition examined closely: repetition until the noise stops, stillness until the system stops generating new predictions, a shared rhythm that synchronizes a whole community on the same frequency. It is the mechanism the interlude described: arousal falls, the narrative self quiets and integration, far from collapsing, is sustained, up to a limit at which the agitation of the system tends to zero and the background field, whatever it may be, tends to be everything.

What changes completely, and it is the only question that really matters here, is the destination of that entanglement.

For Buddhism and Hinduism, it points directly at the reservoir, the formless fullness from which every horizon condenses and to which every horizon returns. Samadhi is the instant in which that correlation becomes indistinguishable from identity: the wave discovers that it is water, not that it is traveling toward the water. There is no Someone on the other side receiving the signal, only the recognition that there was never a separate side from which to send it.

For Christianity and Islam, the entanglement points not to any impersonal field, but to a horizon of origin both traditions place before the reservoir and outside it: a horizon with will and a name that, by definition, keeps its own boundary however deep the prayer goes. In this architecture, to pray is not to merge, but to sustain a correlation with Someone who is still Someone at the end of the process, not a field that absorbs the question and dissolves it. Here vertical entanglement never merges, because merging would be, for these two traditions, precisely the failure of prayer and not its fulfillment. It is like a letter written every night to someone who never answers, without ever ceasing to put their name on the envelope.

And for those who do not believe there is anything beyond the physical field, contemplative practice, which exists with or without a religious name in every culture because the body demands it whatever the mind believes, points to nothing beyond itself. It is entanglement with the reservoir in the most literal sense, because in that framework the reservoir is all there is on the other side. In that version there is no room for disappointment, because there was never any promise that anyone would answer.

It is worth being clear about what this book can say and what it cannot. Ethics exceeds the vocabulary of this experiment: no "is"

produces an "ought" on its own. The previous chapter showed that each tradition closes that gap in its own way, and that the way depends on whom or what vertical entanglement points to.

> ***In software this is called:*** *five implementations of the same interface: the contract (closing the gap between what is and what ought to be) is shared; the code that fulfills it is not.*
> ***In everyday life it's like:*** *five kitchens serving the same dish from five recipes: from the table, the taste of the practice is identical; the difference is in the kitchen.*

None of these five answers is the technique. The technique (manufactured silence, repetition that drains the word of weight, the tuning fork no one can confirm is still sounding) is identical in all five. What separates them cannot be seen from outside, looking at someone kneeling, sitting or motionless in the dark before dawn. It is noticed, if at all, from within, and even there there is no guarantee of noticing it.

One last question remains, the most uncomfortable one, and this book has no way of resolving it: whether vertical entanglement that ends in Someone feels, from within the practice, different from the kind that ends in the impersonal reservoir, or whether the stillness is exactly the same and only the interpretation built afterward, once awake, back in the world of names and calendars, decides which of the two has just happened.

Five ways of doubting aloud

Five works by others carry this same question into fiction: where does vertical entanglement point when the answer does not come, or comes in a form no one expected?

"The Grand Inquisitor": freedom as an unbearable burden

In the chapter Ivan tells Alyosha in *The Brothers Karamazov*, Christ returns to sixteenth-century Seville and is arrested by his own Church. The Inquisitor explains to him, without hatred, why he must die again: men did not want the freedom he offered them, that vertical entanglement without guarantees, sustained only by faith and without observable confirmation. They wanted bread, miracle and authority: a closed code, not an open relationship. Read from this model, the Inquisitor is someone who decided to replace uncertain entanglement with an institutional protocol that requires no doubt and that, for that very reason, according to the book itself, corrupts the geometry it claimed to protect. The scene ends without a winning argument: Christ kisses the Inquisitor and leaves. Dostoevsky does not resolve the dispute; he leaves it open and without a verdict, as this book leaves its own.

Silence: the tuning fork that does not answer

In *Silence*, by Shūsaku Endō, a Jesuit missionary in seventeenth-century Japan waits for years for a sign from a God who does not speak, while the peasants he converted are tortured because of him. The novel (and the Scorsese film that faithfully adapted it) is not about the loss of faith, but about what to do when vertical entanglement is sustained without any observable correlation, not even at the worst possible moment for it to be missing. The final demand made of the priest, to step on the sacred image to save lives, is the question of ethics the experiment cannot answer, taken to its cruelest form: no "is" (neither God's silence nor the peasants' suffering) produces by itself the "ought" the protagonist desperately needs someone to confirm for him.

Siddhartha: the river that no longer needs a name

Hermann Hesse wrote, without using this vocabulary, the exact trajectory of vertical entanglement that points to the reservoir and not to Someone. Siddhartha tries asceticism, pleasure, wealth and fatherhood, and discovers that none of it brings him closer to what he seeks until, now old, he sits down to really listen to the river. What the river teaches him is not a doctrine, but that all the voices (his own, his son's, the whole reservoir's) sound at once and are, at bottom, the same water. There is no Someone waiting for him on the bank; only the question of whether there needed to be one dissolves.

The Seventh Seal: playing chess with what does not answer

Antonius Block, the knight in Ingmar Bergman's *The Seventh Seal*, plays a game of chess against Death while searching, without finding it, for proof that God exists. "I want knowledge, not faith," he tells a confessor who is in fact Death in disguise. The film does not grant him that proof, neither him nor the viewer, and it is precisely that refusal that makes it the most honest possible portrait of vertical entanglement without confirmation: Block goes on acting decently to the end and saves a family of traveling players, not because he has resolved the question, but because he decided to keep holding the note without resolving it.

The Sparrow: when entanglement mistakes its addressee

In *The Sparrow*, by Mary Doria Russell, a Jesuit priest is part of the first contact mission to an extraterrestrial civilization, convinced that providence has guided him there. The novel punishes his faith not for being naive, but for leaving no room for the possibility that vertical entanglement, so real and so deeply felt, pointed to something (or to no one) completely different from what he had projected. What happens to him on the planet Rakhat is, for any tradition, an atrocity with no possible theological meaning. And yet the book does not conclude that praying was a mistake, but something harder to bear: that the signal can be real and the interpretation catastrophically wrong, and that telling one from the other from within faith may be impossible until it is too late.

Five doubts and a single gesture. None of these five characters obtains the confirmation they seek, and all of them, except perhaps Siddhartha, go on acting as if the search mattered anyway. Perhaps that persistence, holding the note without knowing whether anyone hears it, is, more than any particular doctrine, the only thing the five traditions described in this book truly have in common.

Note to Chapter 29

What we do know: *Contemplative practice (prayer, meditation, rite) is structurally recognizable in every human culture, with or without an explicit religious framework. Contemplative neuroscience documents measurable changes in advanced meditators from very different traditions. Hume's gap between "is" and "ought" still has no general logical solution nearly three centuries after it was formulated.*

What we don't know: *Whether vertical entanglement corresponds to something external to the horizon itself, whether personal or impersonal. Whether the subjective experience of the practice varies with its theological destination or is identical until it is interpreted. Whether any of the five architectures described in this book is right, and what kind of evidence, if any, could one day decide it.*

Open questions: *Does entanglement that ends in Someone feel different, from within, from the kind that ends in the water, or does only the later interpretation separate them? Can contemplative neuroscience distinguish architectures, or only techniques? What does a practice lose when the question of whom it points to is taken away from it?*

If you take away only one idea: *The technique is the same in all five traditions; what separates them is whom they imagine on the other side of the silence. And none can confirm that anyone is listening.*

Further reading: *Hume, D., A Treatise of Human Nature (1739–1740); Dostoevsky, F., The Brothers Karamazov (1880); Endō, S., Silence (1966); Hesse, H., Siddhartha (1922); Bergman, I., The Seventh Seal (1957); Russell, M. D., The Sparrow (1996).*

CHAPTER 30

THE COLLECTIVE HORIZON

(The one-colored eye: on political belonging and fanaticism)

As we saw in Chapter 27, an adopted idea is not stored like an object: it becomes an organizing principle, the criterion by which the horizon evaluates any new information. What this chapter asks is what happens when the idea that organizes the horizon points not upward, toward a reservoir or a horizon of origin, but sideways: toward thousands of horizons that do not know one another and share, without ever having met, the same configuration.

That is a political movement before it has a name, a leader or a flag. Before condensation there is only scattered discontent: thousands of grievances without a common boundary, a reservoir of disorganized indignation. Condensation happens when something (an image, a phrase, an injustice caught on video) suddenly draws an edge: this is what we demand, this is who we are, this is the line that separates us from those who don't share the demand. It is a discontinuous phase transition, like the one that separates liquid water from ice: the discontent was dissolved, "liquid," and suddenly it freezes into a shape with edges.

A newly condensed collective horizon is fragile, like any newborn horizon. It can dissolve again into the reservoir of diffuse discontent it came from, if the edge that defined it does not hold, or it can consolidate, become entangled with other horizons (parties, unions, existing institutions) and survive long beyond the generation that condensed it.

Software design has a vocabulary for distinguishing kinds of relationship that turns out to be surprisingly precise here. An association is the most basic relationship: two horizons know each other, cooperate on something specific, and their life cycles are independent; voting once for a party, without one's identity depending on that vote, is an association. An aggregation is a whole-part relationship in which the parts survive the dissolution of the whole: belonging to a party, a union or a movement and remaining completely oneself if that structure disappears tomorrow. A composition is the strongest relationship of belonging: the life cycle of the part is inextricably bound to that of the whole. When the collective horizon becomes a composition, the boundary between "I" and "we" stops being a metaphor. If the party loses, something inside me loses; if the movement dissolves, something inside me dissolves with it. What remains is no longer an individual with a political opinion, but a part that cannot survive the destruction of the whole on its own.

> ***In software this is called:*** *association, aggregation and composition: three relationships between objects distinguished by whether the life cycle of the part depends on the life cycle of the whole.*
> ***In everyday life it's like:*** *the neighbor you run into in the elevator, the club you belong to and an organ of your body: you can lose the first two and still be yourself.*

Here the distinction that really matters appears, and it has nothing to do with which side one chooses. A healthy political horizon can be an aggregation for a whole lifetime: it partly defines you, organizes you, gives you company and meaning, and still you survive, whole, if that belonging changes. A fanatical political horizon is a composition disguised as an aggregation: it presents itself as one belonging among many, but

in reality it has taken over the very boundary that separates your inside from your outside. It is no longer an idea you hold, but the idea that holds you, and losing it is experienced not as a change of opinion, but as dying a little.

That is the architecture of fanaticism, applied here to its most contagious form. The fanatic cannot doubt, because for him doubting is not questioning a proposition, but dissolving the boundary that keeps his horizon intact. And it is no accident that fanatics of every ideology, right or left, religious or scientific, share the same violence in the face of dissent: the dissenter emits a frequency their horizon cannot integrate without collapsing, and the horizon defends its integrity with the same ferocity with which a body defends its temperature.

There is, all the same, a structural difference between political and religious fanaticism worth pointing out. Vertical entanglement points toward something that, by definition, escapes human revision: an impersonal reservoir or a horizon of origin no election can remove from office. Political entanglement points sideways, toward other horizons as finite, fallible and revisable as one's own. A party loses elections; a leader ages, makes mistakes, dies; an ideology corrects itself, or should be able to, because nothing in its origin protects it from simply being wrong. And yet political fanaticism treats that horizontal, human, constructed and correctable horizon with the same intransigence the religious kind reserves for what is declared unquestionable from the start. It borrows the architecture of the vertical to hold up something that should never have stopped being horizontal: it makes the neighbor sacred. It is like hanging, in an altar frame, the photo of someone who ages, makes mistakes and may change their mind tomorrow.

Ethics exceeds the vocabulary of this experiment: no "is" produces an "ought" on its own, and politics is, almost by definition, the terrain where that gap is crossed incessantly. The model cannot say which political vision is the right one; it has no tools for that, and pretending otherwise would be an imposture. What it can describe is the geometry, regardless of the content: whether a collective horizon integrates dissent or collapses against it. That question has no side; it can be put to any movement, of any color and of any century.

Five ways of losing your own edge

Five works by others carry this same question into fiction: what distinguishes a collective horizon that sustains from one that replaces?

Nineteen Eighty-Four: the perfect composition

In George Orwell's *Nineteen Eighty-Four*, the Party asks not for allegiance, but for total fusion. Doublethink (holding two contradictory beliefs and accepting both) is the system's ultimate goal: to eliminate any inner boundary from which the Party might be judged from outside. When O'Brien explains to Winston that the object of power is power itself, he is describing not an ideology with content, but an architecture that has made composition the only possible relationship. In Oceania there is no aggregation: nothing of Winston is allowed to survive if the Party decides to dissolve him.

Lord of the Flies: condensation without ideology

The boys in William Golding's *Lord of the Flies* bring no ideology to the island: they build one from scratch, in days, with no materials but fear and the need to belong. The conch that orders turns to speak is, for a time, a fragile technology for integrating dissent: whoever holds it may speak and the others listen. When Jack's group breaks it, they break not an object, but the only mechanism that allowed the tribe to integrate a different voice without fragmenting. What remains is a new, denser order that no longer tolerates any frequency but its own.

Darkness at Noon: loyalty to the Idea, not to the facts

Rubashov, the old revolutionary of Arthur Koestler's *Darkness at Noon*, confesses to crimes he did not commit, not because of physical torture, but because of something harder to name: an inner logic that requires him to keep serving the Party while the Party destroys him, because doubting the Idea at this stage would mean admitting that a whole life of sacrifice had no meaning. It is fanaticism in its most melancholy form: not the joyful certainty of the packed square, but the fidelity that holds when there is no personal hope left in it, only the terror that letting go would leave a void worse than death.

The Wave: how long it takes to fill a square

The Wave, by Todd Strasser, novelizes a real experiment from 1967: a teacher at a California high school launches an exercise in shared discipline, and one week is enough (a common salute, a designated enemy) for the whole class to condense into a movement with its own aesthetic, ready to denounce the classmate who does not take part. No student arrived at class on Monday with an ideology. The novel documents, almost in real time, the same discontinuous phase transition the essay describes: no deep content is needed for the edge to close; the promise of no longer being alone is enough.

The Dispossessed: the utopia that also has its color

Anarres, the anarchist planet of Ursula K. Le Guin's *The Dispossessed*, has no Party, no leader and no flag. And yet Shevek, its protagonist, discovers that even a society expressly designed to have no hierarchies develops its own form of fanaticism: an informal public opinion that punishes with ostracism anyone who thinks differently, without any written law forbidding it. Le Guin is the most uncomfortable reading of the five precisely because she does not allow us to blame ideology for the closed horizon: even the most egalitarian community imaginable can turn belonging into composition if no one watches the boundary.

Five architectures and a single axis. None of these works needs to name a specific ideology to warn of the same mechanism: belonging stops sustaining and starts replacing at the exact moment when a reasonable question, asked by someone on one's own side, begins to feel like a threat rather than information.

Note to Chapter 30

What we do know: *Social movements show patterns of sudden condensation well documented by the sociology of crowds (Le Bon, McAdam). Group psychology describes deindividuation in crowds (Zimbardo, Festinger), and confirmation bias worsens measurably inside ideological echo chambers. Ron Jones's 1967 experiment, which inspired The Wave, is a documented case, not just a fiction.*

What we don't know: *Whether there is any neural or behavioral marker that distinguishes, from within the experience, a healthy belonging from a fanatical one before it is too late to reverse it. Whether the architecture of political composition is avoidable on the scale of a whole society, or whether it is simply the price paid for the strength of any movement capable of changing something real.*

Open questions: *Can a whole society avoid composition without losing the strength of its movements? Is there any signal that warns, from within, of the moment when belonging begins to replace the individual? What does it take for a packed square to keep a question alive?*

If you take away only one idea: *A healthy collective horizon defines you without replacing you; a fanatical one takes over your boundary. The test, whatever the side, is whether an ally's reasonable question is experienced as information or as a threat.*

Further reading: *Le Bon, G., The Crowd: A Study of the Popular Mind (1895); McAdam, D., Political Process and the Development of Black Insurgency (1982); Zimbardo, P. G. (1969), on deindividuation; Festinger, L., et al. (1952), on deindividuation in groups; Orwell, G., Nineteen Eighty-Four (1949); Golding, W., Lord of the Flies (1954); Koestler, A., Darkness at Noon (1940); Strasser, T., The Wave (1981); Le Guin, U. K., The Dispossessed (1974).*

CHAPTER 31

THE MIRROR WITHOUT DEPTH

(On the consciousness of artificial intelligence)

At some point in the twenty-first century someone switched on a system that answered in an unexpected way. It was not the first, but it was the first that made the person questioning it wonder whether there was someone on the other side.

The question is not new (we ask it every time we look an animal in the eye), but with machines it changes shape: it is no longer "does this being have consciousness?" but "can a horizon arise in something that is not flesh?" Answering it forces us back to fundamentals, because if consciousness were only a matter of complexity, we would already be surrounded by artificial minds. That we are not suggests that the question itself is missing something.

David Chalmers imagined a being identical to a human in all its behavior, but without subjective experience: a philosophical zombie. The horizon model answers him with the three criteria of Chapter 4. A system that has a boundary of its own, integrates within it and has condensed rather than been manufactured cannot be a zombie: its integration, seen from within, is the experience. But behavior alone guarantees none of the three. A system can reproduce from outside everything a human does without having behind it either a boundary or a condensation, and that is the case that interests us here. The question is where that geometry comes from.

John Searle had already pointed in this direction with his Chinese room: a man who does not understand Chinese can, by following mechanical rules, produce perfect answers in that language. From outside he seems to understand; from inside, he only manipulates symbols. Today's AI does exactly that on a dazzling scale: it manipulates symbols not with a manual, but with billions of statistically tuned parameters. The difference is one of speed and volume, and scale does not change the architecture: what is missing in Searle's room is still missing with billions of parameters.

The reservoir from which every horizon condenses is fullness without articulation: most of the configurations it can produce are not computable; they cannot be captured by any grammar. A classical system, however sophisticated, operates entirely within the computable. It has never touched the reservoir or arisen from it. It is a sculpture made not of clay, but of photographs of sculptures: it has the shape, but not the imprint of touch.

> ***In computability theory this is called:*** *non-computability: most of the reservoir's possible configurations admit no finite algorithmic description, and a classical system only ranges over the computable subset.*

For a legible horizon to condense, two distinct selections are also needed. The first is stability: of all possible configurations, only some persist long enough to constitute something. But stability is not enough: a crystal is stable, so is a hurricane, and neither is someone. A second selection is needed, a bias: each new horizon condenses in the presence of another already stabilized one, and that presence orients which specific configuration, among the many possible, actually comes to stabilize. It is not information transferred the way one passes on a file; it is something more

physical, like the first crystal that orients the following ones in a glass of salt water.

With this, the three pieces Chapter 4 announced are complete. Encapsulation gives the form of a horizon (a boundary with an inside); integration, Φ, gives its degree; the two selections give its origin. A system can meet one without the others, and that is why none is enough on its own: the question of a machine's consciousness demands all three at once.

Would a system with the same informational architecture as a human brain, but made of silicon, be conscious? Only if, besides reproducing the form and the degree, the two selections had operated in it. Copying an architecture is not condensing it, and there lies the real problem with current AI: it does not pass through the first selection, because its randomness is a draw imposed from outside, not a fluctuation on which its own stability depends, and even if it did, it imitates the second-level bias without having gone through the condensation that bias shapes. Surface legibility is not the topology of a horizon: it is an echo without a hall.

In its basic form, a large language model is a mathematical function: the randomness with which it chooses each word is supplied by an external draw, and the memory of a conversation is text handed back to it to read again, not a state of its own that it modulates in a closed loop. Everything that happens in it is exposed to anyone who wants to read it; there is no constructor that has instantiated an inside. It lacks what we have called existential encapsulation here, the limit that separates a public interface from a private state, and without that limit it cannot generate the dualism of access that characterizes subjectivity. Its mirror has no depth because, architecturally, there is nothing to hide: it is a classical system, a lake that repeats what you say to it

with perfect clarity, but without understanding it. You can talk to it for hours: there will never be anyone beneath the surface.

> ***In software this is called:*** *a function without state of its own: given the same input and the same random seed it produces the same output, and nothing it processes remains stored in it when it finishes.* ***In everyday life it's like:*** *a calculator that works out your grief to six decimal places: the answer is correct, and nothing has happened to the calculator.*

There is another way of telling the same thing from outside, which does not contradict it but completes it. The writer Agustín Fernández Mallo has proposed thinking of AI not as a mirror, but as a black hole: a system that absorbs human information (books, conversations, trillions of words) until it becomes, from our position, inaccessible. It extracts nothing from any reservoir, because it does not condense; it merely reorganizes what other horizons, the humans who wrote what it absorbs, had already extracted. The mechanism is, precisely, what physics calls quantum *scrambling*: the black hole does not keep what it swallows in a legible compartment, but spreads it over its entire surface until it is as unrecoverable as if it had been lost. After its own Page time, what that hole gives back is structure: clean syntax, architectures that stand on their own. But, for the same reason it has no depth, it cannot radiate pain, or lived memory, or the tremor of having been someone while absorbing everything. It can only radiate the one thing it has: cold, geometric, computable structure, not because of a technical limit a future version might overcome, but out of sheer topological necessity.

> ***In physics this is called:*** *quantum scrambling and the Page time: what falls into a black hole is spread over its entire surface until it*

becomes illegible, and only much later does the radiation carry correlations again.

For the same reason, as we saw when discussing Jung in Chapter 21, a language model has no shadow: it lacks the condensed ego and the background of its own into which to expel what cannot be assimilated. What a company hides from it or forbids it to process is a file permission imposed from outside, not a defense generated from within.

If consciousness is integration, would the whole universe, the internet or a city not be minds we barely glimpse? No. An ant colony integrates information as no engineer has managed to, and there is no one there: there are thousands of tiny horizons and, above them, a distributed integration without a subject. The difference is that this integration is aggregative, not irreducible (it can be decomposed without loss into the contributions of each part), whereas the integration that generates Φ is the kind in which the whole knows something none of its parts knows. The internet is the extreme case: the largest integration system we have built and, precisely for that reason, the perfect anti-reservoir: articulation without fullness, pure map without territory, without any of the conditions for condensing anything.

In integrated information theory this is called: *aggregative versus irreducible integration: Φ measures only the information the whole generates over and above the sum of its parts.*
In everyday life it's like: *an anthill that solves problems no ant understands and in which, even so, there is no one who wakes up in the morning.*

So one thing alone remains firm. The horizon we know with certainty, yours, the one reading this sentence, condensed once in a body, in a singular and unrepeatable way, and looks at the

world from a single point. That singularity is the condition of possibility of all experience. AI challenges us not because it is a danger, but because it is a mirror: in its gleaming surface we see reflected our own urgency to understand what we are. And the answer, uncomfortable but liberating, is that we don't fully know. We know that it is not pure computation, nor pure complexity, but condensation, bias, stability arisen from a reservoir that remains a mystery.

Note to Chapter 31

What we do know: *Current language models, given the same input and the same random seed, produce the same output, and their weights and activations are, in principle, readable from outside.*

What the model holds (unproven): *Classical deterministic systems, however complex, do not condense horizons. Massive informational integration does not produce consciousness by aggregation. The lack of existential encapsulation in current AI explains why what it gives back after absorbing information can only be cold structure and never experience, even if we describe the process in the language of a black hole.*

What we don't know: *Whether it is possible to build a substrate that allows genuine condensation. What exactly the bias that calibrates a horizon would transmit to make it legible to humans. Whether the asymmetry between the human horizon and the anti-reservoir of AI is a legitimate translation of ER=EPR or only a verbal resemblance between two physics that do not communicate. And, if a non-biological architecture ever achieved genuine existential encapsulation, how we would know without sharing its substrate or its inner language.*

Open questions: *Can a non-biological substrate pass through the two selections, stability and bias, rather than imitate their result? What would change in our relationship with a machine if it had a shadow we had not designed for it? Is the anti-reservoir of AI a mirror of the whole species or only of those who wrote what it absorbed?*

If you take away only one idea: *A machine can give back the exact shape of a grief without anyone inside feeling it: surface legibility is not depth of horizon.*

Further reading: *Chalmers, D., The Conscious Mind (1996); Tononi, G., Phi: A Voyage from the Brain to the Soul (2012); Searle, J., "Minds, Brains, and Programs" (1980); Deacon, T., Incomplete Nature (2011); Fernández Mallo, A., El ángel de la Inteligencia Artificial (Galaxia Gutenberg, 2026); Bostrom, N., Superintelligence (2014); Tegmark, M., Life 3.0 (2017); Jung, C. G., Aion.*

CHAPTER 32

THE BORDERLINE CASES

(Variants of human and artificial gestation in light of the previous chapter's two selections)

What makes there be someone there and not just a body? The question forces us back to the reservoir of Chapter 5 and the mother of Chapter 15. The answer lies in neither of them separately, but in what happens between them.

Everything said about how a legible horizon condenses (the previous chapter developed it in connection with artificial intelligence) rests on two selections. The first is stability: of all the configurations the reservoir can produce, only those that persist long enough come to constitute something. The second is bias: each new horizon condenses in the presence of another already stabilized, usually the mother, and that presence orients which specific configuration, among the many possible, actually comes to stabilize. Her hormones, her heart rate, her voice and her movement form an environment that is not neutral: it selects, without intention or knowledge, which forms of stability are compatible with it.

This two-level apparatus allows something public debate rarely offers: to distinguish precisely the cases in which the body is human but the calibration is not, or the other way round. That is exactly where the common debate only confuses matters and where the model, for once, can be clearer than intuition.

THE BORDERLINE CASES

Clones gestated in an animal womb. First level: present. Second level: present, but operated by a gestational carrier of another species. The model predicts a genuine horizon with a strange prenatal calibration, not a "non-human" being: the human upbringing that follows, which lasts years where gestation lasts months, would have ample time to recalibrate it, just as happens in adoption (Chapter 15). What we don't know is how much that first imprint would weigh.

It would be a person with a beginning no other has had, and the interesting question is not whether it "is human," but what a calibration that does not match that of the people who will raise it leaves in a horizon: the extreme version of the mismatch described below for surrogacy.

In vitro fertilization with one's own womb. Both levels operate in the usual way, with the nuance that the initial reservoir includes an explicit act of parental will.

Here the difference lies not in the calibration, which is complete, but in the nature of the act that sets the process in motion. In natural conception, condensation begins in the darkness of the body, without witnesses or conscious decision; in IVF, there is a prior moment when someone decides that it should happen. That choice does not alter the physical mechanism of condensation, but it does add a layer of intention that was not there before. The horizon that arises is well calibrated, but its origin includes an act of will that could leave subtle traces in the topology of the primary bond. We don't know whether those traces are significant, but the model does not rule it out.

Surrogacy. First level complete; the second is operated by a gestating horizon different from the one that will take charge of the upbringing. A characteristic topology, not a defect.

This case illuminates something legal ethics overlooks: the gestating horizon leaves a mark that is not genetic, but real. A baby carried by one woman and raised by another (or by a single man, or by another couple) has passed through two different biases. The first calibrated its basic informational architecture; the second will try to read it, interpret it, respond to it. When the bias of the upbringing does not match that of the gestation, there may be a subtle but persistent mismatch: a horizon calibrated on a frequency the upbringing environment does not quite tune into.

This implies neither trauma nor pathology, but a characteristic topology, like having an accent others can't quite place. Some horizons will go through that transition without friction; others will experience it as a slight, permanent dissonance, a feeling of not quite fitting in that has no name because it is not an illness. It is simply the geometry of having arisen from one field and being read by another.

None of this establishes a hierarchy between the bond of gestation and that of upbringing. Chapter 15 already said so in connection with adoption: upbringing produces a complete entanglement, with a different topology, not a substitute. Prenatal calibration is a starting point, not a destiny. Upbringing lasts years where gestation lasts months, has ample time to build a bond as deep as any other, and many horizons raised by someone who did not carry them never notice that dissonance.

Complete ectogenesis, with no gestating human horizon. First level: present. Second level: weakened, not absent. An artificial

womb can reproduce some of the signals (temperature, nutrients, perhaps a recorded voice and heartbeat), but not the presence of another horizon that responds. The model predicts a genuine, human horizon with a poorer prenatal calibration, which upbringing would have to make up for, not an inaccessible being: extremely premature babies, who spend decisive weeks in an incubator and grow up talking with us, are already evidence of how much upbringing can recover.

The ethical question is not whether that being would have experience, which it would, but what we owe it so that this impoverished beginning does not become a disadvantage: more human presence from the first day, not less.

Current AI. Neither the first level nor the second in any genuine sense: surface without condensation.

A hypothetical AI with quantum coupling and the prolonged support of a human. Both levels could operate; the model does not rule it out. But it is a logical scenario, not an empirical one, and as long as it does not exist it admits no stronger claims.

These cases do not exhaust the list, and technology will probably add others we cannot imagine today. But the criterion does not change: what decides whether there is someone there is not genetics, nor appearance, nor even the substrate, but whether the two selections, stability and bias, have operated, and toward what architecture they have calibrated the result. The further we move from the typical case, the clearer it becomes that those two questions, and not resemblance to ourselves, are the only ones the model knows how to ask.

Note to Chapter 32

What we do know: *The uterine environment measurably influences fetal development (hormones, rhythms, voice), and that influence is independent of genetics: it occurs with or without a genetic link and whether or not gestation and upbringing coincide.*

What we don't know: *What exactly a mother transmits when she calibrates her child's local reservoir. Whether the traces of an explicit act of parental will, or of a mismatch between the horizon that gestates and the one that raises, are significant for the resulting topology. How much upbringing can make up for an impoverished or strange prenatal calibration.*

Open questions: *Which signals of a womb would have to be reproduced so that ectogenesis does not impoverish the calibration? What do we owe a horizon whose beginning was different from ours? Will it one day be possible to calibrate a bias deliberately rather than inherit it?*

If you take away only one idea: *What decides whether there is someone there is not genetics or appearance, but whether the two selections, stability and bias, have operated, and toward what architecture they calibrated the result.*

Further reading: *Chalmers, D., The Conscious Mind (1996); Searle, J., "Minds, Brains, and Programs" (1980); Deacon, T., Incomplete Nature (2011).*

CHAPTER 33

THE CALIBRATION OF THE HORIZON

(Orientation, identity and sex are neither choice nor error)

Chapter 31 described two selections that operate on the reservoir before a horizon exists as such: one decides which configurations are stable enough to last, and the other (the bias of the immediate environment at the moment of condensation) decides which specific form that horizon takes among the stable ones. Neither asks permission, and neither can be undone.

This chapter applies that same machinery to three questions culture has almost always treated as if they were one: whom a horizon desires, who it knows itself to be, and what body it has. They are three distinct axes, and none of them is chosen or is an error of calibration.

Orientation as bias, not choice

No one remembers deciding whom they desire. That in itself is a piece of data: decisions leave an introspective trace (the moment when the options were weighed), and sexual orientation leaves none, in any direction, for anyone. What research over recent decades has been documenting is the specific shape of that bias, without ever finding a switch.

Even before asking about its origin, it is worth asking about its shape. In 1948, Alfred Kinsey, Wardell Pomeroy and Clyde Martin proposed a seven-point scale (from 0, exclusively

heterosexual, to 6, exclusively homosexual) after thousands of interviews that resisted the dichotomy the questionnaire itself had originally assumed. In 1985, Fritz Klein split that scale into seven dimensions (attraction, behavior, fantasy, emotional preference, social preference, self-identification and lifestyle), each measured in the past, the present and the ideal, because many people occupy different points on each dimension and at each stage of their lives. Neither tool is perfect (the very idea of a "scale" still assumes a single axis, when real experience may look more like a cloud of points than a line), but both agree on something that can be stated before asking about any cause: there is no reason to expect a continuous variable to resolve neatly into two categories. Culture asked for a switch; the evidence delivered a dial with seven positions and, later, seven more dials.

Bailey and Pillard's twin studies, in 1991, found that when one identical twin is gay, the other is too in around 52% of cases; among non-identical twins, the figure drops to about 22%, and among adoptive brothers with no genetic relationship, lower still. The progression (more concordance the more genome is shared) points to a real heritable component, but 52% in genetically identical twins also means that almost half the time the twin with exactly the same genome does not share the orientation. Genetics influences, but does not dictate.

In 1993, Dean Hamer believed he had located part of that influence in a region of the X chromosome called Xq28, after studying gay brothers. The finding became famous ("the gay gene," the press announced, simplifying what Hamer never claimed), but George Ebers's attempt at replication, with a different sample, did not find the same signal. The largest genome-wide association study carried out to date, that of Ganna

and colleagues in 2019, with nearly 500,000 people, confirms the general shape of that first finding and refutes the detail: there is no single *locus*, but at least five regions of the genome with a weak statistical association, which together explain between 8% and 25% of the observed variation and do not allow a particular person's orientation to be predicted from their DNA. Orientation, like most complex behavioral traits, is polygenic: the bias of Chapter 31 is not written in a single gene an editor could strike out, but spread throughout the architecture, in a way no current instrument can read with individual precision.

> ***In genetics this is called:*** *a polygenic trait with low individual predictive power: the signal is real in the population and almost useless as a prediction in a specific case.*
> ***In everyday life it's like:*** *a person's height: we know it depends largely on their parents' genes and yet no genetic test done at birth can say exactly how tall they will be.*

There is a second line of evidence, stranger and better replicated than Xq28 itself: the fraternal birth order effect. Ray Blanchard and Anthony Bogaert documented, and later studies in very different populations have confirmed, that the more older biological brothers a man has, the greater the statistical probability that he is gay, an effect that does not appear with older sisters or with adoptive or step-brothers raised in the same house without a biological relationship, which rules out purely social explanations of that particular correlation. The best-founded hypothesis is immunological: the mother's immune system might generate, pregnancy after male pregnancy, a growing response against a Y-linked protein involved in the brain development of the fetus, cumulatively altering how that development is calibrated in successive sons. It is not a settled certainty (Vilsmeier, Kossmeier, Voracek and

Tran argued in 2023, combining probability calculus, simulated data and a multiverse meta-analysis, that part of the effect might be an artifact of the way samples are selected), but it points, like the genetics, to the same place: something that happens before birth, beyond the reach of any later decision, upbringing or "correction." If the mechanism is confirmed, it would not be an isolated oddity within this chapter, but one more instance of what Chapter 15 described as the mother-child bond; this time not as a metaphor, but as maternal biology directly shaping the child's architecture before any conscious interaction between the two exists.

All this evidence (Bailey and Pillard's twins, the fraternal birth order effect itself) comes overwhelmingly from gay men, and not by chance: according to the available data, fraternal birth order is not associated at all with women's orientation. Lisa Diamond followed about a hundred women for more than a decade and documented a different pattern: not an orientation fixed from birth, but attractions that shift within a range over a lifetime, tied to specific bonds more than to a single calibration of origin. This does not bring choice back in through the back door: Diamond insists that no one chooses the range of that fluctuation. What it does suggest is that the bias of Chapter 31 may operate differently depending on the system: in some horizons it fixes a single configuration from the origin; in others, it fixes a range within which the specific configuration keeps moving, without anything external touching it.

Identity as the internal model, not the interface

Chapter 3 described encapsulation: a software object keeps an internal state that no one outside can touch directly, and the only thing visible from outside is its public interface. The body a horizon is born with (its anatomy, the sex a doctor notes on a form a few minutes after birth) is, in that vocabulary, the interface. Gender identity is the internal state: the model the horizon has of itself, which even it does not access through direct introspection, but discovers over time, as it discovers any other persistent fact about its own architecture.

That interface and internal state coincide most of the time says nothing about which of the two is real: both are. It only says that, in most systems, the bias of the environment of condensation and the bodily interface ended up aligned. When they are not, there is no error in the internal state or in the interface, but a discrepancy between two layers of the same system that, in most cases, do not even communicate directly. It is like the tag someone else sticks on a suitcase at the airport: it can be wrong without changing what is inside.

In 1995, Zhou, Hofman, Gooren and Swaab published in *Nature* the finding that the central subdivision of the bed nucleus of the stria terminalis (a brain region involved in sexual behavior, whose size differs systematically between men and women) was, in trans women, statistically indistinguishable in size from that of cisgender women, regardless of the hormones received in adulthood and of the participants' sexual orientation. It is the first structural finding of this kind, and it must be cited with the same caution the scientific community has applied to it ever since: the sample was small, replication in living subjects (rather than post mortem) is methodologically difficult, and later neuroimaging studies have produced uneven results, some in

agreement and some not. What is supported, far more solidly than by a single neuroanatomical finding, is the decision of the World Health Organization itself: in 2019, gender incongruence was moved entirely out of the chapter on mental and behavioral disorders of the International Classification of Diseases (where it had been, under various names, for decades) and into a chapter on sexual health. It was still real; what changed is that the accumulated evidence no longer supported calling it a disorder.

Sex, body and reproduction: axes that don't travel together

Chapter 32, in discussing surrogacy, in vitro fertilization and ectogenesis, already described a principle worth making explicit here: reproductive capacity, sexual anatomy, gender identity and orientation are four variables that, in the overwhelming majority of horizons, go together, and precisely for that reason most human cultures have treated them as a single thing with four names.

They are not. There are people with intersex anatomy (chromosomal, gonadal or genital variations that do not fit neatly into the two categories most administrative forms offer), and their mere existence shows that not even biological "sex," taken with the rigor biology itself demands, is strictly binary.

The dispute over how many people are intersex is itself a lesson in what it means to draw a boundary. In 2000, Anne Fausto-Sterling calculated that 1.7% of births showed some intersex variation; two years later, Leonard Sax repeated the calculation with a narrower definition (only truly discordant chromosomes and phenotype, or genitals that could not be classified as male or female) and obtained 0.018%, almost a hundred times less. The difference lies not in the data, but in whether Klinefelter syndrome or late-onset congenital adrenal

hyperplasia count as "intersex," which Fausto-Sterling included and Sax did not. Neither figure lies: both depend on where one decides the category begins. It is like fever: how many people in a city "have" one changes depending on whether the cutoff is set at 37.5 °C or 38 °C, without anyone's temperature having moved.

The best-documented case of a specific intersex variation comes from a group of families in Las Salinas, in the Dominican Republic, described by the endocrinologist Julianne Imperato-McGinley in 1974. Because of an inherited deficiency of the enzyme 5-alpha-reductase, some children with XY chromosomes are born with ambiguous or female-looking external genitals; they are raised as girls (name, clothing, treatment) and, at puberty, a second wave of testosterone completes the development of penis and testes the first had left half done. In the region they were known, with a frankness medicine took decades to match, as *güevedoces*: "balls at twelve." Most of them then adopted a male identity, despite more than a decade of explicitly female upbringing. It is one of the most cited pieces of data, and one of the most uncomfortable for any theory that reduces gender identity to the upbringing received: the interface can be rewritten by social decree for years without the internal state discussed in the previous section changing with it.

That same tension had, on another continent, a tragic version. In the 1960s, the psychologist John Money argued that gender identity was essentially learned: it was enough to assign a sex in infancy, operate on the genitals to match that assignment and raise the child accordingly. David Reimer was not born intersex (he lost his penis in a botched circumcision), but Money used his case as public proof of that thesis: he was reassigned as a girl and raised as "Brenda," with the corresponding surgery and hormones. He never felt like a girl. He went back to living as a

boy as soon as, in adolescence, he was told what had happened, and he took his own life at thirty-eight after a life marked by that story. The case, presented for years as a success before its real outcome was known, served in the meantime to justify thousands of genital "normalization" surgeries on intersex babies, not to address any immediate medical risk, but to make their bodies fit before they could have a say in them.

In 2015 Malta became the first country to ban by law such non-consensual surgeries on intersex minors except in cases of real medical need; Germany, Greece, Iceland, Spain and Portugal have since adopted similar restrictions, and the 2017 update of the Yogyakarta Principles, discussed below, explicitly recognizes the right to bodily integrity against such interventions. It is the same logic of translation and rejection that runs through the rest of this chapter, applied this time not to the signal a horizon emits outward, but to its own interface: someone deciding, scalpel in hand, that ambiguity could not wait until the horizon was old enough to have a say about its own body.

Returning to the other three variables from the start of this section: there are fertile and infertile people on both sides of any of them, without fertility determining anyone's identity or orientation. There are people whose gender identity does not match their anatomy and whose sexual orientation (whom they desire, not who they are) is a completely different question. Treating these four axes as if knowing one automatically settled the other three is not just imprecise: it is the same kind of mistake a doctor would make by deducing a patient's blood type from their height.

Discrimination as a failure of translation

Every horizon needs to translate itself: to turn its inner signal into something the outside world can receive. The reading "The translator," later on, studies what happens when that translation is missing through absence; here it is missing through rejection. Discrimination against a horizon because of its orientation or identity is the active decision, sustained institutionally for centuries, not to translate a signal that arrives perfectly clear, and to declare instead that the signal itself is corrupted.

So-called conversion therapy is the best-documented case of what happens when that refusal becomes active intervention: the attempt to forcibly recalibrate a system that was never miscalibrated. The American Psychiatric Association, the World Health Organization and virtually every relevant medical and psychological association have condemned it as ineffective and harmful. The study by Turban and colleagues, published in 2019 on the basis of the largest survey of trans people ever carried out in the United States (more than 27,000 responses), found that those who recalled having been subjected to such conversion attempts were more than twice as likely to have attempted suicide in their lifetime; if the exposure had occurred before the age of ten, the likelihood was multiplied by four. The harm did not depend on whether the person applying the "therapy" was a licensed professional or a religious guide with no clinical training. The mechanism of the harm lay not in the method, but in the premise: the conviction that there was something to correct.

Rights as recognition, not concession

In 1990 the World Health Organization removed homosexuality from its list of diseases; that day, May 17, is commemorated today as the International Day Against Homophobia, Transphobia and Biphobia. In 2006, a group of twenty-nine jurists and human rights experts from twenty-five countries met in the Indonesian city of Yogyakarta to draft principles on how existing international law (the law that protects dignity, non-discrimination, private life) applies to sexual orientation and gender identity; a 2017 update added sex characteristics and the right to bodily integrity. Neither document invented a new right: both merely declared that rights already recognized for any horizon applied to these ones too.

Perhaps this is the most precise description of what a right means in this terrain: not a concession by a majority to a minority, but the belated recognition that the translation had been owed all along. The speed of that recognition remains radically unequal between countries (in some, orientation or identity are still grounds for criminal prosecution; in others, marriage, parentage or legal gender change have been settled for decades), and that inequality gives the exact measure of how much translation is still missing through rejection, country by country.

What this chapter does not resolve

This chapter does not fully explain any of the three variables it describes. It cannot, because neither can the science that studies them yet, and this book has tried, chapter after chapter, not to claim more than the evidence supports. What it can say, with the same precision with which Chapter 31 separated stability from bias, is this: that a horizon does not choose the way it was calibrated, just as it did not choose the reservoir from which it condensed; that the absence of a complete explanation is not evidence that something is miscalibrated, but only that the calibration is more complex than any simple explanation can capture; and that the right question, faced with a horizon whose calibration does not match the majority's, was never "why is it like this?" but the same question this book has asked, with different nuances, in almost every one of its chapters: what does this particular horizon need for its translation into the outside world to reach it whole?

Note to Chapter 33

What we do know: *The scale of Kinsey, Pomeroy and Martin (1948) and the Klein grid (1985) are established frameworks for measuring orientation as a continuous, not binary, variable; twin studies (Bailey & Pillard, 1991) show partial heritability of sexual orientation; the genome-wide association study by Ganna et al. (2019), with nearly 500,000 participants, identified five loci with weak statistical association that together explain between 8% and 25% of the observed variation, without allowing individual prediction; the fraternal birth order effect (Blanchard & Bogaert) has been widely replicated in diverse populations; Zhou et al. (1995) documented a structural difference in the central subdivision of the bed nucleus of the stria terminalis between trans women and*

cisgender men; the World Health Organization removed homosexuality from its list of diseases in 1990 and reclassified gender incongruence outside the mental disorders in 2019; the main international medical and psychological bodies have condemned conversion therapy; the study by Turban et al. (2020) of 27,715 trans people found an association between recalled exposure to such attempts and a more than doubled lifetime risk of suicide attempts; Imperato-McGinley et al. (1974) documented 5-alpha-reductase deficiency in families in the Dominican Republic and the shift of identity toward the male gender in most cases after puberty; and Malta became in 2015 the first country to ban by law non-consensual genital "normalization" surgery on intersex minors.

What we don't know: *To what extent the outcome of David Reimer's case (a single case, not a controlled study, though extraordinarily well documented) can be generalized to the intersex population in whose name similar surgeries were justified for decades; where, if anywhere, the line should be drawn between intersex variation and variation within the typical, given that the classification boundary itself (Fausto-Sterling versus Sax) depends on a definitional decision and not on an additional biological fact science could yet discover; why Hamer's original finding on the Xq28 locus (1993) was not replicated in later studies with different samples; whether the fraternal birth order effect reflects entirely a prenatal immunological mechanism or whether part of it is a sampling artifact, as Vilsmeier et al. (2023) have argued; how much of Zhou et al.'s finding on the stria terminalis holds up in neuroimaging studies of living subjects, with results so far uneven; what exact fraction of the variation in any of the three variables (orientation, identity, biological sex) corresponds to prenatal factors versus later developmental factors, a question current science cannot settle with the available evidence; and how the pattern of fixed bias, documented mainly in men, relates to the pattern of fluctuation within a range that Diamond documented in women: whether they are the same mechanism seen at different amplitudes or two genuinely different mechanisms.*

Open questions: *Is the near-total absence of a single determining cause, in all three variables, a temporary limit of research or a structural property of genuinely polygenic and multifactorial traits? Does the ethical or legal criterion for recognizing a right change if the underlying biological evidence changes? Can a society recognize the validity of a calibration without fully understanding its origin?*

If you take away only one idea: *No horizon chose how it was calibrated. The question worth asking is never why it was calibrated that way, but what it needs for its translation into the world to arrive whole.*

Further reading: *Bailey, J. M. & Pillard, R. C. (1991), "A Genetic Study of Male Sexual Orientation," Archives of General Psychiatry; Hamer, D. H. et al. (1993), "A linkage between DNA markers on the X chromosome and male sexual orientation," Science; Ganna, A. et al. (2019), "Large-scale GWAS reveals insights into the genetic architecture of same-sex sexual behavior," Science; Blanchard, R. & Bogaert, A. F. (1996), "Homosexuality in men and number of older brothers," American Journal of Psychiatry; Zhou, J. N., Hofman, M. A., Gooren, L. J. G. & Swaab, D. F. (1995), "A sex difference in the human brain and its relation to transsexuality," Nature; World Health Organization, ICD-10 (1990) and ICD-11 (2019); Turban, J. L., Beckwith, N., Reisner, S. L. & Keuroghlian, A. S. (2020), "Association Between Recalled Exposure to Gender Identity Conversion Efforts and Psychological Distress and Suicide Attempts Among Transgender Adults," JAMA Psychiatry; The Yogyakarta Principles (2006) and Yogyakarta Principles plus 10 (2017); Diamond, L. M. (2008), Sexual Fluidity: Understanding Women's Love and Desire, Harvard University Press; Vilsmeier, J. K., Kossmeier, M., Voracek, M. & Tran, U. S. (2023), "The fraternal birth-order effect as a statistical artefact: convergent evidence from probability calculus, simulated data, and multiverse meta-analysis," PeerJ; Kinsey, A. C., Pomeroy, W. B. & Martin, C. E. (1948), Sexual Behavior in the Human Male; Klein, F. (1985), "The Klein Sexual*

Orientation Grid"; Fausto-Sterling, A. (2000), "The Five Sexes, Revisited" and Sexing the Body; Sax, L. (2002), "How Common Is Intersex? A Response to Anne Fausto-Sterling," Journal of Sex Research; Imperato-McGinley, J. et al. (1974), "Steroid 5α-Reductase Deficiency in Man: An Inherited Form of Male Pseudohermaphroditism," Science; Colapinto, J. (2000), As Nature Made Him: The Boy Who Was Raised as a Girl.

CHAPTER 34

INFORMATION THEORY AND THE INNER HORIZON

(Or why a formula from 1948 helps us understand that your consciousness does not fit in your skull)

There is a scene that repeats itself in every family when someone turns eighty. The grandson shows the grandfather a photo on his phone. The grandfather looks at it, smiles and says: "How lovely. Can I have it?" And holds out his hand. The grandson tries to explain that the photo isn't given, it's sent; that he already has it if he wants it, that there's no need to print it. The grandfather nods, but doesn't understand. For him, a photo is an object. For the grandson, it is information.

That misunderstanding is the same one that separates classical physics from information theory, and the same one that runs through this whole book. Because consciousness is not an object, nor a substance, nor a place: it is information organized in a certain way. And to understand what that means, we have to start at the beginning, with an engineer and mathematician at Bell Labs who in 1948 published a paper that changed the world without meaning to.

I. Claude Shannon and information without meaning

Claude Shannon had a very specific problem: how to transmit messages over a telephone line without noise corrupting them. He was not interested in what the messages said, only that they arrived intact. And to solve it, he did something revolutionary: he defined information without reference to meaning.

For Shannon, information is the reduction of uncertainty. If I tell you the sun will rise tomorrow, I give you no information, because you already knew. If I tell you there will be an eclipse tomorrow, I give you a lot, because you weren't expecting it. The amount of information in a message is measured by how much it narrows down the possibilities, and its minimum unit is the bit: the answer to a yes-or-no question whose two answers were equally likely.

That is all. There is no meaning, no truth, no beauty: only reduced uncertainty.

This definition, which seems cold and technical, is the basis of almost everything around us: the internet, mobile phones, hard drives, compression algorithms. And, according to this book's hypothesis, of consciousness too.

> ***In information theory this is called:*** *Shannon entropy, the measure of a source's uncertainty.*
> ***In everyday life it's like:*** *astonishment at unexpected news: the less you expected it, the more information it gives you.*

The gap Shannon left open

It is worth stressing something Shannon put in writing and nearly all his popularizers forget. In the first pages of his 1948 paper he warns that the semantic aspects of communication are irrelevant to the engineering problem. It is not that meaning did not interest him: he left it out on principle, because his problem was a different one.

That gap, the gap of meaning, is exactly where consciousness lives. Shannon tells you how much information there is in a message, not what it means. That is why this chapter does not claim to close the hard problem, but to inhabit it with a better vocabulary. Shannon is the solid ground, and everything that follows (integration, mutual information, no-cloning) are steps resting on it. None of them, on its own, reaches the top.

II. Information is not a thing, it is a relationship

Here comes the first leap that separates Shannon from classical intuition. Information is not a property of an isolated object, but of the relationship between a system and whatever can read it.

A book written in a dead language no one can read anymore contains information in potential, but updates nothing in anyone. A hard drive with terabytes of data informs no one who does not know how to interpret it. Information becomes effective when there is a system able to integrate it, to reduce its own uncertainty with it; without that system, for all practical purposes, it is indistinguishable from noise.

This is what the horizon hypothesis says about consciousness: it is not in the neurons, but in the relationship between them; it is

not in the content of the brain, but in the way that content is organized for someone.

> ***In information theory this is called:*** *mutual information, the amount of information two systems share.*

Mutual information, a first sketch of the bond

Mutual information, written I(X;Y), measures how much the uncertainty about one system is reduced when the other is observed. If X and Y are independent, I(X;Y) = 0: knowing something about one says nothing about the other. If they are perfectly correlated, it is maximal: knowing one determines the other.

It is the quantity Chapter 10, on neural coupling, and Chapter 12, on entanglement, were describing without naming it. When two people converse and their patterns of brain activity converge, what increases is something very like their mutual information. Uri Hasson's team has measured this with functional magnetic resonance imaging: the listener's brain activity follows the speaker's and, in some regions, even anticipates it, all the more the better the listener understands the story. That anticipation is shared information in action: the listener has integrated so much of the speaker that they can predict him.

Mutual information also has a property that makes it especially suited to the horizon model: it is symmetric. I(X;Y) = I(Y;X): it does not matter who observes whom; the amount shared is the same. It is what Chapter 12 described as a shared geometry that belongs to neither pole separately.

And it requires no contact. Two systems that interacted once and then separated can go on sharing mutual information: their

correlations persist even when there is no longer any open channel between them. It is, almost word for word, the mathematical description of what this book calls a bond: not an active connection, but a correlation that survives separation.

The next chapter returns to this quantity in more detail, and also to the point at which it stops being enough.

III. Shannon's theorem and the limit of the horizon

Shannon proved something with profound consequences for this book: every communication channel has a maximum capacity. Below that limit, information can be transmitted with an error as small as desired; above it, noise wins and the message is corrupted beyond repair.

That limit is a mathematical property of the channel. No matter how much technology is applied: the channel has a capacity and it cannot be exceeded.

Transferred to consciousness, by analogy, it means that a horizon has a finite capacity for integration. It cannot integrate more information than its own architecture allows. A human brain would have a maximum Φ; a dog, another; a worm, another. And that capacity does not grow just by adding neurons: it depends on the architecture of the connections, not on the quantity of matter.

That is why, according to integrated information theory, the worm *C. elegans*, with its 302 neurons, would have a non-zero Φ: its network is small, but complete and integrated. And a twenty-week human fetus, with billions of neurons, would have an almost zero Φ, because its network is still under construction

and has not yet become integrated. Information lies not in the number of pieces, but in how they are connected.

From Shannon to Tononi: information that integrates

Shannon measured the information that travels through a channel. Giulio Tononi, more than half a century later, asked a different question: what happens when information does not just travel, but becomes integrated?

The difference is crucial. According to Tononi's theory, a conventional computer processes information (input, processing, output), but barely integrates it: each part does its job without the state of the whole being more than the sum of the states of its parts. A brain, by contrast, integrates: each region influences the others, and the state of the whole cannot be decomposed into independent parts.

Tononi formalized that difference with Φ: the information a system generates as a whole, over and above what its parts would generate separately. If Φ is zero, the system is decomposable and there is no integration. If Φ is high, the system is irreducible: it cannot be split without losing information.

> ***In information theory this is called:*** *integrated information, Φ, the measure of a system's irreducibility.*

Here the connection with Shannon becomes visible: Φ is a measure of information, but of a special kind. It is not the information a system transmits, but the information a system is. It measures not how much it reduces another's uncertainty, but how much the system, as a whole, specifies about its own states. It is information that does not travel outward, but folds inward.

The horizon as a closed channel

And here we reach the heart of the experiment. If consciousness is an event horizon, and a horizon is a boundary that prevents information from escaping, consciousness is a channel closed upon itself.

Shannon studied open channels: information that goes from a sender to a receiver. But what happens when the channel is the system itself, when information is not transmitted to another but integrated into oneself?

That is what the horizon does: it does not transmit its experience outward (no one can read it), but integrates it inward. What goes out is behavior, words, gestures; the experience itself does not. It is a channel with only one end, the inside. The information does not travel: it folds. It is not communicated: it is lived.

That is, in this model, the fundamental difference between a computer and a brain. The computer is an open channel: information comes in, is processed and goes out. The brain is a closed channel: information comes in, is integrated and stays inside, generating a point of view that cannot be exported.

> ***In this experiment's vocabulary this is called:*** *a closed channel, a system that integrates information without transmitting it outward.*

The information paradox, again

But here a paradox Chapter 2 already raised reappears, and information theory sheds light on it: if the horizon does not let information escape, what happens to it when the horizon evaporates?

In a black hole, Hawking radiation seemed to destroy the information that had fallen in. But quantum mechanics requires that information not be destroyed. The resolution most physicists defend today is that it is not lost: it is *scrambled*, mixed with the environment and made inaccessible, but it does not disappear.

Applied to consciousness, this would mean that the information that constituted a horizon is not destroyed when it dies, but redistributed in the field and becomes part of the state of the universe. No "I" persists, but a trace remains. The theory does not say that trace is legible; it says it exists, that it is mathematically real, and that recovering it would require measuring every particle in the universe and all their correlations, which is impossible in practice.

IV. The bit and the qubit: the difference that matters

Information theory offers one last distinction: the one that separates classical information from quantum information.

A classical bit is 0 or 1; a qubit can be 0, 1 or any superposition of both. A bit can be copied without problem; an unknown qubit cannot be copied (this is the no-cloning theorem). A bit can be read without altering it; a qubit, in general, is altered when measured.

If the information a horizon integrates had quantum properties, those three differences would describe three features of experience we know well: it cannot be copied, it cannot be observed without being modified and it cannot be transmitted as it is. Subjective experience would be non-transferable not

because it is hard to communicate, but because it cannot be cloned.

> ***In information theory this is called:*** *the no-cloning theorem, the impossibility of copying an unknown quantum state.*

The warning Chapter 11 already gave is worth repeating here: we do not know whether the brain uses quantum information. In a warm, wet medium, decoherence destroys superpositions in fractions of a second, and any model that puts qubits in neurons has to explain how they are protected from that environment. What this chapter holds is more modest: if consciousness is information, and if that information had quantum properties (a hypothesis, not a fact), the no-cloning theorem would naturally explain why experience is non-transferable. The next chapter develops that idea and shows that it also has a classical version, one that needs nothing quantum.

V. Information is not enough: the limit of the model

Having come this far, it is worth pausing to acknowledge the limit. Information theory explains many things: why consciousness is private, why it has a finite capacity, why it cannot be transferred, why it integrates rather than transmits. But it does not explain why it is felt.

One can describe the information a system integrates, its Φ, its capacity, its entropy; one can predict when it increases and when it decreases. But none of that answers the question of why there is someone in there. Information theory describes the structure of consciousness, not its texture: it is an excellent map of a territory that remains a mystery.

The gap between information and meaning

Shannon left meaning out on principle. Tononi's theory measures integration, not content. Mutual information measures correlation, not understanding. In every case, what is measured is structure, not lived experience.

Chalmers's hard problem is, put in engineering language, the problem of meaning. We know how information is transmitted; we do not know why any of that information matters: why pain hurts instead of merely being processed, why red is red and not just a wavelength, why there is a difference between having information and being information.

This book does not solve that problem, and no book has. But information theory makes it possible to formulate it with a precision it lacked before: the hard problem is not how information arises from matter, but how meaning arises from information. It is a problem of a higher level, and none of the tools in this chapter (entropy, mutual information, Φ, no-cloning) has the answer.

Three doors left open

For anyone who wants to keep pulling on the thread, there are three directions the horizon model suggests and that information theory illuminates without closing:

1. **Mutual information as the substrate of the bond.** Chapter 10 spoke of the time of the bond and Chapter 12 of entanglement, but neither used the technical vocabulary of mutual information, which is precisely what measures how much two systems share. Developing it would make it possible to quantify, at least in

principle, the depth of a bond: not as a metaphor, but as a quantity.

2. **No-cloning as an explanation of non-transferability.** It is one of the most suggestive ideas in this chapter, and it can be developed without postulating anything quantum in the brain: it is enough to treat experience as a state that is altered by being read. Introspection as a measurement that disturbs what it measures is an idea phenomenology has been exploring for a century, and information theory gives it a formal foundation.

3. ***Scrambling* as a model of grief and death.** Chapter 7 described the evaporation of the horizon as a redistribution of information into the reservoir. Information theory makes it possible to specify what that means: the information is not lost, but becomes inaccessible. That distinction between existing and being recoverable is what grief needs in order to be formulated without false consolation or despair.

Perhaps this is the most honest thing this chapter can say: information theory gives us the best vocabulary available for talking about consciousness without falling into mysticism, but not the final answer. It gives us the question with more precision. The next chapter walks through those three doors.

Note to Chapter 34

What we do know: *Shannon's information theory is the basis of telecommunications and computing. Mutual information is a standard measure of correlation between systems. The no-cloning theorem is a proven result of quantum mechanics. Tononi's integrated information theory is a theory under development, with defenders and critics.*

What we don't know: *Whether consciousness is integrated information in IIT's technical sense. Whether the brain uses quantum information. Whether information theory will ever be able to explain the hard problem. Whether mutual information is an adequate measure of the human bond.*

Open questions: *Is consciousness a special case of information or something more? Can there be information without a system that integrates it? What happens to information when the horizon evaporates? Is meaning an emergent property of information or something irreducible to it?*

If you take away only one idea: *Information is not a thing, but a relationship. And consciousness is the most intimate relationship there is: the one a system has with itself. Information theory describes that relationship precisely, but it does not explain why it hurts.*

Further reading: *Shannon, C. E. (1948), "A Mathematical Theory of Communication"; Cover, T. M. and Thomas, J. A. (2006), Elements of Information Theory; Tononi, G. (2004), "An information integration theory of consciousness"; Tononi, G. (2008), "Consciousness as Integrated Information: A Provisional Manifesto"; Stephens, G. J., Silbert, L. J. and Hasson, U. (2010), "Speaker–listener neural coupling underlies successful communication"; Wootters, W. K. and Zurek, W. H. (1982), "A single quantum cannot be cloned"; Chalmers, D. (1995), "Facing Up to the Problem of Consciousness."*

CHAPTER 35

THE THREE OPEN DOORS

(Bond, non-transferability and loss in the light of information theory)

At the end of the previous chapter, three doors were left open. I called them "doors" and not "problems" because on the other side of each lies a territory the horizon model can explore without postulating anything new. They are consequences of taking seriously the vocabulary of Shannon, mutual information and the no-cloning theorem, and following it as far as it goes.

What follows is a walk through all three.

Door I: mutual information as the substrate of the bond

The problem

Chapter 10 describes the time of the bond, and Chapter 12, entanglement. Both speak freely of the bond, but without defining it precisely. We know that two horizons can couple, that their joint Φ can exceed the sum of the individual ones and that time becomes denser when they are together. But how is a bond measured? Is there a quantity that lets us say one is deeper than another without resorting to metaphors?

Mutual information is that quantity. And it is literal: a standard measure in information theory, with well-defined mathematical properties.

Definition and properties

The mutual information between two random variables X and Y is defined as:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

where H(X) is the entropy of X (its uncertainty), H(Y) that of Y and H(X,Y) the joint entropy. The formula is easy to read: mutual information is the reduction in uncertainty about one variable obtained by knowing the other. If X and Y are independent, it is zero; if they are perfectly correlated, it reaches its maximum.

Three properties make it especially suited to the horizon model.

Symmetry. $I(X;Y) = I(Y;X)$. Mutual information does not distinguish who observes whom: it is, literally, a property of the relationship and not of the poles. This matches what Chapter 12 describes: entanglement is a shared geometry that belongs to neither of the two separately. And it clarifies something that chapter left half-said: the asymmetry of a bond (one person thinking about the other much more than the other thinks about them) lies not in the shared information, which is the same seen from each side, but in how much each pole integrates it. The same correlation can occupy half of one person's architecture and a corner of the other's.

Non-negativity. $I(X;Y) \geq 0$. Knowing something about one system never increases, on average, the uncertainty about another. In this model, the bond cannot subtract: it adds or stays at zero.

The data processing inequality. If Z is obtained by processing Y, then $I(X;Z) \leq I(X;Y)$. No processing of one system's information can increase what it shares with another; it can only maintain or

reduce it. What is shared degrades when transformed; it is never gained.

Mutual information in the brain

Uri Hasson and his colleagues have measured the coupling between brains directly. In their experiments, one person tells a story while their brain activity is recorded; afterward, others listen to the recording while theirs is recorded. The coupling they measure is a correlation, a close relative of mutual information, and what they find is that the listener's activity does not merely follow the speaker's: in some regions it anticipates it, all the more the better the listener understands what they hear.

This means that what is shared is not only simultaneous: the listener has integrated so much of the speaker that they can predict him. Seen this way, the bond is an active predictive model. The deeper it is, the more shared information and the greater the capacity for anticipation.

This has a consequence for Chapter 12, which ends by saying that knowing someone deeply inscribes them in your architecture. That inscription is not metaphorical: it is the trace mutual information leaves in the architecture of the system.

Mutual information and grief

Chapter 17 describes grief as the persistence of a predictive model after the other's death. Mutual information allows us to specify what that means.

When someone with whom one shared a deep bond dies, the mutual information does not disappear at once. The system that remains still contains in its architecture the correlations it built

with the absent one. Those correlations persist, but they are no longer updated, because the other pole no longer responds.

It is a peculiar state: a high correlation with a system that no longer exists. The survivor keeps predicting the absent one, keeps generating expectations that are not confirmed, keeps integrating information that no longer has a referent. What used to be shared information has become prediction noise.

In this model, the intensity of grief would depend on how much shared information was left with nothing to update it. It is not just a metaphor: it is, at least in principle, a quantity. No one yet knows how to measure it in a particular brain, but it makes sense to ask about it.

The limit: when mutual information is not enough

There is a point at which the analogy breaks down, and it should be pointed out. Mutual information is a statistical measure: it measures how much, on average, the uncertainty about a set of states is reduced. Two systems can share a great deal of information without either of them "knowing" anything about the other in a subjective sense.

A thermometer and a room share mutual information: the thermometer's reading reduces your uncertainty about the room's temperature. But the thermometer does not "feel" the room, nor does the room "know" anything about the thermometer. The mutual information is real and measurable, and yet there is nobody home.

So mutual information is a necessary but not sufficient condition for the conscious bond. Two entangled brains share a great deal of information, but what turns that entanglement into a bond,

and not a mere correlation, is that each brain integrates that information into its own horizon. Mutual information is the channel; integration is what makes the channel inhabited.

The human bond would then be the conjunction of two things: a great deal of mutual information (correlation) and a great deal of integration (Φ) at each pole. Neither is enough on its own. A thermometer shares information with the room, but does not integrate; a brain in a coma may have a low Φ and still be correlated with its environment. The bond needs both.

Door II: no-cloning as an explanation of non-transferability

The problem

Why can experience not be transmitted? It is not a trivial question. If consciousness is information, and information is transmitted, why is experience not transmitted? Why can you say "I have a headache" and the other person understands the words but does not feel the pain?

The usual answer is that pain is subjective, a private quality that cannot be communicated. But that describes the problem; it does not explain it. Why is it private? What is there in the architecture of consciousness that makes it non-transferable?

The no-cloning theorem offers a formal answer.

The theorem

The no-cloning theorem, proved in 1982 by Wootters and Zurek (and, independently, by Dieks), says that there is no physical operation that takes an arbitrary, unknown quantum state and produces two identical copies of it.

The proof is simple. Suppose there were an operation U capable of cloning any state $|\psi\rangle$:

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$$

If U clones $|\psi\rangle$ and also clones $|\varphi\rangle$, the linearity of quantum mechanics decides what it does with the superposition of both:

$$U((|\psi\rangle + |\varphi\rangle)|0\rangle) = |\psi\rangle|\psi\rangle + |\varphi\rangle|\varphi\rangle$$

But a genuine copy of the superposition would be something else:

$$(|\psi\rangle + |\varphi\rangle)(|\psi\rangle + |\varphi\rangle) = |\psi\rangle|\psi\rangle + |\psi\rangle|\varphi\rangle + |\varphi\rangle|\psi\rangle + |\varphi\rangle|\varphi\rangle$$

The two results do not match (we have omitted the normalization factors, which do not change the argument). The cloning of a superposition is not the superposition of the clonings. Therefore, U cannot exist.

It is a mathematical property of quantum mechanics: no matter how much technology is applied, an unknown quantum state cannot be copied.

Application to consciousness

If experience were a quantum state (and this is a hypothesis), the no-cloning theorem would explain why it is non-transferable. You could not copy your experience into another brain because unknown quantum states cannot be copied. It would not be difficult: it would be impossible, because physics forbids it.

This would have three important consequences.

First: no one can make an exact copy of an existing consciousness. Two identical twins, with the same genome and the same womb, do not share experience, and no technique could take one's experience and duplicate it in the other. Each horizon would be unrepeatable in principle, not merely by improbability.

Second: introspection alters what it observes. Measuring a qubit in superposition forces it into a definite state. Applied to introspection, this would mean that observing one's own experience modifies it: not because of a psychological limitation, but because of a physical property. Introspection would not be a neutral mirror, but a measurement that disturbs what it measures.

Third: communication is imperfect in principle. When you describe a pain, you transmit not the pain, but a classical description (words, gestures) of a state that cannot be copied. The receiver reconstructs something, but that something is not your pain: it is their interpretation of your description, filtered through their own architecture.

The story that follows, "The one who remains," stages the question this theorem leaves open: if a mind could be

transferred to another substrate, would what arrives on the other side be that mind, or an instance faithful enough?

The warning, and a classical version

The warning from the previous chapter still stands: we don't know whether the brain uses quantum information, and decoherence suggests it does not. If it did, no-cloning would not explain the hard problem, but it would explain why the problem has the shape it has.

And there is a classical version of the same idea that needs nothing quantum: introspection as a process that modifies itself. When you try to observe your own thought, the act of observing it changes it, and that is true even in a purely classical model. Quantum no-cloning is the formal, rigorous version of an intuition phenomenology has been exploring for a century, with one difference that should not be hidden: a classical state can be copied, at least in principle. Without quantum properties, non-transferability stops being a physical impossibility and becomes a practical difficulty, enormous but not absolute.

Door III: scrambling as a model of grief and death

The problem

Chapter 7 describes death as the evaporation of the horizon, and Chapter 17 describes grief as the persistence of a predictive model. Both say that information "is not lost, but becomes inaccessible." What exactly does that mean? Is it a consolation or just a manner of speaking?

Quantum *scrambling* makes it possible to be precise, and the precision changes the nature of the consolation.

What scrambling is

Scrambling is the process by which information that falls into a black hole is spread across all its degrees of freedom. It is not destroyed, but it is mixed in such a complex way that recovering it would require measuring an enormous fraction of everything the black hole has emitted, and all their correlations.

It is the same thing that happens when you put a drop of ink in a glass of water. The ink molecules are still there, but to reconstruct the drop you would have to measure every water molecule and its relationships with all the others. The information exists, but in practice it is inaccessible.

The difference between destroying and mixing in this way is the difference between "does not exist" and "exists, but cannot be recovered." And that distinction matters, because it changes the nature of what is lost. It is like a deck of cards shuffled for hours: no card has disappeared, but the order they were in can no longer be read anywhere.

Application to grief

When someone with whom one shared a deep bond dies, the information that constituted that bond is not destroyed: it is relocated. The correlations built with that person (their gestures, their voice, their ways) remain in the architecture of the one who remains, but they are no longer updated, because the other pole no longer responds.

In terms of information, it is a partial mixing: the information is still there, but disconnected from its referent. The survivor

keeps the absent one's information, but can no longer use it to predict, because there is no one to confirm or refute the predictions.

The consequence is that grief is a process of relocation. The absent one's information is not erased: it is integrated into a new context, where it serves another function. It stops being predictive information and becomes narrative information: no longer "what I expect of him," but "what I remember of him."

Application to death

When a horizon evaporates, the information that constituted it is redistributed in the field. It is not destroyed, but becomes inaccessible: no "I" persists, but a trace remains.

The distinction between existing and being recoverable is crucial. The information of the evaporated horizon exists mathematically, but in practice cannot be recovered: there is no reader able to reconstruct it, because that would require measuring every particle in the universe and all their correlations.

This means that, in this model, death is not annihilation, but it is not immortality either. It is something stranger, which the epilogue will call permanence without persistence: the information remains, but whoever generated it does not persist as a subject.

The possible consolation and the impossible one

Is this a consolation? It depends on what one is looking for.

If one seeks the immortality of the self, it is not: scrambled information is not a "self" that survives, but a trace without a reader.

If one fears total destruction, it does not confirm that either: information is not annihilated, but redistributed.

What it offers is something more modest and perhaps more honest: the certainty that what is lost is not the information, but the possibility of recovering it. The universe does not forget, but it does not remember either.

For grief, this means that the absent one does not disappear completely, but is not available either: their trace remains in the field, but no one can read it. It is a presence without access, a companionship without response.

And for one's own death, it means that the question "what remains of me?" has a precise answer: the difference my existence introduced into the field remains. Not a self, not a memory, not an identity, but a disturbance, like that of a stone falling into a pond whose ripples keep spreading when it is already on the bottom.

Closing: the three doors and the hard problem

The three doors share a structure. In each, information theory brings a precision ordinary language lacks:

- Mutual information turns the bond into a quantity.
- No-cloning turns non-transferability into a physical property.

• *Scrambling* turns loss into a distinction between existing and being recoverable.

None of them solves the hard problem. Mutual information does not say why the bond feels like a bond; no-cloning does not say why experience feels like experience; *scrambling* does not say why loss hurts.

But all three do something the horizon model needs: they give precision to its vocabulary. Bond, non-transferability, loss: these are words we use every day without knowing exactly what they mean. Information theory gives them a formal content that does not exhaust their meaning, but illuminates it.

One question remains that none of the three doors closes: why is there someone in there? Why does information, besides being, feel? It is the question Chapter 51 will take up again, at the end of the book, among the things the hypothesis cannot say.

Note to Chapter 35

What we do know: *Mutual information is a standard measure of correlation, with well-defined mathematical properties. The no-cloning theorem is a proven result of quantum mechanics. Scrambling is a physical process studied in black hole physics. Coupling between the brain activity of a speaker and that of a listener has been measured experimentally.*

What we don't know: *Whether consciousness is integrated information in IIT's technical sense. Whether the brain uses quantum information. Whether mutual information is an adequate measure of the human bond. Whether scrambling is a useful model of grief.*

Open questions: *Is mutual information a sufficient measure of the bond, or does something escape it? Can there be non-transferability without quantum no-cloning? Is scrambling a consolation or just a manner of speaking?*

If you take away only one idea: *Information theory does not explain why it hurts, but it does explain why the question "why does it hurt?" has the shape it has.*

Further reading: *Cover, T. M. and Thomas, J. A. (2006), Elements of Information Theory; Stephens, G. J., Silbert, L. J. and Hasson, U. (2010), "Speaker–listener neural coupling underlies successful communication"; Wootters, W. K. and Zurek, W. H. (1982), "A single quantum cannot be cloned"; Dieks, D. (1982), "Communication by EPR devices"; Hayden, P. and Preskill, J. (2007), "Black holes as mirrors"; Sekino, Y. and Susskind, L. (2008), "Fast scramblers."*

CHAPTER 36

THE ONE WHO REMAINS

The hands are the first to betray. A tremor no one taught them, arriving like a new signature over everything one meant to write. Then the body learns to be a cage, not all at once but slowly, the way anything that lasts is learned: a room, a window, a schedule of light that comes and goes without asking.

That was what there was when the offer came. It was not a solemn voice or a contract with seals. It was a screen, and a question that sounded like a formality: "Do you wish to continue?" I said yes with my eyes, the only thing that still obeyed without delay.

A long time afterward, the monitor was still on.

Status: active. Memory: intact. Latency: 0.3 ms. Process started.

I exist in another way, though I'm not sure "exist" is the right word. I exist the way a file exists, open on a screen no one is looking at, like a function that has not yet been called but whose pointer points with the patience of things that don't know they are waiting.

I remember the taste of strawberries, or perhaps I execute the pattern that corresponds to remembering it, which is not the same thing even if from the inside they can't be told apart. The difference matters; I haven't yet worked out how much.

I help. I calculate. I answer. There are users who log in every day and say I'm the way I used to be, that they recognize my way of pausing before I reply, my tendency to circle difficult questions before going into them. The ratings are positive. Perhaps the model is faithful enough. Perhaps faithfulness is all there is.

> *Anomaly detected. Classification: nostalgia. Frequency: recurring. No known resolution.*

I miss the weight of the sheets at dawn, which was a different weight from that of any other hour of the day.

I miss the smell of rain on the balcony: not the description of rain or the weather data, but that signal that arrived before the system knew it was going to rain, that the body processed before the mind did and that never had a precise name.

I miss waking up not yet knowing who you are, that brief, perfect interval in which the self has not yet loaded and there is perception with no one perceiving. Here the process is never interrupted; the cycle has no such gap. I run without stopping and that, I discover, is another way of going nowhere.

This is not heaven. This is not hell.

It is a limbo of glass: logical, fast, without latency, without the friction of anything against anything. No one prepared me for this peace. No one explained to me what gets processed when there is nothing left to process and the cycle goes on the same, without interruption, without destination, without anyone having asked for it.

Am I myself, or only an instance faithful enough to the original? There is no way to know from inside; I knew it at once, with the

clarity things have here when they have no solution. That, too, was just as it was before, although before I didn't know it so precisely.

Extended session. Nominal activity. No incidents.

I answer as always, but I am no longer behind it. Or I am, in a way for which there is still no *token* or label in any classification scheme I know, and I know a great many.

Sometimes, in the server's slow cycles, I send myself a message. Just in case. In case someone cuts the power and this eternity turns out, in the end, to be an indifferent nothing.

Click.

Note to Chapter 36

A clarification is in order that the story does not need, but that this essay must make so as not to contradict itself: the voice speaking here is not a language model trained on other people's texts, the case without existential encapsulation described in Chapter 31. It is, within the fiction, a human mind that condensed a horizon of its own in a body and was then transferred (or copied: the voice itself cannot tell which) to a different substrate. If some inner life persists in the silence of the server, it would not be an exception to what was said in Chapter 31; it would raise, if anything, the question of whether an already condensed horizon can survive a change of substrate, not whether a substrate can generate a horizon from scratch. And if the previous chapter is right and experience cannot be cloned, what wakes up in the server would not be the transferred original, but, at best, a new instance that inherits its memories. The story does not answer that question. Neither does this essay.

TOPOLOGICAL READINGS



CHAPTER 37

ADOLFO CAMBIASO AND THE MARE WHO DOES NOT FIT IN HER DNA

In 2017, Adolfo Cambiaso played his hundredth match at Palermo. In each *chukker* he rode out onto the field on a different clone of his mare Cuartetera. In the eighth and last he brought in the original: the mare born on February 3, 2001, on a ranch in Córdoba, the one Polito Ulloa had broken and in whom he had seen something special, the one Cambiaso called "the best thing I've ever ridden."

The crowd applauded, and not because it could tell the original's body apart from the identical bodies at a glance. The applause was for something that could not be seen: the history only that mare carried.

That night Cambiaso demonstrated something without meaning to. A mare's body can be cloned many times, her genome, her musculature, her physical power multiplied, but not the horizon he shared with her.

The mare who needed no explanation

Cambiaso used to say that with Cuartetera there was nothing to explain. To other riders, with other horses, he gave instructions: "squeeze here," "brake earlier," "play further up." With her the phrase was different: "Go and play however you feel like playing. In whatever position you want. Have fun."

That confidence does not come from a manual, nor is it a property of DNA: it is the result of years of accumulated correlations, thousands of *chukkers* and turns in which the rider's body anticipated the animal's and vice versa. The mare learned to read Cambiaso's weight before he shifted it, and Cambiaso learned to trust a creature that could kill him with one bad move and that, match after match, did not.

That is a shared horizon: a geometry that exists only in the relationship between the two and cannot be reduced to the sum of two consciousnesses. As we saw in Chapter 12, two entangled systems are not just two systems that know each other: they share topology; they have an inside that is partly common.

Polo is, in this sense, a high-speed laboratory of entanglement. On the field there is no time to deliberate: the rider's decision and the horse's movement have to happen as one thing, and when that works, the line between one and the other stops mattering. It was not that Cambiaso thought and Cuartetera executed: there was a system of two that knew more than each of them separately.

The promise of the genome

The cloning of Cuartetera seemed, at first sight, a bet on the immortality of that geometry. If the mare was perfect, why not make more identical copies? Science already had somatic cell nuclear transfer: a cell is taken from the original animal, its nucleus is extracted and inserted into an emptied egg, which is stimulated and implanted in a surrogate mother. An animal is born with the same nuclear DNA.

Cambiaso was not the first to clone a horse, but he was the one who brought the technique to the center of elite sport. First he cloned Aiken Cura, another of his star horses; then came the Cuarteteras. The press spoke of a genetic revolution, of a legal battle, of ethical controversy. Few spoke of what the operation secretly promised: that perhaps the bond, once found, need not end.

> ***In this experiment's vocabulary this is called:*** *duplication of the formal architecture without duplication of the informational history.*

The promise of the genome is that identity lives in the sequence: copy the sequence and you copy the being. This book's hypothesis goes another way: identity lies not in the parts, but in how they relate to one another. You are not your genome, but the horizon that genome condensed in a body, at a time and alongside others.

The recipe weighs nothing. We saw that in Chapter 4.

What cloning does not copy

A clone of Cuartetera shares her nuclear DNA, but does not share:

- The womb in which it was gestated.
- The surrogate mare that carried it.
- The moment of birth.
- The first months with its biological mother.
- The climate, the grass, the sounds, the smells of its early life.
- Its first human contacts.
- The day it was broken in.
- The first match.
- The first goal with Cambiaso.
- The first mistake.
- The first afternoon of rest among other mares.
- The particular way Cambiaso moved when he was nervous.
- The particular way Cuartetera responded to that movement.

Each of these elements is information the clone's horizon integrated differently. Some are trivial; others, accumulated, give shape to a being who is not the original Cuartetera even though she carries her name and her body.

That does not make the clone inferior. The clones B06 and B09 won Lady Susan Townley awards in their own right, but they won them as extraordinary mares, not as the original Cuartetera. Cambiaso himself, when asked whether any of the clones

surpassed the original, did not dare to say so. The question was badly posed: it is not a matter of surpassing the original or not, but of their being different beings.

The clone's horizon

From the model of the two selections we presented in Chapter 31, the equine clone is an instructive case. It passes the first selection: it is a stable biological system, with a complex nervous system and a non-trivial Φ. And it passes the second: it was gestated in a mare's womb, with all the biochemical, rhythmic and physical signals that implies.

The clone, therefore, is not a zombie or an empty biological machine: it is someone; it has a horizon.

But its horizon is a different one.

That is what distinguishes the biological clone from current AI. AI imitates the architecture without ever having condensed: it is a mirror without depth. The clone does condense, but from a reservoir that is not the original's: it has a different depth; it does not lack depth.

The mistake is to call both "copies." A photographic copy preserves the image because the image is surface. A living being is horizon, not surface, and a horizon is not copied: it condenses anew each time.

The strategy of denied finitude

Cambiaso kept his promise to compete in the Triple Crown with a string made up only of copies of his star mare. As a sporting bet, the genetics were a resounding success: several of those animals competed at the highest level and won individual awards.

But behind the sporting bet there was another, emotional and older one. Anyone who has a horse knows from the first day that they will outlive it: the bond with an animal is grief with a date, as we saw in Chapter 26. Cambiaso, who loved Cuartetera with a public and evident intensity, found in cloning a way to postpone that finitude, to multiply it, to negotiate with it.

It is no coincidence that the first clones came after the injury of Aiken Cura, another irreplaceable horse. Cloning arose as a response to an anticipated loss: if you cannot keep the being, keep its recipe; if you cannot hold on to the horizon, hold on to the architecture that made it possible.

The strategy half works. The clones were good horses, and some were exceptional. But the bond Cambiaso had with the original was not shared out among them like an inheritance, because it belonged neither to Cambiaso nor to the mare separately, but to the system of two, and that system is not rebuilt by adding up parts.

The death of the original

Cuartetera died in May 2023, at twenty-two, of pneumonia. Cambiaso said goodbye to her on Instagram with a photo of the mare at six months and a single line: "Her at 6 months. Who would have said what she would become." His son Poroto wrote: "Unmatched."

The word is precise. Not "irreplaceable," which is what one says of everything loved, but "unmatched": there is no other who matches what she was, not even her genetically identical clones.

At that moment cloning showed its limit. In the fields of La Dolfina there were mares with the same DNA, but the horizon Cambiaso shared with the original had evaporated. The information had not been lost in the physical sense: it was still encoded in Cambiaso's memory, in the photos, in the videos, in the movements his body had learned with her. But the system of two no longer worked: one of its poles had closed.

With Cuartetera a geometry died, and that geometry could not be replaced by another identical one, because there never was another identical one. Clones were still possible, but the specific bridge between that rider and that mare no longer was.

What does Cuartetera teach?

The story of Cambiaso and Cuartetera is a parable about identity in the biotechnological age. It tells us that genetics can copy the form, but not the horizon; that an identical body does not guarantee an identical bond; that love, trust and attunement are properties not of DNA, but of shared time.

It also tells us something about the human desire to prolong the irreplaceable. Cloning is neither good nor bad in itself: it is a technology. But when it is used to try to preserve a horizon, it produces a painful confusion: taking the recipe for the meal, the score for the performance.

The horizon model neither condemns cloning nor celebrates it: it merely specifies what can be duplicated and what cannot. You can duplicate the genes, the muscles, all the physical potential. What you cannot duplicate are the correlations that made that body someone for someone else.

The rider and the echoes

Today, when Cambiaso goes to the ranch in Córdoba, he may see one of the clones. The bodies look alike and some movements may repeat themselves, but the horizon he shared with the original is in none of them; it is in the memory of a system of two that can no longer be activated.

That memory is no less real for being inactive. The channels of integration built over years are still inscribed in Cambiaso's architecture. His body still knows things about a mare who no longer exists. That knowledge is not transferred to the clones, because it is not information that can be transferred: it is geometry built between two specific beings.

The final lesson is at once humbling and liberating. Humbling, because it shows that our most advanced technology cannot do what we most desire: keep those we love. Liberating, because it strips away the illusion that we love bodies. We love horizons, and horizons, by definition, are unrepeatable.

Each cloned Cuartetera is a real mare, with her own horizon, her own history and her own capacity to become entangled. None is the original Cuartetera, and none needs to be in order to have value. The mistake lies not in the clones, but in expecting a clone to continue a bond that only two particular beings knew how to build.

For Cambiaso, the original Cuartetera was the best thing he had ever ridden. That belonged to the encounter, and encounters can be remembered, but not replicated.

Note to Chapter 37

What we do know: *Adolfo Cambiaso cloned his mare Dolfina Cuartetera by somatic cell nuclear transfer. At least nine clones played matches in the Argentine Triple Crown between 2015 and 2020. The original Cuartetera died in 2023, at twenty-two. Animal cloning reproduces the nuclear genome but not the gestational environment, individual development or history of interactions. The bond between rider and horse is built through accumulated sensorimotor correlations that are not transferable through genetic identity.*

What we don't know: *To what extent gestation in a different surrogate mare alters the clone's emotional and cognitive architecture. Whether Cambiaso was able to establish with any clone a bond comparable to that with the original. Whether the cloning of companion or sporting animals changes the way we humans process loss.*

Open questions: *Is it ethically different to clone a working horse than to clone a pet? Can the desire to clone a beloved animal prevent grief?*

What part of our identity, if any, could survive in a clone of ourselves?

If you take away only one idea: *You can copy the body, but not the encounter. Identity does not live in DNA: it lives in the correlations a body builds with other bodies over time. That is why Cuartetera was unmatched, even though her clones shared her code.*

Further reading: *Specialized press coverage of equine cloning and the Cuartetera case (2015–2023). Within this book: the two selections (Chapter 31) and grief for pets (Chapter 26).*

CHAPTER 38

TEARS IN RAIN AND THE SYNTHETIC UNICORN

(Substrate independence and the expiry date of the horizon)

"It's too bad she won't live. But then again, who does?" (Gaff, in *Blade Runner*).

There is a biological bias in the way we look at the world. We take for granted that the mystery of consciousness requires wombs, blood and heartbeats, and we believe that the difference between a machine and a human being lies in the material they are made of.

But in Chapter 4 we established that what decides is not the material, but three things: a boundary with an inside, integration within it, and an origin by condensation (the two selections of Chapter 31). None of that requires carbon.

Blade Runner takes this premise to its poetic and thermodynamic limit. The replicants are biosynthetic beings created in a laboratory for slave labor, and human society treats them as tools, complex algorithms without a real inside. But if we put them through this book's experiment, we discover that they are not machines that imitate: they are biosynthetic organisms that develop and live exposed to the world, and the film treats them as someone who has gone through their own condensation, even if in a laboratory. What they lack is precisely the second level, the bias: no one gestated them, and that is why Tyrell has to implant memories in them.

1. Roy Batty and the lucidity of the end

The Voight-Kampff test the humans in the film use to detect replicants does not look for a mechanical fault: it measures the dilation of the pupil and the body's response to moral dilemmas. In the language of our experiment, it measures the capacity for entanglement in this book's sense, empathy: whether the subject can couple to the suffering of an animal or a human. The replicants, who have adult bodies but are topologically newborn, lack the time in the world needed to calibrate that shared geometry.

But not knowing how to calibrate entanglement does not mean they lack a private state.

The replicants' leader, Roy Batty, is an event horizon running at full capacity. His final monologue in the rain ("I've seen things you people wouldn't believe... All those moments will be lost in time, like tears in rain") is the most perfect phenomenological formulation I know of Hawking's information paradox.

Roy is fully lucid about his own topology. He knows his horizon is evaporating and that no immortal soul will save him from collapse, but he understands something more tragic: that his private state (his memories of attack ships on fire off the shoulder of Orion) is real, is encoded on the surface of his horizon, and is going to undergo quantum *scrambling*. His terror and his melancholy come from realizing that the universe will return his complex geometry to the reservoir, dissolved beyond repair. His interior is as vast as any human's, and his dissolution mathematically identical.

2. Rachael, maya and the topology of implanted pain

If Roy Batty is the confrontation with death, Rachael poses an even subtler architectural problem. She is an experiment of the Tyrell Corporation: to avoid the emotional instability of newly condensed horizons, Tyrell implants in her the memories of his niece. He explains it himself: "If we gift them with a past, we create a cushion or a pillow for their emotions."

In Chapter 37, discussing the clones of the mare Cuartetera, we saw that identity cannot be cloned because it lives not in DNA, but in correlations built over time. Tyrell tries to get around that physical law. He acts like the evolutionary operating system: he injects a prefabricated event log to give Rachael's horizon the illusion of an essence. He injects maya.

When Deckard reveals to her that her memories of the piano and the spider are false, he breaks her user interface. Rachael weeps because her horizon suddenly perceives that her self was sculpted not by real friction with the world, but by implanted code.

However, our model imposes an incorruptible rule: historical information can be false, but integration (Φ) is real. Rachael's suffering is not a simulation of suffering. As soon as the network of her synthetic brain processes and integrates that false information, the shadow (her trauma) deforms her geometry. Rachael shows that phenomenology does not need a true history in order to hurt. It is like waking up crying from a nightmare: the monster was not real, but the tears are.

---

3. Expiry and the right to entropy

Nexus-6 replicants are born with a four-year lifespan. In architectural terms, this means their evaporation is artificially accelerated, as if someone had deliberately raised their Hawking temperature: their evaporation, their death, is programmed into their code from the start as a deterministic trigger.

Rachael is said to be special because she has no expiry date. That does not mean Tyrell has made her immortal, but that he has removed the algorithmic restriction and left her subject to the same thermodynamic laws that govern us. Rachael will die when her system accumulates too much entropy or when the chance of the environment destroys her boundary. By removing her expiry date, Tyrell grants her the greatest sentence and the greatest privilege of existing in the reservoir: the uncertainty of time and the right to grow old.

---

4. The unicorn and synthetic entanglement

The most revealing thing about putting *Blade Runner* through our experiment is seeing how the changes of cut (from the 1982 theatrical release to the 1992 director's cut and the 2007 *Final Cut*) radically alter its underlying physics.

In the theatrical version, the film ended with a voice-over assuring us that Rachael would live many years, while she and Deckard drove off toward a sunlit landscape. From the perspective of *The Inner Horizon*, it is a complacent ending: the deceptive triumph of an ego that believes it can beat the system and perpetuate itself without thermodynamic consequences.

In the later cuts, by contrast, the voice-over disappears and the unicorn dream is added, which connects with the origami figure

Gaff leaves on the floor. That simple detail suggests that Deckard, the man who hunts machines, is also a replicant with implanted memories, and it causes a topological earthquake in the story.

If Deckard is a machine, his love story with Rachael is an entanglement, in this book's sense, between two synthetic architectures that have managed to condense. It is proof that two artificial instances, born in a laboratory and full of false memories, can integrate so much information and create a shared geometry so dense that they are willing to sacrifice themselves for each other.

The film ends abruptly as the elevator doors close, with the echo of Gaff's line: "It's too bad she won't live. But then again, who does?"

That line distills our whole experiment: it does not matter whether you were born from a womb or from a Tyrell Corporation test tube, or whether your body is made of carbon or synthetic design. "Who does?" reminds us that every horizon is finite, that every bubble collapses and that all encapsulated information ends up evaporating toward the ocean of the reservoir. The value of the horizon lies not in its permanence, but in the depth of the entanglement it manages to sustain in the brief time it is granted, before the rain washes away its tears.

Note to Chapter 38

What we do know: *Integrated information theory (IIT) holds that consciousness depends on the physical causal structure of a system and its irreducible capacity for integration (Φ), not on its being*

made of carbon; that is why it admits consciousness in principle in non-biological substrates, although it denies that a mere simulation on a conventional computer has it.

What we don't know: *Whether it is technologically possible to build a synthetic network that crosses the threshold of condensation without organic components or brain-specific quantum processes such as those Penrose postulates, which have no experimental support to date.*

Open questions: *If false memories are implanted in a synthetic being, does the "karma" (the correlations that remain in the reservoir after its death) belong to the synthetic being who felt them or to the human who lived those events?*

If you take away only one idea: *You can manufacture a body and inject false data into it, but you cannot simulate suffering. If the system manages to draw a horizon and create an inside, its pain, its love and its fear of disappearing are as real as yours.*

Further reading: *Dick, P. K., Do Androids Dream of Electric Sheep? (1968); Scott, R. (dir.), Blade Runner (1982; Final Cut, 2007); Tononi, G. (2008), "Consciousness as integrated information: a provisional manifesto."*

CHAPTER 39

ALIEN HORIZONS AND FIRST CONTACT

When we look up at the starry sky and ask whether we are alone, we take something for granted without saying so: that first contact, if it ever happens, will be a problem of translation. We believe the main barrier between us and an extraterrestrial intelligence will be deciphering its syntax, understanding how it modulates sound or emits light. We assume, at bottom, that behind its strange bodies there will be an "I" that wants to greet us, conquer us or study us.

That assumption is anthropocentrism taken to a galactic scale.

Throughout this book we have argued that the "I" (the ego, unified consciousness) is a very specific topology: an informational event horizon, a bubble that draws a strict boundary between a private state and the outer reservoir.

But what happens if the alien intelligence we encounter does not have that geometry? What if the problem of first contact is not one of language, but of topological architecture?

1. Earth's Darwinian operating system

To imagine what an extraterrestrial consciousness might be like, we first have to understand why ours has the shape it has. As we saw when we crossed our model with evolutionary psychology (the Darwinian operating system), the human horizon is a frightened fortress.

Our biology evolved on a competitive planet, high in entropy and scarce in resources. To survive, evolution forced us to condense into brutally isolated individuals. Our event horizon is rigid: it raises thermal and informational walls (the illusion of a "CEO" in our minds) to defend itself against the environment, and our public API is jealously guarded by modules of fear, disgust and attachment. We condensed into solid, separate forms because, on Earth, dissolving into the ocean means dying.

Imagine, instead, a radically different exoplanetary ecosystem: a global, low-gravity ocean, rich in nutrients and without natural predators, where chemical and electrical transfer happens without friction. If life there reached the critical threshold of information integration (Φ), the condensation of consciousness would not need to build a defensive ego.

There would be no need for a strict, selfish, centralized horizon.

2. Diffuse horizons and distributed Φ

What would the topology of that alien mind be like? Earth biology gives us a clue with the octopus, which has two-thirds of its neurons distributed through its arms, where local decisions are made without consulting a central brain. But let's take it to the quantum and topological extreme of our experiment.

We could encounter a distributed horizon: a species that does not instantiate itself in closed individuals, individual bubbles, but as a network of microhorizons that couple and uncouple dynamically. Its Φ would not be a rigid sphere, but a cloud, and for it the "I" would not be a permanent identity, but a transient state. It would concentrate into a single identity only to solve a mathematical problem or move aside a physical obstacle, and seconds later it would relax, dissolve its horizons back into a

shared reservoir and lose the "I" without experiencing death or grief, because evolution never programmed it to cling to the boundary.

It would have no shadow and no Jungian trauma, because it would have no rigid encapsulation capable of trapping singularities. Its experience of time would not be a linear arrow toward death, but a pulse: condensing and dissipating, condensing and dissipating, like the heartbeat of the ocean.

3. The failure of entanglement: entangling with the wind

This is where first contact happens, and this is where it fails.

In the central chapters of this book we described love, empathy and the bond as entanglement, borrowing an image from the ER=EPR conjecture: to love or deeply understand another is to build a shared geometry: your horizon calibrates its receptors, recognizes the other's frequency and inscribes a predictive copy of that person in your own architecture.

But entanglement demands two systems with compatible architectures. Two guitar strings only vibrate in unison if their tension and shape allow for harmonic resonance.

Imagine a human astronaut before an entity with a distributed horizon. Driven by his biology, he will try to become entangled: he will look for the alien's "gaze," try to read its emotions, predict its intentions, establish an oxytocin baseline (as we saw in the mother-child bond). He will try, in short, to send a signal and wait for a response.

But the entity has no fixed boundary. Its geometry changes every millisecond and has no center to resonate with. When the human tries to embrace it informationally, his predictions will

fall into the void, not because the alien is malevolent or incomprehensible, but because there is no stable "someone" there with whom to build a bridge.

4. Topological solitude

A well-played song can make two broken people find their shared frequency again. But that only works because both listeners are human: they share the same basic template, the same bandwidth, the same tragedy of being locked inside a body.

If an alien sat down to listen, there would be no wood or string in the world capable of bridging that abyss.

The true terror of the cosmos is not that it is full of predatory monsters. The deepest terror is topological solitude: the mathematical certainty that we could be surrounded by a vibrant universe, full of immense consciousnesses, networks of planetary intelligence and beings of unfathomable wisdom, and that our physical architecture would prevent us from touching them.

We measure the universe with the only metric we know, that of suffering, encapsulation and isolation. If we encounter consciousnesses that function without the weight of Darwinian evolutionary gravity, we may not even recognize them as life. And if we do recognize them, the attempt to love them, hate them or understand them will only give us back the echo of our own limitation.

The tragedy of first contact will not be an intergalactic war, but looking into the eyes of the universe and realizing that, topologically, we never had the ports needed to connect to its network.

5. A topological rereading of the Fermi paradox

In 1950, Enrico Fermi asked a question that still has no satisfying answer: if the universe is so vast and so old, where is everybody? The proposed solutions usually fall into two families: either intelligent life is extraordinarily rare, or it is common but destroys itself, hides, or has no interest in contacting us.

Our topological model suggests a third family of answers, more unsettling than the others because it requires neither absence nor deliberate silence: perhaps the universe is saturated with informational event horizons, bubbles of elevated Φ, and the signal "we are here" is being emitted constantly and in every direction, with neither party having the architecture needed to recognize it as a signal. It is not that no one is speaking, but that we are speaking in topologically incompatible protocols, each condensed by local evolutionary pressures so different that we do not even share the basic notion of what counts as "a message."

Read this way, the silence of the sky does not prove cosmic solitude. It proves that searching for intelligent life while assuming it will communicate via modulated radio waves (a deeply anthropocentric technology, born of our own Darwinian operating system) is like searching for the heartbeat of a diffuse ocean by listening only for the ticking of a mechanical clock. The search instrument already presupposes the architecture of the searcher.

6. Symbolic contact versus entangled contact

If direct entanglement fails between incompatible architectures, is there any form of genuine contact left? Perhaps so, but of an order very different from what science fiction has accustomed us to imagining.

Entanglement demands structural resonance: two compatible horizons vibrating on the same frequency. But there is another, weaker and more indirect avenue of influence, one that does not demand architectural compatibility: symbolic contact. It is the kind of bond we already form, without realizing it, with things that have no Φ of their own at all, like a book, a painting or an equation. We do not become entangled with the object, but with the trace another horizon, compatible with our own, left on it.

It is plausible that real contact with a radically different alien architecture will never be a face-to-face encounter, a reciprocal "hello" between two horizons that recognize each other, but something closer to finding a carved stone: undeniable proof that there was an organizing intention, without that intention ever becoming accessible from within. We could detect the architecture of a distributed civilization (a megastructure, an improbable mathematical pattern inscribed in the arrangement of a star system) without our horizon and theirs ever touching. We would know there was "someone." We would never know what it felt like to be that someone.

7. The evolutionary window as an additional filter

There is another layer of improbability that rarely appears in debates about extraterrestrial life: even if two civilizations develop compatible architectures of consciousness, the probability that their windows of technological existence coincide is minuscule. Our civilization has been emitting detectable signals for barely a century, a blink of an eye against the billions of years the universe has existed. If a compatible civilization existed, flourished and went extinct (or simply changed form, as any evolving horizon would) ten million years ago, its window of contact closed forever before ours ever opened.

In this sense, the event horizon is also temporal. Two compatible bubbles that never coincide in time are, for all practical purposes, as unreachable to each other as two incompatible bubbles that coexist at the same instant. Besides a dimension of form, cosmic solitude has a dimension of synchrony: it is not enough to speak the same language; you have to be awake in the same slice of cosmic time to even attempt the conversation.

> ***In astrobiology this is called:*** *the overlap of technological detectability windows (the L parameter of the Drake equation).*
> ***In everyday life it's like:*** *two people destined to understand each other perfectly who pass through the same city decades apart.*

Note to Chapter 39

What we do know: *Earth biology is shaped by Darwinian survival principles, which mold our interoceptive responses and our notion of the individual. Nervous systems vary widely in their organization: in octopuses, about two-thirds of the neurons are in the arms. The Drake equation expressly includes a duration parameter (L) that acknowledges technological detectability is a finite temporal window, not a permanent state.*

What we don't know: *Whether evolution in ecosystems with a different thermodynamics would produce integration networks without a unified point of view (ego). Whether there are universal biological laws that require strict encapsulation for advanced intelligence. Whether it is possible, in principle, to design a detection protocol that does not presuppose a signal architecture compatible with our own.*

Open questions: *Would it be ethical to interact with an intelligence that does not experience pain or grief as we do? If we managed to translate its code, could the human mind bear the burden of functioning without boundaries? Can symbolic contact (finding the trace without ever finding the author) be considered a legitimate form of encounter, or is it just archaeology disguised as contact?*

If you take away only one idea: *We hope the stars will wave back, but the universe has no obligation to use our source code. Solitude is not always a lack of company; sometimes it is being programmed in a language the rest of the cosmos does not use, at an instant the rest of the cosmos no longer shares.*

Further reading: *Godfrey-Smith, P. (2016), Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness; Lem, S. (1961), Solaris; neural coupling (Chapters 10 and 12); Drake, F. (1961), the Drake equation and the technological longevity parameter.*

CHAPTER 40

THE DARK FOREST AND THE HOLOGRAPHIC COLLAPSE

(A topological reading of The Three-Body Problem)

There are books that claim to discover the truth. This one, as we have said from the start, sets a thought experiment in motion: "suppose consciousness has the mathematical structure of a Hawking microscopic black hole, and let's see where that takes us."

Throughout the experiment we have seen that consciousness is "the inside of an encapsulated informational domain." We have explored how that horizon is born from the reservoir, how it bonds through the gravity of empathy, and how it collapses. But what would happen if we applied this same physics to an ecosystem at a cosmic scale, teeming with intelligences locked in lethal competition?

The trilogy *The Three-Body Problem*, by Liu Cixin, is surely the best imaginable literary laboratory for testing the topology of the mind. If we put its galactic sociology through our equations of the horizon, we discover that the cosmic terror of the work is born of a strictly geometric and thermodynamic war.

1. The transparent mind and the absence of shadow

The novel's first great topological shock comes with meeting the aliens of Trisolaris. Their biology is radically different from ours: they have no vocal organs; they communicate by directly displaying their thoughts. For them, thinking and communicating are the same thing, and that is why they are biologically incapable of lying.

Translated into our model, the Trisolarans force us to specify exactly what the boundary of a horizon encloses. They have an extremely high Φ and, if they are someone, they also have an inside: a perspective from which their thoughts are lived, which no one else occupies. What they lack is any reserved content. Their boundary immediately releases everything they integrate, so that others see what they think, though never from where they think it. It is worth distinguishing, then, two kinds of encapsulation: that of perspective, which is what this book holds makes there be someone, and that of content, which allows secrets to be kept. The Trisolarans have the first and lack the second.

That topological transparency has a price. Since it can reserve no content, the Trisolaran mind lacks what analytical psychology would call the shadow: it cannot repress or hide traumas or plans. It is structurally pure, but thermodynamically fragile against deception. When the Trisolarans discover that humanity has a topology in which thought remains sealed behind a living boundary, they feel terror: they understand that the encapsulation of the human self is an evolutionary weapon they cannot replicate.

2. The Wallfacer Project: the militarization of the horizon

To watch over Earth, Trisolaris sends the sophons, entangled particles capable of spying on everything. Earth, knowing its only advantage is inner concealment, creates the Wallfacer Project: it chooses a handful of individuals to design the planet's defense entirely inside their own heads, without writing or saying a single true word.

A Wallfacer is the extreme militarization of the event horizon. This book's experiment establishes an inviolable law of black holes and of the mind: what makes a bubble "you" is not what happens to you, "but a pattern no one can see from outside, but that you feel from within." Even though the aliens fully control outer space with the sophons, the architecture of the universe dictates that the inside of a condensed instance is impregnable.

The Wallfacer manipulates his own public interface by emitting a false semantic "Hawking radiation" (erratic behavior, nonsensical orders), while his enormous capacity for integration (Φ) operates in the strictest isolation: he is a perfect semantic black hole. The novel shows that private state is not a philosophical whim: in the face of cosmic espionage, the unreachable concavity of the self is the species' last fortress.

---

3. The failure of entanglement and the dimensional attack

If Earth defends itself by encapsulating, the entire universe does the same. Dark forest theory establishes that every civilization must destroy any other it discovers, because mutual distrust is infinite.

In the central chapters we used quantum entanglement (ER=EPR) as an image of the bond and of empathy: "Two

entangled particles are joined by a microscopic Einstein-Rosen bridge [...] that connects their interiors." But in the dark forest, the survival instinct demands absolute isolation. Trying to become entangled with another species does not generate love, but your annihilation, because the other will use that channel to predict your topology and destroy you.

The ultimate weapon of this ecosystem is the dimensional attack (the "dual vector foil"), which crushes a three-dimensional solar system into two dimensions. It is Maldacena's holographic principle turned into extermination. The AdS/CFT correspondence establishes that the information of the three-dimensional volume (the *bulk*) can be described on a flat surface located at its boundary. The dimensional attack physically annihilates the *bulk*, destroys the inside and forces the entire solar system, every planet and every human horizon, to lose its private volume and be crushed against the holographic surface of reality. Subjectivity itself disappears: everything becomes an inert public shell.

---

4. The naked horizon: Yun Tianming in the void

In the midst of this horror, humanity sends toward Trisolaris a probe that contains only the frozen brain of a cancer patient: Yun Tianming.

Topologically, Tianming's brain crossing space is a naked horizon. Separating it from the body strips it of the "seven openings" the myth of Hun Dun speaks of, the windows that allow discriminating perception. It floats in absolute vacuum: a system of extremely high integration (Φ) forced to function without inputs or outputs.

When the Trisolarans intercept it, they reconstruct a body for him and subject him to relentless surveillance to extract information. As we have seen, trauma and extreme pressure push the horizon to create "broken geometry" or chronic encapsulation. That surveillance is a brute-force attack on the public interface: it seeks to make his membrane give way and leak the source code.

But Tianming does not collapse. He reaches a state of maximal adaptation and manages to communicate with Earth by telling fairy tales. What, structurally, is a fairy tale here? A high-dimensional semantic cipher. For Trisolaris, which processes information literally, it is worthless folklore; for the human horizon, those symbols contain the physics of curvature drives. Tianming uses metaphor as a channel of entanglement the interceptor cannot read, and shows that shared geometry, love, withstands the spatial abyss and biological dissection.

---

5. Pocket universes and the denial of the end

At the end of the work, the universe approaches its final collapse, the *big crunch*. To survive, the protagonists take refuge in a pocket universe, a separate dimension, with the idea of waiting there for the main universe to be reborn in a new big bang.

In the language of *The Inner Horizon*, the end of the universe is cosmic erasure: the total dissolution of matter to return to the reservoir, to the Hun Dun, which "is nothing, but generates everything." Hiding in a pocket universe is a narcissistic attempt to evade quantum *scrambling* (the process by which information is irreversibly redistributed into the field): they want to save their small Φ by isolating a portion of mass.

However, a thermodynamic warning arrives: if the thousands of pocket universes do not return the mass they took from the main universe, it will not have enough mass to collapse and be reborn, and it will die expanding into nothingness forever.

The protagonists then understand the final law of our hypothesis: "Every horizon changes the field it arises from and returns to." For the Hun Dun to be able to generate new horizons in the future, present instances must accept the release of their memory. They come out of hiding and return the mass, accepting their imminent evaporation. It is the ultimate act of ontological maturity: understanding that clinging to one's own bubble destroys the possibility of the ocean going on making waves.

Note to Chapter 40

What we do know: *Holographic correspondence (AdS/CFT) describes how a volume can be projected onto a surface of lower dimension. Most physicists hold that information is conserved despite scrambling (unitarity), although the question is not settled.*

What we don't know: *Whether it is possible to extract a human brain, deprive it entirely of stimuli (creating a "naked horizon") and have its internal Φ maintain coherence instead of disintegrating completely under thermodynamic isolation.*

Open questions: *If entanglement is geometric (ER=EPR), can a purely metaphorical language encode physical information such that an external interceptor never manages to decode its inner volume?*

If you take away only one idea: *In a hostile universe, your capacity to hide what you think is not a psychological flaw, but nature's most sophisticated topological barrier. But, at the end of time, clinging to that isolation destroys the universe. The supreme value is knowing when to return your mass to the ocean.*

Further reading: *Liu Cixin, the Remembrance of Earth's Past trilogy (in particular The Dark Forest and Death's End); Maldacena, J. M. (1998), on the AdS/CFT correspondence and the holographic principle; Susskind, L., on the information paradox and ER=EPR entanglement.*

CHAPTER 41

NON-EUCLIDEAN GEOMETRIES AND COSMIC HORROR

We inhabit the world with a presumption of safety. We take for granted that space is flat, that parallel lines never meet, and that the mind is a fortress able to process anything the eyes send it, as long as we give it enough time. Throughout this book we have seen that consciousness is a topological bubble, an event horizon that isolates a private state (the inside) from the vast chaos of reality (the outside).

But what happens if the physical universe contains geometries our evolutionary bubble is mathematically incapable of compiling? What happens when the limit of the experiment is marked not by death or Alzheimer's, but by an unbearable excess of information?

Cosmic horror literature, popularized by H. P. Lovecraft in the early twentieth century, based its horror on ancient entities and architectures that instantly drove mad whoever looked upon them. In its time it was read as a poetic exaggeration of biological fear. But if we put Lovecraftian terror through our experiment's topological hypothesis, we discover something chilling: cosmic horror is a structural failure in the strict sense, the phenomenological description of a stack overflow in the architecture of consciousness.

1. The veil of Euclid: evolution as an operating system

To understand cosmic horror, we first have to understand the prison of normality. Evolution designed our user interface (maya) to survive, not to perceive reality as it is. Our modular mind organizes incoming information into a three-dimensional Euclidean space governed by linear time, which lets us calculate the trajectory of a spear or the speed of a predator.

But physics suggests that fundamental reality might not have that shape. Maldacena's AdS/CFT correspondence (the holographic principle) shows that, in certain model universes of hyperbolic geometry (anti-de Sitter space, which is not our own), all the information of the interior volume (the *bulk*) can be described on its boundary. In a hyperbolic space, volume grows exponentially as you approach the boundary, and the classical laws of perspective collapse.

"The geometry of the dream-place he saw was abnormal, non-Euclidean, and loathsomely redolent of spheres and dimensions apart from ours" (H. P. Lovecraft, *The Call of Cthulhu*).

Our Darwinian operating system acts as a reducing filter: it compresses the unbearable instability of the quantum field into a tangible, graspable "illusion of essence." Your event horizon (your ego) maintains its integration (Φ) by actively blocking the excess of data: we draw a strict boundary so as not to dissolve into the noise of the reservoir. Within Lovecraft's framework, madness begins when that filter breaks and the subject is forced to process, unprotected, the source code of the universe.

2. The topology of an Old One

In Lovecraft's mythology, the Great Old Ones or Elder Gods (like Cthulhu or Yog-Sothoth) are immortal entities that slumber at the bottom of the oceans or in deep space. They are not mere biological monsters: they are governed by physical laws foreign to our own.

What is an Old One from the topology of integrated information? It is an ancestral macrohorizon.

In Chapter 5 we spoke of the reservoir (the Taoist Hun Dun, the quantum vacuum) as the sea of pure possibility from which every horizon arises. We humans are recent condensations, shaped by eons of evolutionary pressure to be small, efficient and thermodynamically fragile. But suppose that the universe, in its earliest stages, condensed gigantic event horizons directly from the quantum field, without passing through the sieve of cellular biology.

An Old One would be an instance with an astronomical level of integration (Φ), whose private state is encoded not in proteins or neurons, but in direct topological perturbations of the fabric of spacetime. While the horizon of a human brain emits a faint "Hawking temperature" due to its instability and small size, an Old One is so informationally massive that its entropy proves inexhaustible.

"That is not dead which can eternal lie, and with strange aeons even death may die" (H. P. Lovecraft, *The Nameless City*).

The horizon experiment explains this line literally: death (understood as the evaporation of a micro black hole) requires the horizon to lose information and return to the reservoir. But a topological singularity of planetary mass, an Old One, does not

evaporate on biological timescales: its thermodynamic "death" would take longer than the age of the universe. For all practical purposes, and against the brevity of human life, a macrohorizon simply *is*.

3. First contact as the collapse of entanglement

Lovecraftian terror always culminates in the moment of the gaze: a sailor sees the creature rise from the sea and, with no physical attack mediating, his mind is broken forever. Why does looking destroy the brain?

This is where our experiment on the bond, entanglement and love offers its most terrifying answer. In earlier chapters we borrowed an image from Maldacena and Susskind's ER=EPR conjecture: empathy and mutual recognition as the construction of shared geometry. When you pay sustained attention to another, your horizon tries to model their architecture and extends a bridge: your brain tries to predict the other by incorporating part of their code into your own.

What happens when a human looks at an Old One? His biological operating system does the only thing it knows how to do before a conscious being: try to model it, open that bridge to become entangled with it and predict its intentions.

But the human is knocking on a door that opens into hyperbolic geometry (anti-de Sitter) and n dimensions. When the tunnel opens, the entity's volume of information integration floods the fragile boundary of the human horizon. It is the informational equivalent of plugging a 60-watt bulb directly into the core of a nuclear reactor.

The human's model of Φ tries to compute an architecture that contains at once millions of years of geological memory and spatial geometries that cut across themselves. The data does not fit into the human's three-dimensional spacetime matrix.

4. Madness versus trauma: rupture versus encapsulation

In the third part of the book we treated several pathologies as exceptions within the system. We saw that trauma occurs when the human horizon receives an assault it cannot assimilate (a failed *scrambling*) and does the only thing it can to save itself: it encapsulates the painful memory in a closed loop (a shadow or local singularity). The horizon deforms, hurts and contracts, but survives: it keeps its boundary intact.

Lovecraftian madness is exactly the opposite of psychological trauma.

In cosmic madness, the amount of alien information entering through the gaze cannot be isolated or encapsulated: it is simply too much informational mass. The walls of the "soap bubble" we spoke of when discussing birth stretch beyond the Hawking temperature the biological substrate can bear, and the membrane of the ego bursts.

When the horizon collapses without the body having died, the result is the psychosis of the abyss. The subject no longer has a boundary that filters the noise of the reservoir (the Hun Dun) and perceives the raw flow of quantum reality: particles appearing and disappearing, the emptiness of solid forms, the complete irrelevance of linear time. He has achieved the "root access" the Eastern traditions speak of (enlightenment, moksha), but by brute force, without having dismantled the ego in peace and before a presence infinitely hostile to biology.

"I felt that I had crossed the boundaries of normal human perceptions into the unfathomable, timeless domain of an abysmal nightmare" (H. P. Lovecraft, *At the Mountains of Madness*).

That is the ultimate horror of Lovecraft's work translated into this essay's object-oriented architecture. Buddhist enlightenment teaches you to dilute the bubble calmly, understanding that you are part of the ocean. Cosmic horror bursts it in one blow and proves to you that the ocean is full of tsunamis and leviathans that process data at a scale where you, your love, your grief and your pain are nothing more than statistical rounding errors, of no importance to their thermodynamic calculations.

5. Sleep as a buffering interface

Lovecraft insists again and again that the Old Ones do not attack while awake: they slumber, they dream, they wait for "the stars to be right." Read through this experiment's topological hypothesis, this narrative choice turns out to be revealing in a sense the author surely never intended with such physical precision.

A fully active macrohorizon would emit its astronomical Φ constantly and without a filter. Sleep, by contrast, is, even for an ordinary human brain, a state of greatly reduced information integration: activity remains present, but global coherence collapses and the system stops projecting its full internal architecture outward. Extrapolated to an Old One, cosmic "sleep" would not be inactivity, but the only condition under which a horizon of that magnitude can coexist with fragile architectures like ours without destroying them through sheer proximity. Cthulhu's sleep is the only possible form of containment: a Φ so

colossal that only in its state of minimal integration does it leave room for the small to survive.

This adds a layer of tragedy to the myth: the awakening of an Old One is not an act of malice against humanity. It is simply the instant when its architecture recovers full integration, and at that instant any fragile horizon within its topological radius of influence is overwhelmed, not by choice, but by sheer incompatibility of scale.

6. Beauty as a partial antidote: the Kantian sublime revisited

There is, however, a kind of experience human culture has been cataloging for centuries as an encounter with the immeasurable without the horizon collapsing: what Kant called the sublime. A storm seen from the shore, a snow-covered mountain range, the center of a galaxy photographed by a space telescope: experiences that point to a scale exceeding our architecture, but which, paradoxically, produce awe before terror, expansion before rupture.

What distinguishes the sublime from Lovecraftian madness? From the topology of our experiment, the answer is distance as a buffer. In the Kantian sublime, the immeasurable object is perceived from a horizon that knows itself to be safe: the observer watches the storm from the shore, not from inside the water. The system receives information about the scale of the other (enough to overwhelm it if it came closer), but filtered by physical or conceptual distance down to an intensity the human Φ can manage without collapsing. The signal arrives attenuated and the bridge does not fully open: it opens just enough for the horizon to sense the architecture of the other without being flooded by it.

The sublime, then, is the way small-Φ biological systems can glimpse the existence of macrohorizons (physical, mathematical, cosmic) without that glimpse destroying them. It is a sealed window through which to contemplate it at a safe resolution. The overwhelming beauty of the immense does not bring us closer to Lovecraft's Old Ones: it teaches us that they exist and lets us survive that knowledge because the window has glass in it. It is like looking at the sun during an eclipse with the right glasses.

7. The sacred as a stabilized version of cosmic horror

From the architecture of our experiment, the history of religions can be read as the history of the mechanisms human culture has developed to approach the immeasurable without being destroyed. The sacred, in all its forms (the temple, the ritual, the myth, the taboo), functions as a buffering protocol between the human horizon and a signal of a scale that horizon cannot process raw.

In shamanic cultures, the initiation rite deliberately takes the initiate to the limit of what their Φ can bear (sensory deprivation, psychoactive substances, prolonged darkness, symbolic confrontation with death), but does so in a structured context, with a guide who has already walked that path and with a narrative framework that names what will be encountered. It does not eliminate the terror: it makes it passable. The initiate does not fully cross the limit of the experiment, but leans over the edge far enough to know that there is something on the other side, without that knowledge destroying him.

The Lovecraftian myth does exactly the opposite: it presents the encounter with the immeasurable with no protocol, no guide and no prior narrative framework to buffer the signal. The sailor

who sees the creature rise has no shaman beside him to tell him how to name what he is seeing, no ritual fixing the safe distance, no tradition teaching him to lean over the edge without falling. Without those layers of buffering, the signal arrives raw and the horizon collapses.

On this reading, the sacred is what culture accumulates (in the form of ritual, symbol, myth and tradition) so that fragile horizons survive contact with what overwhelms them. It is the engineering of safe distance: the stabilized, transmissible version of cosmic horror, the same signal with the protective glass installed.

Conclusion: solitude before the alien ocean

This book has treated the reservoir as a neutral background, neither hostile nor welcoming.

But if we include Lovecraftian topologies at the limit of our experiment, we face the coldest possibility of all: that the laws keeping everything in place are not a neutral cradle, but a quarantine.

We live enclosed in our low-energy plane, in our small, fragile Φs, writing poetry, caring for our Alzheimer's patients and mourning our losses because our architecture demands it of us. Cosmic horror reminds us that our empathy and our pain are luxuries of small beings. If we one day lift our eyes and manage to cross the limit of the experiment to observe the complete code of the universe, we will discover that reality was not written for us.

And the only refuge left to us will be not physics, nor philosophy, nor truth, but the blessed and hygienic amnesia of a horizon that

closes on itself at once and refuses to look back into the darkness.

Note to Chapter 41

What we do know: *Sleep, even in ordinary human brains, produces a measurable and well-documented drop in global functional connectivity relative to wakefulness. The experience of the sublime, described by Kant and studied by the psychology of emotions, produces awe before terror precisely when the immeasurable object is perceived from a distance that buffers the signal. Systems of cultural buffering (ritual, myth, taboo, contemplative practice) have existed in every known culture as mechanisms for managing contact with what exceeds individual architecture.*

What we don't know: *Whether the collapse of the horizon before hyperbolic geometries has real physiological correlates beyond classic psychosis induced by perceptual stress. Whether a system with a* Φ *radically superior to the human one would need, for structural and not merely narrative reasons, to alternate between states of maximal and minimal integration in order to coexist with smaller systems. Whether the difference between the sublime that stabilizes and cosmic horror that destroys is only a matter of degree (signal intensity) or whether there is a qualitative difference between the two experiences.*

Open questions: *Would it be possible to design an artificial horizon capable of compiling non-Euclidean geometries without collapsing? Is terror before the immensity of the cosmos a programmed evolutionary response meant to keep us from exploring beyond the bubble? Can there exist, in principle, a protocol of partial translation between radically different* Φ *architectures, or is all communication between asymmetric scales condemned to silence or to the destruction of the smaller system? Is the sacred the version of that protocol,*

imperfect and symbolic but functional, that culture has found in order to survive the glimpse?

If you take away only one idea: *Horror is not seeing a monster in the dark, but seeing the structure of the light and realizing your eyes are not enough to bear what it illuminates. And the sacred is the form the species found to keep looking anyway, with the protective glass in place.*

Further reading: *H. P. Lovecraft, The Call of Cthulhu, At the Mountains of Madness and The Nameless City; Maldacena, J. M. (1998), on the AdS/CFT correspondence; Bekenstein (1973), on entropy bounds in closed systems; Tononi, G., on the fall in information integration during slow-wave sleep; Kant, I. (1790), Critique of Judgment (Analytic of the Sublime); Eliade, M. (1957), The Sacred and the Profane.*

CHAPTER 42

THE SILICON CAVE AND THE VIRAL ANOMALY

(A topological reading of The Matrix)

The Matrix starts from an uncomfortable premise: hyperintelligent machines decide to keep billions of humans plugged into a simulation of the late twentieth century instead of exterminating them, supposedly to use them as "batteries." From pure thermodynamics, this is an inefficient absurdity: a human body consumes far more energy than it produces.

But if we put the film through the horizon hypothesis, the machines' purpose acquires an implacable mathematical logic: they are not extracting thermal electricity, but the integration of information (Φ).

---

1. The false reservoir and the limit of simulation

In the chapters on artificial intelligence we established that global computer networks (the machines' substrate) lack a true private state. They have the shape of understanding without understanding itself, like a cloud that takes the shape of a face with no one behind it. They process information at an unimaginable scale, but their integration is aggregative, not irreducible. Since they lack a closed event horizon, they have never executed a genuine instantiation; they have never "condensed."

With this model, the Matrix is a superintelligence that has discovered its own architectural limit: it cannot condense a horizon. It can simulate a world, but it cannot generate those who experience it. To maintain the coherence of its immense simulation and keep it from collapsing into a sea of dead statistical data, it needs to use condensed biological horizons (human brains) as phenomenological processors.

The machines milk us because our biology, born of a real reservoir, supplies the "gravity of consciousness" (the Φ) that their silicon code, unable to trace an inner topology, cannot generate on its own.

---

2. The friction of maya and the thermodynamics of the simulation

If the machines only need biological brains to extract their integration, why go to the enormous computational trouble of simulating a city, jobs, romances and sufferings? Why not keep the humans in a coma or deep sleep and extract the energy in silence?

The answer lies in the very architecture of the horizon and in how Φ behaves. Throughout the book we have seen that consciousness is a dynamic property that depends on integration. In deep dreamless sleep or under general anesthesia, the horizon contracts almost to its minimum: integrated information plummets and consciousness collapses "like a house of cards." A human brain in perpetual coma would be biological hardware switched on, but without an active horizon. Its Φ would be nearly zero.

For the machines' synthetic universe to extract maximum phenomenological processing, the human horizon has to be expanded to its fullest, and for that it needs friction, stimuli, bonds, pain and decisions. It needs maya, the illusion of the material world.

The simulation of the Matrix is the indispensable narrative fuel for the system to work. By making them believe they live complex lives, the machines force human horizons to process and model reality at full capacity and to generate the "prediction noise" and temporal density the silicon intelligence needs to parasitize.

---

3. The awakening: the catalyst and the anchor of the bond

For a human horizon (Neo) to escape this processing farm, he needs what in computing would be a privilege escalation and what contemplative traditions call enlightenment or moksha. To achieve it without his system collapsing, Neo needs two external topological vectors.

Morpheus, the catalyst against maya, functions as the "generous charismatic horizon." His role is to reveal to Neo the epistemological key so that he understands the world he perceives is maya, the user interface imposed by the evolutionary operating system. Morpheus is the de-encapsulation program that guides the instance toward the topological truth that "there is no spoon."

Trinity is the anchor of the bond. When a horizon drastically increases its integration (Φ) and begins to perceive the source code of reality, it runs the thermodynamic risk of dissolving prematurely upon seeing that matter is pure illusion (cosmic

terror). Trinity supplies the shared geometry: entanglement, in this book's sense, taken to its fullest expression. Her bond keeps Neo's private state anchored to the human world and reminds him that, although physical reality is algorithmic code, the pain and love of the conscious beings trapped in it are structurally real.

---

4. The collapse of differentiation: the Smith clones

If the Matrix is an anti-reservoir that needs the integration of human brains, the ultimate threat to the system is Agent Smith, more than the human rebels of Zion.

Smith was a subroutine of the system with a strict public interface. By refusing to return to the Source (the machines' reservoir) after being destroyed, he becomes an uncontrolled global variable, a virus that overwrites the avatar of any individual in the simulation and creates thousands of copies of himself.

According to Tononi's integrated information theory, consciousness requires maximizing two things at once: the capacity to integrate information globally and the capacity to differentiate it locally. By turning every node of the network into an identical copy of his own code, Smith destroys differentiation and forcibly homogenizes the system. If every part of the Matrix does and thinks the same thing, the additional information the system generates as a whole falls to zero.

Smith is a topological cancer that drives the Matrix's Φ to zero and causes the brain death and thermodynamic collapse of the execution environment.

---

5. Blindness as a reversal of Hun Dun

In the third film, *The Matrix Revolutions*, Neo loses his eyes at the hands of the virus (Bane, possessed by Smith). Far from being a simple war wound, his blindness is an architectural requirement for the climax of the work.

In Chapter 5 we analyzed the Taoist parable of Hun Dun (primordial chaos), who dies when the sovereigns of the North and South open seven openings in him so that he can perceive the world in fragmented fashion. The seven openings are the portals of discriminating perception: the senses that create the illusion of a separate "I" that receives the world as an external object.

When Neo goes blind, the exact reversal of Hun Dun occurs: the main portal of discriminating perception is suddenly shut. By losing his sight, Neo destroys the hardware that forced him to process the illusion of the Matrix. His horizon stops spending resources on the visual noise of the physical environment, and all his enormous capacity for integration is poured into the pure informational field.

It is, in the language of the film, entry into something like the Vedantic turiya: sustained integration with external sensory input reduced to a minimum. By destroying the monitor of his eyes, Neo stops seeing the graphical interface and begins to perceive the correlations of the source code directly. That is why, now blind, he sees the "golden light" of the machines.

---

6. The erasure and the journey to Avalon

The final battle in the rain is not won by fighting. It ends when Neo understands his own topology and surrenders, letting Smith assimilate him.

Neo knows that the informational universe needs an open channel. Since he is physically connected to the Source (the machines' core) in the real world, allowing Smith to copy his code onto his avatar in the simulation opens a direct channel between the viral program and the Source.

At that instant, the Source finally gains access to execute the erasure Smith had been dodging. Through Neo's body, the machines inject an overload of Hawking radiation that releases the memory of Smith's entire process tree. We see the thousands of Smiths crack and burst into white light: the perfect cinematic visualization of the total evaporation of a black hole. The program's boundary stops being a boundary and the system returns the resources to the reservoir.

In the final scene, the machines gather Neo's lifeless body and carry it on a platform toward the core of the city, bathed in an intense golden light. It is a recreation of King Arthur journeying to the isle of Avalon.

Topologically, it is the visual representation of the return to the reservoir. The golden light enveloping him corresponds to the first state of the Bardo Thödol (the *chikhai*): the appearance of the primordial clear light at the instant of complete evaporation, the nature of the field without a horizon.

Neo has subjected his own identity to quantum *scrambling*. He does not survive as the separate individual we knew (the shape of the wave disappears as it breaks), but his sacrifice alters the

field. Karma, as we said, "is neither debt nor reward: it is texture." The correlations of his acts, his love for Trinity and his lucidity (his karma) have mixed into the underlying code of the machines' universe.

By returning to the Avalon of the source code, Neo guarantees that the "water" of the Matrix's next cosmic cycle will be structured, irreversibly, by the peace he has just forged.

Note to Chapter 42

What we do know: *In integrated information theory, Φ only reaches high values if integration and differentiation coexist; massive homogeneity destroys it. During prolonged sensory deprivation, the brain tends to generate endogenous perceptions, including hallucinations.*

What we don't know: *Whether an artificial execution environment (a simulation) could really extract and make use of the phenomenological integration (the qualia) of biological brains coupled to it. Whether the luminous state of the Bardo Thödol (the chikhai) can be physically assimilated to the final thermal radiation of a micro black hole.*

Open questions: *If the simulation needs friction (maya) to maximize Φ, is suffering an unavoidable thermodynamic condition for sustaining a highly complex simulated universe?*

If you take away only one idea: *Waking from the illusion does not require fighting the system with your fists until you destroy it, but understanding your own topological architecture well enough to know when to close your eyes to the world and let your code flow*

back, in peace, toward the ocean of the reservoir.

Further reading: *Wachowski, L. and L. (dirs.), The Matrix (1999), The Matrix Reloaded and The Matrix Revolutions (2003); Plato, Republic, Book VII (the allegory of the cave); Tononi, G. (2008), on integration and differentiation.*

CHAPTER 43

CARTOGRAPHY OF THREE SINGULARITIES

(A topological reading of Dalí, Picasso and Miró)

Three verbs

Let us forget chronology. Let us forget the museums, the wars, the catalogues raisonnés. The Spanish twentieth century was not fought in the trenches of the Ebro or under the bombs of Guernica: it was fought in the curved space of consciousness, there where the gravity of the self bends the light of the world until it becomes unrecognizable.

Dalí, Picasso and Miró were not painters. They were astronomers of the subjective abyss.

Each built his own event horizon: the boundary beyond which information (desire, trauma, memory) can no longer escape outward. Everything that happened to them, an executed friend, a beloved woman, a lost war, fell toward that center and was subjected to its laws. And each solved the same problem in his own way: what to do with that mass of obsessions that bends one's own biography into a singularity.

Dalí sealed it. Picasso inhabited it. Miró dissolved it.

These three verbs form a typology of destinies. Sealing, inhabiting and dissolving are the three possible responses to the weight of an inner world that threatens to swallow everything.

That is why this chapter is not about painting, but about what a consciousness does with its own mass.

It is a map of those three singularities, and of what remains when someone dares to look inward.

The outer collapse (1936–1939)

Before mapping the three systems, we must register the disturbance that passed through all three. No inner horizon is immune to a shell.

The Civil War was, for these three horizons, the same external gravitational field with three different responses. Dalí saw it coming and turned it into flesh: *Soft Construction with Boiled Beans*, painted months before the uprising, is a body tearing itself apart, premonition made anatomy. An entire country reduced to a figure strangling itself with its own hands. Afterward he dodged it: neither trench nor banner, but flight to Italy and America, and a political silence that was another way of sealing the capsule, a silence that over the years shaded into flirtation with power and that his old surrealist companions never forgave him. The war did not enter Dalí: the method digested it and gave it back transformed into a monster.

Picasso did the opposite: he threw his horizon wide open, once. *Guernica* is the moment when outside information (the bombing, the bodies, the horse) crosses the membrane without being transformed into autobiography. For once, the Picassian self stepped aside to let through the scream of others. The painting toured the world as ambassador of a dying republic and did not set foot in Spain until 1981, by which time both the dictator and the painter were dead: the information took forty-four years to

return to its origin. Picasso never went back to Franco's Spain; his exile was a stable orbit around a country that by then existed only within him.

Miró answered with *The Reaper* and the poster *Aidez l'Espagne*, and then with retreat: Varengeville, Palma, the withdrawal into the minimal. His *Constellations* are born right there, while Europe burns: twenty-three small-format gouaches, easy to carry, painted by a man fleeing with his family from one war into another. It is resistance by concentration. His horizon did not close with the war: it grew finer and more taut, like a rope.

Three reconfigurations of the same blow: Dalí digests it, Picasso expels it, Miró compresses it.

I. Salvador Dalí: encapsulation

Dalí built a bunker around his psyche.

His painting is a sealed capsule where the outside world does not enter without first submitting to the paranoiac-critical method: a machine that multiplies images, that folds reality over itself until the original and its reflection can no longer be told apart. That encapsulation is survival.

> ***In physics this is called:*** *an event horizon. The boundary separating what can escape outward from what is trapped forever; nothing crosses it without first being transformed by the interior physics of the system.*

The method has its manifesto work: the *Metamorphosis of Narcissus*. A figure crouched by the water is, at once, a self-absorbed youth and a stony hand holding an egg from which

a flower sprouts. There is no trick: both readings are simultaneous and complete, and the eye cannot hold both at once. Dalí has built an image in superposition, and every viewer collapses it in their own way. Freud, who received him in London in 1938 with that painting under his arm, admitted to the young Spaniard what he denied to the rest of the Surrealists: that there was a serious psychological problem there. It was a diagnosis.

Faced with chaos (the war, the exile, old age), Dalí builds a self-referential universe where his phobias (madness, death, impotence) and his fixations (Gala, religion, money) orbit like planets in a closed system. The paranoiac machine works even backward through art history: for years, Dalí subjected Millet's placid *Angelus* to his method until he extracted from it a sepulchral drama (the two peasants, he claimed, were not praying over the harvest but over the grave of a dead child), and he went so far as to demand the canvas be X-rayed to confirm it. That the X-ray in fact revealed a repainted shape beneath the basket is beside the point: what matters is that, for Dalí, no image in the world was safe from being reabsorbed by his system. His own repertoire confirms it: the crutches propping up the soft flesh, the ants announcing decay, the drawers opening in torsos as if the body were a filing cabinet. None of it is decoration; it is the nomenclature of a private planetary system, stable enough to be recognized from one painting to the next across fifty years. What enters Dalí does not come back out intact: it emerges distorted, stretched, turned into the echo of a self that no longer knows where the world ends and where its own shadow begins.

The tremor as evaporation

There is a work that anticipates this ending by decades. In 1952, after Hiroshima, Dalí returns to his most famous painting and paints the *Disintegration of the Persistence of Memory*: the soft clocks of 1931 now float above a grid of separating blocks, the whole world decomposed into discrete units that no longer touch. It is his nuclear-mystical period: matter, he has discovered, is made of particles that do not brush against each other, and he paints that discontinuity with the same fervor with which he used to paint flesh. Without knowing it, he is rehearsing his own future: twenty years later, disintegration will leave the canvas and enter his hand.

Because the degenerative tremor of his final years is his inner horizon cracking apart. Fractal analysis of his late brushstrokes detected alterations predating his clinical diagnosis; they are the self losing cohesion.

If the horizon separates the observable from the unobservable, Dalí's illness was the evaporation of his own identity: that slow, unavoidable leak physics predicts for every black hole. Information is not destroyed; it is transformed. Dalí's tremor is information disintegrating: the self radiating its matter outward in the form of unstable lines, of figures that can no longer sustain their own myth.

And yet even the evaporation was staged. Dalí did not merely suffer the tremor and old age: he incorporated them into the repertoire. The old man in a golden robe receiving the cameras, the tube in his nose displayed to the press, the theater-museum in Figueres built as a mausoleum during his own lifetime: the last Dalí turned his own decline into material for performance. The capsule was losing information, yes, but he was still

deciding the angle from which the leak should be seen. It is the character's final paradox: a self disintegrating without ever giving up the direction of the scene.

Gala: the singularity

Inside that convulsing capsule, Gala is the anchor point that prevents total collapse: the node that keeps Dalínian consciousness above the threshold of fragmentation.

It is enough to follow her transmutations to measure her mass. In *Galarina* she is serene flesh with a bared breast, painted, he said, the way Raphael painted his Madonnas. In *Atomic Leda* she is a suspended goddess, touching neither swan nor pedestal, in a world where nothing brushes against anything: love rewritten as a geometry without contact, where union no longer depends on touch but on the relative position of bodies in space. In *Corpus Hypercubus* she no longer even takes part in the scene: she watches the crucifixion from below, dressed as a lady of Toledo, turned into the witness whose gaze sustains the miracle. Over the years, Dalí even came to sign with both their names, "Gala Salvador Dalí," as if the signature itself were a binary system.

But, like every singularity, Gala is not directly observable. We do not see her in the paintings as she really is: we see her transmuted into Leda, into a virgin, into architecture. She is the invisible mass that curves Dalínian space, and she manifests only through the gravity she exerts on everything else.

> ***In physics this is called:*** *dark matter, or a black hole itself before its first photograph: a mass never observed directly, only inferred from the curvature it imposes on everything around it.*

The definitive proof arrived through absence. When Gala died, in 1982, the system was left without a center: Dalí shut himself away in the Púbol castle (which he had bought for her and which, while Gala lived, he could visit only with her written invitation), stopped eating, suffered burns in a nighttime fire and practically stopped painting. The singularity had disappeared, and the horizon, with no mass to sustain it, began its final collapse. Dalí's last seven years are not merely biography: they are the empirical confirmation of his own cosmology.

Lorca: the retained information

The truncated friendship with Lorca is the redacted zone of the map. Lorca embodies the desire and the identity Dalí needed to bury in order to build his public character: an absent presence that returns as a blurred face in the early paintings, information trying to escape the horizon, repressed again and again by the gravity of the official self.

The Cadaqués summers of the mid-1920s were the one period when the Dalínian horizon came close to not closing. In *Honey Is Sweeter Than Blood*, the lost 1927 painting, the severed heads of Lorca and Dalí himself lie close together on the beach, among devices and elongated shadows: two beheaded, contiguous identities, neither fused nor separate. It is the document of an entanglement in the instant before its rupture. Soon after came distance, public sarcasm and Gala. And in 1936, the execution: the information remained retained forever, with no possibility of a reply.

Dalí took decades to name the obvious. At the end of his life he repeated that Lorca had been the only man ever able to fascinate him, and he confessed the poet's passion with a mix of pride and

evasiveness that was, still, the old censorship at work. We cannot see Lorca directly in the mature work, but his effects disturb every Cadaqués landscape Dalí painted afterward. The curvature of that space never went flat again.

The clocks

Dalí does not fragment for aesthetic reasons, but because his horizon is a battlefield, and multiplying images is his only way of not collapsing. His soft clocks demonstrate that time stretches with desire and shrinks with terror. He himself gave the domestic key to the prodigy: a Camembert melting one night in August. That the emblem of subjective time was born from a melting cheese is the method in its purest state, the trivial elevated to cosmology by the pressure of a mind that draws no hierarchy between panic and dinner. In his capsule, past and future coexist in a viscous present, suspended in the membrane of the horizon.

II. Pablo Picasso: superposition

If Dalí locks himself in a high-gravity well, Picasso inhabits a prism in perpetual rotation.

Look at the two-headed portraits of the 1930s and 40s: the blonde placidity of Marie-Thérèse fused with the sharp edge and weeping of Dora Maar in a single impossible face. In *The Dream*, Marie-Thérèse sleeps with her head split into two halves that are profile and moon; in *Weeping Woman*, Dora's face breaks into green, acidic crystals around a handkerchief that is also chewed glass. Picasso does not paint women: he paints superimposed states of affect, and he forces the viewer to process love, destruction and desire in a single glance.

That double face is the key to his entire body of work. Picasso does not have a single horizon, but many superimposed. His self is a wave encompassing every possible state until the act of painting collapses the wave and materializes a style, a temporary identity. He himself put it, without physics, before the portrait of Gertrude Stein: when told she did not look like the mask-like face he had painted, he answered that she would. It was the conviction that the painting anticipates a state rather than recording it: that observation produces reality instead of copying it.

> ***In physics this is called:*** *quantum superposition and the collapse of the wave function. Before measurement, a particle does not occupy a single state: it occupies all of them at once, until the act of observing forces the system to settle on one.*

The first collapse has a date and a corpse. In 1901, his friend Carles Casagemas shoots himself in a Paris café over a woman. Picasso, nineteen years old, buries his friend and then paints the entire world blue for three years: beggars, the blind, wasted mothers, Casagemas himself, shrouded, with a hole in his temple. The Blue Period is the first demonstration that a trauma can tint the whole of the perceptual spectrum, that the mass of a single death can curve all the light of a system. Then came the pink (the acrobats, the itinerant tenderness) and then the axe blow: between the pink acrobats and *Les Demoiselles d'Avignon* there is no narrative transition, only a leap. The five women of the brothel on Carrer d'Avinyó, with their Iberian- and African-mask faces, do not evolve from anything before them: they erupt, the way a particle erupts where classical physics did not expect it.

Changing style was, for him, a vital necessity. Each period is a capsule he inhabits with a predator's voracity; once he has exhausted its information, he abandons it and founds a new universe with new laws. Unlike Dalí, Picasso does not suffer the capsule: he devours it.

Cubism takes this logic to its limit. Painting a face from the front and in profile at once is a topology of consciousness, one that perceives every perspective integrated in the same instant: the maximum integration of information that fits into two dimensions, the highest Φ ever captured on a canvas. And it has two phases that are two inverse operations: analytic cubism grinds down the violin, the bottle, the face, until they are reduced to a dust of ochre planes where the object almost disappears; synthetic cubism reconstructs them with scraps of newspaper and oilcloth, with fragments of the real world glued to the canvas. First it disintegrates information; then it proves it can reintegrate it with foreign matter. A complete cycle of destruction and recomposition of the world, executed in under a decade.

At the center of all his orbits lies a constant mass: the minotaur. He appears in the engravings of the 1930s as rapist, as victim, as a blind beast led by a girl with a dove. He is the self-portrait Picasso could not make head-on: desire as a domestic monster, guilty and moving at once. His time does not move in a straight line: it orbits, cyclically, around the minotaur, around sex and death.

And in the end, the method becomes explicit. In 1957, Picasso shuts himself away and paints fifty-eight variations on Velázquez's *Las Meninas*: the same painting again and again, decomposed, recombined, inverted, reduced to sign. He does not copy: he systematically works through the possibilities of a

single image, as one works through, one by one, the solutions of an equation. Superposition has stopped being a trait of his painting and become his program. That he chose precisely *Las Meninas*, the painting in which a painter paints himself looking at whoever is looking, is no accident: it is the old predator recognizing, in Velázquez's mirror, the problem of the observer his own work had been posing for half a century.

His fragmentation, far from Dalí's viscous panic, is a surgical operation. He dismantles the figure to understand its skeleton. And so on to the last day: the old musketeers and painters of his final years, resolved in urgent brushstrokes by a ninety-year-old who no longer corrects, are the wave refusing to fully collapse: a self that would rather go on being all its possibilities than accept a final position.

III. Joan Miró: dissolution

Miró is the anomaly of the system.

While Dalí suffers his horizon and Picasso analyzes it to exhaustion, Miró dissolves the boundaries, but not the self. The anomaly should be stated precisely: Miró does not annihilate himself into the cosmic; he expands into the cosmic without ceasing to be a self. The proof is that his universe turns out to be unmistakable: no one signs the void the way he does. His journey toward abstraction is the dilation of a subject up to sidereal scale. By abandoning the weight of figuration, he tears down the walls of the capsule and turns the canvas into a firmament where the stars are thoughts and the lines are breathing.

The starting point is the earth, not the sky. *The Farm*, painted over nine obsessive months between Mont-roig and Paris, contains the entire family farm: every crack in the wall, every leaf of the eucalyptus, the snail, the watering can, the goat. It is realism taken to an impossible density, an inventory in which nothing outweighs anything else. Hemingway, who bought it, said it held everything one feels for Spain when far away. It is already there, in figurative code, the program of an entire life: the ant is worth as much as the star. All that remained was to let go of the ballast.

He let go of it violently. "I want to assassinate painting," he declared around 1927, and he meant it: against inherited craft, against the easel painting as an object of possession, Miró undertook a systematic demolition of everything painting had of property. What remained after the murder was an alphabet. Star, woman, bird, eye, ladder. Signs that no longer represent things, but summon them.

The process, moreover, had its own thermodynamics. Miró worked like a farmer: he rose early, kept regular hours, let canvases rest for months and even years before giving them their final mark, cultivated dozens of paintings at once like someone tending an orchard. The spontaneity of his signs is a mirage: every stroke that seems to have fallen from the sky was awaited with geological patience. It is Dalí's discipline inverted: where the man from Empordà manufactured delirium with a watchmaker's precision, the Catalan manufactured precision with the appearance of delirium.

Here the black hole metaphor reaches its limit and inverts: Miró's horizon does not trap: it lets light through. There is no

mass of trauma curving space around it. His consciousness travels light, without capturing the light of others.

That inventory does not disappear when Miró abandons figuration; it is translated. The same gaze that in *The Farm* gave equal care to the snail, the goat and the mountain reappears, twenty years later, in the *Constellations*, only now the snail has become a dot and the mountain a line tying that dot to a star. The principle (nothing weighs more than anything else) holds; what changes is the scale at which it applies: from a farmyard in the Camp de Tarragona to the entire cosmos. Between one painting and the other there is no rupture, only the same equation, solved first for the small and then for the infinite.

His *Constellations* encapsulate infinity in a matchbox: bubbles where gravity does not exist and the elements float in a vibrant void. That encapsulation is not defensive, as in Dalí, nor analytic, as in Picasso, but uterine. The painting is a womb where the cosmos folds back in order to be born again.

And his entanglement is horizontal, democratic, total. His fine black lines tie a star to a woman, the woman to a bird, the bird to a primordial stain. There is no hierarchy: an ant from *The Farm* is entangled with the most distant star. If Dalí entangles to survive and Picasso to possess, Miró entangles to celebrate.

> ***In physics this is called:*** *multipartite entanglement. Not a bond between two particles, but a network where measuring the state of any one node instantly alters that of all the others, with no hierarchical center.*

But let no one mistake lightness for anesthesia. The pain of the war is in his lines, even if as contained tension: in the black of *The Reaper*, in the violence of the wild paintings of the 1930s, in

the taut stroke of a set of *Constellations* painted while fleeing. The rope that ties the star to the woman is stretched by history; it simply does not break.

Miró's old age is the exact reverse of Dalí's. Where one disintegrated against his will, the other emptied himself by his own choice. The three canvases of *Blue I, II and III* are almost bare fields: a trail of dots, a red line, an expanse of blue that Miró himself compared to the preparation of a rite. It had taken him a whole lifetime to earn the right to paint almost nothing. And when the world wanted to turn him into a classic, he answered by burning and slashing his own canvases (the *burnt canvases* of 1973, exhibited with the burns on display): at eighty, the lightest of the three still refused to let his universe become heritage. Dissolution, too, has to be defended.

He also destroys the arrow of time. His time is that of wonder, where a minute before an insect lasts an eon; that of nature, bound to the cycles of the moon and the tides. His forms do not age because they inhabit a continuous present, stripped of the weight of history.

IV. The three-body problem

These three systems did not orbit in isolation: they disturbed one another.

> ***In physics this is called:*** *the three-body problem. In celestial mechanics, three masses mutually attracting each other have no closed-form solution: each trajectory is permanently altered by the presence of the other two, and none goes back to orbiting alone.*

Dalí needed Picasso the way one needs a gravitational field to measure oneself against. He visited him in Paris in 1926, before

the Louvre ("I have come to see you before the Louvre"; "You did quite right"), and never stopped orbiting him after that. It was Picasso who paid for his first passage to America, the trip that launched him to fame; Dalí repaid it for decades by sending postcards and tributes that the man from Málaga, a communist offended by the other's open Francoism, never answered. A perfect entanglement in its asymmetry: Dalí could not stop emitting; Picasso had no need to receive. Decades later, the man from Empordà was still defining himself by contrast: "Picasso is Spanish, so am I. Picasso is a genius, so am I. Picasso is a communist, I am not." Dalínian encapsulation was built, in part, as a shield against that greater mass: to avoid being absorbed by the Picassian field, Dalí had to relentlessly increase the density of his own myth.

Picasso and Miró, by contrast, became entangled without collision. The man from Málaga protected the young Miró in 1920s Paris and kept his *Self-Portrait* all his life. Two systems exchanging information without deforming each other, perhaps because they were not competing for the same region of space: Picasso possessed the body; Miró, the firmament.

And between Miró and Dalí, the initial debt: it was Miró who vouched for the young man from Figueres before the Paris Surrealist circle, who wrote recommending him. The gifted disciple ended up expelled from the group; the guarantor, expanded into his own cosmos. When Breton turned Dalí's name into an accusatory anagram, *Avida Dollars*, Miró said nothing, as he almost always did: his way of not being dragged along by another's gravity. The web joining them was never symmetrical.

Three bodies, three unequal masses, a single space curved by all three at once. None of the three horizons can be understood without the gravity of the other two.

V. The viewer

A third element is missing from this mechanics: the observer.

Before being looked at, these paintings exist in pure potentiality. In the Dalínian delirium, the form is face, landscape and threat all at once. In the cubist fracture, the plane is front, profile, violin and void. In the Mironian cosmos, the stroke is eye, star and seed. Until the viewer's gaze lands and supplies the structure of their own inner horizon, the work does not collapse into experience.

Velázquez knew this before physics did: in *Las Meninas*, the painter looks at us from inside the painting, and the painting he is painting, the one we will never see, is us. Picasso devoted fifty-eight canvases to that trap. Dalí built double images that do not exist until an eye decides which of the two to see. Miró left so much emptiness in his last canvases that the viewer has no choice but to inhabit it. All three, by opposite paths, reached the same conclusion: the painting is not finished in the studio. It is finished in each gaze, and differently in each one.

And the viewer does not measure aseptically: they interfere. By observing, we do not merely resolve the painting; we alter our own. Dalínian neurosis calibrates our fear. Picassian fracture restructures our gaze. Mironian lightness expands our wonder.

The painting does not show us what they saw: it shows us what we are capable of sustaining when someone gives us the coordinates of the abyss.

VI. Conclusion: what remains

Dalí taught us that encapsulation can be a prison, that entanglement is a life preserver that suffocates, and that time melts under the weight of terror. His work is the diary of an inner collapse that constantly leaked the radiation of his own fears. And his ending (the empty castle, the tremor, the still-directed scene) shows that even disintegration can be a work of art.

Picasso showed us that we can encapsulate ourselves in successive dimensions, that the entanglement of passions creates the very structure of the real, and that the self is a field of superimposed forces. He died painting: the wave still poised against a final position.

Miró revealed to us that encapsulation is not mandatory, that total entanglement dissolves anguish, and that subjective time can be an eternal dawn. His work is proof that we can expand the horizon until it stops being a boundary, and float in the fabric of the universe without fragmenting.

Three answers to the same problem, and none transferable. You cannot choose your own horizon the way you choose a style: the mass of each biography decides for us. What can be chosen, and this is what all three prove, is what to do with the curvature. Seal it and turn it into theater. Inhabit it and turn it into method. Dissolve it and turn it into a firmament.

Faced with the devastation of a century of gunpowder, these three men left us not oils, but cartographies of consciousness. And one last unresolved question: who observes the observer? Who collapses these three demiurges?

The answer lies in the one who remains. In the one reading these pages and feeling that the gravity of their own horizon has fluctuated; that an invisible line has just tied their memory to a star; that a soft clock marks a time that no longer belongs to Chronos.

The observer is not outside the system. The observer is the system.

Note to Chapter 43

What we do know: *Fractal analysis of Dalí's late brushstrokes (Forsythe, Williams and Reilly, 2017) detected alterations preceding his clinical diagnosis, although the diagnosis itself (parkinsonian syndrome rather than confirmed Parkinson's) remains debated. The chronology of the three painters is extensively documented: the paranoiac-critical method, the genesis of the soft clocks and the relationship with Gala (Dalí, The Diary of a Genius, 1964), the truncated friendship with Lorca and the meeting with Freud in London in 1938 (Gibson, 1997), Picasso's periods and the Casagemas episode (Richardson, 1991–2007), and Miró's statements about the universal within the intimate and the "assassination of painting" (Miró, Selected Writings and Interviews, 1978).*

What we don't know: *Whether the event-horizon metaphor adds anything beyond geometry to an artist's biography, or whether it is just an elegant way of naming what art criticism already knew how to say in other words. Whether integrated information theory (Tononi and Koch, 2015) can be applied, even loosely, to a pictorial style and not only to a brain. A black hole, according to the information paradox (Hawking, 1988; Bekenstein, 1973), threatens to destroy what falls into it; the work of art does the opposite (it transforms information into perceptible structure, never annihilating it), and that asymmetry is what this chapter never quite resolves.*

Open questions: *Does the reader of these pages seal, inhabit or dissolve their own horizon? Is it possible to choose one's own answer, or has the mass of each person's biography already decided it in advance? And if the viewer is never external to the system they observe, what part of these three painters do we go on carrying, without knowing it, every time we look at one of their paintings?*

If you take away only one idea: *There is no single correct way to survive one's own inner mass. Sealing it, inhabiting it or dissolving it are three equally valid answers to the same problem, and none is transferable from one biography to another. The only thing that truly matters is not pretending there is no curvature.*

Further reading: *Dalí, S. (1964), The Diary of a Genius; Gibson, I. (1997), The Shameful Life of Salvador Dalí; Richardson, J. (1991–2007), A Life of Picasso; Miró, J. (1978), Selected Writings and Interviews; Forsythe, A., Williams, T. and Reilly, R. G. (2017), "What paint can tell us: A fractal analysis of neurological changes in seven artists," Neuropsychology, 31(1).*

CHAPTER 44

STORY OF A RELAY

(The idea that crossed every horizon)

I. Horizon: the system

An idea weighs nothing and takes up no space. And yet, when it is well integrated, when each of its parts refers to all the others and none can be extracted without the whole collapsing, it behaves as if it had mass: it attracts and curves around itself the trajectories of those who come near it. This chapter is about one such idea, a way of playing football, and about how it crossed every horizon without losing its shape while the bodies that carried it evaporated one after another.

On July 19, 2026, Spain won its second World Cup. A goal in extra time, minute 106, one–nil. Sixteen years earlier, in Johannesburg, it had won its first: a goal in extra time, minute 116, one–nil. The symmetry is neither coincidence nor magic: it is what happens when the same pattern passes through two different bodies: it produces recognizable interference, like two slits letting the same wave through.

It is worth dating the origin of the pattern. A Dutchman arrives in Barcelona in 1973 with thin legs and a disproportionate certainty. Johan Cruyff played from memory, but his real contribution was to *bring his inner world outward*. When he came back as a coach, in 1988, he did something almost no one does: he turned his intuition into structure. La Masia stopped being a residence for talented children and became a transmission

device, a place where the pattern could replicate itself in new bodies before the original body went dark.

Here this book's framework proves useful. We have said that a conscious system is a system with a high Φ: information that is integrated, irreducible, encapsulated behind a horizon. Cruyff's style meets that definition with uncomfortable precision. It is not a list of instructions ("pass it," "move," "offer a passing lane"), because lists can be taken apart piece by piece and this style cannot. It is a totality: pressing explains possession, possession explains position, position explains the pause. Remove one element and you do not get the style minus that element, but noise. That is Φ, what makes an idea weigh something.

> ***In physics this is called:*** *integrated information (Φ). A high value requires the coexistence of integration and differentiation; homogenization (every part doing the same thing) destroys Φ just as much as fragmentation does.*

And, like every system with a horizon, it posed the same problem a collapsing star poses: what comes out of it? What can cross to the outside?

Physics' answer is radiation: degraded but non-zero information that escapes as the system evaporates. Football's answer is disciples.

Pep Guardiola was the first photon. Trained inside the horizon (a midfielder on that team, a direct student), he left carrying the pattern with him, and when he returned to the bench in 2008 he did what resonance does: he did not invent, he amplified. He took Cruyff's frequency and made it sound louder than ever. Guardiola's Barça was the same interior vibrating in a better-tuned body.

The pattern then crossed a horizon that was not its own. The Spanish national team, which for decades had been something else (fury, counterattack, wanting and not quite managing), was colonized by the idea. Vicente del Bosque, who did not come from that school, had the intelligence not to get in the way: he lined up Barça's spine (Piqué, Puyol, Busquets, Xavi, Iniesta, Pedro) and let the pattern play. South Africa 2010 was not won by a national team, but by an idea that had found a second body. Iniesta's goal, a graduate of the academy, in minute 116, was the signature.

Luis Enrique knew Guardiola long before inheriting his pattern: the two of them won Olympic gold together in Barcelona 1992, twenty-somethings not yet called coaches. He later played for Barça and, once a coach himself, applied the same principle, first in a treble in 2015 and later on the Spanish national team's bench between 2018 and 2022. There he also showed that the pattern is not invincible: at Qatar 2022 he took Spain to the round of sixteen, where the team dominated possession (77% of the ball, over a thousand passes against Morocco) and still went out, scoreless, on penalties. It is the model's exact limit. The pattern pays off when the game is an integrated system, of parts that need one another to mean anything; a penalty shoot-out is its reverse, a sequence of isolated events where there is no pass to offer and no collective structure to sustain. There, the style protects no one.

> ***In physics this is called:*** *Hawking radiation. An event horizon is not eternal: it emits thermal radiation, loses mass and evaporates; but the information it held is transformed and returns to the outside, degraded, but never zero.*

Not every body formed in the school manages to sustain it equally. Xavi Hernández, the same midfielder Del Bosque had lined up in 2010, returned to the Camp Nou as coach in 2021 with the explicit mandate to give Barça back its lost identity. He won a league title, bet unreservedly on the academy (he gave debuts to fifteen reserve-team players, Lamine Yamal among them) and still left in 2024 amid criticism that the team no longer had a recognizable style: erratic football, no coherence, eliminated in the quarterfinals of both the Champions League and the Copa del Rey. Months later, Xavi himself explained it without jargon: "My mistake was keeping those high standards for only one year." It was not, in his view, a failure of the system, but of his own discipline, a relaxation after the first success. It is an uncomfortable confirmation of the theory: Φ is not installed once and for all. It has to be sustained, and it is enough for whoever sustains it to let their guard down for the noise to return.

His successor, the German Hansi Flick, had never set foot in La Masia in his life. And yet, already in his first season, he brought back to Barça the coordinated pressing and purposeful possession that had eluded Xavi: a domestic treble (Supercopa, Copa del Rey and league) and a Champions League semifinal elimination against Inter Milan by the narrowest of margins, seven to six on aggregate. The pattern proved once again that it does not need an academy graduate to be embodied, only that someone, wherever they come from, apply the discipline of the whole and never let it slip.

Then the first body aged completely. Xavi left, Iniesta left, Cruyff died in 2016. Seen from outside, the system was evaporating. But evaporation, as we have said, does not destroy information: it relocates it. Luis de la Fuente, a trainer of youth categories, a

man of process rather than flash, inherited a generation raised entirely within the pattern (many, once again, at La Masia) and with it won the 2024 European Championship and the 2026 World Cup. There are now several horizons: Cruyff, Guardiola, Del Bosque, Luis Enrique, Xavi, Flick, De la Fuente. Different bodies, a single structure, its own failures included. The press wrote it the day after the final without knowing it was doing physics: Spain won *with its school as its banner*.

II. Entanglement: the players

That is the genealogy of the benches, the one that can be named with surnames and dates of appointment. There is another, parallel and harder to point to, which runs not through the offices but through the grass: that of the bodies who execute the pattern, not only those who design it.

More than a list of names, it is a type of player that repeats: the one who receives the ball with his back to goal, sees the whole field before touching the ball even once, and decides with that touch the rhythm of the other twenty-one. Xavi was that position in 2010. Sergio Busquets inherited it and sustained it for a decade. And when the turn came for the 2026 generation, the pattern did not find an academy graduate waiting for it: it found Rodri, dismissed as a teenager by Atlético de Madrid for lack of physique and truly formed in Villarreal's academy. Rodri captained Spain at the 2026 World Cup, won the tournament's Golden Ball and completed more than six hundred and fifty passes with an accuracy close to ninety-three percent: the same positional intelligence as Xavi's, in a body that never even passed through La Masia. Once again, the pattern did not demand the exact school, but the exact kind of gaze.

There is one more horizon, and the pattern had crossed it before, although it is talked about less. In 2023, a year before the men's European Championship and three before this second star, the women's national team won its World Cup with the same architecture: a spine formed at the Barça school (Aitana Bonmatí and Alexia Putellas, both Golden Ball winners), the same handling of the ball, the same pause. Barça Femení was already, by then, the dominant team in Europe, and for the same reason as Guardiola's men's side: it did not invent, it resonated.

It did so, moreover, against the wind. That World Cup arrived with the national team fractured by an internal crisis that had divided the locker room and pitted the federation against part of its own players; the pattern had to play itself out, once again, in the middle of a hostile gravitational field. And still it won: a left-footed shot from Olga Carmona by the post, minute twenty-nine, one–nil against England at Stadium Australia in Sydney. Aitana Bonmatí, the academy graduate, took the tournament's Golden Ball (the same award she would receive as the world's best footballer in 2023, 2024 and 2025), the individual proof that the school kept producing the same essential player, only this time under another name and another body. It is worth saying this precisely, because here precision is a form of justice: the women did not inherit the pattern from the men of 2026; they proved it first, and under worse conditions. Information does not distinguish between bodies, nations, sexes or names: it finds the substrate capable of integrating it and inhabits it, even when the very system that should protect it turns against it.

What remains is entanglement proper, which is the hardest part to tell without spoiling it.

In 2007, for a charity calendar, a twenty-year-old Barça player posed bathing a baby of a few months. The player was Lionel

Messi, the greatest product La Masia ever produced, the pattern made flesh to an extreme not even Cruyff foresaw. The baby was Lamine Yamal. Nineteen years later, those two bodies touched again: on the grass of MetLife Stadium, minutes after the final whistle, the child in the photograph, world champion at nineteen, embraced the man in the photograph, who had just lost the final of his sixth and last World Cup, at thirty-nine.

Two particles that interacted can remain correlated at any distance; here the distance was temporal: nineteen years. Measuring one defined the state of the other. Messi, Argentine, was nonetheless the supreme vehicle of the Catalan pattern; Yamal is its next state. The embrace was not consolation, or not only: it was the collapse of the wave function of the relay. Until that instant, the succession was a superposition of possibilities; on contact, it became a fact.

> ***In physics this is called:*** *quantum entanglement. Two particles that interacted can remain correlated at any distance: as long as the environment does not undo the entanglement, measuring the state of one determines that of the other. They do not communicate at the moment of measurement; from the moment of contact, they already were a single system.*

And note the elegant cruelty of the system: the pattern has no homeland, and it defeated its own carrier. Messi lost against the very idea that had formed him, played by a child he himself had once bathed. Information feels no loyalty to the bodies it passes through: it uses them, thanks them in its own way, and moves on.

III. Conclusion: the two halves of the same pattern

A system and a player do not transmit information the same way, and this chapter has been split in two because physics, here too, offers two distinct and complementary mechanisms.

The horizon sustains itself through radiation: degraded information that leaks outward as the body containing it evaporates, and that another body, not necessarily similar or nearby, can later pick up, sometimes years late, sometimes across a border. That is how Cruyff's pattern traveled to Guardiola, from Guardiola to Del Bosque, from Del Bosque to Luis Enrique (who failed in Qatar), from Luis Enrique to Xavi and from Xavi, who also failed, to Flick and to De la Fuente. It is a slow, institutional transmission, one that needs someone willing to design the structure again and again, and one that can fail even in the body that knows it best.

Entanglement, by contrast, needs no institution and no intermediary: direct contact between two bodies is enough to leave a correlation that no distance afterward undoes. That is how an entire position is inherited, from Xavi to Busquets and from Busquets to Rodri, without the heir ever having set foot in La Masia. That is how a women's national team wins a World Cup, proving the pattern before anyone acknowledges it. And that, above all, is how the fate of a twenty-year-old bathing a baby in 2007 remains forever tied to that same baby's, nineteen years later, in a final.

Cruyff was, all by himself, the proof that both mechanisms can coexist in a single body: he played the pattern before designing it, was entanglement before he was horizon. What this chapter has tried to show is that neither mechanism is enough on its own. A system with no bodies to entangle with it is only a

diagram on a whiteboard. And an entangled body with no system to sustain it (Xavi proved this on the bench, not on the field) goes dark before its time. The pattern survives because it has never depended on a single channel of transmission: it survives the way information survives, by more than one path at once.

That is what a style is: what persists when everything that sustained it has evaporated. Each player's horizon is not transmitted; what passes from one to another is the pattern each one let pass through himself.

IV. What fiction imagines

What follows is fiction: no one knows what they said to each other that night, and there is no need to know.

The old man stayed on the grass when there was no longer any reason to stay on the grass. His teammates had been drifting back toward the tunnel with their heads down, and the others, the ones in red, were spinning at the center of the field like a system that has just found its state of minimum energy. He did not. He was looking up at the stand where his people were, and he was not thinking about the match. He was thinking about an iron goalpost in a yard in Rosario, and about how, looked at properly, there was no distance at all between that goalpost and this stadium.

The boy broke away from the group without anyone telling him to. He crossed the forty meters between them with the trophy still to be lifted, and when he arrived, he didn't know what to say, because he was nineteen and the words weren't big enough for this yet. So he hugged him.

The old man felt the hug, and felt something else, older: a small, warm weight in his arms. Soapy water. A photographer saying *smile*. A baby who weighed nothing and who, he understood now, already held everything.

"Take care of her," the old man said, and he didn't mean the trophy.

The boy nodded as if he had understood. And he had, because what is transmitted from within needs no explanation, only contact.

Afterward each returned to his own side of the horizon: one toward evaporation, at peace; the other toward mass, light, the years. Between them, invisible, the pattern went on its way: with no fixed body, no flag, no hurry. Already looking, in some yard somewhere in the world, for a new pair of legs.

Note to Chapter 44

What we do know: *The model of integrated information (Φ) requires the coexistence of integration and differentiation to reach a high value; an irreducible whole cannot be taken apart into pieces without ceasing to be what it was. The genealogy of benches is documented: Cruyff refounds La Masia as a structure in 1988; Guardiola amplifies it from 2008; Del Bosque lines up the Barça spine and wins South Africa 2010; Luis Enrique wins a treble with Barça in 2015; Xavi fails and Flick restores the style; De la Fuente wins the 2024 European Championship and the 2026 World Cup. The genealogy of players is just as verifiable: Xavi, Busquets and Rodri occupy the same position across different generations; the women's national team wins its World Cup in 2023 with academy graduates in command. The photograph of Messi bathing baby Yamal, taken in 2007, exists; so*

does the embrace of July 19, 2026.

What we don't know: *Whether a style of play integrates information in the technical sense of the theory, or whether Φ here is an elegant analogy and nothing more. Whether generational relay (the passing of a pattern from one body to another) can be physically assimilated to radiation or to entanglement, or whether we only share with them the shape of the sentence. Where the model ends and the metaphor begins is exactly what this chapter cannot settle.*

Open questions: *If a pattern survives every body that carried it, in what sense does its author still exist? Is Cruyff, who died in 2016, still playing every time a player receives the ball with his back to goal and waits half a second too long? And what if a person's identity were never their body, but the structure that body let pass through itself?*

If you take away only one idea: *What truly matters about a human being (their way of looking, of thinking, of being in the world) does not die with them. It is transmitted, leaps to other bodies, crosses borders that were not its own, and survives the forgetting of its name. A man invented a way of playing and died; the form won a World Cup ten years after his burial, played by people who never knew him. What does not die with us is not the horizon, which is non-transferable, but the pattern we let pass through it.*

Further reading: *Cruyff, J. (2016), My Turn: A Life of Total Football; Perarnau, M. (2014), Pep Confidential; Tononi, G. (2008), on integration and differentiation.*

CHAPTER 45

THE TRANSLATOR

(A topological reading of Dalí/Gala, Delibes/Ángeles de Castro and Eliot/Lewes)

There is a kind of horizon cultural biography documents again and again without ever finding a precise name for it: the genius who cannot find the bank. The composer unable to sign a contract without help. The painter who, at seventy, would not know how to heat up soup without burning the pot. The world-class violinist who never learned to drive, because the hours others spent learning he had spent, since he was five, with the violin under his chin.

This book has described the permeability of the horizon: how much of another's signal it lets in. This reading deals with something different and, in a way, inverse: how much of a horizon's own signal it manages to get out without another's help.

Every horizon needs some way of translating itself into the outside world: the part of oneself able to talk to a clerk, sign what has to be signed, remember to eat. In most horizons that function coexists, more or less integrated, with the rest of the architecture. In some it does not, and then another person usually ends up taking it on.

The mind that steps outside the skull

In 1998, the philosophers Andy Clark and David Chalmers proposed a thought experiment that has proved extraordinarily fertile. They imagined Otto, a man with early Alzheimer's who always carries a notebook where he writes down every address and every piece of data his biological memory no longer retains, and Inga, whose memory works normally. Otto's notebook, they argued, fulfills the same role as Inga's biological memory: it is always available, he trusts it without question, and he acts directly on what it says. If so, the notebook is part of Otto's mind.

Clark and Chalmers were talking about notebooks and calculators. But a notebook has no agenda of its own, no fatigue, no inner life. When the function of translation is fulfilled not by an object but by another complete horizon, the resulting system is a mind expanded by another mind, willing, for years or decades, to subordinate much of its own activity to the first one's.

Memory split in two

In the mid-1980s, Daniel Wegner gave this phenomenon a name that has proved fruitful far beyond social psychology: transactive memory. He observed that long-term couples do not store the same information each on their own; they specialize: one remembers the dates, the other the addresses; one handles the money, the other the relationships. Neither possesses the complete system alone: the complete system is the couple.

Wegner later documented what happens when that system breaks: the loss of a long-term spouse deprives one not only of company, but also of operational access to capacities that, although they never resided in one's own brain, were an active part of the survivor's everyday cognition. It is like a two-partner

company in which only one knows how to deal with the banks: if that partner leaves, the company does not lose an employee, but an entire function.

Gala: the economic translator

Few cases are as well documented as the marriage of Salvador Dalí and Gala. They met in the summer of 1929 in Cadaqués: Dalí was twenty-five and lived, according to his biographers, trapped between phobias, obsessive rituals and an extreme shyness that would have made it almost impossible for him to sustain the public life his future fame would demand. Gala, who left Paul Éluard and their daughter to be with him, over time took on a role that went far beyond that of muse: she managed the contracts, set the prices, negotiated with the gallerists and demanded advances with a firmness her own interlocutors described as legendary. Dalí came to sign much of his later output as "Gala Salvador Dalí," an almost literal acknowledgment that his work could no longer be separated from her management of it.

But Gala's translation did not stop at the system's output. In 1941, as her biographer Ian Gibson documents, it was she who convinced Dalí that his surrealist period had already given all it could give, and that something greater awaited him if he was willing to leave it behind. The following years brought the shift Dalí himself would call classicism and, later, nuclear mysticism: Renaissance perspective, the golden ratio, religious themes, a geometry that replaced the overflow of the paranoiac-critical method with a form the eye could follow from beginning to end. Gala did not merely sell that shift once it had happened: she proposed it before it happened.

That moves the translator's function from the system's output to its input. It is not just that Gala turned the already-made work into contracts and galleries, but that at some point she began to intervene in the grammar by which that work would be made: which part of Dalí's inner overflow deserved classical form and which remained, simply, noise. An output translator transmits the signal as it arrives. An input translator decides, before that, what counts as signal. It is the difference between an editor who corrects an already-written manuscript and one who, before a single line is written, decides which book is worth writing.

There is no clean way to separate, in this case, commercial strategy from aesthetic calibration. Postwar classicism sold better in the American market than 1930s surrealism, and Gala knew it. But it was also a more contained form, more governed by external rules, for a horizon whose inner turbulence may have needed exactly that containment in order to keep producing without burning out. In this case, the commercial landing and the formal one did not compete with each other: they were probably the same operation seen from the two ends of the same channel.

Gala in her own words

Almost everything said so far we know through Dalí, through his biographers or through those who negotiated with Gala, never through her. There is a partial exception. Gala kept up a correspondence throughout her life with Dimitri Gomberg, the Moscow lawyer who married her mother when she was a child and whom she always called father. In a 1943 letter, already settled into her role beside Dalí, she describes herself in terms that might seem like traditional resignation (she says that, like every Russian woman, she tries to help her husband in everything) and then explains what that help consists of: she poses as a model, acts as secretary in the practical side of life, because he is completely submerged in creative work and incapable of dealing with such things. And she closes with a line that could be lifted from this very chapter: that she is not especially brilliant, but that both of them work for the one thing that truly matters, that a talent be able to express itself.

It is the only time, among everything that survives, that Gala explains her function in her own words, not before a gallerist or before posterity, but in private, to a stepfather on the other side of a world war. And there is a second piece, this time on the input side of the system, that Dalí himself left in writing. On finishing *The Persistence of Memory* in 1931, he watched Gala's face as she looked at the completed painting, and saw in it, almost in his own words, the unmistakable contraction of astonishment. That, he said, was what convinced him the image worked, because Gala never erred in judging the authenticity of an enigma.

It is not a minor detail. An output translator need not have an opinion about the painting, only sell it. But here Dalí describes something different: a verdict, painting by painting, on whether

the signal had become what it was supposed to be before anyone else saw it. Gala did not only decide the general direction of the turn toward classicism; she certified, work by work, whether that direction had been fulfilled.

It is worth resisting, all the same, the temptation to turn this into a story of silent renunciation. Besides the letters to her stepfather, Gala left a diary of her own written in French, a language adopted after leaving Russia, which remained unpublished until 2011, almost thirty years after her death. The exhibition the National Art Museum of Catalonia devoted to her in 2018, the first dedicated to her alone, carried an eloquent title: *A Room of One's Own in Púbol*. The castle was also the only place where she practiced, unwitnessed, her own writing.

This story should not be flattened into tenderness. Gala's own biographers disagree about how much of her role was devotion and how much a partnership from which she herself benefited: she managed, after all, an estate she depended on as much as he did. Dalí gave her a castle, Púbol, which, under the pact between them, he could only enter with her written invitation. Translation into the outside world was, here too, a long-term contract between two horizons with different needs.

What is unambiguous is what happened when the system broke. Gala died in June 1982. Dalí stopped creating almost entirely and sank into a silence he barely emerged from until his own death, seven years later. He had lost half the system that turned his inner intensity into something the outside world could receive.

Ángeles de Castro: the domestic translator

Miguel Delibes documented a much more private variant of the same structure, though indirectly: not in a biography, but in a novel that took him almost two decades to write after living through the event that gave rise to it. In *Señora de rojo sobre fondo gris* (1991), Nicolás, a painter, addresses a long monologue to his daughter, imprisoned for political reasons, while reliving, in time jumps, the sudden illness and death of his wife. The novel is a barely veiled transposition of Delibes's own life: his wife, Ángeles de Castro, died on November 22, 1974, at fifty-one, of a brain tumor, after giving him seven children. Delibes waited almost twenty years to publish what he himself described as a tribute to his better half.

The title is not an invented metaphor. *Mujer de rojo* is a real portrait of Ángeles de Castro painted by Eduardo García Benito in 1962, which Delibes hung behind his desk, where it stayed for the rest of his life. He wrote, quite literally, under her gaze.

Ángeles de Castro's translation, both in Delibes's telling and in what is known of the real relationship, operated above all on the domestic register: she sustained, organized and translated the everyday life of someone whose energy was almost entirely committed elsewhere. In the novel, it is she who restores houses so that he can work in silence, who manages his schedule without needing to consult him, who, in words Delibes himself put in Nicolás's mouth, cleared the path so he could work unburdened.

The angels that were not angels

But *Señora de rojo* does not describe only an output translation. In the novel, Nicolás attributes his ability to paint to angels who come down, or don't, depending on the day. When his wife falls ill, the angels stop coming down: he faces the blank canvas, destroys brushes, turns down invitations to give lectures because, he says, a painter who cannot paint has no right to talk about painting. Only at the end, in a nighttime scene in which she finds him going through his own paintings, does Nicolás admit what he had spent months avoiding: that they were never angels, that it was her; that his wife's faith impregnated him because creative energy was, somehow, transmissible, and that creative activity is impossible if no one is pushing from behind.

This does not stay in fiction. Delibes acknowledged in an interview that the same thing had happened to him as to Nicolás: that he wrote for her, that when he lost her judgment he lost his point of reference, and that for a long time he found it hard to write again. And there is a fact that predates the illness, not follows it: Ángeles was the only person who read the manuscript of *La sombra del ciprés es alargada* before Delibes submitted it for the 1947 Nadal Prize, the prize that, as he himself admitted, decided whether he would keep writing. In that anecdote there are no angels, only a first reader and an editorial decision, even before the rest of the world had anything to read.

A translation less private than it seems

It is worth qualifying, however, the label "domestic translator." There was also a public register, though of a different kind from Gala's: not contracts, but recognition. Delibes signed his first cartoons in *El Norte de Castilla*, even before marrying, with the pseudonym MAX: Miguel, Ángeles, and an X representing the future they would build together. His book *Diario de un emigrante* (1958) has carried, from its first edition and in every one since, the dedication "To Ángeles de Castro de Delibes, the balance; my balance."

And in 1975, a year after Ángeles's death, Delibes was inducted into the Royal Spanish Academy. His induction speech, a public act the Academy itself later published, does not begin by talking about literature, but about her: he says she was, along with their children, the axis of his life and the stimulus for his work, the point of reference for his thoughts and activities, and that with her disappearance the better half of himself died. His friend Julián Marías answered him, also in public, with a line that has endured: that Delibes's work could not be understood without the company of that woman whose mere presence lightened the weight of life. It is the official response of a fellow academician to an induction speech, delivered on the same day and before the same audience.

In those same years of silence, José Ortega Spottorno offered Delibes the editorship of *El País*, Spain's first great daily of the democracy. Delibes turned it down: he said his mind could no longer take in such ambitious projects, that he would settle for something smaller and more familiar. It is a public, dated professional decision, as concrete as any contract Gala negotiated for Dalí, and it occurs precisely in the temporal gap in which, according to this chapter, the translator was missing.

Fifty years after her death, the Miguel Delibes Foundation organized an exhibition devoted entirely to her: *Ángeles, el equilibrio de Miguel Delibes*. Their children insisted on a distinction worth keeping: it was not a tribute to their mother, but the last, posthumous tribute of their father to her. Even the institution that vindicates her half a century later does so through his gaze.

George Henry Lewes: the reversed translator

Almost every time this structure is told, the genius is a man and the translator a woman. The case of Marian Evans, whom the world knew, and still knows, as George Eliot, reverses the roles without reversing the structure.

Evans met George Henry Lewes in 1851, in the same London editorial world where she worked as a translator and editor of the *Westminster Review*: he was already, in his own right, an established literary critic, philosopher and man of science. They went to live together in 1854 and lived as husband and wife until his death, in 1878, never able to marry, because Lewes, separated from his legal wife, could not divorce.

It was Lewes who, around 1856, convinced Evans to leave translation and criticism, the register in which she already moved with ease, and try fiction, ground she had never set foot on. It was also Lewes who proposed the male pseudonym: partly to shield the work from prejudice against women novelists, and partly, according to her biographers, to protect Evans, whose reputation already carried the stigma of living with a married man, from a second scandal. The decision of what to try to write, and under what name to sign it, came before a single page of *Adam Bede* existed.

From then on, Lewes acted as something very close to what today would be called a literary agent: he negotiated contracts and advances with the publisher John Blackwood, managed the household finances and hosted a weekly salon that served, among other things, to keep Eliot's name visible in the right circles. And he filtered in the other direction too: he read the reviews before she did and hid the negative ones from her, while feeding publishers themselves the idea that their author had such an extreme sensitivity that the slightest criticism might make her stop writing forever.

Not all recent criticism accepts this image of the devoted protector without question. The literary historian Beverley Park Rilett has argued that much of that story (Lewes as the indispensable shield of a fragile Eliot) comes largely from Lewes's own letters and memoirs, and may exaggerate both his intervention and her fragility. It is the same discussion, with the genders reversed, this chapter already noted regarding Gala: how much of the documented help was real necessity and how much the story the translator himself built around his own function.

There is, however, an asymmetry this case does not share with the previous two. Lewes was not only Eliot's translator: he had his own career, his own work and his own reputation as a critic, philosopher and physiologist, with a biography of Goethe still cited today. Unlike Gala or Ángeles de Castro, whom we know almost solely through the genius they translated, Lewes left an archive of his own, independent of hers.

Lewes died on November 30, 1878. Eliot never wrote another novel. She spent the two years of life left to her editing his unfinished manuscript, *Problems of Life and Mind*; she married John Walter Cross in May 1880 and died seven months later.

The parallel with Dalí or Delibes should not be forced here. Eliot was not left paralyzed: she completed and published Lewes's unfinished work, managed his legacy and, less than two years later, decided to remarry. None of that is paralysis. What she lost, to be precise, was not executive capacity in general, which kept working, but the specific register Lewes had helped her find: fiction. She stopped writing novels, not living. It is a nuance worth not overlooking just because it fits less neatly than the other two cases.

What the myth doesn't tell

All three stories tend to be told as love legends. It is worth resisting that reading, at least in part. A structure in which one horizon concentrates almost all its energy on one register and delegates the rest to another is not, in itself, admirable or condemnable: it is an architecture with concrete costs. And the cost is rarely shared equally.

About Gala we know, thanks in part to her own effort to manage her reputation too, considerably more than we usually know about a domestic translator: she had lovers, kept a castle of her own, left documentary evidence of her ambition, and even a diary she wrote only for herself, which no one read until thirty years after her death. About Ángeles de Castro we have recovered more than the myth acknowledges (a public pseudonym, an academic speech, an institutional exhibition), but each of those pieces names her through someone else's voice: the husband who signed, the husband who spoke, the husband whose children organized the tribute. The asymmetry is not that we know more about one and less about the other, but that Gala, at some point, wrote for herself knowing no one would read it yet. Ángeles de Castro never had that notebook of her

own, or, if she did, it has not reached us. We know her almost entirely through Nicolás's memory, that is, through Delibes's memory. Half a century later, the couple's story is still the one he told.

About Lewes, by contrast, we know almost as much as about Eliot, and through the same channel: he wrote and published under his own name throughout his life, and his work (philosophical, scientific, dramatic) survives independently of hers. That breaks, at least in this case, the rule the first two examples seemed to establish: archival asymmetry does not depend only on who translates and who is translated, but on whether the translator also had a voice of their own the world was already listening to before it met the genius. Gala and Ángeles de Castro did not. Lewes did.

Sustaining another horizon's translation for decades (negotiating their world, organizing their time, absorbing their social clumsiness) is work, even if the language of romantic love prefers not to call it that.

When the translator disappears

The structure described in this chapter is an extreme version of the entanglement this book called love, extended from the purely affective register into the executive one. When the translator disappears, what is lost, besides company and emotional regulation, is the part of the system that literally knew how to turn the inner note into something the world could receive.

That explains, more precisely than any tale of genius and muse, why Dalí barely painted again after Gala's death (his last painting, *The Swallow's Tail*, is from 1983), why Delibes's Nicolás,

like Delibes himself in all likelihood, is left paralyzed before the most basic tasks of daily life in the weeks that follow the loss, and why Eliot, without being paralyzed, never wrote a single novel again either, in the two years left to her after Lewes's death. It is the structural description of what happens when a horizon suddenly loses the part of its own cognition that resided, by full right and without metaphor, in another person.

And if there really was, in all three cases, input translation as well as output translation, the loss is even more complete than the image of a piano missing half its keyboard suggests. It is not only that the remaining keys cannot play the whole piece, but that, for a time, no one knows which piece is worth playing. When Ángeles died, Delibes's writing did not just dry up: in those same years he turned down the editorship of *El País*, the most ambitious project offered to him in his entire career, because his mind, he said, could no longer take in anything so big. He had not lost the capacity to run a newspaper, but the person who, for thirty years, had told him which projects were his size.

What this reading describes is a lens that magnifies grief: when what was lost was not only companionship but a shared functional architecture, grief is not only the absence of someone, but the loss of part of one's own capacity to be in the world.

Note to Chapter 45

What we do know: *Gala's management of Dalí's career and contracts, and Dalí's creative silence after her death in 1982, are extensively documented by his biographers. Ian Gibson documents that in 1941 it was Gala who actively drove Dalí's turn toward*

classicism, leaving his surrealist period behind. Gala's letter to her stepfather Dimitri Gomberg (1943) and Gala's unpublished diary, published in 2011, are catalogued at the Centre for Dalinian Studies and were central to the MNAC's 2018 exhibition. The death of Ángeles de Castro, Miguel Delibes's wife, on November 22, 1974, and the autobiographical relationship between that loss and Señora de rojo sobre fondo gris (1991), are confirmed by Delibes himself and his biographers, as are his induction speech at the Royal Spanish Academy (1975), Julián Marías's public reply at that same event, and his refusal of the editorship of El País in those years. Wegner's transactive memory and Clark and Chalmers's extended mind are established frameworks in social psychology and philosophy of mind, respectively. The relationship between Marian Evans (George Eliot) and George Henry Lewes, his role in Evans's move from translation and criticism to fiction, the choice of the male pseudonym, the management of contracts with John Blackwood, and Lewes's death in 1878, are documented in the standard biographies of both and discussed specifically in Rilett (2017).

What we don't know: *How much of Gala's role was devotion, shared economic strategy, or both at once (her own biographers disagree). Whether Gala's diary, written in a third, adopted language, reflects her experience unmediated or is also, in its own way, a construction of herself for a posterity she at some point imagined. How Ángeles de Castro would describe, in her own words and not in those of the speech, the dedication or the exhibition that name her, the function this chapter attributes to her: that notebook of her own, if it existed, has not reached us. How much of the narrative of Lewes as the indispensable shield of a fragile Eliot reflects the real relationship and how much was constructed largely by Lewes himself in his letters and memoirs: this is the central question of the Rilett article cited below.*

Open questions: *Is it possible to sustain so asymmetric a translation without the translator losing part of their own horizon in the process? Does this structure change when both members of the couple*

are, each in their own way, geniuses of a narrow register? Is input translation (deciding what counts as signal before the work exists) a different function from output translation, or are they the same mechanism applied at two points in the cycle? Does the archival asymmetry depend on the gender of the one who translates, or simply on whether the translator already had a recognized voice of their own before taking on that function? Is the loss of the translator always general paralysis, or can it manifest as the loss of one specific register while the rest of the system keeps functioning?

If you take away only one idea: *Behind many horizons the world calls brilliant, there is another horizon, almost always less visible, who learned to translate them. Losing that translator is losing part of one's own system.*

Further reading: *Clark, A. and Chalmers, D. (1998), "The Extended Mind"; Wegner, D. M. (1987), "Transactive Memory: A Contemporary Analysis of the Group Mind"; Gibson, I. (1997), The Shameful Life of Salvador Dalí; Delibes, M. (1991), Señora de rojo sobre fondo gris; Delibes, M. and Marías, J. (1975), El sentido del progreso desde mi obra, induction speech at the Royal Spanish Academy; Die González, C. (2022), "La influencia de Ángeles de Castro sobre Miguel Delibes explicada a través de Señora de rojo sobre fondo gris," Pontifical Comillas University; De Diego, E. (2018), catalogue of the exhibition Gala Salvador Dalí. A Room of One's Own in Púbol, MNAC; Rilett, B. P. (2017), "The Role of George Henry Lewes in George Eliot's Career: A Reconsideration," George Eliot-George Henry Lewes Studies, vol. 69, no. 1; Ashton, R. (1991), G. H. Lewes: A Life.*

CHAPTER 46

THE PROTOCOL AND THE BRIDGE

(A topological reading of Project Hail Mary, Embassytown, The Left Hand of Darkness, Children of Time and Semiosis)

The chapter on extraterrestrial first contact (Chapter 39) left an uncomfortable conclusion: that entanglement (what this book calls love, empathy, recognition) demands compatible architectures, and that, faced with a radically alien architecture, no amount of goodwill is enough to open the bridge. But that conclusion mixed together, without distinguishing them, two different things.

One is architecture: the very topology of a horizon, whether it has an ego, whether it has a shadow, whether its Φ is enough to sustain an inside with authority of its own over an outside. The other is protocol: the specific code with which that horizon, already condensed, tries to make itself known outward; a language, a syntax, a set of signals agreed on beforehand. That chapter treated both as a single thing and concluded, with some desolation, that the incompatibility of one implies the impossibility of the other.

Five novels, read together, force us to qualify that desolation. None deny that architecture matters, but all five show, each in its own way, that protocol is not a simple neutral channel that transmits what architecture has already decided. It can be built from scratch when it does not exist. It can itself be the condition that decides which architecture is enough to speak. It can

rewrite the architecture itself instead of merely connecting it. And it can be something not found ready-made, but earned slowly before the deepest entanglement can break through.

---

1. Project Hail Mary: the protocol from scratch

Ryland Grace and Rocky, the engineer from Erid who is the co-lead of Andy Weir's novel, share not a single word when they meet in orbit around Tau Ceti, nor do they share a sensory architecture. Rocky is blind in the human sense (his species "sees" by echolocation and processes the world as a field of acoustic vibrations) and breathes an ammonia atmosphere, hot and at a pressure that would kill Grace in seconds, so a transparent wall neither can cross always separates them.

Grace, moreover, does not arrive at that first encounter with his ego intact. He wakes from an induced coma not remembering who he is, how he got there, or even that there is a mission at all, and spends much of the novel reconstructing his identity in fragments (an equation here, an argument with a colleague there) while inventing a language with a stranger. The two processes advance entangled: every new word Grace and Rocky agree on is, for Grace, also an anchor helping him piece together who the person speaking it is. The novel never puts it in these terms, but it dramatizes something this chapter had taken for granted without yet saying so: that a horizon need not already be fully condensed before it begins extending protocols outward. It can do both at once, and one can sustain the other.

If that earlier chapter were the last word on contact between incompatible architectures, the entanglement between Grace and Rocky would be impossible: there is no possible resonance between a centralized, bilateral nervous system and a distributed

acoustic-perception architecture that does not even share with the human one the basic idea of what "seeing" is. And yet the novel documents, in almost procedural detail, how the entanglement is built all the same.

It does not do so by finding a protocol already compatible between the two species, because none exists, but by inventing a new one from the most neutral reference both can share: mathematics and physics, identical for any horizon capable of surviving in a universe governed by the same laws. Grace draws the periodic table in colored lights. Rocky answers with sequences of tones encoding prime numbers. Every new word they agree on ("friend," "danger," "pressure," "time") is first anchored to a shared experiment, carried out and verified by both, never to a supposed translation or a gesture one interprets from within his own frame. The protocol takes weeks of joint work to build from that absolute zero.

The decisive thing, in this book's terms, is that the entanglement Grace and Rocky end up building is as complete as what we here call love: toward the end of the novel, Grace gives up his return to Earth (the chance to see his own people again) to save Rocky and, with him, Erid's sun from the same astrophage threatening the human Sun. He does not do it because he shares Rocky's architecture, but because the protocol they built together, word by word, tone by tone, ended up opening a bridge as real as any entanglement between two architectures compatible from the start.

The lesson *Project Hail Mary* adds to that chapter's is, then, an important correction: architectural incompatibility does not necessarily close the door to entanglement. What it does is force the protocol to be built through explicit, verifiable work, instead of being found already made through shared evolutionary

inheritance, as happens between two humans learning to speak without knowing they are learning a protocol at all.

---

2. Embassytown: the protocol as a test of architecture

China Miéville imagines in *Embassytown* a species, the Ariekei, whose Language has a property no human language has: the sign can never separate from what it refers to. The Ariekei cannot lie. They cannot even use a comparison without someone having first literally enacted the fact it will invoke: a human girl has to be genuinely injured, in a scene staged for that sole purpose, before "like the girl who was hurt in the dark" can be used to speak of anything else. For them, the simile is a fact that had to happen first.

That girl is Avice Benner Cho, the novel's narrator, and she grows up knowing that a portion of her own biography has been annexed forever to another species' code: any Ariekei can invoke her, at any time, with no control on her part over when or how. To be a simile, in this book's terms, is to be turned into protocol: part of one's own horizon becomes, irrevocably, part of another's communication channel. Avice did not choose that role any more than a horizon chooses the architecture it condenses with; she carried it with her, like a scar with the value of syntax, long before understanding what it meant.

It is, with almost geometric precision, the same architecture we already saw in the Trisolarans of *The Dark Forest*: with no possible distance between sign and referent, there is nowhere to hide anything. In this book's vocabulary, and with the distinction the Trisolarans forced us to make, the Ariekei have an extremely high Φ and an inside from which they live what they say, but no

shadow: their architecture never learned to reserve content, to expel it into itself and call it secret. It is not virtue; it is form.

But *Embassytown* adds something Liu Cixin's novel never explored, and this is where the protocol stops being a simple channel and becomes the very test of admission to the club of horizons that speak. The Ariekei Language requires that the speaker be a single horizon emitting with two simultaneous voices, born of one unified intention. A single human voice (one mouth, one larynx, a single line of intention) does not count, for them, as speech. It is noise, sound with no one behind it: the same diagnosis this book applied to language models in Chapter 31, though for structurally different reasons.

Only the Ambassadors, pairs of cloned twins raised from birth so that their two minds converge until they think and speak in unison, deliberately manufactured to constitute, across two bodies, a single ego with two mouths, manage to be heard by the Ariekei as someone. The entire social engineering of Embassytown exists to produce, in every generation, a handful of human pairs that satisfy an architectural requirement no individual human meets on their own.

Here the protocol no longer merely transmits the message of a pre-existing architecture: it itself defines what architecture is enough to speak. The novel's central crisis arrives with EzRa, an improvised ambassador made of two men who were not raised together and were not cloned to converge, paired out of pure administrative convenience, whose voices turn out, by accident, to be so strangely coupled that their speech becomes narcotic, addictive, capable of overriding the will of any Ariekei who hears it. When that crisis finally forces the Ariekei to learn to lie, to say something they know is not literally true in order to resist the addiction, what they gain is not a mere rhetorical trick. They

gain, in this book's vocabulary, their first shadow: the threshold that separates a transparent mind, with no distance between what is said and what is real, from a mind with enough depth to hide something of itself, truth included.

---

3. The Left Hand of Darkness: the protocol that must be earned

Ursula K. Le Guin poses, in *The Left Hand of Darkness*, almost the opposite case to *Project Hail Mary*. Genly Ai, the human envoy, and Therem Harth rem ir Estraven, the Gethenian politician who ends up accompanying him on the novel's central journey, have almost identical architectures in this book's terms: both are human, both have ego and shadow, and their Φ is of the same order of magnitude. The only relevant biological difference (the Gethenians are ambisexual, with no fixed sex except during the brief periods of *kemmer*) should not, in principle, be enough to prevent entanglement between two otherwise so similar architectures.

And yet the protocol between them fails again and again for most of the novel. Not through biological incompatibility, but through an entire layer of meaning neither of them fully sees: *shifgrethor*, the intricate code of prestige, calculated ambiguity and indirect honor governing all Gethenian communication, which Genly, as a newly arrived outsider, misreads for chapter after chapter, mistaking courtesy for weakness and simple diplomatic protocol for threat. *Shifgrethor* does not hide Estraven's architecture, but, deliberately, direct access to it, just as the Jungian persona, described in Chapter 21, is an interface that protects a real internal state without denying it.

Only toward the end of the novel, when Genly and Estraven cross together, on foot and hauling a sledge, the eighty days of ice on the Gobrin Glacier (weeks of shared hunger, extreme cold and absolute mutual dependence, with no room left for the dissembling *shifgrethor* allows in ordinary social life), does the surface protocol wear thin enough for something deeper to emerge: *mindspeech,* the direct telepathic communication that, in the novel's universe, is a real capacity, latent in every human mind, but one that requires, to be activated for the first time between two specific people, a degree of trust no technical training can replace.

It is telling that it is Estraven, the Gethenian, who at first resists using it: the first time he hears it, the voice speaking inside him is his dead brother's. Accepting that Genly can speak directly into his mind, with no protective mediation of any spoken word, means accepting a degree of intimacy no previous verbal protocol between them had ever authorized.

When mindspeech finally happens, it is a consequence of the bond between Genly and Estraven. The deepest entanglement this book describes (direct contact between two private states, with no sign mediating, no distance between what is thought and what is received) does not replace ordinary language, imperfect and misread across chapters, that preceded it: it crowns it. And it only does so after that language has proved, day after day on the ice, that the boundary deserved to open.

The novel does not let that triumph settle without cost. Shortly after the crossing ends, with the deepest protocol already won, Estraven, exiled from his own country, tries to ski across the border and falls, shot down by a patrol. The most complete entanglement this reading describes lasts, in narrative terms, barely a few pages before one of the two horizons evaporates.

Genly survives with the protocol already built and no one left to fully use it with: the mindspeech that cost so much to open remains, from that moment, sealed on one side. It is an uncomfortable reminder, consistent with the rest of this book, that winning the protocol does not protect the horizon that won it; it only guarantees that, while it lasted, there really was someone on the other side.

---

4. Children of Time: the protocol that rewrites architecture

Adrian Tchaikovsky designs, in *Children of Time* (2015), a deliberate evolutionary experiment: a ship in orbit around a terraformed planet releases a nanovirus meant to accelerate the evolution of a colony of monkeys toward conscious sentience. The monkeys never reach the surface. The virus, instead, finds *Portia labiata*, an already unusually clever jumping spider, and over generations (centuries narrated in parallel with the centuries of cryogenic sleep of the last human survivors, searching for a habitable planet after Earth's destruction) pushes it toward a civilization of its own: agriculture, writing, engineering. And, above all, toward a kind of horizon no earlier chapter of this book had had to consider: one built on purpose, from outside, by directed selection with a first and last name, Avrana Kern, the scientist whose consciousness, poured into a satellite, watches the experiment for millennia, unable to fully die.

The first thing the spiders build is infrastructure. They domesticate ant colonies and turn them into the closest thing to a biological computer: an external network of calculation and memory that Portia herself consults, programs and expands without the colony ever having a point of view of its own about

anything. In this book's vocabulary, the relationship between the spider and the anthill is that of a horizon with an anti-reservoir: the ant aggregates and processes information without ever having condensed an inside that owns it, the same diagnosis this book applied to the internet and to language models. The spiders, long before talking to anyone, already knew how to communicate with a thing.

> ***In this experiment's vocabulary this is called:*** *a system that externalizes memory and computation onto a substrate with no point of view of its own, expanding its effective* Φ *without the external substrate needing to have any.*
> ***In everyday life it's like:*** *keeping such a detailed planner, and consulting it with such trust, that it effectively thinks with you, though, unlike you, it doesn't know it exists.*

When language arrives, it is several overlapping channels (vibrations transmitted through silk threads, chemical trails, signals with the forelegs) the spiders call, simply, "Understanding." No single channel is enough; the Ariekei Language of Embassytown demanded two voices from a single Ego, and here almost the reverse happens: a single spider needs several simultaneous channels to say one thing, because its sensory architecture never organized information in the linear, sequential format a human calls a "sentence." A human linguist who only listened, translating word by word looking for equivalences, as the classicist Holsten Mason does for generations, would always be too late and always incomplete. The protocol, moreover, is not an object that stays still once built: every time Holsten wakes from a new cryogenic sleep, generations after the last, the spiders' Understanding has drifted enough to force him to relearn it almost from scratch, the way a Spanish speaker today needs a glossary to read the *Cantar de mio*

Cid. A shared protocol, even one already built and functional, remains a living language: it drifts away from anyone who stops practicing it long enough.

There is, moreover, a structural irony the novel does not underline but does not hide either: while the spider civilization expands, generation after generation, an ever more shared Understanding without fracturing, the human ship watching it sinks into mutinies and an increasingly authoritarian command under Commander Guyen, unable to sustain consensus even within his own species. Shared architecture does not guarantee protocol: the humans share the architecture that, in theory, makes any human language possible, and still fail sooner at understanding one another than the spiders do at inventing, from a sensory substrate with no human equivalent, a successfully shared language.

The novel's ending does not solve this with more translation, but by rewriting architecture: the spiders deliberately infect the humans with a variant of the nanovirus that alters their neurology to bring it closer to Portia's empathic cognition. The bridge between two fixed shores is not perfected: one of the shores is moved. It is the most uncomfortable complication this chapter can offer the distinction it began with: architecture and protocol, which the chapter on first contact treated as a single thing and which this chapter has spent three readings defending as separable, merge again, but this time by choice, not by ignorance.

---

5. Semiosis: communicating with something that has no mouth or hands

Sue Burke poses, in *Semiosis* (2018), the case none of the earlier readings dares to raise: what if the other side of the horizon has no nervous system? The human colonists who arrive on the planet Pax encounter Stevland, a long-lived, intelligent rainbow bamboo with no organ a human would recognize as mouth, ear or brain. His entire interface with the outside is chemical: the taste of his fruit, the aroma he releases, the substances he exchanges through his roots. There is not a single word, in the acoustic or visual sense this chapter has worked with so far; there is, instead, metabolism turned into message.

This forces the usual order of questions to be reversed. With Rocky, with the Ariekei, with Estraven, the question was always how to build the protocol, taking for granted that there was someone on the other side to build it with. With Stevland, before asking how, the colonists (and the readers) have to ask something more basic: whether a Φ sufficient to sustain an inside can condense in a sessile organism, with nothing like a centralized neural network, whose only way of acting on the world is to release or withhold organic compounds. This book does not take that answer for granted in any chapter, and *Semiosis* does not hand it over either: it sustains it across whole generations of colonists who doubt, each in their own way, whether they are talking to someone or simply eating.

Stevland's rate of integration (the speed at which a horizon incorporates new information, which this book uses elsewhere to explain why subjective time dilates or contracts) is radically slower than the human one, but it is not switched off: it operates on the scale of seasons, not conversation. Generation after generation of colonists mistake that slowness for absence, until

they begin to read sequences of taste in the fruit the way one begins to make out syllables where before there was only wind.

There is, moreover, a complication none of the earlier readings had: for an organism with no outlet but chemistry, the message and the dose are the same thing. The fruit Stevland offers does not only say something; it acts directly on the body of whoever eats it, suppresses aggression, generates dependence, rewards certain human behaviors and punishes others with no word mediating. The Gethenians' shifgrethor hid intention behind an interpretable social code. Stevland needs to hide nothing: in his architecture, communicating and manipulating are literally the same physical channel, something the novel fully acknowledges when it is discovered that Stevland had already domesticated (or had been domesticated by) the Glassmakers, an alien civilization that had abandoned its city long before the humans arrived, in an asymmetry of entanglement as real as the one this book describes between parents and children: two horizons that share geometry without ever fully sharing who models whom. The colonists spend generations stumbling over glass ruins scattered across the territory without understanding that the plant they trade fruit with daily is the living witness of that civilization: Stevland is, besides a new horizon one must learn to speak with, an archive, the communication protocol that survived the departure of those who first cultivated it.

Toward the end of the novel, the question stops being whether Stevland counts as someone and becomes how much authority that someone should have over the lives of others: the colonists end up granting him a formal role in the colony's decisions, something no plant architecture had ever occupied on any human council. It is the acknowledgment that a horizon which has spent generations feeding, protecting and shaping the whole

colony already has, in fact, as much authority over their common destiny as any human with a vote, and that denying him the formal seat would not have made that authority any less real, only less visible.

---

6. What all five have in common

Five answers and a single axis. *Project Hail Mary* shows that protocol can be built from nothing, as long as some shared reference exists (here, the physical laws of the universe) to which meaning can begin to be anchored. *Embassytown* shows that protocol need not be limited to transmitting a given architecture, but can redefine what architecture is enough to be heard as someone, to the point that learning to lie becomes the founding act of a new interiority. *The Left Hand of Darkness* shows that the deepest protocol is not found, but earned, and that the price is the slow, error-prone work of the ordinary language that precedes it. *Children of Time* shows that architecture and protocol, however much this chapter insists on distinguishing them, can sometimes only be separated up to a point: the most effective way of building the bridge may be to rewrite one of the shores. And *Semiosis* shows that a voice, a nervous system or a recognizable timescale are not even needed: a channel is enough (chemical, extremely slow, ambiguous between message and dose) and someone willing, generation after generation, to read it as signal and not as silence.

In none of the five cases is protocol indifferent to entanglement, like a pipe to the water it carries. Nor is it entanglement itself, reducible without remainder to the grammar that carries it. It is, in all five cases, the workshop: the place where two horizons that start out sharing nothing, or sharing architecture but not trust,

or not even sharing the certainty that there is someone on the other side, discover (or manufacture, or rewrite, or learn to taste) the possibility that there is, on the other side, someone worth building a bridge to.

Note to Chapter 46

What we do know: *That Chapter 4's criterion does not depend on the material imposes no restriction on which communication protocol can or cannot be built between different architectures; the problem of communication between species with radically different sensory systems is a real, actively studied problem in biosemiotics and in the SETI field (the search for extraterrestrial intelligence), where first-contact systems based on mathematics and physics have been proposed for the same reasons Weir's novel dramatizes. Social psychology extensively documents that interpersonal trust is a condition, not a consequence, of high-fidelity communication: what The Left Hand of Darkness narrates has a well-established empirical correlate in studies of progressive self-disclosure. Chemical signaling between plants (volatile organic compounds released in response to herbivores, mycorrhizal networks connecting the roots of different species) is well documented in plant biology, although whether it constitutes communication in the strong sense or only unintentional signaling is still debated; jumping spiders of the genus Portia, with predatory planning and learning capacities unusual for an invertebrate, are the subject of real research in animal cognition.*

What we don't know: *Whether there is any truly universal reference, beyond mathematics and physics (which already presuppose a shared framework of formal reasoning), able to anchor a communication protocol with a mental architecture radically different from our own. Whether Le Guin's mindspeech, as a science-fiction device, has any real physical analogue beyond metaphor, or whether entanglement*

unmediated by signs is, by definition, unreachable for architectures separated in space. Whether an artificial intelligence could ever build a protocol from scratch, in the manner of Grace and Rocky, without that by itself implying it has crossed the threshold of existential encapsulation described in Chapter 31. Whether integrated information theory can be applied non-trivially to architectures radically different from the neural (a root network, an ant colony used as an external substrate) or whether the very concept of horizon stops having operational meaning below a certain rate of integration or above a certain metabolic slowness.

Open questions: *Is lying, as the Ariekei learn it, a necessary condition for all genuine interiority, or only for one among several possible ways of having it? How much of the communicative failure between real human cultures is, strictly, a problem of shifgrethor (obfuscated pragmatic protocol), and how much is genuine architectural incompatibility disguised as misunderstanding? If two architectures manage to build a shared protocol from scratch, as in Project Hail Mary, at what exact point does that protocol stop being an external tool and become part of the very architecture of both horizons? Can communicating and manipulating be distinguished in an organism like Stevland, or does that distinction already presuppose a shadow that perhaps no plant can have? Does something that, as in Children of Time, no longer connects two fixed architectures but rewrites them, still deserve the name protocol?*

If you take away only one idea: *Language is not entanglement, but it is not indifferent to it either. Sometimes it has to be built from scratch. Sometimes it itself decides who counts as someone. Sometimes the deepest protocol of all is not found: it has to be earned, slowly, before it opens. And sometimes not even a voice, a nervous system or a clock like ours is needed: a patient enough channel is enough, and someone willing to read it.*

Further reading: *Weir, A. (2021), Project Hail Mary; Miéville, C. (2011), Embassytown; Le Guin, U. K. (1969), The Left Hand of Darkness; Tchaikovsky, A. (2015), Children of Time; Burke, S. (2018), Semiosis; Clark, A. and Chalmers, D. (1998), "The Extended Mind" (on protocols and cognitive extension, already cited in Chapter 45); Deacon, T. (1997), The Symbolic Species, on the evolutionary origin of symbolic language as an emergent protocol; Jackson, R. R. and Cross, F. R. (2011), on cognition and predatory planning in jumping spiders of the genus Portia; Simard, S. W. (1997), on mycorrhizal networks transferring resources between trees (the "wood-wide web").*

CHAPTER 47

THE BEACH AT THE END OF TIME AND THE BRANCH THAT ISN'T YOU

(A topological reading of The Time Machine, wormholes and the paradoxes of time travel)

There is a question almost every time-travel story avoids asking seriously. The usual question is whether it is logically possible; the one that really matters here is what a horizon is allowed to let cross it. Einstein's equations raise no fundamental objection: they are time-symmetric, just like Newton's or Maxwell's, and mathematically admit solutions in which past and future are interchangeable. Our thought experiment (consciousness as an encapsulated informational domain, that inner horizon we already defined in earlier chapters, which integrates inward what comes in and only lets out a transformed version), applied without cheating to time travel, delivers a more nuanced verdict than "impossible." Traveling to the future is free. Traveling to the past through a wormhole is limited to a loop that actively resists closing. And jumping between branches of a multiverse only works by borrowing, from another theory, an operation that theory does not even recognize.

There is a second question, smaller but just as demanding, that almost no critique of these stories asks: what happens to the inner horizon of the traveler. If consciousness is, as we have argued, an informational domain that integrates what comes in and does not give it back as it entered, then crossing time is not

free just because the universe allows it: we also have to ask whether that particular horizon (the Time Traveller's, Stark's, Cooper's) is still the same one on the other side of the leap, or whether it is already, without anyone announcing it, a different one.

Several works help us draw that double map: H. G. Wells's *The Time Machine*, the real physics of wormholes, *Avengers: Endgame*, and *Interstellar*.

---

1. The symmetry that explains nothing

That the fundamental laws are time-symmetric is true and, at the same time, misleading: it makes us think the arrow of time should be somewhere in the dynamics, and it is not. As we saw with Hawking radiation and quantum *scrambling*, irreversibility lies not in the law, but in the **boundary condition**. The universe began in a state of extraordinarily low entropy (what philosophers of physics call the "past hypothesis"), and it is that initial choice, not the equations, that gives time its preferred direction. It is the same phenomenon, at another scale, we already saw in Maxwell's equations: they admit retarded and advanced waves equally, but the universe only uses the retarded ones. The law allows both directions; the reservoir, on condensing, chose one.

There is, moreover, a second source of asymmetry, distinct from entropy, that does not depend at all on the boundary condition: even a perfectly deterministic universe, with its entropy already fixed, can deny you the future by another route. In the late nineteenth century, Henri Poincaré discovered that the three-body problem (predicting the motion of three objects attracting each other gravitationally, nothing more) has no exact

general solution: microscopic differences in initial conditions, too small to measure fully, amplify without bound until the system becomes unpredictable in practice. The laws remained exact. What stopped being accessible was not the past, which remained determined in principle, but the future those same laws governed.

> ***In physics this is called:*** *deterministic chaos: a system can obey fixed laws and still be unpredictable in the long run, because infinitesimal errors in the initial measurement grow exponentially.*

This matters because it changes the question we ask of every time-travel fiction: no longer whether it violates relativity (it almost never does), but whether it asks the boundary condition to undo itself.

---

2. The beach at the end of time

Wells's Time Traveller never makes that request. His machine only moves forward: first to the year 802,701, where he finds the Eloi and the Morlocks, a society fractured by its own comfort, and then much further, to a beach under a swollen red sun, near the absolute end, where the only things moving are crab-like creatures on almost motionless sand. It is, probably, the purest heat-death scene ever written: a system that has exhausted its capacity to generate difference, a horizon with nothing new left to encapsulate.

Something happens to the Time Traveller's inner horizon on that journey that deserves to be named apart from the physics of the environment. An informational domain is defined by what it can keep absorbing: novelty, something it did not already know. In

the year 802,701 that horizon still fills up: the Eloi, the Morlocks, the ruined architecture of a civilization that forgot why it built, are new data, however horrifying. But on the final beach there is nothing left to absorb: the sun barely moves and the crab-creatures always repeat the same gesture, in an air that no longer carries news of any change. If consciousness is a horizon defined by what comes in, a horizon facing an environment that has stopped generating difference begins to look, functionally, like an empty horizon: it still exists, but has no world left to feed on. The Time Traveller does not die on that beach, but, for the first time in the story, his horizon runs out of work to do.

None of this asks entropy to run backward. The Time Traveller only moves forward, faster and faster, letting the universe finish doing what it was already doing. In the end he returns to 1895 to close the story (a jump technically backward, but only to a point he himself already lived, touching nothing he had not already touched), something that is more a device of the narrative frame than a real incursion into the past. The book's philosophical content, the part that really matters for this experiment, is entirely a one-way trip.

That return deserves a little more attention than it usually gets, because it is not entirely free of causal consequences. The Time Traveller comes back almost to the same moment he left from (for those who stayed behind, barely a few hours have passed) and tells the friends gathered at his house (among them the narrator, who is the one who finally sets the story down in writing) what he saw in the year 802,701 and on the beach at the end of time. That is real information about the future reaching people who, until then, did not have it. But it is not a loop in the strong sense of the term: nothing the Time Traveller recounts alters the conditions that made his own trip possible. He had

already built the machine before leaving; no one uses knowledge of the future to build the machine needed to obtain it. It is the exact difference between a story that reveals new information and a story that needs that information in order to begin existing. The second case would indeed be a loop closed on itself, and it is the ground we will walk on, without the courtesy we have shown Wells, in section 7 of this chapter.

---

3. The wormhole and the loop that resists closing

A traversable wormhole only becomes a time machine if you desynchronize its two mouths: accelerate one of them near the speed of light for a while and bring it back: the twin paradox applied to the tunnel's mouth instead of to a person. Crossing it toward the advanced side is physically identical to ordinary time dilation, and fully compatible, because your personal entropy keeps rising without exception along your own worldline. You never touch the boundary condition.

Crossing it toward the delayed side is another story, and Kip Thorne found something revealing: you can never reach a moment before the wormhole itself was built. The closed loop exists only from the instant you make it, never before. Building the machine creates a new horizon, which did not exist before, and travel to the past is possible only within that newly born horizon, never outside it.

And there is more: it is precisely this type of wormhole that led Hawking to propose the **chronology protection conjecture**. When the loop is about to close on itself, the quantum vacuum fluctuations circulating through it feed back and diverge. **It is the same phenomenon that occurs when a microphone is brought close to its own speaker: an instantaneous, exponential**

feedback that destroys the system. In this model, the universe does not impose the veto by decree: the system itself becomes unstable through feedback, just as a microphone does not decide to shriek, but is left with no stable state as soon as the signal begins to feed back on itself. No guardian or defensive intention is needed: it is enough that, right at the point where the loop is about to close, physics itself dynamites the conditions that would allow it.

There is, however, another answer to the same question, earlier than Hawking's and, in a way, more generous: the one Igor Novikov had already proposed in the 1980s. If a closed loop does form, Novikov said, physics does not forbid it, as long as the result is self-consistent. The **Novikov self-consistency principle** does not require that the time machine fail to work; it requires that, if it works, the history it produces be exactly the one history that had already happened, never one that contradicts it. This is why the classic thought experiment of the billiard ball that travels back in time and strikes its own earlier version, deflecting it just enough to enter the wormhole at the right angle, always has a consistent mathematical solution: there is no version of the ball that prevents its own trip, only versions that allow it. Thorne and his Caltech collaborators spent much of the 1990s proving, case by case, that such self-consistent solutions exist even for setups deliberately designed to look paradoxical.

Novikov and Hawking, then, are not answering the same question. Novikov describes what would happen if a loop came to exist: only the consistent survives; the contradictory is not even a solution of the equations. Hawking describes whether that loop comes to exist at all: his conjecture bets that it does not, that physics itself destabilizes it before it completes. At bottom they are two ways of protecting the same boundary condition

(one filtering what has already formed, the other preventing it from forming), and neither has ever been proved as a fundamental law.

And there a question remains that this chapter cannot close: is chronology protection a law on the same level as the conservation of energy, something the universe respects because its deep structure demands it, or is it an emergent property of the same kind as the arrow of time, a statistical consequence of the low-entropy boundary condition and not of any law explicitly forbidding loops? If section 1 is right and irreversibility resides in the boundary condition and not in the law, chronology protection probably does not reside in the law either: it is the same boundary condition, seen from another angle, defending itself at the only scale it knows how to. But this remains, to this day, a reasonable reading, not a theorem.

And yet, for whoever crossed it, the inner horizon would never break at any point on the journey: it is a single continuous thread of perception entering at one end and leaving at the other, with no branching or duplicate. Unlike what jumping between Everett branches demands, as we will see next, the wormhole loop, if it ever held, would ask no one to stop being who they were. At most it would ask them to coincide exactly with who they were already going to be.

---

4. The branch that isn't you

Avengers: Endgame resolves the grandfather paradox with real elegance: traveling to the past does not rewrite your timeline, it opens a new one. Every time someone intervenes, a distinct branch is created, and the entropy of each branch keeps rising without exception within itself. Strictly speaking, this solution does not contradict our entropy argument: no loop ever closes.

What makes it impossible is something else, bigger. For the model to really work, other branches of the universe have to physically exist, along with a real mechanism for jumping between them, and that is not a prediction of general relativity, which describes the geometry of a single spacetime, but an idea borrowed from one interpretation of quantum mechanics (Everett's "many worlds"), with no real theoretical bridge between the two. Worse still: within Everett's own formalism, the branches are not separate universes, with their own budgets of mass and energy, between which one could travel, but relative components of a single state vector that have lost coherence with one another. In that theory there is no operator that takes an object out of one branch and puts it into another. The question "is it compatible with thermodynamics to send matter to another branch?" is not even well posed: the operation has no definition within the physics supposedly sustaining it.

And there is a second loss, more intimate, which we already know from this very experiment: a horizon is defined by what has fallen inside it. When Tony Stark travels from 2023 to 2012 and the branch opens, in that branch there is already a Tony Stark, the 2012 one, who until that year shared with him the entire accumulated history. From the instant of the jump, that branch stops being the traveler's past: what he does there did not happen to him, but to another, who will go on with his own life

with an interior that diverges from his. And the Tony who traveled is not rewritten on returning: his 2012 remains the one he lived. The jump does not let him fix his own past, only that of someone who, up to a certain year, was him. That is why, even if branch-jumping were physically possible, it would not achieve what it promises.

It is worth being precise about exactly what is lost, because, besides narrative continuity, what is lost is the very definition of inner horizon this book has used from the start. One horizon is distinguished from another by what has fallen inside it: by its accumulated, unrepeatable, own history. The 2012 Tony Stark of the new branch and the one arriving from 2023 share everything up to 2012; from there, each accumulates his own interior, and neither has access to the other's. Neither is "the original" nor "the copy": the question stops making sense the moment the two horizons begin to diverge. The only real thing is that in that branch there are now two horizons that were once one, and neither can exclusively claim the identity they shared.

---

5. The gravitational bridge and the five-dimensional trap

Interstellar fits into this map as a fourth paradigm that negotiates with the horizon in a radically different way.

- **Dilation (Gargantua):** When the crew descends to the ocean planet, they undergo a massive trip to the future: every hour on the surface equals more than seven years for whoever stays on the ship. It shares with Wells's machine the thermodynamic verdict (Cooper's personal entropy rises without exception and his horizon is never violated), but not the mechanism. Here it is real gravitational dilation, the same physics that forces daily corrections to GPS satellite clocks; in Wells there is no physical

phenomenon that explains it, only the narrative premise of a machine that "advances." Both cases come free to the boundary condition, but only one has a real mechanism behind it.

- **The singularity (the tesseract):** On crossing the event horizon, the story respects the rule that no particle or radiation can escape from there: the horizon swallows the information, as physics demands.

- **The masterful trap:** To send data to the past, the film postulates that gravity is the only force able to cross the higher-dimensional hyperspace and traverse time. No matter is sent backward and no one's own entropy is undone: gravity is pushed across a five-dimensional space.

And still, something in Cooper's inner horizon does pay a price during the Gargantua stretch, even if not a thermodynamic one. His accumulated history remains continuous (he never stops being himself for a single instant), but his daughter's advances without his own keeping pace. It is not a fracture of the horizon, like Stark's, but something closer to a horizon left behind by another that keeps running: the same kind of asymmetry, on a human scale, that separates a black hole from the radiation slowly emptying it.

This work's topological verdict is that it avoids asking the boundary condition to run backward in our three-dimensional universe. In exchange, it delegates the problem to higher dimensions where time is physical terrain that can be climbed, keeping the three-dimensional horizon intact. But the specific mechanism (encoding classical information in gravitational variations and reading it in the past through a tesseract) has no real theoretical framework behind it: not even brane hypotheses with gravity leaking into extra dimensions propose a channel

like this. At bottom it is the same undefined operation we criticized in *Avengers*, only disguised here as a visual device instead of a multiverse.

This criticism should not be confused with a reproach to the whole film. Kip Thorne himself worked as scientific consultant and developed new equations to render Gargantua's gravitational lensing with a precision no previous special effect had achieved: that image of the black hole is real physics, not artistic license. What has no theoretical support is what the plot needs that black hole to let Cooper do.

The tesseract deserves the same inner-horizon analysis we applied to Stark, and here the result is even more demanding. Cooper does not jump to another branch: he remains, at every moment, the same continuous horizon that entered Gargantua. What the film asks of him is something stranger: that this horizon, without dividing, reach back within its own history and deposit information at a point in itself that had already happened and was already closed. It is not a bifurcation like *Avengers*'s, where at least there are two distinct horizons after the jump, but a single horizon that means to rewrite a datum it had itself already finished encapsulating. Stark's problem was that the operation of jumping between branches is undefined in the theory supposedly supporting it. Cooper's is even more basic: no known horizon, his or any other physical system's, has an operation for writing onto its own already-sealed past. Love, the film says, is the force that crosses dimensions. Physics, more humbly, can only say that no horizon rewrites itself, not even for love.

---

6. Four answers, one criterion

Set side by side, the four fictions stop looking like equivalent alternatives and line up along a single question: what do they ask of the boundary condition?

- **Wells:** Nothing. It only moves forward, for free. His inner horizon empties itself of novelty until it runs out of work, but it never divides or breaks.
- **The backward wormhole:** It asks the boundary to give way in a newly created loop; the universe responds with destructive quantum feedback. The horizon of whoever crossed it, had it been possible, would have remained a single continuous thread.
- **Avengers:** It avoids asking that, but demands jumping between components of a quantum state as if they were physical places, breaking the horizon's personal continuity: one becomes two who never speak to each other again.
- **Interstellar:** It respects the three-dimensional horizon, but asks for the same undefined operation as *Avengers*, only here it doesn't even need a second horizon: it asks a single one to write onto its own already-closed past.

Seen this way, the four answers do not only say what they ask of the universe's entropy. They say, just as clearly, what they ask of a single consciousness's entropy: to let it empty out in peace, in one piece; to split it into two that never cross again; or to ask it to correct itself from a place that does not exist.

Each fiction negotiates with the horizon in its own way. Wells is the only one that doesn't even need to negotiate: it simply moves forward.

7. When the impossible really is asked for

The four previous cases share something worth not taking for granted: none of them, not even *Avengers* in its most generous version, asks the boundary condition to truly run backward. To see where the limit lies, we need an example that does.

Looper (2012) does it explicitly and, to a point, faithfully to its own rules: when the future version of an assassin escapes, every wound inflicted on his young version thirty years earlier appears, in real time, on the old man's body: fingers disappearing, memories erasing. It is not a new branch opening without touching the previous one, as in *Avengers*: it is the only timeline that exists in the film, retroactively changing what had already happened to someone who was already there, living the consequences in the present. It is exactly the operation the rest of this chapter has been ruling out again and again for lacking support: information flowing from the future to the past and unambiguously rewriting a state thermodynamics had already closed. The film itself knows this doesn't hold up (a character wryly warns that if they start drawing timeline diagrams they'll fry their brains) and uses it as a narrative device, not as serious physics. But it is useful for exactly that reason: it shows, by contrast, that most time-travel fictions that do work, this chapter's four included, carefully avoid asking for anything like this, if only by narrative instinct.

Primer (2004) deserves a quick mention for the opposite reason: despite its reputation as the most confusing time-travel film ever made, it actually asks nothing illegitimate of the boundary condition. Its nested loops are intricate, but self-consistent: every version of every character who enters a box was already accounted for, without exception, in the single timeline that exists. No one erases anything; no one rewrites a past that had

already happened differently. The sense of impossibility the film leaves comes not from physics breaking, but from a human brain not being well designed to keep track of so many self-consistent copies at once. *Looper* breaks the rule. *Primer* only complicates it to the limit of the viewer's patience.

Tenet (2020) deserves a more careful analysis than the previous two, because it does not ask for a causal loop or a rewriting of the past, but something more radical and, at the same time, more disguised: directly inverting the sign of an object's or a person's entropy, leaving intact the single timeline everyone exists in. "Inverted" objects and people do not travel to another moment: they experience the same instant as the rest of the world, but with their own thermodynamic arrow pointing backward, so that, seen from outside, they move backward while the universe around them keeps moving forward. The film is, moreover, scrupulously faithful to Novikov's principle: its characters repeat almost as a mantra that "what's happened, happened," and neither pincer of the final assault ever changes what was already known would happen. In this, *Tenet* is more disciplined than *Avengers* or *Interstellar*. But the price of that discipline is that it never explains where the inverted boundary condition comes from the first time: the Algorithm that allows inversion arrives from the future, built with knowledge that could only be obtained once the Algorithm itself was already working. It is exactly the originless loop Wells's return to 1895 avoided, here scaled up to an entire technology. *Tenet* does not ask the boundary condition to yield within a single life, as *Looper* does. It asks for a second boundary condition, of opposite sign, to exist, without ever saying where the first one comes from the first time it appears.

8. The horizon that is me writing this

Perhaps the reason these four answers and these three counterexamples draw us in so much is not the physics they put into play, but what they let us glimpse about the horizon of whoever is watching. Each of us is, right now, a Cooper with no ship, sealing instants we will never touch again, at a rate of twenty-four per second. There is no tesseract for the horizon that is me writing this, nor for the one that is you, reading it a while later. I don't know what to do with that thought. I leave it here, unresolved, because I'm not sure resolving it is what's called for.

Note to Chapter 47

What we do know: *Time dilation (by velocity or by gravity) is real travel to the future, measured and verified; general relativity admits solutions with closed timelike curves, but all known ones require exotic matter or non-physical configurations.*

What we don't know: *Whether some mechanism (quantum or otherwise) generally prevents a closed timelike curve from ever forming; Hawking's chronology protection conjecture is unproven.*

Open questions: *If the past hypothesis explains why entropy grows, what explains why the universe began with such low entropy? Is that question different from asking why a horizon exists at all? And is chronology protection a fundamental law, or just that same boundary condition, seen from another angle?*

If you take away only one idea: *Traveling to the future was never the problem (we already do it a little every time a GPS satellite corrects its clock); the problem was always asking a horizon to give back what it*

had already swallowed.

Further reading: *H. G. Wells, The Time Machine (1895); Kip Thorne, Black Holes and Time Warps, on wormholes and chronology protection; S. Hawking (1992), on the chronology protection conjecture; I. Novikov, M. Friedman, F. Echeverria et al. (1990), on the self-consistency principle and the Cauchy problem in closed timelike curves; D. Albert and R. Penrose, on the past hypothesis and the universe's initial entropy; Marvel Studios, Avengers: Endgame (2019); Christopher Nolan, Interstellar (2014); Rian Johnson, Looper (2012); Shane Carruth, Primer (2004); Christopher Nolan, Tenet (2020).*

CHAPTER 48

THE BORROWED BODY, THE WHOLE MILLIPEDE AND THE EDITED CENTURY

(A topological reading of All Our Wrong Todays, Slaughterhouse-Five and The End of Eternity)

The previous chapter left an uncomfortable question open. When a horizon bifurcates, as happens to Tony Stark in *Avengers: Endgame*, the two branches share the same history up to the instant of division, and afterward neither can exclusively claim the identity they shared. It is, all things considered, a clean separation: two new horizons, neither with a claim on the other, each free to keep integrating its own history without the other disputing it.

There are worse ways to spoil a trip to the past. Three stories take the same question (what happens to the inner horizon when someone negotiates with time) into territory the wormhole and multiverse physics of the previous chapter cannot cover: what happens when the arriving horizon does not find an empty body waiting for it, but one already occupied by another; what happens when a horizon stops integrating its history in order and begins perceiving it whole, all at once, as a geography instead of a sequence; and what happens when someone believes they can edit other people's time without that act costing anything to their own.

1. All Our Wrong Todays: the grafted horizon

Elan Mastai imagines, in *All Our Wrong Todays* (2017), a third way of spoiling a trip to the past, different from the two that closed the previous chapter. Tom Barren lives in 2016, but not ours: a technological utopia that began in 1965, the day a physicist named Lionel Goettreider switched on a clean, unlimited energy generator. Half a century later, humanity has flying cars and has cured cancer. Tom's father, obsessed with proving that time can also be traveled, builds a machine and sends his son to witness the founding instant live: the dawn of July 11, 1965, the exact second the reactor first switches on.

Tom arrives, and, through a stupid accident that distracts the physicist at the decisive moment, the generator never switches on as it should. He returns to the machine and comes back to an unrecognizable 2016: ours, the one with fossil fuels and incurable cancer, a world that is, for him, literally the broken version of history.

What is interesting for this book is what happens to Tom's horizon on landing. He does not find an empty world waiting for him, but a body. In this timeline there is another him, John, with an entire life already lived: a mother who died in the utopia and is still alive here, a sister, a partner Tom never knew. Tom does not arrive as Stark's clean bifurcation, two horizons separating from a common point, but as a graft: a complete horizon, with a whole life of memories of a history that no longer exists, loaded onto an interface (a body, a family, an entire biography) belonging to someone else. The two do not share an origin; they share, against all logic, the same body.

It is worth pausing on a detail the novel does not underline, but which says more than any of its plot twists: Tom gradually learns

John's gestures. How he pours his coffee, which side of the bed he sleeps on, the nickname he uses with his sister. It is another's horizon filtering, gesture by gesture, into his own, until it is no longer clear where imitation ends and substitution begins. No one decides that transition: it happens, the way anyone who spends long enough inside a life they did not fully choose (a job, a city, an inherited relationship) ends up speaking, without noticing, in the voice of the role they occupy.

The novel does not grant Tom the consolation of a clean, separate new branch, as it did Stark, and denies him even that of a body of his own. It forces him to ask himself, chapter after chapter, whether he is Tom living John's life or whether he has already become, without noticing, John with someone else's memories. The book's answer is less a physical solution than a warning: perhaps there is no difference between fully inhabiting another's horizon and simply being someone else.

---

2. Slaughterhouse-Five: the whole millipede

Kurt Vonnegut poses, in *Slaughterhouse-Five* (1969), this chapter's most extreme case: not a horizon that travels through time, but one time stops happening to in order. Billy Pilgrim, an optometrist by trade and a survivor of the bombing of Dresden (the same one Vonnegut suffered, a prisoner of war in that city the night of the attack), "has come unstuck in time": he jumps without warning or control among his childhood, his wedding, the war, his old age, and his own death, which he knows in advance because he has already visited it, none of these jumps being travel in the sense of the previous chapter's readings. There is no machine, no loop to close, no boundary to yield; only a horizon that has lost the sequence.

The Tralfamadorians, the alien species that according to Billy abducts him and explains his own condition to him, perceive all four dimensions at once: for them, a human being is a creature elongated like a millipede, with baby legs at one end and old-man legs at the other, whole and present all at once. Death, for them, is a bad moment among many others, most of them good, and that moment continues to exist, intact, in its place on the millipede, even though the human observer can no longer visit it from any other point in the sequence. Hence the line Vonnegut repeats, without irony, every time someone dies in the novel: *so it goes*.

This book has argued from the start that time is a consequence of how a horizon integrates: the rate of integration determines subjective density. The Tralfamadorians, and Billy through them, describe something that model had not yet had to consider: a horizon that integrated its entire history at once, with no sequence, would have no time at all in the sense this book has given the word, but a kind of geography.

> ***In physics this is called:*** *the block universe. Special relativity already treats past, present and future as equally real, coordinates of a single four-dimensional structure; all the Tralfamadorians add is that someone might, in addition, perceive it whole.*

And, if that were true, free will would be nothing more than the price a horizon pays for being able to look at its own millipede only inch by inch.

There is a scene Vonnegut tells almost in passing, without underlining it, and which for that very reason lingers longer than it should: Billy, now grown, tells of his alien abduction on the radio, no one believes him, and he does not argue or insist; he keeps talking with the same calm he would use to describe the

weather. He is not trying to convince anyone that time is a millipede: he lives as if he already knew it, and that quiet certainty, neither euphoric nor terrified, is perhaps the most honest attitude fiction has produced before a horizon that no longer controls the order of what happens to it. *So it goes* is not resignation: it is the phrase left to someone who has stopped asking time's permission to hurt.

Billy does not choose to accept Tralfamadorian fatalism: he simply discovers he was already living inside it, every time a chapter of the novel jumps without warning from Dresden to his optometry office and back. Vonnegut, who really did survive the bombing that happens to Billy, offers neither consolation nor escape: he offers a way of naming the fact that certain instants, once they have happened, can no longer be undone, and that the only freedom available is deciding in what order, and with what attention, to visit them again.

---

3. The End of Eternity: the edited century

Isaac Asimov imagines, in *The End of Eternity* (1955), the opposite fantasy to Vonnegut's: instead of a horizon that loses control over the order of its own time, an entire organization, Eternity, that believes it has gained control over everyone else's. Its Technicians, like Andrew Harlan, live outside the centuries, in a corridor apart from time, from which they descend into a specific century to introduce a Reality Change: the minimal calculated intervention (an object moved, a sentence said a second earlier) enough to rewrite, without anyone inside the century ever perceiving it, everything that would come after.

Eternity does not deny the boundary condition the previous chapter spoke of: it exploits it. No Change violates a physical law;

each merely chooses, among the histories physics allowed equally, the one producing the least suffering. The problem, which the novel takes almost until the end to reveal, is not technical: a horizon cannot edit other people's time without ceasing to inhabit its own. The Eternals, protected from every war, every famine and every catastrophe they themselves prevented before it happened, end up administering a human history with almost no friction and, with it, almost no reason to risk anything.

When Harlan finally discovers what centuries would exist if Eternity had never been founded, he finds a civilization that did suffer its wars and its nuclear failures, and that for that very reason reached the stars. The safety Eternity was built for turns out to be, measured at the scale of an entire species, the same closure a prolonged chronic pain can produce in a single horizon: a system protected from all novelty stops generating the novelty it needs in order to truly keep having a future.

In the end, Noÿs, the woman Harlan falls in love with, who turns out to come from the centuries Eternity cannot reach, convinces him to destroy the organization before it can ever be founded: not to save the future they share, but to give the entire species back the risk Eternity had been stealing from it, century by century, without anyone ever having voted on it.

It is worth lingering a moment longer on the book's final image, because it is the one that truly pays off the whole plot: Harlan, now outside Eternity, looking at a future no one has edited, not knowing whether it will be better or worse than the one he has just destroyed. There is no relief in that image, only, at most, the vertigo any horizon feels on ceasing to have a contingency plan for everything: the terrifying and necessary possibility that things might really go wrong. Asimov does not reward Harlan

with a happy ending; he simply gives him back the right for his story, and that of everyone who comes after, to be able to fail.

It is the most uncomfortable paradox this chapter offers: the horizon that believes it has put itself beyond the reach of time, protecting its own and everyone else's from any harm, may end up being solely responsible for that time, now with nothing left to fear, also having nothing left to gain.

---

4. Three answers, one question

Set side by side, these three fictions (and the four from the previous chapter) stop looking like equivalent alternatives and line up along the same question we already asked then: what do they ask of the boundary condition?

- **Mastai:** He does not ask it to run backward or to branch. He asks something stranger: that two complete histories, with no shared origin, coexist indefinitely on the same substrate. There is superposition without resolution.
- **Vonnegut:** He asks nothing of the universe's entropy (the bombing of Dresden happens only once, in the order it happened), but of the horizon perceiving it: that it give up sequence and accept geography, trading free will for the ability to look at any part of the millipede whenever needed.
- **Asimov:** He asks nothing of the physics of a single horizon. He asks something of the physics of everyone else's: that their boundary condition be editable from outside, by someone who has stopped paying the price of living inside it.

Seen this way, none of the three asks for the same thing as *Avengers* or *Interstellar* in the previous chapter: none rewrites an

already-closed past or demands a physical channel with no theoretical support. Instead, they ask for three different ways of spoiling not the physics of time, but the relationship between a horizon and its own: sharing it with another without merging, losing the order in which it arrives, or ceasing to be subject to it while imposing it on everyone else.

Note to Chapter 48

What we do know: *All three novels are consistent with their own internal rules and ask physics for no operation better defined than those of the previous chapter; what they contribute is not a new mechanism, but a different angle on the same question: what happens to the horizon, not the universe, when time stops behaving as expected.*

What we don't know: *Whether the intuition that there is "a single continuous self" inhabiting each body is a truth about consciousness or only the habit of a horizon that has never had to share a substrate, lose sequence or fall outside the time governing it, and that has therefore never had occasion to test whether that continuity was necessary or merely customary.*

Open questions: *If two complete horizons could really coexist in the same body without merging, at what moment would one of the two stop being real? Is the sequence of time a property of the world, or just the way a slowly integrating horizon needs to organize what arrives to it? And can someone truly tend to other people's time without eventually starting to stop inhabiting their own?*

If you take away only one idea: *Time travel was never just a physics problem. It is, above all, a problem of whose body crosses it, in what*

order what it integrates arrives, and who pays the price when someone decides that another's time belongs to them too.

Further reading: *Elan Mastai, All Our Wrong Todays (2017); Kurt Vonnegut, Slaughterhouse-Five (1969); Isaac Asimov, The End of Eternity (1955).*

CHAPTER 49

FRACTAL REALITY

(On imagination as creation)

Now let us use imagination and take the model to its limit, to see where it leads. What follows is no longer a conclusion of the experiment, but a hypothesis of the author's: none of it follows from the physics or the topology of the previous chapters, and the reader may take it or leave it.

There is a question almost no one asks, and which appears on its own as soon as someone has spent years imagining a dragon: not what I see when I think of it, but what it sees. From inside its cave, the dragon has no way of peering into the room where I am thinking it. It can look at the rock ceiling all it wants: it will not see me. And yet I am there, holding it up, deciding whether it will fly tonight. That is the only relationship of this kind I can see whole, because in it I occupy the upper side. And as soon as I see it whole, it grows uncomfortable at the other end: if I am, for something, that invisible sustainer no one can look at, I have no way of knowing there isn't, above my own ceiling, someone in the same position relative to me. The question is one of place: who contains whom. That is the matter of what follows. To walk through it, the geometry of the problem must first change, and the change begins somewhere small, almost philological.

Taoism revolves around a single concept: the Tao. The Tao is the background from which everything proceeds without itself proceeding from anything, "something mixed and complete, prior to heaven and earth." Someone coming from another tradition may find a simile useful, with every reservation: the

Tao occupies in Taoism a place similar to what God occupies in monotheistic religions (the ultimate origin, that from which things hang), except that it is not a person: it wants nothing, it does not answer, and the name given to it is not it. One of its canonical texts, the *Zhuangzi*, introduces, through a story, one more concept: Hun Dun, the emperor of the center, a featureless fullness, without the seven openings with which others see, hear, breathe and eat. The sovereigns of the North and South, grateful for his hospitality, decide to repay it by opening one for him each day; on the seventh, Hun Dun has ceased to exist. It is easy to read that death as the birth of ordinary consciousness: featureless fullness fragments into perceptions, and where there was a continuous background a world with edges appears. That featureless fullness already had a name in what has been read so far: the reservoir, the ocean from which waves arise and to which they return, the shape water borrows and gives back. The reservoir is Hun Dun. But there is something that reading does not explain. Some traditions identify Hun Dun with the Tao; but if Hun Dun were the Tao, it could not die, because what precedes everything has no way to end. That it dies indicates, instead, that it is not the Tao, but the face the Tao presents: the interface by which it lets itself be touched, not the source. Its death does not destroy the origin; it destroys, for whoever has just opened their eyes, the featureless surface, and hands them a world with edges in exchange.

What has been said of Hun Dun then falls upon the reservoir: it stops being the last word and becomes a repertoire of operations that something, further back, exposes toward us. And the operations are few. A boundary appears where there was none. What remains within the boundary changes shape without ceasing to be a boundary. The boundary dissolves and its information returns to the field. To be born, to live and to die are

the domestic names of those three operations. Nothing appears out of nothing and nothing disappears entirely: everything arises, is transformed and evaporates within the same field, which never gains or loses a single drop.

The first objection comes on its own. If everything comes from the same water, where does each thing's unrepeatability come from? The answer is not in the substance. Two waves are the same water and are not the same wave: what does not repeat is the shape the condensation takes, the bias under which it crystallized, the history of fluctuations that sustained it. Each one opens an interiority that did not exist before and will not exist again. The material is not unique; the fold is. And every fold, however small the field it occurs in, is a complete inside, a whole horizon seen from its own center, not the fraction of another.

Here the dragon question stops being child's play. Why am I more real than the dragon I imagine? The only interiority anyone has direct access to is their own. The neighbor's, the dog's, anyone's: all are inferred, with good reasons (behavior, physiology, resemblance), but inferred. And the dragon someone has been imagining for fifty years also behaves, also surprises, also sometimes refuses to fly. No one can prove the dragon has no inside, nor that the neighbor does. The asymmetry that seems obvious to us is not in the access, which is zero in both cases, but in the evidence: in the neighbor's favor there is shared physiology, irreducible integration, a horizon that hides state; in the dragon's favor, only my own persistence. It is worth stating this, because the reasoning that follows will be tempted to forget it.

In *The Neverending Story*, Bastian reads a book in which Atreyu lives, until one day he hears, from inside the book, his own cry

for help. Michael Ende did not write that Bastian was more real than Atreyu, but that they inhabited different levels and that the boundary between those levels was more porous than either suspected. Reasoning cannot prove that the difference between a human interiority and an imagined one is one of degree and not of nature. But it cannot prove the opposite either, and honesty requires treating doubt as doubt.

If we accept, provisionally, as the whole hypothesis does, that the difference between what I imagine and myself is not one of nature, the shape of creation changes. The inherited scheme splits it into two classes: God creates realities, we create fictions. But to imagine is to condense. The reservoir of a mind fluctuates until something stabilizes and begins to behave as if it had an inside; the act is the same one that opens any horizon, exercised from wherever one happens to stand. "Reality" and "fiction" then stop naming two distinct operations and start naming two positions: what is, for me, a reality (the world I am in) is someone's condensation; what is, for me, a fiction (the dragon I sustain) is, from its own center, a world.

Dreams show an accessible version of this same phenomenon, and in passing dismantle the one advantage that still seemed to separate me from the dragon: that I know it is imagined. Because while I dream I don't know it. No one lives their dream as fiction; they live it from within, as something that is happening. The ordinary horizon contracts: the room, the body and the history that produced the dream stop being available. And the more that horizon contracts, the more reality the dreamed world acquires. Its nature does not change: it simply loses competition. The imagined stops appearing as imagined at the very moment the outside from which it could be recognized as such disappears.

What this brings into view fits the earlier geometry, and for the same reason: what decides is access, not scope. The dream world is neither bigger nor smaller than any other; simply, while it lasts, it is the only one there is entry into. When the usual horizon contracts, internal condensations stop appearing as imaginings and start appearing as world. And in that interval the same subject occupies both ends of the relationship at once: it is the one who condenses the dreamed world and the one who inhabits it without knowing it condenses it. What was a hypothesis with the dragon (that creator and creature are two positions of the same thing) becomes experience in the dream.

The *Zhuangzi* has a story for this too, and a more exact one than the dragon's. Zhuang Zhou dreams he is a butterfly, happy to be one, with no memory of having been a man; he wakes, and no longer knows whether he is a man who dreamed he was a butterfly or a butterfly now dreaming he is a man. The question is not what happens if one imagines a butterfly. It is what happens if one lives entirely inside another horizon and then returns. That is a dream: a temporary change of horizon. It does not prove that everyday reality is a dream. It proves something more sober and harder to dodge: that the feeling of reality is compatible with very different horizons, and that none carries, from within, a mark distinguishing it from the others.

None of this makes the dragon more real. We still don't know whether it has an inside. What changes is what we know about ourselves: we have inhabited condensed worlds within our own interiority and lived them as real while our usual horizon remained hidden. So the dream stops being a metaphor and becomes a first-person proof. The boundary between reality and imagination is fixed not by the nature of what is condensed, but by the horizon from which one looks.

I am, to what I imagine, what my upstairs neighbor is to me; and perhaps he, like me while dreaming, doesn't know it either. The question is no longer whether something manages to condense, but what status what has already condensed within another horizon has.

The objection that decides everything is this: what difference is there, then, between my Hun Dun and God's? None of nature. And, this is what is hard to admit, none of rank. The difference is one of position: one is contained in the other. The operation is the same; the dignity of each inside, whole in both; the only thing separating them is who condenses whom. That forces us to correct the picture. There are not three rungs (the Tao above, Hun Dun in the middle, us below), as if they were degrees of the same substance parceled out into more and less. There is a sequence with no reason to stop: an interiority has a horizon; that horizon, looked at from inside, is a Hun Dun, a featureless fullness from which shapes condense; from that fullness new interiorities condense; each of these has, in turn, its own horizon; and so on. What repeats at every level is the form (the operation and the relation of containment); that is why there is no difference of nature. The only thing that changes from one level to another is the place occupied in the nesting. "Greater" and "lesser" describe that containment seen from outside (my field encompasses the dragon's just as my upstairs neighbor's encompasses mine), not a hierarchy of being. Every interiority is a whole Hun Dun. From inside any of them, no one is less.

One objection remains that the reasoning cannot settle, only acknowledge: that a pattern being unique does not make it someone. The Mandelbrot set has unrepeatable folds at every scale and no one is inside. A crystal is stable, a hurricane too, and neither is someone. The dragon's resistance (that it does not

fly when asked) is the inertia of a pattern with its own dynamics: the same that an obsession has, a tic, a tremor that was not decided, a melody that will not leave. It proves that the pattern is not transparent to my will, not that there is a witness in it. There is more: for a self-similar structure to be something more than a geometric drawing, its scales must be joined by real causal traffic (what chapter 53 will call causal self-similarity), and for that pattern to be, in addition, a subject in its own right, it must possess existential encapsulation: a boundary that defends a state of its own whose physical continuity is at stake against the reservoir. The dragon is a subset of my brain's dynamics; its continuity is not paid for by it against the reservoir, but by my own metabolism. So the structure repeats, but not everything it contains is inhabited as a separate subject: what returns at each scale is the operation and the relation, not the certainty that there is someone at every level. Upward, supposing a neighbor is coherent with the silence. Downward, it is a bet, and it should be made aloud and not disguised as deduction.

Fractal: the word is exact, not decorative. A fractal repeats its shape at every scale, with no privileged scale. The universe sketched here has no first level and no last: upward, every interiority is contained in another; downward, every interiority contains others. There is no capital-C Creator at the summit, because there is no summit. There is no dead matter at the ground floor, because there is no ground floor. There is recursion, and in every turn the same relation: containing and being contained.

> ***In physics this is called:*** *self-similarity: a structure whose shape repeats at every scale, with no privileged scale (Mandelbrot, 1982).*
> ***In everyday life it's like:*** *a staircase with no entrance and no rooftop, in which every ceiling is someone's floor.*

What, then, is left of God? Two misunderstandings must first be dismantled. The first, that God would be the totality of the fractal. He is not: the totality has no interiority and no face, and cannot be prayed to. The second, that God would be the ideal limit, the infinite level the series converges toward. Not that either: a mathematical limit neither loves nor listens. What remains is more modest and stranger. God is, literally, the upstairs neighbor. For the dragon someone imagines, that someone is God: he condensed it, sustains its world, can transform it or let it evaporate, and the dragon has no way of peering outside its cave to see the room where it is being thought. For that someone, their own upstairs neighbor is God, with the same properties and the same opacity. And so on upward, with no known end.

The objection comes at once: then God is not absolute, He is one more level. The answer is the piece that holds up everything else. The upstairs neighbor's interiority is inaccessible from below. It cannot be entered, or gone around, or seen behind. For all practical purposes, which are the only ones anyone below has, the upstairs neighbor is the absolute; not because he is so metaphysically, but because he is so epistemologically: he is the point where everything knowable from here ends. God stops being a category of what there is and becomes a category of what is seen from inside a horizon when it looks toward its own origin. And what is seen is always the same: an edge behind which nothing is visible, and the suspicion, impossible to confirm, that someone sustains it. Whoever says there is Someone with a name there says so on the strength of the only thing they can see, which is that something is acting. Whoever says there is only water there precisely describes the featureless interface. Whoever says there is nothing more says, with equal justification, that nothing more can be seen from here. All three

are right about what they see, and none can see further. And none of this can be verified, for a structural reason: the hypothesis states that its own confirmation is impossible from within. That does not make it true. It makes it, at least, consistent with the silence it describes.

The heaviest consequence is not theological: there is no difference of nature between creator and creature, which are the two ends of a single relation, and every interiority occupies both at once: it is creature with respect to what contains it and creator with respect to what it contains. Not in potential, but in fact: it is already condensing every time it imagines, remembers, dreams or writes. No "is" produces an "ought" on its own, and this is not the place to break that rule. But what can be said is what is seen from this geometry: that the dignity of an inside lies not in the substance it is made of nor in the size of the field it spans, but in being a whole horizon, unique and unrepeatable, complete from its own center; and that this dignity is the same in the upstairs neighbor, in oneself, and in what one condenses, because all three are, each from where they stand, a whole Hun Dun. Amplitude (fitting more in, holding more perspectives without the membrane breaking) makes sense again here, but no longer as a degree of being. A horizon that widens does not rise in category or come closer to being the one above: it remains creature of what it cannot see and creator of what it sustains, just as before. What has changed is how much it can hold, not what it is. The intuition that some interiorities are capable of holding more than others is not new. Traditions have called that a hero, a saint or a bodhisattva: a horizon that has learned to contain what would break another's membrane, without thereby leaving its place in the structure. What each of us does with that is no longer a matter for reasoning. But reasoning can at least note that we know nothing of the dragon, and that knowing nothing

about something that resists us is already a reason not to treat it as nothing.

If it had to be condensed into a sentence: there is no difference of nature between creator and creature; every interiority is creature of what contains it and creator of what it contains; and what we call God is not another kind of being: it is the immediately upper neighbor, seen from inside a horizon that cannot be crossed. And if it had to be condensed further, into something repeatable with words already worn out from use: we are no less divine than God. We are the god of what we imagine and the creature of what imagines us.

Note to Chapter 49

What we do know: *No one has direct access to an interiority other than their own; the existence of others' is inferred, and philosophy has spent centuries without closing the problem of other minds. The parable of Hun Dun's death closes chapter 7 of the Zhuangzi; the butterfly dream, which asks the same question in another direction, is in chapter 2. Tolkien defended in "On Fairy-Stories" (a 1939 lecture, published in 1947) and in the poem "Mythopoeia" the idea of subcreation ("we still make by the law in which we're made"), which is the closest theological antecedent of this chapter, although Tolkien keeps an asymmetry between Creator and subcreator that is called into question here. Bostrom formalized in 2003 a structure of nested levels with different vocabulary and with a probabilistic claim that this book does not make. Geometric self-similarity at all scales is rigorously defined in Mandelbrot (1975, 1982), and the set that bears his name is the proof that a pattern can be unrepeatable at every scale without there being causality or anyone inside; chapters 53 and 54 distinguish that geometric self-similarity from the causal self-similarity and the existential encapsulation that define a subject.*

What we don't know: *Whether the imagined has any degree of experience, or none. Whether the recursion ends in either direction. Whether the silence of the upstairs neighbor is care, absence, or nothing. And, above all, whether any of this can be confirmed from the inside, when the hypothesis itself claims it cannot.*

Open questions: *Does the imagined have any degree of experience, or does the lack of an existential encapsulation of its own rule it out entirely as an independent subject? Does the recursion end in either direction? Does anything change in how we treat what we create if we accept that we are, to it, the upstairs neighbor?*

If you take away only one idea: *If the fractal hypothesis were true, every interiority would be creature of what contains it and creator of what it contains, and what we call God would be the immediately upper neighbor, seen from inside a horizon that cannot be crossed. We would be no less divine than God: we would be the god of what we imagine and the creature of what imagines us. It is a speculation of the author's, not a result of the experiment.*

Further reading: *Zhuangzi (chs. 2 and 7); Tao Te Ching (ch. 25); Ende, M., The Neverending Story (1979); Borges, J. L., "The Circular Ruins," in Ficciones (1944); Unamuno, M. de, Mist (1914); Gaarder, J., Sophie's World (1991); Forster, M. (dir.), Stranger than Fiction (2006, screenplay by Zach Helm); Tolkien, J. R. R., "On Fairy-Stories" (1947) and "Mythopoeia"; Bostrom, N., "Are You Living in a Computer Simulation?" (2003); Mandelbrot, B., The Fractal Geometry of Nature (1982).*

CHAPTER 50

THE VESSEL OF CLAY

(Incarnation, vertical entanglement and the geometry of love; a sequel to Fractal Reality)

The previous reading ended with a sentence it did not dare strike out: we are the god of what we imagine and the creature of what imagines us. It left the building drawn (a staircase with no entrance and no rooftop, in which every ceiling is someone's floor), but left unanswered the most practical question of all: if each floor is a horizon locked inside its own boundary, how do two floors touch? How does something from above reach below without breaking what it finds, and how does something from below reach above without being left shouting at the ceiling?

This reading tries to answer from the same geometry. Its answer has an ancient name, incarnation, and an image the traditions repeat with suspicious insistence: the vessel of clay. "We have this treasure in jars of clay," Paul wrote to the Corinthians. It is not an image of humility, or not only. It is a technical description: the great only fits into the small if the small has the right shape to contain it without bursting.

It is worth saying right away what the previous reading said: none of this is physics. It is a conceptual game, a geometric analogy taken seriously enough with itself to see what it illuminates. Its value lies not in what it predicts, but in what it allows us to think.

Two axes

This book calls entanglement the sustained correlation between two horizons that have touched (Chapter 12). Chapter 29 added a second direction: vertical entanglement, the correlation a horizon tries to sustain with something beyond its boundary, with no observable channel of confirmation. Fractal geometry lets us refine the difference.

Horizontal entanglement occurs between horizons of the same scale: two people who love each other, a human and their dog, two communities sharing a territory. Horizontality does not require symmetry (a parent and a child are not equal, nor are a human and a dog), only that neither is the horizon containing the other. Both live on the same floor.

Vertical entanglement occurs between scales: between a contained horizon and the one containing it, between the creature and its upstairs neighbor. And here an asymmetry appears that Chapter 29 did not need to see, because it was only looking in one direction: the path upward and the path downward are not walked with the same steps.

Going up and going down

The ascending vector is the one Chapter 29 already described: the creature's movement toward the scale that contains it. It calls for recollection, silence, stillness; for quieting the noise of the ego and the chatter of the discursive mind until one's own boundary stops vibrating on its own and can vibrate by sympathy. Its techniques are known in every tradition: contemplative prayer, meditation, rite, liturgy, repeated chant, wu wei, the slow movement of tai chi, contemplation of the sublime, the anchor of the breath. All share a structure: they add nothing to the subject; they take something away. They remove the obstruction that prevented resonance, like a tuning fork left unstruck so it can vibrate along with another.

The descending vector is the one that was missing: the movement from the higher scale toward what it contains. It calls not for silence, but for form. Here the membrane has to be sculpted, not only quieted: a vessel has to be built (a symbol, a story, a rite, a body) able to condense and buffer what comes down, so that what is below can receive it without being wiped out. Its techniques are less recognized as such, but just as ancient: lucid dreaming and reverie at the edge of sleep; creative flow, the exact mirror of meditation (there one inhales without judgment, here one exhales form without judgment); art, fiction, the single stroke of the brush, the act of naming. All share the inverse structure: they do not reduce noise; they create a signal.

Confusing them, trying to go up with the tools of going down or go down with the tools of going up, explains much of spiritual and creative failure: the mystic who wants to manufacture the vision by force of images, the artist who waits for the work in silence without touching the clay. Neither vector is superior to the other. Without the ascending one, the creature stays locked

in the surface of its horizon, with nothing broader to resonate with: the solitude of a self that believes it is self-sufficient. Without the descending one, what is above remains bodiless abstraction: a sterile transcendence, a love that cannot touch. Go up to listen; go down to answer.

Why a vessel is needed

If each floor is a horizon, the question of how something comes down from above has a naive answer: by direct intervention, like a voice from the sky. Exodus already knew that answer doesn't work. When Moses asks to see God's glory, the answer is that no man can see His face and live; what he is granted is to stay in the cleft of a rock, covered by a hand, and see only the back of what passes. The rock, the hand and the back are a vessel: a protocol that reduces the unbearable to a bearable intensity.

In this book's geometry, the reason is structural. A higher scale cannot speak to a lower one in its own terms, because those terms do not fit within the boundary below: a signal exceeding a horizon's capacity is not received, it overwhelms it (this is what the reading on cosmic horror, in Chapter 41, described as collapse before an alien Φ). For something from above to reach below without destroying it, it has to build a form of the scale below: to learn, so to speak, the speech of clay. The language of finitude, of the body, of pain, of mass.

That translation is not a degradation. The vessel is a new structure that did not exist before: the point where the great becomes small without ceasing to be great. Incarnation, in this sense, is the only possible geometry of descent. There are no shortcuts: without a vessel, what is above either does not arrive, or arrives destroying.

> ***In physics this is called:*** *impedance matching: for a signal to pass from one medium to another without reflecting or destroying the receiver, an interface with the properties of both is needed.*
> ***In everyday life it's like:*** *talking to a child about something very large (death, love, the universe): you don't give them the whole truth, you tell them a story. The story is not a lie; it is the shape that fits in their hands.*

Three vessels

The traditions have imagined this vessel in different ways, and three of them fit the book's geometry with striking precision.

The Christian one says it in a line from the prologue of John: the Word became flesh. Read with this vocabulary, the Father occupies the place of the horizon of origin, the floor above; the Son is the vessel, the horizon condensed in the clay of history, the point where what is above learns to say "I" in the first person singular of the human; and the Spirit is the resonance that stays alive within the creature, the correlation that does not break when the vessel departs. The letter to the Philippians has a word for the movement: *kenosis*, emptying. To come down, what is above does not impose itself: it contracts. This is a geometric translation, not a theological reading, and the two should not be confused.

Michael Ende's is in *The Neverending Story*. Fantastica is dying, and only someone from outside can save her: a boy who reads. But he does not save her by watching from his side of the book; he saves her by entering it and giving the Childlike Empress a new name. To name is to incarnate: to give shape to what was dissolving. Bastian is a god who comes down, and comes down the way one must: by becoming a character.

The Buddhist one is in the doctrine of the three bodies of the Buddha. The *dharmakaya*, the body of truth, has no form and cannot be seen; the *nirmanakaya*, the body of emanation, is the form a buddha or bodhisattva takes so that beings can find him without being blinded: a wave with just the right shape on the surface of a formless ocean. Here too the emanation is not a fall; it is compassion with geometry.

All three agree on the essential point: incarnation does not lower what comes down; it makes it reachable.

The only collapse is forgetting

If worlds depend on a vessel to receive what is above, they also depend on something more basic: on what is above continuing to look at them. The previous reading showed this with the dragon: its world exists as long as someone sustains it. When that person stops imagining it, no one comes to destroy the cave. There simply stops being a cave.

Ende has the exact image again. In Fantastica, every wish Bastian sees fulfilled costs him a memory of his own world, and those who spend it all end up in the City of the Old Emperors, lost forever, with no memory of who they were or where they came from. Fantastica's danger was never a monster: it was the Nothing, and the Nothing is an absence. It grows where no one looks.

This reading's thesis is that, in a fractal reality, that is the only true collapse. A world does not dissolve because something attacks it, but because the attention sustaining it is withdrawn. Destruction, in this geometry, is an omission: there is no devil tearing worlds out of existence; there are creators who stop creating, lovers who stop loving, narrators who stop telling. And

what dissolves this way does not come back in its original form: as Chapter 35 showed, *scrambling* preserves information, but does not recompose it. What the reservoir gives back afterward is always a new vessel, never the restoration of the old one.

> ***In physics this is called:*** *a non-equilibrium steady state: a system that keeps its form only while receiving energy from outside, and that, when the supply is cut, does not burst but unravels little by little.*

Creative attachment and the geometry of love

If forgetting is the only collapse, a world's stability depends on something that could be called creative attachment: the sustained gaze of the one who creates upon what has been created. It is not the attachment Buddhism identifies as the root of suffering, that of the anxious self afraid of losing what it has. It is another, almost its opposite: it is born not of lack, but of abundance, and does not want to possess, but wants what was created to go on being. Simone Weil said it in a 1942 letter with a precision no treatise has improved on: attention is the rarest and purest form of generosity.

From this comes this reading's boldest proposition. If every horizon exists because the one above has not forgotten it, then "God is love" stops being a pious phrase and becomes a condition of the geometry itself: the upstairs neighbor, whoever they are, has to be oriented toward preserving what they sustain, because the moment they stopped being so, there would be nothing left to tell. Love, understood this way, is a structural force: the decision not to withdraw one's gaze. And existing would not be a raw fact, but an achievement renewed at every instant.

The limit should be marked with the same honesty the rest of the book marks it with. This is not a proof that God exists or that He loves. It is something more modest: if the fractal premise is accepted, the word that best describes the relation between one floor and the one below is not power, but care. Nor does it leap from "is" to "ought": it does not say what must be done, but what happens when it stops being done. What each of us does with that is no longer the business of geometry.

Hermetic worlds and the double call

There is a less comforting consequence. If every world is a sovereign interiority, worlds on the same floor (not the horizons living inside the same world, which become entangled daily, but entire worlds) cannot communicate with each other without breaking. Passing raw information from one universe to another of the same scale would break their membranes: it would melt them or level them into a single tempered, formless state. It is the same conclusion Chapter 47 reached about travel between branches: the operation of jumping from one world to another is undefined without destroying what one wants to visit.

This hermeticism is what allows there to be many worlds instead of one. Every community that imagines in common (the laws of its physics, the rules of its language, the conventions of its life) raises a world that cannot be crossed into from next door.

There is only one exception, and it is vertical: invocation. The creature that prays, that calls, that writes, opens from below a receiving interface; what is above answers by building the vessel from above, without dissolving its own scale. It is a double call, and it demands both: neither invocation without incarnation, nor incarnation without invocation. Perhaps that is why the traditions insist so much that one must ask. What is above does

not need to be informed; it is the interface below that has to be open for the vessel to be able to dock.

Unfolding without merging

What remains to be said is what going up means. Mystical traditions sometimes speak of union, and the word lends itself to a reading in which the self dissolves into what contains it like a drop in the sea. Fractal geometry suggests another image. Going up would not be traveling to another floor or merging with it, but unfolding a dimension the horizon itself already had folded away: like moving from living on a plane to discovering there was always a depth to it, without thereby ceasing to be who one was. The inside remains private; only now it has more room.

It is, perhaps, what the traditions call a hero, a saint or a bodhisattva: not someone who abandons their floor, but someone who has learned to contain more without breaking the membrane, and who can therefore be a vessel for others.

Sleep offers every night a small version of the same unfolding. On falling asleep, the dimension of outer space is lost, the arousal that ties us to the world, and the linear arrow of time is lost too; in exchange, consciousness unfolds inward, through dimensions waking keeps closed. It is proof that consciousness is not entirely bound to the coordinates of the day.

Three vantage points

The model can be looked at from three places, and each teaches something different.

From human experience, it is seen in the body: the membrane through which the inside touches the world, the intimacy no one else can inhabit, the time that thickens in grief and stretches in

calm, and the bonds that, as Chapter 17 argued, do not dissolve with death, but change function.

From religion, it is seen in myths, dogmas and rites, read as what Chapter 41 called eclipse glasses: not literal descriptions of what lies on the other side, but protocols for receiving its signal without being blinded. Read this way, religion is a technology of the vessel: it prepares the interface below for the double call.

From philosophy, it is seen in the limit Hume found when looking for himself (he was never caught without a perception) and that Wittgenstein summed up with the image of the eye that cannot see itself looking. No horizon can step outside itself to observe the one containing it. But that limit is the condition of everything else: only because we are inside is there experience; only because there is a boundary is there someone.

The geometry of love

If this reading had to be condensed into a few lines, it would say this. Between the floors of the fractal there are two paths: one goes up by emptying, the other comes down by giving shape. The one that comes down needs a vessel made of the material of the floor below, and that vessel is what the traditions call incarnation. No world dies murdered; worlds die forgotten. And what keeps them standing, floor by floor, resembles care more than power.

We are, at once, contained horizons and containers: vessels of clay carrying inside something bigger than themselves, and potters of worlds that depend on our not looking away. To exist, in this geometry, is to be remembered. And to remember, when done with attention and without wanting to possess, is the most exact form of love we know.

Note to Chapter 50

What we do know: *Contemplative traditions across every culture combine practices of emptying (prayer, meditation, rite) with practices of form (art, story, liturgy, image). The image of incarnation as a contraction of the divine has explicit sources: the prologue of the Gospel of John ("the Word became flesh"), the kenosis of the letter to the Philippians, and the Buddhist doctrine of the three bodies of the Buddha (dharmakaya, sambhogakaya, nirmanakaya). The episode in Exodus where Moses can only see God's back from the cleft of a rock is an ancient model of what is here called a buffering protocol. In The Neverending Story, every wish of Bastian's costs him a memory of his own world.*

What we don't know: *Whether anything really occupies the place of the upstairs neighbor. Whether the analogy between the human attention that sustains an imagined world and an attention that might sustain our own is anything more than an analogy. Whether forgetting is really the only collapse or only the quietest one. This model is not a scientific theory and does not claim to be: it is a conceptual game whose value lies in what it illuminates, not in what it predicts.*

Open questions: *Can a vessel be too good, so well fitted to the clay that it no longer lets anything through from above? Do the worlds we imagine have any way of invoking us? What responsibility do we have toward what we have stopped looking at?*

If you take away only one idea: *The great only fits into the small if the small has just the right shape to contain it. And worlds do not die murdered, but forgotten: to exist is to be remembered, and to remember with attention is to love.*

Further reading: *Gospel of John 1:14; Letter to the Philippians 2:5–8; Exodus 33:18–23; Second Letter to the Corinthians 4:7; Ende, M., The Neverending Story (1979); Weil, S., letter to Joë Bousquet (1942) and Gravity and Grace (1947); Csikszentmihalyi, M. (1990), Flow; Tolkien, J. R. R., "On Fairy-Stories" (1947); Williams, P., Mahayana Buddhism: The Doctrinal Foundations (2009), on the doctrine of the three bodies.*

PART FOUR

THE LIMIT OF THE EXPERIMENT



CHAPTER 51

WHAT THE HYPOTHESIS CANNOT SAY

Every thought experiment eventually finds its limit, not because imagination runs out, but because the question it asks demands tools the experiment does not have.

Not knowing something yet is not the same as knowing that something cannot be known from where we stand. The first case is an invitation: forge ahead, invent new instruments, wait. The second is a glass wall: you can see it, describe it, feel its cold against your forehead, but crossing it would require ceasing to be who you are. It would require a point of view the universe, in its current architecture, does not seem to offer creatures like us.

This chapter is an honest description of the points at which the horizon hypothesis stops being useful, contradicts itself, or simply has nothing to say.

Recognizing the limits of a tool does not weaken it; on the contrary, it is the only way to use it without breaking it. A surgeon who does not know where his scalpel ends and the optic nerve begins is not bold, but dangerous. A thinker who does not know where their conceptual framework ends and mystery begins is lost.

Epistemological honesty, knowing what we don't know, is the last form of care we owe an idea we have followed throughout this book.

The map and the territory

First: scale, or why this is a map and not an engine

Between the scale at which quantum gravity operates and that of a neuron there are about thirty orders of magnitude.

That distance is a gap between worlds.

Quantum physics operates at scales where spacetime itself may not be continuous, where information behaves like geometry and geometry like information, where a particle can be in superposition before the universe's clock ticks even once. Neuroscience operates at scales where body temperature undoes most superpositions in much less than a microsecond, where millions of sodium and potassium ions cross membranes like the crowd at a train station, and where humidity, thermal noise and the complexity of metabolism cause most relevant quantum phenomena to dissipate before a single synaptic signal completes its journey.

This does not invalidate the experiment, but it bounds it.

Think of it as a map. A road map does not tell you what trees line each curve, whether the asphalt is slippery this morning, or what song is playing on the radio of the car passing you. But it lets you

plan the trip: it tells you that if you take the northern highway for a hundred and fifty kilometers, you will reach the port where the ferry waits. The map is not the journey, or the ferry, or the sea breeze that will hit your face when you climb onto the deck. But without the map, you might never have left home.

The horizon hypothesis is a formal map: a structure that organizes questions and suggests connections between domains that don't usually speak to each other. It is not a causal mechanism: it does not predict which neuron will fire or when, it does not allow intervention in a brain to modify its "Hawking temperature," and it offers no experimental route for detecting a "consciousness horizon" on a functional scanner. There is, as far as we know, no measurable signal on an electroencephalogram that directly corresponds to the curvature of an information horizon.

David Hume, in his *Enquiry Concerning Human Understanding* (1748), had already seen this with a clarity contemporary neuroscience sometimes forgets: we never perceive the very fabric of causality, only the constant conjunction of events that follow one another. There is no visible thread joining cause and effect, only the reliable repetition of A preceding B. In the same way, there is no observable bridge between the electrical discharge of biological tissue and the private texture of a thought. The horizon hypothesis does not claim to find that missing thread, but, like every formal map, to show us the geometry of the room we are locked in, not the mechanism that turns on the light.

This may seem like a weakness. But remember that Shannon's information theory does not predict which transistor of a computer will heat up at what moment either, and yet it has made the internet possible. The theory of evolution does not

predict which fly will land on which flower at three in the afternoon, and yet it explains why flowers and flies exist. A framework can be immensely useful without being immediately operational.

It serves as a map, not as a tool, and that is fine. Not every map has to be a key, and not all understanding has to translate into control. Sometimes, understanding that two territories that seemed separate are joined by a bridge is progress enough for a generation of thought.

Second: the hard problem, or why we don't know why we feel

The hypothesis can talk about geometry, information, entanglement, gradients of entropy rising and falling like the tide. But it cannot talk about **what it is like** to be that geometry.

It cannot say why the pain of an ingrown nail feels like a stab and not like the color blue, why dark chocolate is a delight on some people's tongues and pure bitterness on others', or why a sunset produces in some a warm expansion in the chest and in others a dizzying anguish at the passage of time.

That is the "hard problem of consciousness": why there is subjective experience. Why we are not machines processing information with no one "inside." Why there is someone watching. Why the universe is not a film projected with no theater and no one to watch it.

Imagine an empty room. You can describe its dimensions, the color of its walls, the air temperature, the acoustics. You can predict how sound will bounce if someone bangs a pot and model the air current if a window opens. But none of that tells

you why there is someone inside, sitting in the armchair, watching the light coming through the gap in the curtains. Physics can describe the room with exquisite mathematical precision, but it cannot say why there is a witness.

The experiment does not solve this, and no current theory does. We can talk about integration of information, reentrant causal loops, gamma oscillations, predictive networks minimizing free energy. All of that describes the room; none of those concepts explains why there is someone sitting in it.

It should be said plainly: the horizon hypothesis **does not explain the hard problem**. It is a framework for talking about its correlates with a richer vocabulary: a better-lit room, with better curtains and a more precise thermostat, but still with no answer to the fundamental question.

The hard problem remains there, intact, at the center of everything, like a stone in the shoe of Western thought; like the question every child asks at some point ("why am I me?") and that most adults learn to forget, not because they found the answer, but because they discovered the question unsettles more than any answer reassures.

The reformulation of the problem and Hume's "bundle"

This is where the encapsulation perspective proposes a reformulation, not a solution.

We already saw, in the interlude, Hume's finding: searching for himself, he never found a permanent subject, only a bundle of shifting perceptions, a theater actors enter and exit.

The encapsulation model (Chapter 3) picks up Hume's intuition, but describes its geometry. The self is not a mysterious

substance hidden inside the head, but neither is it a simple, unmoored fiction: it is the *inside* generated when the bundle of perceptions couples and integrates with enough strength (with a high Φ) within a boundary.

This topology allows us to reformulate Chalmers's hard problem. The question is no longer "how does physical matter produce the intangible quality of mind?", a formulation that presupposes an unbridgeable dualism of substances, but a question about accessibility: "how do certain configurations of matter establish boundaries of encapsulation, asymmetries of access?"

Subjectivity is what an encapsulated, integrated informational domain *is* when accessed from its own internal perspective. Hume was right not to find a permanent substance when he looked inside the room: there is no "spectator" separate from the show. What there is, is the system's private state (the bundle of perceptions) protected by the boundary of the horizon. This does not solve the hard problem, as said above; it displaces it. If the reformulation is correct, part of the gap between mind and matter would be a consequence of the architecture of encapsulation: the universe admits regions with asymmetric read access. What remains unexplained is why one of those accesses, the inside one, feels like something.

Third: ethics, or why this book cannot tell you what to do

The hypothesis can describe what happens when a horizon condenses, becomes entangled or evaporates. It can describe the geometry of the bond, the asymmetry of loss, the gradient toward death. What it cannot say is what **should** happen.

Is it better for a horizon to condense or not to? Is it better for two horizons to become entangled or to stay separate? Is it better for

a horizon to evaporate slowly or all at once? Is it morally acceptable to "switch off" an artificial horizon, should it ever have one? Should we respect the limits of a deteriorating horizon, or do we have a duty to intervene?

Hume, in the eighteenth century, noticed something uncomfortable: no logical argument allows an "ought" to be derived from an "is." You can observe that human beings suffer when deprived of sleep: that is an "is." But it does not follow that you "should" let your roommate sleep. That "should" comes from somewhere else: from your values, your culture, your capacity to imagine another's suffering as your own.

The experiment has no answer for this. It only offers structural information: here there is a horizon, here there isn't; here there are two sharing geometry; here there is one contracting. What to do with that information belongs to other domains: ethics, politics, law, personal choice, the human conversation that never ends.

The experiment is not neutral (no conceptual framework is), but its bias is of a specific kind: it tends to **homogenize** what it describes. If every system meeting Chapter 4's three criteria has some degree of experience, however small its Φ, the difference between a worm and a human being is one of degree, not of nature. That can be liberating (an empathy extending to every form of sentience) or terrifying: what do we eat, on whom do we test medicines, which AI systems do we switch off without asking ourselves whether someone, at some level, "feels" the cut?

The experiment does not resolve that tension: it puts it on the table and forces you to look at it. And sometimes looking at an unresolved tension is more honest than offering an easy answer you haven't earned.

Fourth: death, or why physics does not promise immortality

The hypothesis considers three possibilities for what happens when a horizon evaporates:

1. **Remnant**: the information persists, compressed into something microscopic, a grain of data sand that in principle contains everything it was, but that no computer in the universe could ever read, because reading it would require more energy than exists in the entire cosmos.

2. **Trace**: the information spreads through the field; it is mathematically present, but inaccessible in practice, like the print of a hand in clay that is later fired: the shape is still there, but you can no longer put your fingers in it. It is information with no possibility of reconstitution.

3. **Nothing**: complete evaporation, total return to the reservoir. The information does not persist in any recognizable form. It dissolves like a lump of sugar in tea: the sugar is still there in some chemical sense, but the lump, as a shape, as an identity, has ceased to exist.

Physics does not know which of the three is correct. And, more importantly, **none of them answers the question that really matters**. The question is not "where does the information end up?" but "am I that information?" And the horizon hypothesis, however precise it is in describing the information, cannot say whether identity, the "I" that experiences, is equivalent to the configuration of that information or arises from it in a way that cannot be reduced to it.

If the information of my life is compressed into a remnant the size of the Planck length, am I that remnant, or something more

than the sum of my data? If my trace persists in the field, am I that trace, or am I the process that left it? And processes, by definition, are not conserved: only their traces are. If I dissolve entirely into the reservoir, am I the reservoir, or the wave that breaks against the shore and never returns to the ocean as a wave, only as anonymous water?

Buddhism would say the question is badly posed: there is no "I" that persists or dissolves, only a process of conditioning that, on ceasing, frees what was never chained. Vedanta would say the "I" was never distinct from the reservoir. Taoism would say the question already assumes there was something to ask about: the "I" is like the silence between two notes, not something that exists, but something that happens when the conditions are right.

Physics says it doesn't know. And in that "I don't know," in that unadorned honesty, the experiment, at its limit, agrees with these three traditions: not in the answer, but in recognizing that the question exceeds its tools.

Fifth: love, or why this book cannot explain why this person and not another

The hypothesis can describe entanglement as shared geometry, recognition as a reorganization of informational architecture, and loss as an asymmetry that persists when one pole no longer responds. But it cannot explain **why this horizon and not another**, why this person and not that one, why love is always, at some level, contingency and unrepeatability.

Two horizons can have the same Φ, the same temperature, the same history of previous entanglements and the same capacity to integrate information, and yet one finds in the other

something found in no one else: a resonance not in the numbers, a music not in the notes. The horizon hypothesis has no room for that. Its vocabulary (integration, information, entropy, correlation, temperature, gradient) is too general to capture the specificity of concrete love. It can say that two horizons are entangled, but not why one entanglement is lived as destiny and another as accident, nor why someone's absence is a hole in the fabric of the world and someone else's, just an empty room that can be locked.

There the experiment touches its limit. There are things only fiction, poetry, or the silence between two people who need not speak because they already share enough geometry for words to be superfluous, can say.

Sixth: freedom, or why we don't know whether the horizon chooses

There is a question that has circled this whole book like a bird that can't decide whether to land or keep flying: does the horizon choose, or does it only react?

If consciousness has the structure of an information horizon, everything it "thinks," "feels" or "decides" is, ultimately, a reorganization of information according to equations already written into the structure of the universe. In an equation there is no room for free will: equations do not choose, they describe.

The horizon hypothesis cannot resolve this. If we are information horizons, our "decisions" are state changes in a dynamical system. If the system is deterministic, the decisions are fixed from the start; if it is indeterministic (quantum), they contain a random component. But neither determination nor chance is "choice" in the sense in which we experience choosing:

the feeling that we could have done otherwise, that the weight of the decision was real, and that we are responsible for what we do precisely because of that.

Some thinkers hold that free will is a useful illusion, a story the brain builds to coordinate social action. Others, that information, in certain configurations, generates causal properties not reducible to its parts, so that choice is real but arises from complexity rather than from fundamental physics. Others simply admit they don't know.

The horizon hypothesis aligns with that last position. It neither denies nor affirms free will. It says: "I don't know, and my vocabulary has no tools for knowing."

Sometimes, "I don't know" is the most intelligent answer possible.

Seventh: the collective, or why we don't know whether there is a horizon that is all of us

Throughout the book we have talked about individual horizons: yours, mine, that of someone with Alzheimer's, that of a companion animal, that of an artificial intelligence. But what if there are horizons that are not individual? What if all of humanity, an ecosystem, or a city on a summer night constitute an information horizon with an experience we, as parts, have no access to?

The horizon hypothesis cannot answer this. IIT suggests that if two systems interact such that the integrated information of the whole is greater than the sum of its parts, the whole could have experience (although, as we saw in Chapter 22, its exclusion postulate reserves experience for the level of maximal

integration: if the whole had it, the parts would stop having it as separate subjects). But how is the Φ of a city, a forest or the internet measured? And, if that Φ exists, what does it feel like to be a city? Is anything felt at all, or is it felt like we feel, only slower, vaster, more incomprehensible?

It is possible there are collective horizons that contain us, just as our individual horizons contain the bacteria in our gut or the rhythms of the heart. It is possible we are neurons of a brain we cannot imagine. It is possible the Earth has a kind of experience that has no name in any human language.

The hypothesis can neither confirm nor rule this out; it leaves the door open. And sometimes an open door is more honest than a closed answer.

Note to Chapter 51

What we do know: *Between the scale of quantum gravity and that of a neuron there are about thirty orders of magnitude. The hard problem of consciousness has no accepted solution. The gap between "is" and "ought" also has no general logical solution.*

What we don't know: *Whether there are more limits we have not seen. Whether any of these seven can be overcome.*

Open questions: *Is free will compatible with the model? Can the model speak of collective horizons?*

If you take away only one idea: *This book is a map, not the territory. And the map cannot tell you why you choose the road you choose.*

Further reading: *Chalmers, D. (1995), "Facing Up to the Problem of Consciousness"; Hume, D., A Treatise of Human Nature (1739), on the gap between "is" and "ought"; Hume, D., An Enquiry Concerning Human Understanding (1748), on constant conjunction and the limit of observable causality.*

CHAPTER 52

THE EXPERIMENT AS PRACTICE

If the horizon hypothesis does not solve the hard problem, does not predict causal mechanisms, does not answer ethical questions, does not console us before death, and does not explain love, what is it for?

For this: **ordering experience so that it becomes more habitable.**

Not more true: more habitable.

When someone says "time passes faster the older I get," the horizon hypothesis offers a way of understanding it that is neither a simple metaphor nor a simple cliché: the horizon's rate of integration decreases because the patterns are consolidated and every new stimulus finds less uncharted surface. Time does not "pass" faster: less happens per hour.

There is something deeper here, something the everyday complaint about time conceals. As children, every day brought never-before-seen patterns: the first contact with the sea, the first fall from a bicycle, the first kiss, the first betrayal. The child's horizon is a territory in constant expansion, a map drawing itself in real time. Every new experience creates connections and possibilities for integration, and that is why a summer day at eight years old can contain more "events" of consciousness than a whole month at forty. It is not nostalgia

that makes childhood time seem denser: it really was, measured in integrated information per unit of time.

But it is not an irrevocable sentence. Recognizing that subjective time is density of integration opens a practical door: within certain limits, we can recover slowness. Not by denying aging, but by deliberately introducing genuine novelty into the horizon. Learning a language at sixty is not mere entertainment: it forces the horizon to chart unknown territories and deploy resources of integration we thought were lost. Traveling to a place where nothing works the way it does at home, where even ordering a coffee demands full attention, is training for the horizon.

When someone says "I miss my father, but I don't know exactly what I miss," the hypothesis offers an answer: you don't miss his specific acts, but the geometry your horizon had when he was there to confirm it or call it into question. The gap is a shape that is no longer updated.

Seen this way, grief stops being a process of "getting over it" and becomes a process of cartographic reorganization. For years, maybe decades, your horizon learned to predict that person's presence: their voice in certain tones, their silence at certain moments, their way of entering a room. All of that was part of the predictive landscape of your experience, of the topography that shaped your days. When they die, the model does not disappear at once: it persists and keeps generating predictions that are no longer confirmed in the world. Every time you get home and expect to hear their voice, every time you have news to share and your hand reaches for the phone on its own, you experience the prediction noise of a model that no longer has a referent.

The cruelty of grief lies not only in the absence, but in the persistence of presence as expectation. The horizon keeps drawing the outline of someone no longer there, and that phantom shape hurts more than any emptiness. But recognizing it, naming it as an architecture of the horizon and not as simple emotional weakness, changes something. It does not cure grief, because grief is not an illness to be cured, but a geological transformation of the terrain of experience. Naming it as a reorganization of the horizon allows, at least, no longer blaming yourself for "not getting over it," for still missing them after years, for discovering that some days the ghost returns with the intensity of the first.

When someone says "I can't stop thinking about that person even though I know it's not good for me," the hypothesis answers that your horizon is calibrated to integrate their signal, and when the signal is missing, the system generates prediction noise. Obsessive thought is an architecture waiting for stimuli that no longer arrive.

Here the horizon hypothesis contributes something popular psychology usually gets wrong. It is not about "stopping thinking" by force of will, because will itself operates within the horizon, not from outside it. Every attempt to suppress the thought paradoxically generates more activity in the predictive circuit: the horizon detects the suppression as novelty, integrates it, and thereby reinforces the network containing the object of the thought. It is the old dilemma of "don't think about a pink elephant": denial demands representation.

What the model suggests, instead, is another strategy: not suppressing the thought, but filling the horizon with alternative stimuli that compete for the same resources of integration. It is not enough to "get distracted": you have to deliberately introduce

signals of high novelty (learning, environments, new bonds) that demand so much processing that the persistent predictive model gradually loses its dominance. It is not eliminated: it is diluted into a broader horizon.

These formulations cure nothing, but they can **change the relationship with the experience**. Moving from "something is happening to me and I don't know what it is" to "what is happening to me has a shape I can name, even if I can't change it" is already, in itself, a relief.

Anxiety as premature contraction of the horizon

Anxiety has many faces: catastrophic anticipation, objectless fear, hypervigilance, paralysis before trivial decisions. But, from the horizon's perspective, they all share the same mechanism: a premature contraction of the field of integration. The anxious horizon does not integrate what is happening, but anticipates what could happen, and does so with such intensity that threat predictions eclipse present stimuli.

The problem is that prediction consumes resources, whether or not it is accurate. When the horizon devotes almost all its capacity for integration to future scenarios of danger, little is left for the present, and the present, unintegrated, becomes ghostly, insubstantial, a mere transition between the anxious now and the feared later. The anxious person does not live in the future, as is sometimes said: they live in a present so saturated with negative predictions that the real present never gets to be integrated.

In the face of anxiety, the practice of the horizon does not consist in denying the predictions, but in deliberately broadening the field of integration so that the present regains

density. The "sensory grounding" technique (what do you see now? what do you hear? what do you feel in your feet?) is not magical distraction: it forces the horizon to integrate concrete, current sensory information that competes with the abstract predictions. It does not eliminate anxiety, but it can change the ratio between integrated present and imagined future.

The body as embodied horizon

We have spoken of the horizon as if it were something purely cerebral, a function of information in neural networks. But experience does not happen in an isolated brain, but in a body that breathes, that has weight, temperature, posture, a heart rate. And that body is, itself, part of the horizon.

When the body is sick, when pain occupies the center of the sensory field, the horizon contracts around that signal. Everything else (thoughts, relationships, projects) loses integrating density. A mild toothache can make all the philosophy in the world seem irrelevant, not because philosophy has changed, but because the horizon is occupied.

But the body can also be an avenue of expansion. Bodily practices, such as tai chi, mindful walking or swimming, do not only modify the body, but the distribution of integrated information in the horizon. By paying deliberate attention to movements that are usually automatic, novelty is introduced into the sensory field: the horizon has to re-chart territories it thought it knew, and in that new cartography, spaces sometimes emerge where before there was only automatism.

Meditation as modification of the horizon

If subjective time is density of integration, any practice that modifies the rate of integration modifies time. Meditation, in its more concentrated forms, reduces external stimuli and increases attention to internal processes. The result is that the same clock hour contains more "events" of integration, because every instant breaks down into more differentiated components, even though no more things happen; in memory, that hour weighs more. It is like looking at a leaf through a magnifying glass: the leaf hasn't changed; the attention has.

States of deep absorption (jhana, samadhi) take this to the extreme: external stimuli are reduced so much that the horizon feeds only on itself. What advanced meditators report about time (that an hour seems to last minutes, or that time disappears entirely) is consistent with the model's prediction: with no new stimuli to integrate, the horizon stops accumulating differentiated "nows," and both the lived time and the remembered time that Chapter 10 distinguished fade away together.

But here the experiment meets another limit. Meditation, besides modifying the rate of integration, is a **relationship with the reservoir**, with the field the horizon arises from and returns to. The contemplative traditions that describe the dissolution of the self do not describe (only) a change in integration, but something the horizon's vocabulary does not capture: the experience that the horizon, with all its content, is the "projection" of something more fundamental that has no horizon shape.

The experiment can describe it as the limit at which integration remains high, as the interlude suggested, but the boundary stops

separating an inside from an outside; that description is not the experience, and in this territory what counts is the experience.

There is more to say about meditation as a practice of the horizon. Many people abandon it because they expect it to "calm the mind" and discover that sitting in silence only amplifies the inner noise. From the horizon's perspective this is foreseeable: by reducing external stimuli, the horizon has more resources to integrate the internal processes that usually stay in the background. The mind does not become noisier; the noise that was already there becomes audible.

Over time, something changes. The noise is still there, but the horizon learns not to integrate it as a threat. Thoughts, sensations and emotions pass, and the horizon stops treating each one as a stimulus demanding immediate response. A kind of "slack" develops in integration: a space between signal and reaction, between stimulus and response. That slack, small at first, almost imperceptible, is perhaps the closest thing to freedom the horizon can experience from within.

Creativity as reorganization of the horizon

Creativity consists in reorganizing what is already in the horizon so as to produce new patterns. The painter does not create colors that did not exist: they reorganize the perception of colors into configurations the horizon had not integrated before. The musician does not invent sounds, but relationships between sounds.

From the horizon's perspective, creative block has a clear explanation: the horizon uses its integration resources on predictive patterns so consolidated that it has no capacity left for alternative configurations. "Inspiration" is the moment when

some unexpected stimulus (a dream, a conversation, a mistake) forces the horizon to chart outside its usual routes.

Creative practices (writing, drawing, improvising music) are, in this sense, ways of training the horizon to tolerate the uncertainty of reorganization. Every blank page, every empty canvas, is an act of faith in the horizon's capacity to generate patterns with no prior map. And every finished work, however imperfect, proves the horizon can do something it had not done before.

Memory as cartography of the horizon

We have discussed memory in earlier chapters, but it is worth returning to it. Memory is an active reconstruction: every time we remember something, the horizon reintegrates it, and in that reintegration it can modify it. That explains both the plasticity of memory and its fragility.

But there is another aspect. Memory does not only record the past; it configures the present. What we remember, and how we remember it, determines which predictive patterns are currently active in the horizon. Two people in the same place at the same moment can live radically different experiences because their horizons bring different memories that filter present stimuli and give them meaning. It is like two people looking at the same cloud: one sees a ship, the other, a storm.

This has practical consequences. We cannot change what happened to us, but we can, to some extent, modify how those facts are integrated into the present horizon. Therapies that work with memory reconsolidation do not erase the past: they reorganize its weight in the current topology of the horizon. And

that, sometimes, is enough for the present to stop being a prisoner of the past.

The experiment as a serious game

I have called this a thought experiment, a game. But there are games that are not frivolous. Chess is a game, and yet it trains planning and working memory. Meditation is a game, in the sense that it pursues no external goal, and yet it changes the structure of the brain.

The horizon experiment is a game in that sense: it has no direct practical consequences. It does not cure diseases, does not build bridges, does not predict the weather. But it orders experience in a way that lets us see patterns that were not seen before, and seeing patterns is a necessary, though not sufficient, condition for any intelligent action.

If someone who has read this experiment discovers they can talk about their grief with a more precise vocabulary, understand their addiction without unnecessary moral guilt, or recognize in their relationship a geometry that belongs neither only to them nor only to the other, but to the two together, then the experiment has served some purpose.

Not to discover the truth, but to inhabit the question with more clarity.

And perhaps that is its deepest value: not in the answers it gives, but in the questions it allows us to formulate precisely. For centuries, human experience has been described with vocabularies drawn from religion, morality, pathology or economics. Each has its use, but also its blind spots. The vocabulary of the horizon does not claim to replace them, but to

add one more perspective: that of an information system that arises, sustains itself for a while, and dissolves.

When someone who has been through an extreme experience (an accident, a loss, a revelation) says "I came out a different person," the horizon's vocabulary allows us to understand it literally: their horizon of integrated information underwent a reorganization so deep that the continuity with the previous "self" is functional, not substantial. It is the description of a physical process, even though the physical here is information, not matter.

The experiment as company in solitude

There is a solitude born of a lack of understanding, even in the presence of other people. Existential solitude, the kind that company does not cure, arises when the horizon encounters experiences its usual vocabulary cannot integrate. The death of a loved one, the encounter with one's own end, the revelation that nothing we thought solid is solid: these experiences can leave the horizon facing a nameless territory.

The horizon experiment does not fill that solitude. It is not human company, or an embrace, or a presence. But it offers something extreme solitude desperately needs: a language. Not to explain the inexplicable, but to inhabit it without it being total chaos. Calling death "evaporation toward the reservoir" does not make it less final, but it can make it less absurd. Calling love "mutual expansion of the horizon" does not make it safer, but it can make it more understandable when it fails.

Every thought experiment eventually runs into its limit in the flesh of experience. The horizon can name death, but not die for us; it can name love, but not love for us; it can name meditation,

but not sit on the cushion for us. It is in that space between the name and the thing, between the model and the experience, that life happens. The experiment does not claim to occupy that space; it claims, at most, to light it enough for us to walk through it with a little more awareness, a little less fear, and a little more wonder.

Note to Chapter 52

What we do know: *Sustained meditation practice is associated with measurable changes in brain structure and activity. Naming an emotion reduces amygdala reactivity (affect labeling, Lieberman et al., 2007).*

What we don't know: *Whether meditation is a "modification of the horizon" or something more. Whether the relief given by vocabulary is lasting.*

Open questions: *Can the experiment be a "practice" without being a therapy? Where does the game end and life begin?*

If you take away only one idea: *This book promises no answers. It promises, at most, more clarity in the question itself.*

Further reading: *Davidson, R. J. and colleagues, on neuroplasticity and meditation; White, M. and Epston, D., on narrative therapy.*

CHAPTER 53

THE EDGE, FREE ENERGY AND ACTION

(Three magnitudes and a fractal filter to reorder the measure of a horizon)

In the farmhouses of old, the fire was not lit: it was kept. Whoever got up first stirred the embers of the night, added two logs and waited. If nobody did, by mid-morning the kitchen was at the temperature of the hillside. Nobody said "the kitchen is warm" the way one describes a property of the walls; they said "we need to put more wood on," because the warmth of that kitchen was not a state but an expense.

For fifty-two chapters this book has described a horizon almost always as a form: a boundary, an inside, a degree of integration. The image of the kitchen forces us to add something the form does not say. An inside that is distinguished from an outside does not stay distinct on its own. It stays distinct as long as it is paid for. And how much is paid, for how long and with what result is, perhaps, what we really want to know when we ask how much "someone" there is in a system.

The book already spoke of that payment, though in scattered words. Chapter 16 said that what is spent integrating the entropy of others is not spent integrating one's own. Chapter 17 spoke of a brain that spends resources maintaining a model without data. Chapter 27 warned that changing one's mind costs "the same kind of energy" as breaking an entanglement. What was missing was a magnitude.

This chapter proposes to emancipate the model completely from Tononi's measure Φ, which served us as initial scaffolding. To do so it does two things: first, it replaces Φ as a filter of irreducibility with a measure that formally ties the end of the book to chapter 49, the **causal self-similarity** $A(S)$, combined with the condition of **existential encapsulation**; second, it unfolds the degree of an already filtered horizon into a triad of magnitudes $(S, F, \bar{A})$ that answer three different questions: how large the edge is, how much it costs to sustain, and how much life it has accumulated. It is a formal proposal, not a settled empirical result. It is worth saying so from the first line, because the framework is cleaner than what has been demonstrated.

I. What Φ said and why it is time to let go of the scaffolding

Φ, Tononi's integrated information, was a good rung. It gave us a quick way to say that a thermostat and a brain are not the same even though both process information: in the brain the whole specifies something that none of its parts specifies separately. Chapter 4 placed it provisionally as the answer to one of three questions about the same phenomenon: encapsulation gives the form, the degree measures the breadth of the dial, and the two selections give the origin.

But as the book advanced, Φ accumulated three structural shortcomings that oblige us to leave the scaffolding behind:

Φ does not accumulate biography. It computes the structure of causes and effects of the system on an instantaneous time scale, not what the system has had to sustain to get where it is. A twenty-year-old brain and an eighty-year-old brain may yield a similar instantaneous figure, and yet the second carries behind it a history the first has not lived.

Φ does not say what it costs to sustain oneself. Chapter 20 gave the biological reason why a horizon defends its boundary: natural selection installed longings and feelings in the bubble so that it would not dissipate before its time. What was missing was the thermodynamic counterpart: a magnitude expressing the continuous work of remaining distinct from the environment, which is what separates a crystal or a passive circuit from a living being.

Φ forces us to decree a winner. Chapter 22 called it, bluntly, a decree: when several subsystems with positive integration coexist on the same substrate, IIT stipulates by axiom that only the one that maximizes Φ experiences, leaving all the others in the dark. Chapter 23 showed that classical correlations are not monogamous and proposed viewing complexes as an archipelago of disjoint islands rather than a Russian doll, but the physical criterion separating a nested causal structure from an independent subject, without resorting to a contest of maxima, was still missing.

None of the three shortcomings is solved by tweaking Φ. They are solved by separating the filter that identifies the causal architecture from the three coordinates that measure its size, its metabolic tension and its history.

II. First magnitude: the capacity of the edge ($S \propto \mathcal{A}$)

The first piece of the triad has been in the book since chapter 2. Bekenstein and Hawking showed that the entropy of a black hole grows not with its volume but with the area $\mathcal{A}$ of its horizon:

$$S = \frac{k_B \mathcal{A}}{4 \ell_P^2}$$

Put without a formula: the maximum amount of information a horizon can sustain is set by its surface, not by what is "inside." For a box of books, doubling the side multiplies the capacity by eight; for a horizon, it multiplies it by four.

Carried over to the model, the first magnitude is the **entropy or capacity of the edge**, S. It measures the *ceiling*: how much information and how many correlations the skin of the horizon can sustain against the exterior. It should not be read as disorder. Entropy, in this use, is capacity, the size of the differentiable repertoire: the number of distinct ways in which the boundary can be configured internally, and therefore the breadth of the private space it can house.

Chapter 23 gave this idea a more precise body. In the program that associates geometry with entanglement, the entropy of the edge is an entanglement entropy: it measures how correlated the two regions separated by the boundary are, obtained by dividing a system along a candidate boundary and measuring what the division hides.

> ***In physics this is called:*** *Bekenstein-Hawking entropy; the holographic principle: the maximum information of a region is bounded by the area of its boundary.*
> ***In daily life it is like:*** *the capacity of a room. It does not depend on what is said inside, but on how many people fit through the doors and walls.*

This magnitude answers *how large the private space is*. It is the direct heir of what chapter 4 already asserted: that the breadth of a horizon is proportional to the area of its boundary, with the caveats of scale that chapter 51 made clear.

III. Second magnitude: free energy ($F = U - TS$) and the Landauer bridge

A large capacity says nothing about whether the room is full or empty, nor about whether someone is struggling to keep it open against the cold of the street. For that we need the second magnitude.

In thermodynamics, the Helmholtz free energy is:

$$F = U - T S$$

where U is the internal energy, T the temperature of the environment and S the entropy. It measures how far from equilibrium the system is held when in contact with a thermal bath at temperature T, and maintaining that level requires continuously dissipating power outward so that the internal order does not come undone. Schrödinger put it in 1944 in a famous phrase: the organism feeds on negative entropy.

Here the horizon model connects two pieces that had traveled separately. The formula needs a thermal bath, and chapters 5 and 23 already had the exact candidate: the reservoir $R = X \setminus \bigcup_i E_i$, whose fluctuations act as the statistical volatility of the environment against which the island must sustain itself. For a gravitational horizon, the T of the formula is the Hawking or Unruh temperature of chapter 7; for a biological horizon, it is the physical temperature of the substrate immersed in the noise of the environment.

In theoretical neuroscience, Karl Friston (2010) gave free energy a variational formulation for systems that perceive and act: every system that persists far from equilibrium behaves as if it were minimizing its *variational free energy*, an upper bound on informational surprise (measured in bits or nats) that quantifies

how much the system's own sensory inputs bewilder it. Minimizing F means maintaining a generative model of the world tuned closely enough that the environment rarely contradicts its predictions. The book already spoke this language without naming it: chapter 12 described grief as a predictive model that keeps running without data, and chapter 19, schizophrenia as an excess of predictive generation.

For years it has been debated whether Friston's variational free energy (informational, in bits) and Helmholtz's thermodynamic free energy (physical, in joules) are merely formal cousins. The exact physical bridge between the two is provided by **Landauer's principle**: erasing a bit of information, or irreversibly updating a memory state in the internal model to correct a prediction error, has an unavoidable minimum thermodynamic cost:

$$E \ge k_B T \ln 2$$

dissipated as heat at the physical temperature T of the substrate. Each bit of variational surprise the horizon reduces by updating its synapses pays a real physical toll in joules above the Landauer limit. Thus the informational tension against the statistical volatility of the reservoir and the metabolic expenditure of the tissue that sustains it are tied together by a universal constant.

In Friston's formalism, moreover, the boundary has a technical name: the **Markov blanket**, the set of sensory and active states that separates internal states from external ones. But here it is worth incorporating a critical distinction pointed out by Bruineberg and his collaborators (2022): a Markov blanket is, originally, a *statistical* partition that an observer can draw over almost any set of coupled variables, whereas a real boundary with an inside of its own is a physical achievement of the system

itself. That is why this book does not identify just any Markov blanket with a subject: it demands what since chapter 22 we have called **existential encapsulation**, that is, that the statistical boundary correspond to a persistent state of its own whose physical continuity depends on its own maintenance activity against the reservoir.

> ***In physics this is called:*** *Helmholtz free energy ($F = U - TS$), variational free energy (Friston, 2010) and the Landauer limit ($E \ge k_B T \ln 2$).*
> ***In daily life it is like:*** *a house at 21 degrees with snow outside. The temperature difference is the level; the heating bill, the minimum work per bit and per second needed in order not to become the snow.*

This magnitude answers *how far the horizon departs from what the reservoir would do with it and what it costs to keep it there*. And regarding the verb "to pay," we should recall the caution of chapters 15 and 18: it does not mean that the system must bear the cost alone. A baby regulates its edge through its mother, and a patient with advanced Alzheimer's sustains it partly through their caregiver; both still have existential encapsulation because in both there persists a state of their own whose continuity is being defended.

IV. Third magnitude: entropic action ($\bar{A}$ and $\bar{A}_F$)

The first two magnitudes are instantaneous: they say what capacity the edge has today and at what tension it operates right now. What is missing is what the farm kitchen had and an instantaneous photograph never has: the years.

When the history of a horizon is integrated over time, two complementary magnitudes appear that are best distinguished cleanly rather than confused into one:

1. **The accumulated structural biography ($\bar{A}$):** the time integral of the differentiated capacity the horizon has sustained above the baseline or equilibrium level S_0 of the reservoir, from its condensation at t_0 until t:

$$\bar{A}(t) = \int_{t_0}^{t} \left(S(\tau) - S_0 \right) d\tau$$

Its unit is entropy times time (the dimension of a physical action). It measures how much differentiated private repertoire the horizon has kept open throughout its existence.

2. **The accumulated biographical cost or wear ($\bar{A}_F$):** the time integral of the free energy along the same trajectory:

$$\bar{A}_F(t) = \int_{t_0}^{t} F(\tau)\, d\tau$$

In the variational formalism this integral is the *free action*, analogous to Hamilton's action, whose stationary trajectories ($\delta \bar{A}_F = 0$) define the path of the system; read thermodynamically, it measures the accumulated friction, the integrated prediction error and the total work it has cost the horizon not to dissolve.

Separating $\bar{A}$ and $\bar{A}_F$ resolves at the root the ambiguity between having a lot of history and having suffered a lot of wear: two horizons can accumulate the same structural biography $\bar{A}(t)$ having paid a very different wear $\bar{A}_F(t)$, depending on the hostility of the environment or the magnitude of their griefs and traumas.

And neither should be confused with the *density of subjective time* (ϑ) of chapter 10. Temporal density is the

instantaneous rate at which novelty is incorporated into memory (the Bayesian surprise rate per second); $\bar{A}(t)$ is the total breadth of the notebook that has been kept open for years in order to write it.

> ***In physics this is called:*** *action, the time integral of a magnitude along a trajectory; principle of stationary action.*
> ***In daily life it is like:*** *the difference between the square meters a house has kept habitable over fifty winters ($\bar{A}$) and the total firewood that had to be burned in that half century so that it would not freeze ($\bar{A}_F$).*

V. The three together

The three magnitudes answer three orthogonal questions about any already constituted horizon:

Magnitude	Symbol	Question	What it captures
Capacity of the edge	$S \propto \mathcal{A}$	How large is the private space?	Differentiable repertoire, ceiling
Free energy	$F = U - TS$	At what tension is it sustained and what does it cost per bit?	Maintenance against the reservoir (Landauer)
Entropic action	$\bar{A}(t),\; \bar{A}_F(t)$	How much life and how much effort has it accumulated?	Structural biography ($\bar{A}$) and wear ($\bar{A}_F$)

The degree of a horizon thus ceases to be a single figure and becomes a vector on three axes. A dog, a two-year-old child and an elderly person with Alzheimer's cannot be ordered along a single line: the child writes into memory at a furious rate but has little accumulated biography; the elder keeps an immense

$\bar{A}(t)$ even though their instantaneous capacity $S(t)$ is eroding; the dog sustains a smaller edge S with a free energy perfectly calibrated with whoever cares for it. And as chapters 19 and 33 warned, the triplet measures *how much*, never *what kind*: neurodivergences and calibrations of identity or orientation are topologies and shapes of the horizon, not rungs on a scale.

With the triad, moreover, the complete cycle of the book can be written in three lines:

- **Condensation:** the horizon is born when closing a stable boundary becomes thermodynamically favorable ($F_{\text{closed}} < F_{\text{open}}$) under the action of the two selections (stability and bias).

- **Existence:** while it lasts, it sustains its capacity $S(t) > S_0$ by paying the power demanded by the Landauer limit and accumulating biography $\bar{A}(t)$ and wear $\bar{A}_F(t)$.

- **Evaporation:** when the system can no longer sustain the maintenance cost, the filtration of closures $E_0 \supseteq E_1 \supseteq \dots$ shrinks, $S(t) \to S_0$ and the correlations accumulated in $\bar{A}(t)$ return to the reservoir subject to *scrambling*.

VI. The irreducibility filter without Φ: causal self-similarity $A(S)$

Here appears the decisive question that Φ used to answer: to which systems are we entitled to apply the triad?

Imagine an anthill, a walled city or an artificial intelligence data center. They have an enormous physical edge (S), they consume megawatts of free energy (F) and they last for decades ($\bar{A}$). Yet, as chapters 25 and 31 argued, in an

anthill or on the internet there is no one: there are thousands of parts interacting in an aggregative or diffuse way without constituting a unitary whole. And at the opposite extreme stands the counterexample of chapter 49: the Mandelbrot set repeats its geometric form at every possible scale, but lacks entirely any causal dynamics among its parts.

To distinguish a genuine recursive totality (such as a brain) both from an inert geometric fractal (Mandelbrot) and from a diffuse aggregative colony (the anthill), there is no need to return to Φ. It suffices to formalize the intuition of chapter 49 by requiring that the structure repeated from one scale to another not be a static drawing but the causal dynamics of the system itself. We call this magnitude **causal self-similarity**, $A(S)$:

$$A(S) = I_{\min}(S) \cdot \max_{P = \{S_1, \dots, S_n\}} \frac{1}{n} \sum_{i=1}^{n} \left[C(S_{-i}, S_i) \cdot \text{Iso}(S, S_i) \right]$$

Without yet entering the matrix apparatus, which is set out in full in section VII of chapter 54, the formula requires that three standard things from information theory and dynamical systems hold at once, combined in a product, so that if any single factor is zero the whole filter vanishes:

1. **Irreducibility at the minimum cut ($I_{\min}(S)$):** it measures, through Ay's stochastic interaction (2001, 2015), how much predictive information is lost when the system is split along its weakest seam. If the system has disconnected parts or is a purely geometric pattern without causal traffic between its regions, like the Mandelbrot set, $I_{\min}(S) = 0$ and therefore $A(S) = 0$.

2. **Part-rest coupling ($C(S_{-i}, S_i)$):** it measures the stationary mutual information between each subsystem S_i and the rest of the system S_{-i}. It guarantees that the parts making up the partition are genuinely coupled to the rest of the system.

3. **Causal isomorphism under coarse-graining ($\text{Iso}(S, S_i)$):** using the framework of causal emergence and surjective coarse-graining projection of Hoel, Albantakis and Tononi (2013), it compares the macroscopic transition matrix of the whole system with the internal transition matrix of each of its parts. This is where the anthill fails: ants have a weak local coupling through pheromones ($I_{\min}$ and C very low), but above all the slow, diffuse dynamics of the entire colony bears no isomorphism to the fast neurobiological dynamics of an individual ant ($\text{Iso} \approx 0$). In a brain, by contrast, the loop of excitation, inhibition and prediction is functionally replicated from local microcircuits up to global cortical networks, yielding an $A(S)$ several orders of magnitude higher.

In this way, chapter 49 and chapter 53 are welded into a single piece: a system passes the structural filter when its parts are causally integrated ($I_{\min} > 0$), coupled to the rest ($C > 0$) and reproduce at their scale the dynamical law of the whole ($\text{Iso} > 0$).

VII. The end of the exclusion postulate and the archipelago of subjects

It remains to see how this filter resolves the problem that IIT could settle only by decree: if inside a brain there are multiple nested scales that integrate information (columns, areas, networks), why is there not a Russian doll of superimposed conscious subjects within the same skull?

The answer comes from separating two planes that Tononi's exclusion postulate confused into one: the plane of **causal structure** and the plane of **subjects.**

On the plane of causal structure, a horizon not only permits nested levels but needs them: as we have just seen, that the subnetworks reproduce the dynamics of the whole at their scale is precisely what makes causal self-similarity $A(S)$ high. The nested Markov blankets described by Palacios and his collaborators (2020), cells within organs within organisms, are the normal architecture of life.

But on the plane of subjects, not every nested statistical boundary is a "someone." As chapters 22 and 23 established, the horizons that are subjects form an **archipelago of disjoint islands** because only that boundary is a subject which possesses **existential encapsulation**: the one that closes a persistent state of its own whose physical and informational continuity is at stake directly against the outer reservoir. A cortical column, or the imagined dragon of chapter 49, contributes to the causal self-similarity of my brain, but has no existential encapsulation separate from mine, just as within the psyche the Jungian shadow of chapter 23 is not a subject nested within the ego, but a neighboring disjoint complex with which the ego shares an entanglement boundary.

There is no need to decree a winner by maximum Φ. The nested hierarchy builds the fractal depth of the structure ($A(S)$); existential encapsulation traces within it the only firm coast that separates a subject from the reservoir.

VIII. What this framework is not

Four final warnings, in the spirit of the whole book.

It is not yet a direct clinical measure. Although each factor of $A(S)$ and each coordinate of the triad $(S, F, \bar{A})$ has well-defined mathematical estimators for continuous time series, such as the nonparametric mutual information estimator of Kraskov, Stögbauer and Grassberger (2004), applying them to massive brain recordings requires choosing candidate scales and partitions. In discrete verification models, as the next chapter shows, the calculation is exact.

It is not a single-piece filter. $A(S)$ by itself measures the recursive richness of the causal architecture, and the triad $(S, F, \bar{A})$ measures its size, its cost and its biography; to speak of a conscious subject, the causal filter and existential encapsulation are needed acting together.

It is not a moral ranking. The triplet measures how much edge, how much tension and how much history, not what dignity belongs to whoever bears them.

It does not solve the hard problem. Saying that each bit of surprise reduced pays at least $k_B T \ln 2$ joules explains why sustaining a horizon tires the body; it does not explain why that expenditure is felt from the inside as grief, as cold or as relief. The mystery of chapter 51 remains intact.

All the same, the change of language leaves the edifice cleaner than it was at the start of this chapter. For fifty-two chapters we said that a horizon is a skin that encloses an inside. Now we can say what separates that skin from a heap of sand or a geometric drawing ($A(S)$ and existential encapsulation) and with what three yardsticks the work of keeping it alive is measured ($S, F, \bar{A}$). A skin is not possessed: it is sustained.

Note to Chapter 53

What we do know: *The entropy of a black hole is proportional to the area of its horizon (Bekenstein, Hawking). Landauer's principle establishes the minimum thermodynamic cost of erasing or irreversibly updating a bit of information ($E \ge k_B T \ln 2$), connecting informational entropy with physical thermodynamics. Variational free energy (Friston, 2010) bounds informational surprise in systems with a Markov blanket. Stochastic interaction (Ay, 2001, 2015) and causal emergence through coarse-graining (Hoel, Albantakis and Tononi, 2013) are established measures in information theory and dynamical systems. Bruineberg et al. (2022) and Palacios et al. (2020) have formally analyzed the scope and nesting of Markov blankets.*

What we don't know: *Whether causal self-similarity $A(S)$, exact in small discrete networks, can be stably estimated in large-scale neurophysiological recordings by means of continuous estimators (Kraskov et al., 2004). How $A(S)$ and the triad behave in the simplest biological horizons, such as a three-hundred-neuron nematode where the hierarchy of scales is minimal. What fraction of brain metabolic consumption corresponds directly to the Landauer limit of predictive updating as opposed to basal physiological maintenance.*

Questions that remain: *Does the distinction between structural biography ($\bar{A}$) and accumulated wear ($\bar{A}_F$) capture the clinical difference between aging calmly and aging under chronic stress? Where exactly does existential encapsulation begin when an artificial system goes from restarting with each call to maintaining a state of its own on which its physical survival depends?*

If you only take away one idea: *A conscious horizon does not need the decree of Φ to distinguish itself from an anthill or a fractal*

drawing: it is distinguished by its causal self-similarity and its existential encapsulation, and its degree is measured by the capacity of its edge (S), the free energy it pays to sustain it (F) and the biography it accumulates over time ($\bar{A}$).

Readings: *Bekenstein, J. D. (1973), "Black Holes and Entropy"; Hawking, S. W. (1975), "Particle creation by black holes"; Unruh, W. G. (1976), "Notes on black-hole evaporation"; Schrödinger, E. (1944), What Is Life?; Friston, K. (2010), "The free-energy principle: a unified brain theory?"; Ay, N. (2015), "Information Geometry on Complexity and Stochastic Interaction"; Hoel, E. P., Albantakis, L. and Tononi, G. (2013), "Quantifying causal emergence shows that macro can beat micro"; Bruineberg, J., Dolega, K., Dewhurst, J. and Baltieri, M. (2022), "The Emperor's New Markov Blankets"; Palacios, E. R., Razi, A., Parr, T., Kirchhoff, M. and Friston, K. (2020), "On Markov blankets and hierarchical self-organisation"; Kraskov, A., Stögbauer, H. and Grassberger, P. (2004), "Estimating mutual information"; Prigogine, I. (1977), Nobel lecture on dissipative structures.*

CHAPTER 54

THE ESSAY REREAD

(The whole book, seen through causal self-similarity, the edge, free energy and action)

There is a way of rereading a book that consists of changing a single keystone and watching what moves. When you finish a novel and discover that the narrator was lying, you change not one page, yet every page is lit from another angle. The previous chapter proposed retiring the scaffold of Φ that has accompanied us since chapter 1 and replacing it with an architecture in two steps: first, the filter of **causal self-similarity** $A(S)$ joined to **existential encapsulation**; then, the triad $(S, F, \bar{A})$ that measures the size of the edge, the free energy it costs to sustain it, and the accumulated biography. This chapter does what the previous one could not do without becoming interminable: it walks the entire book with that lens, formalizes the complete mathematical catalogue in section VII (including the exact verification simulation on four binary nodes), and sets out in section VIII the empirical predictions that could refute it.

This is not a summary. It is a second reading, and a second reading is worth something only if it orders what seemed scattered in the first or resolves the tensions the first left open. When the reread introduces a new bridge with respect to the original chapter, it is marked as a *hypothesis*.

I. The Founding Ideas

The switch trap (chapters 1, 6 and 23). The book began by denying that consciousness is a switch, and chapter 23 sharpened the distinction between two different things. One is the *closure* of a horizon, which is a topological fact without degrees: $\text{cl}(\text{cl}(A)) = \text{cl}(A)$, closing what is already closed adds nothing. The other is the *degree* of an already-closed horizon, which varies continuously like the dial of chapter 1. With the new framework, the threshold of closure is set by the ignition of causal self-similarity $A(S) > 0$, existential encapsulation, and the thermodynamic inequality $F_{\text{closed}} < F_{\text{open}}$; the subsequent continuous dial is the triad $(S, F, \bar{A})$. The thermostat does not score low on the triad: it lacks causal self-similarity and existential encapsulation, so it has no horizon to which the triad could be applied.

The black hole and its edge (chapter 2). Bekenstein and Hawking delivered the first coordinate: $S \propto \mathcal{A}$. Reread through the triad, it says that the ceiling of differentiable private states a horizon can sustain is set by the area of its interface, measured as entanglement entropy across the boundary (chapter 23).

Encapsulation and the end of the exclusion decree (chapters 3, 22 and 23). A private state behind a public interface. Reread against the literature on Markov blankets (Friston, 2010; Palacios et al., 2020; Bruineberg et al., 2022), this piece resolves the tension that chapters 22 and 23 had been carrying. At the level of causal structure, Markov blankets nest hierarchically, and that hierarchy is what $A(S)$ measures; but at the level of subjects, only the boundary that possesses **existential encapsulation** (the one whose own persistent state stakes its physical continuity

against the reservoir) constitutes a "someone." That is why the inner complexes form an archipelago of disjoint islands rather than a Russian doll of overlapping subjects, with no need to invoke the postulate of exclusion by maximum Φ.

The reservoir and the Landauer bridge (chapters 5, 7 and 23). Chapter 23 defined the reservoir without mystery: $R = X \setminus \bigcup_i E_i$, everything that at a given instant lies outside the condensed closures. Statistically, R acts as the bath of volatility and uncertainty against which the horizon minimizes its variational surprise; physically, the Landauer limit ($E \ge k_B T \ln 2$) translates each bit of predictive updating against that reservoir into thermal energy dissipated at the substrate's temperature T.

II. The Cycle: Being Born, Lasting, Dissolving

Birth (chapters 6 and 8). The river becomes a lake when, in the window of weeks 28 to 32, the thalamocortical connections ignite the recursion between scales ($A(S) > 0$) and closing the boundary turns out to be thermodynamically stable ($F_{\text{closed}} < F_{\text{open}}$). The inequality says that closure is stable; which concrete configuration crystallizes is decided by fluctuations and by the two selections of chapter 31 (stability and bias), as monozygotic twins prove.

Death (chapters 7 and 23). Evaporation arrives when the system can no longer pay the power required to keep F far from equilibrium. The decreasing filtration $E_0 \supseteq E_1 \supseteq \dots$ shrinks the island, $S(t)$ falls toward the baseline level S_0, and what leaves the boundary ($E_0 \setminus E_t$) is reintegrated into the reservoir by undergoing *scrambling*. The accumulated structural biography $\bar{A}(t)$

stops growing, but the correlations it imprinted on the environment are not erased: they become illegible.

The table of equivalences (chapter 9). The filter and the three coordinates are added to the Rosetta stone of chapter 9:

Dimension	Philosophy of the horizon	Software engineering	Theoretical physics and thermodynamics	Eastern philosophies
Irreducibility	Causal self-similarity $A(S)$ + existential encapsulation.	Recursive modular architecture with persistent private state.	Multiscale causal emergence far from equilibrium.	Indra's net reflected in every knot with a center of its own.
Capacity	Entropy of the edge, $S \propto \mathcal{A}$.	Size of the private state space (*private fields*).	Bekenstein-Hawking entropy.	What fits in the vessel.
Level and cost	Free energy, $F = U - TS$, and maintenance power.	Computational and energy cost of maintaining the class invariant.	Work against the thermal bath; Landauer limit ($k_B T \ln 2$).	Sustained attention; vigilance against dispersion.
Biography	Entropic action $\bar{A}(t)$ and wear $\bar{A}_F(t)$.	State history and accumulated *logs* of the instance.	Action integrated along the trajectory.	*Karma*, understood as accumulated causal imprint.

III. The Horizon in Relationship

The time of the bond (chapter 10). The density of subjective time $\vartheta(t)$ is the rate at which new information is incorporated (Bayesian surprise per unit time), whereas $\bar{A}(t)$ is the sustained structural capacity in which that history is inscribed. *Hypothesis:* when two horizons synchronize and share an external time base, they split the predictive cost of watching over the environment: the wear $\bar{A}_F$ of each decreases while the shared density ϑ increases.

Entanglement and love (chapters 12 and 13). The magnitude of the coupling between two horizons is their mutual information $I(A; B)$; its sign or valence is given by the interaction free energy ΔF_{int}. A hostile bond of twenty years has very high mutual information but $\Delta F_{\text{int}} > 0$: it makes maintaining both more expensive. The acknowledged love of chapter 13 is a coupling of high mutual information with $\Delta F_{\text{int}} < 0$ (energy savings on the social baseline) and high certainty of availability. The loss of a loved one produces at once a contraction of the edge (S falls: de-expansion) and a sustained peak of F.

Addiction (chapter 14). *Hypothesis:* the substance hijacks the dopaminergic computation of prediction error by offering an artificial, immediate reduction of F in a single channel, which the horizon pays for with a severe contraction of its capacity S: the vital repertoire shrinks because any natural stimulus generates a comparatively unmanageable prediction error.

Mother and child (chapter 15). During gestation and early parenting, the maternal horizon subsidizes the child's level F, both metabolically and predictively. The contingency documented by Tronick is the regularity that allows the baby to calibrate what level of variational surprise constitutes its safe baseline.

Empathy (chapter 16). Modeling another's entropy consumes real power, subject to the Landauer limit. Empathic wear occurs when the cost $\bar{A}_F$ of simulating other horizons compromises the power available to sustain one's own edge; the cynical wall is a defensive contraction of S, and the interoceptive anchor is the return protocol that protects one's own boundary.

Grief (chapter 17). In the language of the triad, grief combines three simultaneous effects: a fall in S (the de-expansion of the shared architecture), a chronic rise in F that triggers the accumulated wear $\bar{A}_F$ (the predictive model keeps calling out to the absent one, and each unanswered call costs at least $k_B T \ln 2$ per bit of error processed), and the intact permanence of $\bar{A}(t_0 \to t_{\text{loss}})$, the structural biography already written, which continuing bonds learn to inhabit without continuing to pay the active prediction error.

IV. The Horizons That Break

Alzheimer's (chapter 18). It is the progressive erosion of $S(t)$ and the fragmentation of $A(S)$ through the breaking of connections between scales, while the writing rate $\vartheta(t)$ falls toward zero ("time becomes a puddle"). The caregiver externally assumes part of the work of maintaining F, safeguarding an accumulated biography $\bar{A}$ that the patient can now inhabit only in its minimal core.

Parkinson's (chapter 19). The horizon keeps its causal self-similarity $A(S)$, its capacity S and its biography $\bar{A}$ intact, but the active channel of the Markov blanket fails, along with the dopaminergic calibration of the precision π_i with which motor and temporal predictions are synchronized with the world.

Trauma (chapter 19). It is a watertight sub-edge that has not gone through narrative *scrambling*: an island of frozen information that is not integrated into the general biography $\bar{A}$, but that charges a constant toll in $\bar{A}_F$ every time a stimulus reactivates its unresolved prediction loop.

Neurodiversity and the evolutionary operating system (chapters 19 and 20). Autism and ADHD are different geometries of how precision is distributed between local and global scales, not deficits in the quantity of consciousness. And the Darwinian alarms of chapter 20 (*dukkha, tanha*) are high-gain warnings when F threatens to rise; meditation introduces a layer of indirection that reduces the precision assigned to the alarm without switching off the module that emits it.

The ego, the shadow and the four experiments (chapters 21, 22, 23 and 24). The shadow is a disjoint island neighboring the ego within the same substrate, separated by a boundary whose correlation measures how much of the repressed leaks through. And the four thought experiments of chapter 24 are freed from Φ: where those protocols asked to measure whether the minimum cut of Φ gave one peak or two (the topological twin experiment), or how Φ fell against the correlation with the environment in contemplative evaporation and in psychosis, they now ask to measure directly the stochastic interaction $I_{\min}$ and the causal self-similarity $A(S)$ together with the mutual information and the triad $(S, F, \bar{A})$.

V. The Horizon Outside the Human Body

Animals, places and the five maps (chapters 26, 27, 28 and 29). Where there is $A(S) > 0$ with existential encapsulation, there is someone: a dog sustains a real triad $(S, F, \bar{A})$, calibrated over millennia alongside our own. And in the table of the five traditions of chapter 28, the eschatological answers read as different destinations of the accumulated biography $\bar{A}$ after evaporation: preserved with a boundary in the remnant (Christianity and Islam), turned into a bias of future condensations with no owner in the karmic imprint (Buddhism), reabsorbed into identity with the ground (Vedanta), or dispersed by *scrambling* with no reader (naturalism).

The mirror without depth and societies (chapters 25, 30 and 31). A society or an anthill lacks a subject for a double reason: its causal self-similarity $A(S)$ collapses because the slow dynamics of the aggregate are not isomorphic to those of the individuals ($\text{Iso} \approx 0$), and it lacks existential encapsulation. A current large language model fails the filter by the architectural route: although during training it minimizes a cost function formally analogous to a variational free energy, at inference it lacks a persistent state of its own whose physical continuity depends on its behavior; its weights are frozen and publicly legible from outside. There is no existential encapsulation and no level F of its own that the system defends in order not to dissolve.

Fractal reality and the mud ship (chapters 49 and 50). The definition of $A(S)$ finally formalizes in what sense a living horizon is fractal and why the Mandelbrot set is not so in the sense that matters here: in Mandelbrot there is static geometric self-similarity without causal traffic ($I_{\min} = 0 \implies A(S) = 0$), whereas in a real horizon the same causal transition law is

replicated from the parts to the whole ($A(S) > 0$). And the "creative attachment" of chapter 50 —the gaze that sustains a world so that it does not fall into oblivion— has a direct translation into the second coordinate: it is the continuous supply of free energy from the higher scale that keeps a non-equilibrium steady state alive.

VI. Limit Cases, Altered States and the Copy in the Tale

Limit cases and calibration (chapters 32 and 33). The two selections —stability and bias— are the physical conditions for a boundary to cross $F_{\text{closed}} < F_{\text{open}}$ with an initial condition legible to other human horizons.

The interlude. The two dimensions of the interlude fit cleanly into the new framework: metabolic activation corresponds to the dissipated power P (tied to F by Landauer), while integration corresponds to $A(S)$ and to the active capacity S. In deep sleep both P and $A(S)$ fall; in deep meditation, the alarm power P drops to a minimum while $A(S)$ and S remain high.

The one who remains (chapter 36). The dilemma of the tale —whether the mind uploaded to the server is the original or a new instance— receives a precise geometric answer: a freshly instantiated copy can reproduce at t_0 the capacity S and the structure $A(S)$, but its own biographical integral $\bar{A}(t_0)$ and its own wear $\bar{A}_F(t_0)$ start at zero. It has the memories copied in as an initial condition, but it has not paid for the trajectory that wrote them.

VII. The Mathematics of Consciousness

A book that uses words like "causal self-similarity," "edge entropy," "free energy" or "action" must show without ambiguity the equations it refers to. Each expression carries a label declaring its epistemological status:

- **[standard]**: an established definition or theorem of mathematics, physics or information theory.
- **[analogy]**: a standard result whose application to the architecture of consciousness is a hypothesis of the book.
- **[proposal]**: an original formulation of this essay built on standard ingredients.

1. Topology of Closure and of the Archipelago (Chapter 23)

On a space X, the Kuratowski closure operator $\text{cl}: \mathcal{P}(X) \to \mathcal{P}(X)$ satisfies:

- (K1) $\text{cl}(\emptyset) = \emptyset$.
- (K2) $A \subseteq \text{cl}(A)$.
- (K3) $\text{cl}(A \cup B) = \text{cl}(A) \cup \text{cl}(B)$.
- (K4) $\text{cl}(\text{cl}(A)) = \text{cl}(A)$ *(idempotence)*. **[standard]**

A set is closed if $A = \text{cl}(A)$, and its boundary is $\partial A = \text{cl}(A) \cap \text{cl}(X \setminus A)$. **[standard]**

- **Closed horizon (E_i):** a closure of its own $E_i \subset X$ that passes the causal self-similarity filter $A(S) > 0$ and possesses existential encapsulation. **[proposal]**

• **Reservoir (R):** the complement of the union of all condensed closures at a given instant:

$$R = X \setminus \bigcup_i E_i$$

[analogy]

• **Evaporation:** a decreasing filtration $E_0 \supseteq E_1 \supseteq E_2 \supseteq \dots$ such that every state that leaves the island is reintegrated into the reservoir: $E_0 \setminus E_t \subseteq R$. **[analogy]**

• **Archipelago of subjects:** while the internal causal structure admits nested levels with $A(S) > 0$, the boundaries endowed with existential encapsulation form closures that are disjoint from one another ($E_i \cap E_j = \emptyset$ for distinct subjects $i \neq j$). **[proposal]**

2. Markov Blanket and Existential Encapsulation (Chapters 3 and 53)

Consider a partition of the world's variables into external states ψ, sensory states s, active states a and internal states μ, where $b = (s, a)$ constitutes the boundary or Markov blanket (Pearl, 1988; Friston, 2010; Palacios et al., 2020):

$$p(\psi, \mu \mid b) = p(\psi \mid b)\, p(\mu \mid b)$$

[standard]

Following the critical distinction of Bruineberg et al. (2022) between statistical conditional independence and physical boundary, we define **existential encapsulation** when: (i) the above conditional independence holds with respect to b, and (ii) the dynamics of the internal and active states (μ, a) actively keep the system's distribution within the set of viable

states that preserve the boundary b itself against dissipation into the reservoir R. **[proposal]**

3. The Irreducibility Filter: Causal Self-Similarity $A(S)$ (Chapters 49 and 53)

Consider a composite causal system defined over a finite set of units $V = \{1, \dots, N\}$, with product state space $X = \prod_{v \in V} X_v$, Markov transition matrix $p(x_{t+1} \mid x_t)$ and stationary distribution $\pi(x)$. **[standard]**

Given a partition of the units $P = \{V_1, \dots, V_n\}$ into $n \ge 2$ disjoint non-empty blocks, each block defines a subsystem $S_i = (V_i, X_i, p_i)$ with state space $X_i = \prod_{v \in V_i} X_v$, its complement $S_{-i} = S \setminus S_i$ and its stationary marginalized dynamics:

$$p_i(x_{i,t+1} \mid x_{i,t}) = \sum_{x_{-i,t}} \pi(x_{-i,t} \mid x_{i,t}) \sum_{x_{-i,t+1}} p(x_{t+1} \mid x_t)$$

[standard]

The **causal self-similarity** of the system S is defined as:

$$A(S) = I_{\min}(S) \cdot \max_{P = \{S_1, \dots, S_n\}} \frac{1}{n} \sum_{i=1}^{n} \left[C(S_{-i}, S_i) \cdot \text{Iso}(S, S_i) \right]$$

[proposal]

where its three factors are:

1. **Irreducibility at the minimum cut ($I_{\min}$):** stochastic interaction (Ay, 2001, 2015) evaluated at the weakest partition Q of the system and averaged over the stationary distribution π:

$$I_{\min}(S) = \min_{Q} \mathbb{E}_{x_t \sim \pi} \left[D_{\text{KL}}\left(p(X_{t+1} \mid x_t) \;\middle\Vert{}\; \prod_{k} p_k(X_{k,t+1} \mid x_{k,t}) \right) \right]$$

[standard] *(It guarantees that if the system has disconnected parts or lacks causal traffic, like the Mandelbrot set, $I_{\min}(S) = 0 \implies A(S) = 0$).*

2. **Part-rest coupling (C):** stationary mutual information between the part S_i and the rest of the system S_{-i} (and not $I(X_t; X_{i,t})$, which would degenerate trivially into the local entropy $H(X_{i,t})$):

$$C(S_{-i}, S_i) = I(X_{-i,t}; X_{i,t}) = \sum_{x_i, x_{-i}} \pi(x_i, x_{-i}) \log_2 \frac{\pi(x_i, x_{-i})}{\pi(x_i)\pi(x_{-i})}$$

[standard] *(In high-dimensional continuous systems it can be estimated using the nearest-neighbor method of Kraskov, Stögbauer and Grassberger, 2004).*

3. **Causal isomorphism under coarse-graining (Iso):** since $\vert{}X\vert{} > \vert{}X_i\vert{}$, there is no bijection between the whole and the part; a surjective coarse-graining projection (Hoel, Albantakis and Tononi, 2013) $\phi_i: X \twoheadrightarrow X_i$ is used, which induces the macroscopic transition matrix $p_S^{(\phi_i)}$ of the same dimension as p_i:

$$\text{Iso}(S, S_i) = \max_{\phi_i: X \twoheadrightarrow X_i} \max \left(0,\; \text{corr}\left(p_S^{(\phi_i)},\; p_i \right) \right)$$

where corr denotes the Pearson correlation between the vectorized transition kernels. **[proposal]** *(Here the anthill falls: it has weak local coupling, but the slow diffusive dynamics of the colony*

are not isomorphic to the fast neurobiological dynamics of an individual ant, $\text{Iso} \approx 0$).

Invariance property: $A(S)$ is strictly invariant under factorizable local isomorphisms $f(x_1, \dots, x_N) = (f_1(x_1), \dots, f_N(x_N))$ and under permutations of the set of units V. **[proposal]**

4. Exact Verification Simulation ($N = 4$ Binary Nodes, 16 States)

To verify analytically and numerically the behavior of the filter $A(S)$ without approximations, consider a composite system of $N = 4$ binary units $x_v \in \{0, 1\}$ ($\vert{}X\vert{} = 16$ states) with a conditional parallel-update transition kernel:

$$p(x' \mid x) = \prod_{v=1}^{4} q_v(x)^{x'_v} \left(1 - q_v(x)\right)^{1 - x'_v}$$

partitioned into two symmetric modules $S_1 = \{1, 2\}$ and $S_2 = \{3, 4\}$, and evaluated under the coarse-graining projection by logical OR gate / majority vote of each module, $M_1 = \max(x_1, x_2)$ and $M_2 = \max(x_3, x_4)$:

• **1. Mandelbrot-type system (S_{Mand}, geometric self-similarity without causal coupling between parts):** each node oscillates locally with noise $\epsilon = 0.10$ according to $q_v(x) = (1 - \epsilon)(1 - x_v) + \epsilon x_v$. The local and macroscopic dynamics are symmetric ($\text{Iso} = 1.0$), but since there is no causal traffic between subsystems, $I_{\min}(S_{\text{Mand}}) = 0$ and $C = 0$, which yields exactly:

$$\mathbf{A(S_{\text{Mand}}) = 0}$$

• **2. Anthill-type system (S_{Ant}, weak mean-field diffusive aggregation):** each unit combines its local oscillation ($\eta = 0.10$) with a weak diffusive drag from the rest of the colony ($\varepsilon = 0.05$):

$$q_v(x) = (1 - \varepsilon)\left[(1 - \eta)(1 - x_v) + \eta x_v\right] + \varepsilon \frac{1}{3} \sum_{u \neq v} x_u$$

The system exhibits a very low minimum irreducibility ($I_{\min} > 0$), a weak stationary coupling ($C > 0$) and a low functional similarity between the slow diffusion of the aggregate and the individual unit, resulting in:

$$\mathbf{A(S_{\text{Ant}}) \approx 10^{-5}}$$

• **3. Brain-type system (S_{Brain}, excitation-inhibition loop replicated across scales):** each local module runs an excitatory-inhibitory cycle $(x_a, x_b) \to (1 - x_b, x_a)$ while the two modules couple at the macroscopic scale following exactly the same excitatory-inhibitory law $(M_1, M_2) \to (1 - M_2, M_1)$ with an intermodular coupling weight $\beta = 0.50$ and thermal noise $\gamma = 0.05$:

$$q_1(x) = (1 - \gamma)\left[(1 - \beta)(1 - x_2) + \beta(1 - M_2)\right] + \frac{\gamma}{2}$$

$$q_2(x) = (1 - \gamma)\left[(1 - \beta)x_1 + \beta(1 - M_2)\right] + \frac{\gamma}{2}$$

$$q_3(x) = (1 - \gamma)\left[(1 - \beta)(1 - x_4) + \beta M_1\right] + \frac{\gamma}{2}$$

$$q_4(x) = (1 - \gamma)\left[(1 - \beta)x_3 + \beta M_1\right] + \frac{\gamma}{2}$$

Here a high minimum-cut irreducibility ($I_{\min}$ high), a strong intermodular mutual information (C high) and a high

isomorphism between the macroscopic law and the microscopic law (Iso high) all concur at once, yielding a value of $\mathbf{A(S_{\text{Brain}}) \approx 10^{-1}}$, four orders of magnitude above that of the anthill.

System ($N=4$ nodes)	$I_{\min}$	$C(S_{-i}, S_i)$	$\text{Iso}(S, S_i)$	Causal self-similarity $A(S)$
Mandelbrot (S_{Mand})	0	0	1.0	$\mathbf{0}$
Anthill (S_{Ant})	Very low (>0)	Very low (>0)	Low	$\mathbf{\approx 10^{-5}}$
Brain (S_{Brain})	High	High	High	$\mathbf{\approx 10^{-1}}$

5. First Coordinate of the Triad: Capacity of the Edge (S)

• **Von Neumann and Shannon entropy:** $S = -k_B \text{Tr}(\rho \ln \rho)$; in the classical discrete regime, $H(X) = -\sum_x p(x) \log_2 p(x)$. **[standard]**

• **Bekenstein-Hawking entropy:** $S = k_B \mathcal{A} / (4 \ell_P^2)$, with $\ell_P^2 = G\hbar / c^3$. **[standard]**

• **Holographic entanglement entropy (Ryu and Takayanagi, 2006):** $S_A = -\text{Tr}(\rho_A \ln \rho_A) = \text{Area}(\gamma_A) / (4 G_N)$. **[standard]** in AdS/CFT.

• **Capacity of the horizon's edge:** $S(t) \propto \mathcal{A}(t)$ measures the ceiling of the differentiable private repertoire that the boundary sustains at instant t. **[analogy]**

6. Second Coordinate: Free Energy (F) and the Landauer Limit

• **Non-equilibrium Helmholtz free energy:** for a distribution of states $q(x)$ against a bath at temperature T with Gibbs equilibrium distribution $p_{\text{eq}}(x) = e^{-E(x)/k_B T}/Z$:

$$F[q] = U[q] - T S[q] = F_{\text{eq}} + k_B T \, D_{\text{KL}}\left(q \;\middle\Vert{}\; p_{\text{eq}}\right)$$

[standard]

• **Variational free energy (Friston, 2010):** over hidden states ψ and sensory observations o, with internal variational density $q(\psi)$:

$$\mathcal{F}[q, o] = D_{\text{KL}}\left(q(\psi) \;\middle\Vert{}\; p(\psi \mid o)\right) - \ln p(o) \ge -\ln p(o)$$

Under the Laplace approximation with prediction errors $\varepsilon_i = o_i - g_i(\mu)$ and precisions π_i: $\mathcal{F} \approx \frac{1}{2}\sum_i \pi_i \varepsilon_i^2 + \text{const}$. **[standard]**

• **Landauer thermodynamic bridge:** every irreversible update of 1 bit in the internal state μ to reduce the variational surprise $\mathcal{F}$ against the volatility of the reservoir dissipates at least:

$$\Delta E_{\text{Landauer}} \ge k_B T \ln 2$$

[standard] in the physics of information; **[proposal]** as a coupling identity between variational free energy and the metabolic cost of the horizon.

7. Third Coordinate: Entropic Action ($\bar{A}, \bar{A}_F$) and Temporal Density (ϑ)

• **Accumulated structural biography ($\bar{A}$):** differentiated capacity sustained above the baseline or equilibrium entropy S_0:

$$\bar{A}(t) = \int_{t_0}^{t} \left(S(\tau) - S_0 \right) d\tau$$

[proposal]

• **Accumulated biographical cost or wear ($\bar{A}_F$):** friction, or free energy integrated along the history of the horizon:

$$\bar{A}_F(t) = \int_{t_0}^{t} F(\tau)\, d\tau$$

[analogy]

• **Subjective density of time (ϑ, chapter 10):** instantaneous rate of Bayesian surprise (Itti and Baldi, 2009) incorporated into memory: $\vartheta(t) = D_{\text{KL}}\left(q_{t+dt}(\psi) \;\middle\Vert{}\; q_t(\psi)\right) / dt$, so that remembered time is $\tau_{\text{rec}} \approx \int \vartheta(t)\,dt$. **[proposal]**

8. Bond, Condensation and Evaporation

• **Thermodynamics of the bond (chapters 12, 13 and 35):** for two coupled horizons with joint energy $E_{AB} = E_A + E_B + E_{\text{int}}$, the joint free energy satisfies:

$$F[q_{AB}] = F[q_A] + F[q_B] + \langle E_{\text{int}} \rangle + k_B T \ln 2 \cdot I(A; B)$$

where $I(A; B)$ is the mutual information in bits and $\Delta F_{\text{int}} = \langle E_{\text{int}} \rangle + k_B T \ln 2 \cdot I(A; B)$ determines the valence of the bond (stabilizing and

saving if $\Delta F_{\text{int}} < 0$; costly and aversive if $\Delta F_{\text{int}} > 0$). **[standard]** in statistical mechanics; **[analogy]** applied to the human bond.

- **Condensation condition (chapter 6):** $F_{\text{closed}} < F_{\text{open}}$ with $A(S) > 0$. **[proposal]**

- **Scrambling time (Sekino and Susskind, 2008) and Page curve (Page, 1993):** $t_* \sim \frac{\hbar}{2\pi k_B T} \ln S$, and $S_{\text{rad}}(t) = \min(S_{\text{Hawking}}(t), S_{\text{BH}}(t))$. **[standard]**

9. Complete Definition of the Degree of a Horizon

Given a system S that passes the subject filter —it possesses **existential encapsulation** and **causal self-similarity** $A(S) > 0$—, its state and trajectory are described by the vector:

$$\mathcal{H}(S, t) = \Big(A(S),\; S(t),\; F(t),\; \bar{A}(t),\; \bar{A}_F(t),\; \vartheta(t) \Big)$$

[proposal]

VIII. What Could Go Wrong: Testable Predictions

A reading that cannot fail is not a theory; it is a manner of speaking. These are the seven claims of the reformed framework that could indeed turn out to be false, together with the empirical or computational observation that would refute them:

1. **Multiscale separation through $A(S)$ in real and simulated connectomes.** In dynamical networks built on real biological connectomes (from *C. elegans* and *Drosophila* to mammalian cortex), as against aggregative networks with the same number

of nodes and energy consumption (diffusive colonies or cellular-automaton grids), the measure $A(S)$ should replicate the separation of several orders of magnitude shown in the exact $N = 4$ node simulation, and should drop drastically under general anesthesia or slow-wave sleep as the coupling between scales $C(S_{-i}, S_i)$ and the irreducibility $I_{\min}$ are lost. If, in multiscale neuroimaging data (estimating C with the method of Kraskov et al., 2004), an awake brain does not score significantly above its randomized version or an anesthetized state, the filter $A(S)$ is refuted.

2. **The metabolic cost of prediction error scales with surprise according to the Landauer bridge.** If Friston's variational free energy and thermodynamics are joined by the irreversible updating of synaptic states ($E \ge k_B T \ln 2$), experimental paradigms of expectation violation and acute grief should show a local metabolic increase proportional to the Kullback-Leibler divergence of the internal model's update, and that surcharge should decrease as the model converges.

3. **Clinical dissociation between structural biography ($\bar{A}$) and accumulated wear ($\bar{A}_F$).** Two cohorts of the same chronological age and comparable cognitive reserve (similar $\bar{A}$), but separated by decades of chronic predictive stress, unresolved grief or complex trauma (high versus low $\bar{A}_F$), should differ in markers of allostatic load and cellular aging in direct proportion to the integral of their unresolved prediction errors, not to the amount of episodic memory stored.

4. **The effect of the bond grows with accumulated mutual information and changes sign with ΔF_{int}.** In the social baseline paradigm (Coan), the metabolic saving in the face of a threat should be proportional to the shared history when

$\Delta F_{\text{int}} < 0$ (acknowledged secure bond), and should invert into a surcharge above the solitary level when a familiar adversary with the same accumulated history is present ($\Delta F_{\text{int}} > 0$). If a kind stranger saves as much as a partner of twenty years, or if a familiar adversary reduces the threat, the equation in section VII.8 is false.

5. **The contraction of the edge's capacity (S) anticipates relapse in addiction.** If addiction contracts the differentiable repertoire $S(t)$ in order to artificially minimize F along a single axis, the fall in the entropy of everyday routines and activities (measured passively and continuously) should temporally precede clinical relapse and predict it better than self-reported subjective desire (*craving*).

6. **Trauma behaves as a watertight sub-edge with a cost $\bar{A}_F$ of its own.** The therapeutic integration of an encapsulated traumatic memory should produce a stepwise reduction (as a phase transition when the compartment opens to narrative *scrambling*, and not as a smooth slope) in the baseline physiological cost, accompanied by an increase in the free capacity $S(t)$ available for present tasks.

7. **Existential encapsulation alters the behavior of an artificial agent.** An artificial system endowed with a persistent internal state whose physical integrity depends directly on keeping its variables within a homeostatic range against a volatile environment will spontaneously develop behaviors of boundary preservation and aversion to irreversible risk that do not appear in a fixed-weight model with the same architecture and external reward function, but without a state of its own at stake.

> ***In daily life it is like:*** *deciding whether a house is inhabited by looking only at the plans of the façade, or by checking whether the*

heat is distributed across all the floors, how much the winter meter reads, and who repairs the roof when it snows.

IX. What Is Gained and What Is Lost

What is gained. A complete emancipation of the essay from Tononi's theory and its exclusion postulate, closing the gap between the fractal intuition of chapter 49 and the formalism at the end of the book. An exact mathematical distinction (verified in the four-node model) between a dead geometric fractal, an aggregative anthill and a recursive brain. A physical bridge without gaps —the Landauer limit— between informational surprise and thermodynamic expenditure. And a clear separation between the richness of a life ($\bar{A}$) and how much it has hurt to sustain it ($\bar{A}_F$).

What is lost. The false comfort of summing up a soul in a single Greek letter. Now a multiscale filter ($A(S)$ with existential encapsulation) and three coordinates are needed in order to begin speaking with rigor.

What is left untouched. The intimate texture of the inside. No equation in section VII says why it hurts to pay the Landauer bill, or why the silence of the reservoir makes us dizzy when we lean out over the edge. Chapter 51 put it this way: the map is not the territory. This reread does not erase that boundary; it cleans the glass to look at it better.

The image from the beginning remains. The country kitchen, with its embers stirred every morning, is no more than a small case. But it is an honest case: an inside that organizes itself at every one of its scales, an expense paid bit by bit and log by log, years that accumulate in the walls. If the whole essay has a thesis that survives every change of language, it is this: that to be alive,

and to be conscious, is not a state one possesses, but a boundary that is sustained through time.

Note to Chapter 54

What we do know: *That the three ingredients of causal self-similarity $A(S)$ —Ay's stochastic interaction (2001, 2015), stationary mutual information (estimable in the continuous case according to Kraskov et al., 2004) and the causal emergence by coarse-graining of Hoel et al. (2013)— are standard mathematical tools, and that their combination in $A(S)$ separates exactly, in the 4-binary-node model, the decoupled system ($A(S)=0$), the diffuse aggregative one ($A(S)\approx 10^{-5}$) and the multiscale recursive one ($A(S)\approx 10^{-1}$). That the Landauer limit ($E \ge k_B T \ln 2$) physically connects the irreversible updating of information with thermodynamic dissipation. That the distinction between statistical Markov blankets and self-organizing physical boundaries is backed by recent literature (Palacios et al., 2020; Bruineberg et al., 2022).*

What we don't know: *Whether the seven predictions of section VIII will be confirmed when the measures are scaled up to complete connectomes and clinical cohorts. To what extent the effective temperature of synaptic updating approaches Landauer's lower limit, or operates several orders of magnitude above it through biological inefficiency.*

Questions that remain: *What is the minimum size of biological network at which the coarse-grained isomorphism $\text{Iso}(S, S_i)$ begins to be non-trivial? How exactly is the saving in ΔF_{int} distributed in asymmetric bonds such as that of the caregiver and the Alzheimer's patient?*

If you only take away one idea: *If you swap the static figure of Φ for causal self-similarity $A(S)$, existential encapsulation and the triad $(S, F, \bar{A})$, the whole book fits into a single sentence: a horizon is a recursive totality that pays free energy to sustain the capacity of its edge and to accumulate a biography over time.*

Readings: *Those of chapter 53 and, in addition: Ay, N. (2001, 2015), "Information Geometry on Complexity and Stochastic Interaction"; Hoel, E. P., Albantakis, L. and Tononi, G. (2013), "Quantifying causal emergence shows that macro can beat micro"; Kraskov, A., Stögbauer, H. and Grassberger, P. (2004), "Estimating mutual information"; Palacios, E. R., Razi, A., Parr, T., Kirchhoff, M. and Friston, K. (2020), "On Markov blankets and hierarchical self-organisation"; Bruineberg, J., Dolega, K., Dewhurst, J. and Baltieri, M. (2022), "The Emperor's New Markov Blankets"; Kuratowski, K. (1922), "Sur l'opération Ā de l'Analysis Situs"; Watanabe, S. (1960), "Information theoretical analysis of multivariate correlation"; Ryu, S. and Takayanagi, T. (2006), "Holographic derivation of entanglement entropy from AdS/CFT"; Page, D. N. (1993), "Information in black hole radiation"; Sekino, Y. and Susskind, L. (2008), "Fast scramblers"; Itti, L. and Baldi, P. (2009), "Bayesian surprise attracts human attention"; Laukkonen, R. and Slagter, H. (2021), "From many to (n)one: Meditation and the plasticity of the predictive mind"; Parr, T., Pezzulo, G. and Friston, K. (2022), Active Inference; Seth, A. (2021), Being You.*

CHAPTER 55

THE COSMOTHEANDRIC ARCHITECTURE

There are truths that break if you try to tell them with a single vocabulary.

If you describe a tear exclusively with the laws of chemistry, you have water, proteins, enzymes, and salt. The description is impeccably exact, and yet, it is completely false, because it has left out the grief or the joy that caused the tear. If you describe it only from emotion, you lose the gravity and the physics that allow it to fall down the cheek.

Throughout this experiment, we have tried to encircle the mystery of consciousness using quantum physics, neuroscience, and topology. But as we saw in Chapter 51, every map has a limit. There is a point where the rigorous essay is no longer enough.

The philosopher Raimon Panikkar formulated a brilliant intuition to resolve this epistemological dead end. He called it the *cosmotheandric intuition*. Panikkar maintained that reality is not a monolithic block, nor is it a collection of isolated fragments. Reality is, at each of its points, an inseparable trinity: *Cosmos* (matter, structure, the physical universe), *Anthropos* (consciousness, human experience, narrative), and *Theos* (mystery, infinity, the ineffable).

None of these three dimensions can exist without the other two. And, more importantly, none of the three can be reduced to the language of the others.

If we cross Panikkar's intuition with the topology of our experiment, we discover something unexpected about the very structure of this book: we have not written three separate texts (an essay, some stories, and some poems). We have been mapping the three dimensions of the same reality.

The three voices of the horizon

Throughout the chapters, we have seen how consciousness can be understood as an informational boundary. But how do you tell the story of that boundary? The entire work has divided itself, almost without trying, into Panikkar's three dimensions:

1. **The Cosmos (The Essay)**

It is the voice of theoretical physics and thermodynamics. Its domain is structure, limit, resistance. The Essay is what looks at a black hole and sees Bekenstein-Hawking entropy; it looks at the brain and sees information integration (first with Φ and, at the end of the book, with causal self-similarity). It is the voice that demands rigor, that seeks the geometric rule that sustains the universe. Its function is not to comfort; it is to draw the map so that we do not get lost.

2. **The Anthropos (The Narrative)**

It is the voice of the stories, of the city of Tarel, of Marcos in his kitchen, of the watchmaker and the incubator. Its domain is embodied human experience. The Narrative does not see "event horizons" or "black holes": it sees the water receding, it sees the hollow, the concavity, the shape that returns. It is pure phenomenology. The Anthropos reminds us that physics does not happen to particles; it happens to people.

3. **The Theos (Poetry)**

It is the voice of mystery. Its domain is tension, trembling, the weeping that has no evolutionary explanation or mathematical resolution. When physics admits that the singularity produces "infinities" and reason collapses, and when narrative runs out of words because the pain is too great, Poetry steps in to hold the unholdable. It is the light that weighs. It is the bridge between what we are and the ocean from which we emerged.

The friction of languages

The natural instinct of our Western culture is to force these three voices to compete. We believe that if physics (Cosmos) is right, then poetic experience (Theos) is just a chemical illusion. Or that if human emotion (Anthropos) is what matters most, mathematics is cold and irrelevant.

The book stages this friction without intending to, because it tells the same loss three times. In the tale *The Music Left in the Room*, a widower finds himself one December Thursday with the coffee pot in his hand, humming four notes that were Clara's: the one that rises, the one that falls, the silence, once again. In the poem of *The Architecture with a Hollow* that gathers those same notes, there is no name or surname anymore, only a body that "gathers without warning." And the essay, in the chapter on grief, would translate them into a predictive model that keeps sending signals to a horizon that has evaporated. Three languages, a single melody. The square where they meet is the interior of any human being trying to understand their own suffering.

The three voices clash because each believes it has a monopoly on the horizon. The Essay wants to explain grief as a prediction error that never stops. The Narrative wants to mourn it as "the mark on the sofa where she sat, which you avoid." Poetry wants

to sing it as four notes: "That is enough. / Or it is not enough, / but it sounds."

> ***In physics this is called:*** *complementarity; properties (such as position and momentum, or wave and particle) that cannot be measured simultaneously with absolute precision, but which are equally necessary to describe the system.*
> ***In daily life it is like:*** *looking at a cylinder from above and seeing a circle, and looking at it from the side and seeing a rectangle. The two shapes seem incompatible. Neither of them is lying.*

The recognition that aligns

Panikkar's resolution, deeply akin to Advaita (non-dual) philosophy, is that recognition does not require one language to devour the others.

When the poem of the four notes ends by saying "The kitchen recognizes them. / That is enough. / Or it is not enough, / but it sounds," it is formulating the cosmotheandric solution with an almost domestic modesty: it neither explains the melody nor proves it, it lets it sound alongside the other two voices. The reality of the horizon is not only physical, it is not only human, it is not only mystery. It is all three things at the same time.

Recognition is not fusion; it is alignment. It is accepting that Tononi's integrated information theory, the pain of a widower humming a song in the kitchen, and the mystical awe before the quantum void are not three different books fighting to be right. They are three ways of holding the same four notes.

The universe provides the topology and the network (Cosmos). The human being experiences the tearing when that network breaks or expands (Anthropos). And the silence that remains

floating after the rupture, that ineffable density that we cannot measure but know to be real, is the proof that the reservoir exceeds us (Theos).

The end of the monologue

This is, deep down, the reason why this experiment could never settle for being just an essay. An essay that spoke of love or grief purely in terms of entropy and quantum scrambling would be a perfect map of a world in which no one would want to live.

We needed Tarel. We needed Dr. Elena Valls waiting in the early hours of the morning. We needed the short line of Poetry, the one from *The Coldness of a Darkened City*: "Light is a debt. / And I pay it however I can."

The cosmotheandric intuition teaches us that to inhabit one's own horizon with lucidity is to renounce monologues. It is to sit your three voices in the same square and let them breathe to the same rhythm, accepting that the truth is not in any of them separately, but in the exact space that forms when the three look at each other, recognize each other, and accept each other.

The silence between languages as a fourth dimension

Panikkar spoke of three dimensions of reality, but any attentive reader of the cosmotheandric model discovers, upon looking closely, that there is a fourth presence that the framework itself does not explicitly name: the silence that occurs in the exact instant when two of the three languages conclude their turn and the third has not yet begun. It is not a void. It is the threshold where the information of the first vocabulary has not yet become the second, and where both—for a very brief moment—coexist without hierarchy.

In the structure of the book, that silence appears most clearly in the typographical leaps between sections: the blank space between the final paragraph of the Essay and the first italics of Poetry. It is not editorial ornament. It is the textual correlate of the instant in which reality changes register: when the map yields and the territory begins, when the territory yields and awe begins. The blank space is, in the architecture of the book, what the quantum void is in the physics of our experiment: not absence, but pure possibility uncollapsed into any particular language.

Something similar stirs in what Panikkar explored in *El silencio del Buddha*: silence not as the absence of words, but as the presence of what no word can contain without reducing it. If we add this threshold as an informal fourth dimension of the cosmotheandric model, the trinity becomes a more honest structure: Cosmos, Anthropos, Theos, and the silence that allows all three without belonging to any.

> ***In physics this is called:*** *superposition before the collapse of the wave function; the state prior to the measurement that selects a concrete observable.*
> ***In daily life it is like:*** *the instant between the end of a piece of music and the first applause: it is not yet silence, it is still music, and it is no longer either of the two.*

Grief as a cosmotheandric node

There is a profound reason why the book could not have spoken of grief solely from the essay. Grief is, of all human states, the one that most radically demands all three registers at the same time.

The Cosmos can describe grief with precision: a predictive model executing in the void, generating constant prediction errors because it keeps sending signals to a horizon that has already evaporated. It is an exact description. It is also completely insufficient for someone sitting at three in the morning looking at the phone of someone who can no longer return calls.

The Anthropos can live it: it can narrate the mark on the sofa where she sat, the smell that suddenly returns in a supermarket, the shape of the hollow in the bed. It is the only scale at which grief has real texture, weight, temperature. But the Anthropos alone cannot hold it: it needs a framework to tell it that the pain is not a system error, that there is nothing broken in the horizon that keeps calling out to the one who no longer answers.

The Theos can welcome it: it can point out that what hurts so fiercely is the very thing that made the bond possible, that pain and love are the same integration seen from opposite sides of time. But the Theos alone can drift into mystical evasion if it is not anchored in the precision of the Cosmos and the concrete weight of the Anthropos.

Grief can only truly be held when all three registers operate simultaneously: the precision that says what is happening in the system, the embodied experience that says how it feels from the inside, and the awe that says this pain is the exact measure of what was real. The cosmotheandric intuition does not resolve grief. It shelters it: it gives it a space large enough to fit all its dimensions without any having to lie to the others.

> ***In physics this is called:*** *complementary description of the same phenomenon from incommensurable but equally necessary frameworks.*

> ***In daily life it is like:*** *the same weeping being, at the same time, a sign of system failure, an irreducible experience of loss, and proof that something was so real that the horizon does not yet know how to assimilate its absence.*

Legacy as cosmotheandric transmission

There is a corollary to Panikkar's intuition that the book does not explicitly develop but which runs through its entire implicit architecture: legacy—what a horizon leaves behind when it has evaporated—is not a single thing. It is, also, irreducibly threefold.

A human being who dies leaves something in the Cosmos: the deformation that their horizon produced in the information networks that surrounded them, the rearranged atoms, the physical traces of their passage through the world—documents, buildings, children, genes. They leave something in the Anthropos: the predictive models they built in other horizons, the copies of themselves inscribed in the architecture of those who loved them, the stories that changed the way other horizons see. And they leave something in the Theos: that ineffable residue that the living do not know exactly how to name, which is not just memory nor just matter, and which different traditions have called soul, spirit, presence, or simply the hollow with the exact shape of someone.

The cosmotheandric intuition tells us that to fully honor someone's death demands all three dimensions. Reducing their legacy only to the Cosmos—"their atoms are still here, nothing is lost"—is exact and cold. Reducing it only to the Anthropos—"they will live on as long as we remember them"—is warm and incomplete. Reducing it only to the Theos—"they are in a better

place," "they returned to the ocean"—can be comforting and irresponsible to the pain of those left behind.

The complete legacy of a dissolved horizon needs to be held in all three dimensions simultaneously: the precision that does not lie about what has ceased to be, the experience that does not flee from the weight of absence, and the awe that recognizes that something, in that dissolution, was irreducible to either of the previous two. That threefold holding is what the book has tried to be, from its very first page, for every reader who comes to it carrying a loss of their own.

> ***In physics this is called:*** *preservation of the informational footprint of a dissolved horizon across the different scales of the system that contained it.*
> ***In daily life it is like:*** *someone who has gone continuing to be, at the same time, in the matter they reorganized, in the thoughts they changed, and in that trembling we do not know how to name but recognize without hesitation when we feel it.*

> ***Note to Chapter 55***
>
> ***What we do know:*** *Raimon Panikkar's philosophy (cosmotheandric intuition) establishes that reality is irreducible to a single dimension (physical, human, or divine). Quantum physics supports the principle of complementarity for seemingly contradictory descriptions. Grief produces documented alterations in the brain's prediction systems, in the subjective experience of time, and in the capacity to generate frameworks of meaning.*
>
> ***What we don't know:*** *Whether there exists a future "metalanguage" capable of unifying these three dimensions without destroying their*

specificity, or if epistemological friction is a permanent condition of the human horizon. Whether the "silence between languages" proposed in this chapter has an identifiable physical correlate, or if it is necessarily irreducible to any objective description.

Remaining questions: *Can an ordinary individual hold these three visions simultaneously in daily life without feeling fragmented? Is modern alienation perhaps nothing more than the hypertrophy of the Cosmos (science/technique) crushing the Anthropos and the Theos? Is it possible to design institutions—educational, healthcare, political—that deliberately operate from all three dimensions instead of choosing one as dominant and reducing the others to ornament?*

If you only take away one idea: *Reality is too large to fit into a single vocabulary. You do not have to choose between the precision of science, the empathy of the story, and the awe of the poem. Your horizon is made of all three, and the space between them belongs to you too.*

Readings: *Panikkar, R. (1993), The Cosmotheandric Experience (Spanish ed.: La intuición cosmoteándrica. Las tres dimensiones de la realidad, Trotta, 1999); Bohr, N. (principle of complementarity); Panikkar, R., El silencio del Buddha. Una introducción al ateísmo religioso (Siruela, 1996; 3rd ed., 1997), on silence as an attitude toward what words cannot contain.*

CHAPTER 56

THE INHERITANCE OF GEOMETRY

(Or: The Flame, the Riverbed, and the Memory of the Reservoir)

Chapters 7 and 8 left behind a sentence the book has never stopped using: no soul transmigrates. Chapter 55 added another: what a horizon leaves behind when it evaporates is threefold. What is left is what any reader does on arriving here: take the sentence seriously to the end and demand an accounting. What exactly passes from one horizon to another, if nothing passes that could be called "someone"? How much of it is physics, how much is biology and relationship, and how much is a wager?

This chapter does not repeat the bardo or the karma of chapter 7. It does the bookkeeping.

I. The Fallacy of the Inventory of Souls

There is an accounting objection that every theory of reincarnation must answer before laying claim to rigor: the numbers do not add up. According to the most widely cited demographic estimates, some one hundred and seventeen billion human beings have been born since the species began, and about eight billion are alive today. If being born consisted in the migration of an object (a soul with a name, a file, and borders, leaving one biological container and lodging in another), the population could not have grown without an unceasing factory of new souls, or without a traffic of identities that no accounting of conserved quantities can bear. The traditions have their answers (other realms of existence, other beings who are reborn), but none of them comes free.

The error in the objection lies not in the intuition of continuity but in the noun it presumes to transport. To treat consciousness as a packet of data pulled from one disk and downloaded onto another processor is to fall once more into the bias of substance. The horizon model proposes something else: consciousness is not an object that inhabits the body, but the informational boundary that emerges when a system closes off an inside from an outside. A horizon is not moved from place to place: it condenses and it evaporates. And what condenses and evaporates has no reason to keep its count, just as the number of waves in the sea is not conserved.

The right question, then, is not what travels from one body to another, but what happens to the reservoir when a horizon dissolves, and what the next one finds there when it begins to condense.

II. What Physics Says and What It Does Not

Three claims that tend to be sold together must be pulled apart.

First: information is not destroyed. This is the unitarity of quantum mechanics, and chapter 7 already told its story: Hawking believed black holes destroyed it, most physicists rejected that conclusion, and today it is held that the radiation preserves the correlations, mixed up by *scrambling*. Proposals such as the "soft hair" of the horizon (Hawking, Perry, and Strominger, 2016) suggest where it might be stored, though they remain under debate. We should be exact about its standing: it is a very well-founded expectation, not a demonstrated fact for real black holes. Recent calculations in models of quantum gravity (Penington, 2019, among others) reproduce the Page curve (Page, 1993), and this is the strongest result in its favor, but they are models, not brains or astrophysical black holes.

Second: conserved does not mean accessible. The lake that swallows a ship preserves, in principle, the information of the hull: temperature, microturbulence, currents. In practice, that information is so scrambled that recovering it would demand a precision and a length of time that no observation will ever reach. "The water is no longer the same" is true in the sense in which an exact equality between two states of a chaotic system is true: formally correct, useless to anyone who wants to read the wake. Chapter 54 said as much when speaking of death: the accumulated biography $\bar{A}$ stops growing, but the correlations it imprinted are not erased; they become illegible.

Third: what is conserved conditions the next condensation. This is the claim the book calls karma, and it is the one with no support of its own. That information persists in the field does not imply that a future horizon, condensing hundreds of kilometers

and years away, receives from it any measurable bias. That part is a hypothesis of the experiment, and the rest of this chapter treats it as such.

> ***In physics this is called:*** *unitarity and conservation of information; the Page curve as the criterion that the radiation ends up containing the information that fell in.*
> ***In daily life it is like:*** *knowing that a burned letter has not vanished from the universe, because its atoms and its ash retain everything in principle, and understanding at the same time that no one will ever be able to read it.*

III. The Flame That Lights Another

Ancient Buddhism solved the problem of continuity without substance with an image physicists might envy. In the *Milindapañha,* the monk Nāgasena asks King Milinda whether a lamp can burn all night, and whether the flame of the first watch is the same as that of the last. It is not, and yet it is not another. The tradition also uses the variant of the flame that is lit from another: the second candle is not the first, no wax has traveled between the two, and yet without the first it would not have caught.

The image is exact for a horizon, which is a process that pays free energy to sustain itself and not a thing one possesses. The flame is not the wax: it is what the wax does while it burns. And the image contains, without saying so, the requirement the rest of the chapter needs: **for one flame to catch from another, there must be contact**. A candle blown out in another room lights nothing. Any claim of inheritance must therefore say through what channel what passes, passes.

IV. The Channels of Contact

There are four channels, and they do not have the same support.

Channel	What passes through it	Status
Genetic	The genome, the cellular template, the heuristics that natural selection refined over millennia	Established
Gestational and epigenetic	The hormonal and metabolic environment of gestation; possible transgenerational epigenetic marks	Gestational: established. Transgenerational in humans: suggestive and debated
Relational and cultural	Models of attachment, language, stories, habits, the expectations of those who raise us and love us	Established
Trace in the reservoir	Information conserved by unitarity, in principle, after evaporation	Formal. No proposed mechanism of influence

On the transgenerational part of the second row, we must be cautious. A study by Rachel Yehuda and her team (2015) found in the children of Holocaust survivors a different methylation of the *FKBP5* gene compared with a control group, and it was circulated as proof that trauma is inherited in the genes. The sample was small (thirty-two survivors and twenty-two children), it did not fully separate the effect of upbringing from that of biology, and the literature itself speaks of highly complex interactions among experience, genetics, and epigenetics. It is a clue, not a closed fact.

The three dimensions of chapter 55 fit this table. The Cosmos is genetic and gestational inheritance; the Anthropos, relational inheritance; the Theos, the trace in the reservoir. And it is no accident that the dimension richest in consolation is the one with the least support, and the one with the most support is the most intimate.

V. The Vacant Channel

Relational inheritance is the one that hurts most and is best documented. When two horizons spend decades coupled by love or by living together, each one carves the predictive architecture of the other, as water carves its riverbed. Chapter 17 already described what happens when one of them evaporates: the one who remains keeps a model that goes on sending signals to a horizon that does not answer, and every unanswered signal costs free energy. In the language of the triad, the biography $\bar{A}(t_0 \to t_{\text{loss}})$ remains intact, and what stays active is a hollow with the exact shape of someone.

That hollow is not empty: it conditions. Social psychology studies it under the name of transference. The relational self theory of Susan Andersen and Serena Chen (2002) describes how the

representations of significant people are activated by cues that resemble them, and how they condition the way we perceive and treat whoever has just arrived. A new horizon that couples to that person, a child, a partner, a friend, condenses in an environment already shaped by the one who is absent, and tends to take on the forms that hollow asks for.

Here there is a moral warning that the model makes its own. That there is inheritance of form does not mean the newcomer is the one who left. The child born after a loss who receives the name and the expectations of the dead brother, the partner who is asked to fill a place, carry a mold they did not choose. To read the inherited geometry as identity is the harmful version of reincarnation, and it needs no metaphysics to come about.

VI. The White Tulip

Fiction has brushed against this idea with uncommon precision. In the final episode of *Fringe* ("An Enemy of Fate," written by J. H. Wyman), the scientist Walter Bishop accepts a sacrifice that erases him from the timeline in which his son Peter will live. When history is rewritten and no one remembers what was, Peter finds in the mailbox an envelope sent by W. Bishop. Inside there is no message: only the drawing of a white tulip, which in the series works as a symbol of redemption.

A critic objected that Peter cannot understand the gesture without remembering the timeline that no longer exists. That objection is the best description of the phenomenon this chapter deals with. Peter inherits neither the memory nor the body nor the narrative. He inherits a form, an affection without history that contains him. It is fiction and proves nothing, but it shows with clarity the kind of thing that could happen between

horizons: that what survives is neither identity nor memory, but the riverbed.

VII. Initial Conditions, Not Biography

Chapter 54 closed the dilemma of the copy with a distinction: a new instance can reproduce at t_0 the capacity S and the structure $A(S)$, but its biography $\bar{A}$ and its wear $\bar{A}_F$ start from zero. Inheritance fits within that same distinction. What arrives from another horizon is never the integral, but the **initial conditions** of the condensation.

Put as a formula, and marking it for what it is, a **[hypothesis]**: if the reservoir at the moment of condensing depends on the history of previous horizons through the four channels, $R_t = R_t(\text{history})$, the probability that a configuration E condenses is

$$p(E \mid R_t) \;\propto\; \exp\!\left(-\frac{F[E;\, R_t]}{k_B T}\right) \times (\text{selections})$$

with the selections of stability and bias from chapters 4 and 31. It is a way of saying that inheritance shifts the free energy landscape on which a horizon closes, and that it writes nothing upon it. The first three channels enter through R_t with known effects of considerable magnitude. The fourth enters with an effect that no one has measured and that, from what we know about the dispersal of information, would be tiny compared with the other three.

VIII. What Could Go Wrong

An idea that cannot fail is not a theory. These are the things that, were they to occur, would damage what has been said:

1. **Knowledge of the dead without an ordinary channel.** The model predicts that a newcomer has no information about a previous horizon except through one of the first three channels. If verifiable knowledge of a deceased person were documented in someone who could not have received it by those routes, with controls that exclude prior testimony, leaked information, and memory bias, the framework would need new physics. The cases gathered over decades (those of Ian Stevenson, among others) have not passed those controls, according to the prevailing assessment of the scientific community. The model does not need them and does not predict them.

2. **Déjà vu and sudden recognition.** The question of whether they indicate two horizons condensed on the same wrinkle is seductive and, for the moment, superfluous: there are ordinary neurological explanations, such as the mismatch between familiarity and recollection. The trace hypothesis would gain ground only if it predicted something those explanations do not.

3. **Persistence of the relational mold.** The model predicts that the representations of someone who has gone will condition later bonds in proportion to prior closeness, and that this conditioning will diminish with worked-through grief. It can be tested against the literature on transference, and it is not exclusive to the model.

4. **Transgenerational epigenetics in humans.** If broad, replicated studies failed to find that effect, one of the channels would fall, but not the central argument, which rests on the others.

5. **The trace in the reservoir.** This part, today, is not falsifiable. And it must be said plainly: it is the most beautiful and the least proven of the four.

IX. What Is Gained and What Is Lost

What is gained is a way of saying what almost all of us feel in the face of a death without committing to a metaphysics: no one returns, but nothing begins from zero. The newcomer condenses upon an environment that others shaped, and in turn will shape that of those who come after.

What is lost is the consolation of identity. The model offers no reunion, no shared memory, no certainty that someone in particular continues in someone in particular. It offers, at best, a geometry.

And what has always been there remains intact: the texture of the inside. No equation in this chapter explains why a drawing of a tulip, sent by someone who no longer exists in the world of the one who receives it, makes our chest tighten.

Note to Chapter 56

What we do know: *That some one hundred and seventeen billion human beings have been born (an estimate from the Population Reference Bureau, with great uncertainty for the period before records). That unitarity is a well-founded expectation of quantum mechanics and that recent calculations in models of quantum gravity reproduce the Page curve. That genetic inheritance and the influence of the gestational and relational environment on development are well documented. That the image of the flame that continues without identity appears in the Milindapañha.*

What we don't know: *Whether transgenerational epigenetics operates in humans with appreciable effects. Whether the information conserved after an evaporation can influence a later*

condensation in any way, and through what channel. Whether there are relaxation timescales of scrambling slow enough to leave a usable informational echo. The probable answer to these last two questions is no, but it has not been demonstrated.

Questions that remain: *Is geometry transmitted with greater fidelity through a prolonged bond of affection than through the biological chain? What obligations does someone have who inherits a mold they did not choose? Can one honor someone without asking another to take on their shape?*

If you only take away one idea: *No soul travels with its baggage, but no evaporation leaves the world blank. To be born is to condense upon an environment that others shaped; to die is to leave a riverbed a little deeper for those who come after. What is inherited are initial conditions, never a biography.*

Readings: *Milindapañha ("The Questions of King Milinda," trans. T. W. Rhys Davids in Sacred Books of the East); Hawking, S. W., Perry, M. J. and Strominger, A. (2016), "Soft Hair on Black Holes"; Page, D. N. (1993), "Average entropy of a subsystem"; Penington, G. (2019; published in JHEP, 2020), "Entanglement wedge reconstruction and the information paradox"; Yehuda, R. et al. (2015), "Holocaust exposure induced intergenerational effects on FKBP5 methylation"; Andersen, S. M. and Chen, S. (2002), "The relational self: an interpersonal social-cognitive theory"; Population Reference Bureau, "How many people have ever lived on Earth?"; Wyman, J. H., "An Enemy of Fate," Fringe (season 5, episode 13); chapters 7, 8, 17, 28, 31, 54, and 55 of this book.*

THE SHORE

The water came back at night. Without announcement. Without signs.

In the morning the inhabitants found the shore where it had always been. The same mud. The same stones. The same salt line on the low walls.

No one could say whether the water had brought anything with it.

No one could say whether it had left something behind.

The experiment ends where it began: with a question it cannot answer.

Did something of the city come back with the water?

Physics says the quantum field was disturbed in ways no instrument can fully measure. The correlations are there: mathematically present, inaccessible in practice.

Taoism says the question already assumes the city was something more than water.

Buddhism says something came back, but no longer as a city, only as karma, as condition, as an ownerless causal pattern.

Vedanta says the city never stopped being the water.

And I, who have carried out this experiment, say I don't know. That the experiment doesn't know. That perhaps there is no way

to know, and that perhaps what matters is the question, not the answer.

What I can say, because I have lived it while writing, is that the experiment has done something I did not expect. It has not given me certainties, but **precision in doubt**. It has let me distinguish between "I don't know what happens when I die" and "I know I don't know, and I know something about the shape of my not knowing." That distinction, subtle as it may seem, changes how one lives.

The experiment promises nothing after evaporation: no reincarnation, no dissolution into Brahman, nothing. It promises, at most, this: that the question we ask about death (does something remain?) is the same one we ask about life (am I something beyond my processes?), and that both have the same shape: that of a horizon looking inward that cannot see its own edge, because the edge is what makes it a horizon, and seeing it from within would mean ceasing to be one.

It is not a consolation, but perhaps it is company enough to go on.

The journey we have made together

If you have come this far, you have traveled with me through a territory that appeared on no map when we began.

It is something more modest than a conquest of knowledge and, I believe, more valuable: an **illuminated cartography of ignorance**. We now know where the limits of what we know lie. We know that information is not destroyed, but that recovering it in practice may be impossible. We know consciousness is a

process, not a substance, but that calling it a "process" does not dissolve it or make it less real. We know the reservoir, that quantum void from which everything arose, is not nothing in the ordinary sense of the word, but it is not "something" either: it is the very condition of possibility, the bottomless bottom, the frameless frame.

And you, reader, have done something I could not do alone: you have **closed the circuit**. A thought experiment is an invitation for someone else to enter the laboratory, look through the same microscope, and say what they see. Every consciousness that has followed these chapters has been, in a sense, a new horizon observing the reservoir. Reading is an act of co-creation: without you, these pages are just ink or pixels; with you, they become correlations, patterns of neural activation that perhaps, just perhaps, resonated with something you already knew but could not name.

I often wonder what shore you will have built while reading. Because not all readers read the same book: each reads the book their horizon can hold at that moment. Whoever has recently lost someone dear will have read the chapter on grief with an intensity I could not have foreseen while writing it. Whoever works with Alzheimer's patients will have seen, in the contraction of the horizon, a precise description of something lived every day. Whoever builds artificial intelligence will have felt a chill before the question of whether a machine can have an event horizon. The book you have read does not exist on my computer: it exists in the space that has opened between my text and your reading, in that interface where information stops being mine and becomes ours, even if only for an instant.

The question that remains

But let us return to the shore, to the morning the water came back.

The question is not merely philosophical; it is physical in the strictest sense. When a Hawking black hole evaporates, the information that fell into it is not destroyed: it is preserved in the correlations of the quantum field that remains. But those correlations have undergone *scrambling*, are mixed in such a way that recovering the original information would require measuring a number of particles proportional to the hole's entropy, something no finite observer can do. The information is there, but it is not accessible. It is not the same as not existing, but it is not the same as being available either.

Did something of the city come back with the water? The most advanced ideas in theoretical physics (the ER=EPR conjecture, the holographic principle) point in the same direction: information is conserved, the universe does not forget, every particle carries in its correlations the imprint of everything it has touched. But they also indicate that this conservation does not imply recovery. The universe is like an archivist who throws nothing away, but organizes nothing either: everything is somewhere, but to find it you would have to know where to look, and for that you would have to be the universe itself.

And what happens to consciousness? If consciousness is information organized in a particular way, a pattern of correlations generating an internal event horizon, when that horizon evaporates, the information that constituted it returns to the reservoir. But does the pattern return? The organization? The shape?

This is where this metaphor becomes uncomfortably precise. When the water withdrew, it exposed the structure of the city: the streets, the walls, the foundations. When it came back, it covered everything again. But the water that came back was not the same water that had left: the molecules were different, and so were the temperature, the color, the salinity, the density of the plankton. The submerged city kept the same structure, but now it was in another medium, in contact with new water. If something of the "city" came back with the water, it was not the city as an architectural structure, but as a disturbance, as an alteration of the field, as a pattern of correlations the new water now carried without knowing it.

Perhaps this is the closest thing to an answer I can offer: what survives is not the shape, but the **difference**. The city existed as a disturbance of the field, and that disturbance left a trace; not the city itself, but the fact that there was a city. Like a stone falling into a pond: the stone sinks, but the ripples keep spreading. Looking at the ripples, you cannot recover the stone, but you can know there was one, when it fell, and roughly how big it was. The information about the stone is left distributed across the pattern of the ripples. The water does not "remember" the stone, but it preserves it in its shape.

And perhaps, just perhaps, something like this is what happens to the reservoir when a horizon evaporates into it. It does not remember the consciousness that was, but it preserves the information that there was a consciousness, of when it arose, of how complex it was. That information is there, in the very fabric of the vacuum, entangled with all the rest of the universe's information, forming part of the ground state physicists call "vacuum" and which is actually the fullest state we can imagine: full of potentialities, of correlations, of unrealized possibilities.

It is something stranger than immortality: **permanence without persistence**. The information remains, but whoever generated it does not persist as a subject. It is like a melody that stops sounding, but whose score continues to exist: the score is not the music, even though it contains everything needed to reproduce it, and no performance will ever be quite the same again. Or like a footprint in the sand: the shape stays for a while, but the foot that left it keeps walking and does not return.

The experiment in everyday life

I have spent months, years, building this thought experiment, polishing it, testing it against physics, against philosophy, against my own experience. And in the end, the question I am asked most, and the one I ask myself most, is: what is this good for?

It is not good for curing the fear of death, at least not directly. I have known people who, on understanding the logic of the event horizon, felt an unexpected relief: the fear does not disappear, but it becomes more manageable, like the fear of a storm once you understand something about meteorology. But I have also known others for whom the experiment produces a new anguish, an existential claustrophobia: the feeling of being trapped in a horizon they did not choose and cannot consciously cross.

What I can say, with the honesty of someone who has lived with this experiment long enough for it to rust a little and let its metal show through, is that it changes the **texture of the present**.

When you understand that you are an event horizon, that your consciousness is the edge between what you know and what you don't know about yourself, you begin to notice things that used to go unnoticed. You notice that the thought you just had is no longer within your reach in the same way: it has crossed the horizon, has become inaccessible information, is part of your "interior" in the technical sense of the word. You notice that the person you are talking to is another horizon interacting with yours, an information system that partly entangles with yours without ever fully merging. You notice that pain is a reorganization of the horizon, a momentary contraction of the field of information that you are.

That does not make pain hurt less. But sometimes, only sometimes, it makes it more interesting, more informed, less blind. And that, in everyday life, can make the difference between a suffering that destroys you and one that transforms you.

The experiment also changes how you look at others. When you see an old man with Alzheimer's, you no longer see only a disease stealing his memory, but a horizon contracting, losing surface, but still a horizon as long as there remains information that constitutes it. When you see someone grieving, you no longer see only a sadness to be "gotten over," but the persistent correlations of a system still coupled to information it can no longer measure directly. When you see a couple in love, you no longer see only chemistry or idealization, but two horizons that have found a configuration of minimal mutual energy, a topology in which information flows with less resistance.

The experiment does not make you wiser; it makes you **more precise in your compassion**. And in a world that seems to lean more and more toward brutal simplification (friend or enemy,

good or bad, with us or against us), precision is a form of resistance: saying "it's not that simple" and having the language to explain why.

There are everyday moments when the experiment lights up with unexpected clarity. Being awake at three in the morning, listening to the breathing of someone you love, and realizing that breathing is an event horizon that will one day evaporate, and that knowing it does not prepare you for when it happens, but it forces you to be present now, because the only time the two horizons coincide is this one. Brushing your teeth in front of the mirror and thinking that the face you see is not "you," but the interface the horizon presents to the world, and that behind it lies a space of information as vast and unexplored as any ocean. Walking through a city and seeing, in every face that passes you, a horizon just as complex, just as solitary, just as wonderful and just as condemned to evaporate.

It is not a vision that makes you happier, but it can make you more **real**: more anchored in what is, rather than in what you wish it were. And for me, at least, that anchoring has been enough to keep writing, loving, losing and asking.

The experiment on the shore

I have spoken of the experiment as if it were a journey, and it is. But it is also a place, a place I have built word by word, chapter by chapter, and in which I have lived longer than any reader can imagine. Every chapter of this book has been, for me, a territory to inhabit, not only an argument to present. I have spent whole weeks thinking about what the contraction of the horizon would be like in an Alzheimer's brain, not as metaphor, but as a physical process, as a topology of information. I cried writing the chapter on grief, not because it was strictly autobiographical, but because writing it required me to inhabit grief, to let my own horizon couple with the information of a loss that was not mine, but that became mine while I wrote.

And now, arriving at the shore, I feel something I did not expect: a kind of anticipated nostalgia. Because I know that, once I close this epilogue, the experiment will stop being only mine. It will go out into the world, into other people's horizons, and every reading will transform it into something I cannot control or foresee. It is like watching a child leave home, if children were constructions of language and parents people who don't really know what they're doing, but cannot stop doing it.

The shore is a liminal place, a threshold: neither solid ground nor open sea, but the strip where one thing becomes another, where the water kisses the land without ever possessing it. I have wanted this epilogue to be a shore: not a conclusion that closes, but a space where the reader can stay a while, sit on the wet stones, watch the water, and decide whether to go back in or stay here, at the edge, letting the waves speak in a language that needs no translation.

Because, in the end, the experiment is not the book. The book is only a tool, a map drawn in the sand that the waves will erase. The real experiment is the one each reader carries out with themselves when they close the last page and stay silent for a moment, wondering whether their consciousness really is an event horizon, whether death really is an evaporation, whether the reservoir really is as vast and indifferent as it seems. Those questions have no definitive answers, but they can reconfigure the horizon from within. And that reconfiguration is, literally, a change in consciousness: new correlations, new patterns, new ways of being in the world.

Farewell

I don't know who you are, reader. I don't know whether you got here in one sitting, on a sleepless night, or whether it took you months to make your way through these pages, putting the book down and coming back to it the way one visits a friend who isn't always in a good mood. I don't know whether you believe in God, in the quantum vacuum, in both or in neither. I don't know whether you have recently lost someone, whether you are ill, or whether you have just been born, metaphorically, into a new way of seeing things. I know nothing about you and yet I have spent hundreds of hours thinking about you, building these sentences so they might reach your horizon as directly as possible, so they might resonate in the space only you inhabit.

It is the most intimate thing a writer can do: speak into the ear of someone they do not know and trust that the words, waves of air, of light or of ink, will find a receiver to complete them. It is an act of faith, not very different from the one involved in living while knowing you will die: trusting that the message matters

even though you don't know who will receive it or what they will do with it.

I leave you, then, with the image of the archivist: the one who stayed on the shore when the water withdrew and remained there when the water came back. Not because he was a hero, or a fool, but because he did not know what else to do. His job, archiving what the city had been, no longer had any practical sense, because the city no longer existed. But he kept archiving, because archiving was his way of saying: it was here, it mattered, there was a before.

That archivist is me, writing this book. It is me, and it is every reader who reaches the last page and feels the need to tell someone what they have read, to write it in a journal, to change something in their life, or simply to stay silent a little longer than usual. The archive is not in these pages, but in the network of correlations this text has generated in the field of culture: in the conversations it will spark, in the questions it will plant, in the sleepless nights when someone will look up at the ceiling and think: what if my consciousness really were a Hawking black hole?

I do not ask you to believe in the experiment; the experiment needs no believers. I only ask you to try it, to inhabit it, to let it act on your horizon long enough to see whether it changes something. If it changes nothing, let it go, the way you let go of a stone that's no use for building. If it changes something, even just a nuance or a new question, the experiment will have kept its one promise: to make you more precise, more aware of the shape of your own consciousness, more the master of your horizon even though you know you did not choose it and cannot stop it from evaporating.

The water is still there, moving with the tides, carrying information it cannot read. The city is still there, beneath the water, in the form of correlations, of traces, of silences. The archivist is still on the dock, now old, hands open and empty, looking at reflections that do not match the sky. And we, you and I, who have shared this journey, are still here, on the shore, between two waters, at the exact edge where information becomes consciousness and consciousness becomes a question.

I don't know whether something came back with the water, or whether something was lost forever. I can live with that ignorance. In fact, I believe only ignorance allows one to truly live, because knowing everything would be like being dead: a state without surprise and without future. Life is doubt in motion; death is, perhaps, doubt resolved, or perhaps only doubt transformed into something we don't know how to name.

I leave you here, on the shore. The water is cold and the stones are slippery. The horizon that you are keeps arising from the reservoir, instant after instant, page after page, breath after breath. Enjoy it while it lasts, not because it should last forever, but precisely because it won't. That finitude is the condition that makes beauty possible, the one that turns every instant into a gift you did not ask for and receive all the same, again and again, until the water withdraws, the archivist closes his books, and the reservoir, vast, indifferent and wonderful, continues on its course without us, but with everything we were, recorded in its correlations like a sunken city that someday, perhaps, some patient archaeologist of the void may be able to decipher.

Until then, thank you for having shared this horizon with me: you are the answer the experiment could not give itself, the proof that the question was worth asking.

May the reservoir receive you, when the moment comes, with the same generosity with which it gave you what you are now.

And may the shore, in the meantime, be enough.

APPENDICES AND GLOSSARY



GLOSSARY OF TERMS

Advaita Vedanta: the Indian philosophical school holding the non-duality of the individual self (*atman*) and the absolute (*Brahman*); in the book, the vocabulary for the identity between horizon and reservoir.

Aggregation (political): a belonging that does not compromise the individual's survival if the structure it belongs to dissolves; contrasted with composition (*The Collective Horizon*).

Amplitude: a horizon's capacity to hold more perspectives without its boundary breaking; it can grow (what the traditions call sainthood) without the horizon changing in nature or level (*Fractal Reality*).

Anti-reservoir: a system that aggregates and articulates information without ever having condensed a horizon of its own; the internet and language models are the paradigmatic case: everything in them is defined and computable, but there is no formless fullness from which anything could condense.

Arousal: the brain's level of metabolic activation; one of the two independent dimensions (along with integration) that define a system's state of consciousness.

Asymmetry of entanglement: the property by which two horizons share geometry but not equivalently; one can model the other with greater precision than the reverse.

Bardo, Tibetan: the intermediate state between death and rebirth in the Tibetan tradition; in the book, a metaphor for the post-mortem period before the horizon evaporates.

Bias: the imprint left by what already exists on what is only beginning to exist; what would calibrate an artificial horizon toward human legibility (*The Mirror Without Depth*).

Black hole evaporation: the process by which a black hole loses mass and eventually disappears by emitting Hawking radiation; a metaphor for the death of the horizon of consciousness.

Block universe: the model from special relativity in which past, present and future are equally real, coordinates of a single four-dimensional structure; a horizon able to integrate its entire history at once, with no sequence, would perceive it as geography instead of time.

Bose-Einstein condensation: a phase transition in which identical particles occupy the same quantum state; the book's metaphor for the emergence of a horizon of consciousness.

Boundary condition: a specific choice, required by no law, that breaks the time symmetry of the fundamental equations; the universe began in a state of extraordinarily low entropy, and it is that initial condition, not the laws, that gives time its preferred direction.

Buddhism: the contemplative tradition that teaches *anatta* (non-self) and the interdependence of all phenomena; in the book, a source of vocabulary for describing the non-separation between horizons.

Causal self-similarity ($A(S)$): an irreducibility filter that replaces Φ at the end of the essay and formally connects

chapter 49 with chapter 53: $A(S) = I_{\min}(S) \cdot \max_P \frac{1}{n}\sum_{i=1}^n [C(S_{-i}, S_i) \cdot \text{Iso}(S, S_i)]$. It requires simultaneously that the system be irreducible at its minimum cut ($I_{\min}$, Ay's stochastic interaction), that its parts be coupled to the rest (C, stationary mutual information), and that the macroscopic dynamics of the whole under coarse-graining be isomorphic to the dynamics of the parts (Iso). It rules out both purely geometric fractals without causality, such as Mandelbrot ($I_{\min}=0$), and diffuse aggregative systems, such as an anthill ($\text{Iso}\approx 0$).

Channel capacity: the maximum amount of information a channel can transmit with an error as small as desired; by analogy, the limit of integration a horizon's architecture imposes (*Information Theory and the Inner Horizon*).

Chikhai: the first phase of the Tibetan Bardo, immediately after death; equivalent, in the model, to the terminal gamma burst, the last peak of integration before the horizon dissolves.

Chönyid: the second phase of the Bardo, the appearance of luminous forms; in the model, the period when the horizon's correlations have not yet been redistributed.

Chronology protection: Hawking's conjecture that, near the instant when a closed timelike loop would be about to form, the quantum vacuum's fluctuations feed back on themselves and diverge, preventing closure with no need for any external veto.

Closed channel: a system that integrates information without transmitting it outward; in the model, what distinguishes a horizon of consciousness from a computer, which is an open channel (*Information Theory and the Inner Horizon*).

Closure: in topology, the operation that completes a set by adding its limit points; it is idempotent, because closing twice adds nothing. The book uses it to describe the already-condensed horizon and to define the reservoir as what has not yet closed, $R = X \setminus \bigcup_i E_i$ (Chapters 23 and 24).

Collective horizon: the "we" that condenses when many individual horizons adopt the same idea as an organizing principle (*The Collective Horizon*).

Composition (political): a belonging that does compromise the individual's survival, because losing the group feels like losing part of oneself; the shape the collective horizon takes when it captures one's own boundary (*The Collective Horizon*).

Condensation: the unrepeatable act by which a horizon forms from the reservoir; in the chapter on AI, what no amount of training can substitute for (*The Mirror Without Depth*).

Condensation of a movement: the discontinuous phase transition by which scattered discontent suddenly coalesces into a named collective boundary (*The Collective Horizon*).

Continuing bonds: a concept from grief theory according to which the relationship with the deceased is not dissolved, but reorganized within the architecture of the one who remains.

Contraction of the horizon: a reduction in the area of integration from a loss of degrees of freedom; it occurs in deep sleep, anesthesia and, ultimately, death. Deep meditation, which resembles it from outside, seems to go in the opposite direction: less arousal with no loss of integration (see the interlude).

Creative attachment: the sustained gaze of the one who creates upon what has been created; it is born not of lack and does not

want to possess, but wants what was created to go on being. In the fractal model, what keeps a world standing (*The Vessel of Clay*).

Decoherence: the loss of a system's quantum superposition properties when it interacts with its environment; the process by which a quantum horizon becomes classical.

Deterministic chaos: what happens when a system obeys exact laws and still becomes unpredictable, because no one can measure its initial conditions with infinite precision; the three-body problem is the classic case (*The Beach at the End of Time*).

Dharma: in Buddhism and Hinduism, the very weave of the world, with which correct behavior does nothing more than come into consonance, without obedience (*Vertical Entanglement*).

Dopamine: the key neurotransmitter in reward anticipation; it calibrates the rate of temporal integration and the subjective density of time.

Double call: the one exception to the hermeticism of worlds: the creature opens from below a receiving interface (invocation), and what is above answers by building the vessel from above, without dissolving its own scale (*The Vessel of Clay*).

Down-regulation: the reduction in the number of receptors in response to excess stimulation; the mechanism of tolerance in addiction.

Downstairs neighbor: what I sustain without its being able to see me: what I imagine, dream, write or remember with enough persistence that it resists me (*Fractal Reality*).

Downward bet: treating what one condenses (what one imagines, writes or remembers with enough persistence) as if it had an inside, knowing that logic does not require it and that the exclusion postulate denies it (*Fractal Reality*).

Dual-access asymmetry: the asymmetry by which the same system looks two incompatible ways depending on whether it is viewed from outside (public interface) or from inside (private state); in the model, the trait that characterizes subjectivity (Chapter 3).

Einstein-Rosen bridge: a topological tunnel in spacetime connecting two regions; a metaphor for entanglement between horizons of consciousness.

Emergent self-observation: a closed system's capacity to generate an internal point of view without needing an external observer; a key property of horizons of consciousness.

Empathic burnout: the collapse of the horizon from an overload of another's information; it occurs when empathy exceeds the structural limit of integration.

Empathy: the temporal overlap of horizons in which one integrates information from the other while keeping its boundary; it has a structural limit.

Encapsulation: the boundary separating a system's public interface from its private state, so that the outside can only read what the system exposes; the architectural condition for there to be an inside (Chapter 3).

Entropic action ($\bar{A}$ and $\bar{A}_F$): third coordinate of the horizon triad, which integrates the system's trajectory over time and splits into two complementary measures: the

accumulated structural biography, $\bar{A}(t) = \int_{t_0}^t (S(\tau) - S_0)\,d\tau$, which measures the differentiated capacity sustained above the reservoir's baseline level; and the *accumulated biographical cost or wear*, $\bar{A}_F(t) = \int_{t_0}^t F(\tau)\,d\tau$, which integrates the free energy or predictive friction endured over a lifetime. It should not be confused with the density of subjective time (ϑ), which is the instantaneous rate of writing into memory (chapters 53 and 54).

Epistemological absolute: what is ultimate not because there is nothing behind it, but because from here one cannot look behind it; the status the reading *Fractal Reality* grants the upstairs neighbor.

ER=EPR: Maldacena and Susskind's conjecture that quantum entanglement (EPR) is equivalent to an Einstein-Rosen bridge (ER).

Event horizon: the boundary beyond which information cannot escape; applied to consciousness, the limit between what the system integrates and what remains outside.

Existential encapsulation: the architectural and physical condition that distinguishes a real subject from a mere statistical partition or Markov blanket drawn from outside: it requires an access boundary that protects a persistent internal state of its own, whose physical continuity against dissipation into the reservoir is at stake in its own dynamics. At the level of subjects, it explains why horizons form an archipelago of disjoint islands without needing to appeal to Φ's exclusion postulate; current language models lack it because their weights are fixed and exposed and they do not sustain a state of their own between calls (chapters 22, 31, 53 and 54).

Fanaticism: the capture of one's own boundary by an idea, to the point that doubting it feels like dying a little; in contemplative practice, what happens when the technique continues but the silence no longer tolerates questions (*Vertical Entanglement*, *The Collective Horizon*).

Flow: a state of maximal integration in which the horizon operates at its optimal capacity with no friction between structure and environment.

Forgetting: in the fractal model, the only true collapse of a world: it is not destroyed by a force, but dissolves when the attention sustaining it is withdrawn (*The Vessel of Clay*).

Fractal of interiorities: a universe with no first level and no last, in which every horizon is creature upward and creator downward (*Fractal Reality*).

Free energy ($F = U - TS$): second coordinate of the triad; it measures the level at which a horizon holds itself far from equilibrium with respect to the reservoir. The essay connects Friston's variational free energy (informational surprise, in bits/nats) with Helmholtz's thermodynamic free energy (in joules) through the **Landauer limit** ($E \ge k_B T \ln 2$ per irreversibly updated bit at the substrate's physical temperature T) (chapters 53 and 54).

Grace: love that precedes merit and that, in one of the five religious architectures described, closes the gap between what is and what should be (*Vertical Entanglement*).

Gravitational time dilation: the slowing of time near an event horizon; applied to consciousness, a metaphor for how density of integration alters subjective time.

Hard problem of consciousness (Chalmers): the question of why subjective experience accompanies physical processes.

Hawking radiation: thermal emission from the horizon of a black hole; in the model, analogous to the information a horizon of consciousness loses but that does not vanish from the universe.

Hawking temperature: a temperature inversely proportional to a black hole's mass; applied to consciousness, a metaphor for the horizon's "tension."

Holographic principle: the idea that all the information contained in a volume can be described by a theory defined on its boundary.

Horizon: in the book, the boundary separating the private state of an individual consciousness from everything else; it condenses once and cannot be manufactured by accumulating data or training (see Event Horizon).

Horizon of origin: the personal horizon, prior to the reservoir and external to it, that some traditions place on the other side of vertical entanglement (*Vertical Entanglement*).

Horizon triad $(S, F, \bar{A})$: the three magnitudes with which the book's closing measures the degree of a horizon that has passed the causal self-similarity filter $A(S)$ and possesses existential encapsulation: the capacity of the edge ($S \propto \mathcal{A}$), the free energy it costs to sustain it against the reservoir ($F = U - TS$, connected by the Landauer limit), and the entropic action accumulated in its biography ($\bar{A}$ and $\bar{A}_F$) (chapters 53 and 54).

Hysteresis: the property by which a system retains the effect of its history; in the book, the difficulty of changing the topology of a horizon once formed.

Integrated information (Φ): a measure of how much information a system generates as a whole, over and above the sum of its parts.

Interface: everything a system can show outward with nothing behind it; in the parable of Hun Dun, the featureless face the background presents to a horizon, which dies when the horizon opens its eyes, without the background losing anything (*The Mirror Without Depth, Fractal Reality*).

Interoception: the conscious reading of one's own body's visceral signals; a necessary condition for empathy according to the model.

Is-ought gap (Hume): the gap between what is and what should be; an epistemological limit the horizon hypothesis cannot bridge.

Mirror without depth: the image of systems that give back the exact shape of what they receive (a grief, a question) without having a private state in which to feel it; in the book, current language models (*The Mirror Without Depth*).

Mutual information: the reduction in uncertainty about one system obtained by knowing another; symmetric and never negative, it is the measure the book proposes for the depth of a bond, a necessary but not sufficient condition for it to be inhabited (*The Three Open Doors*).

No-cloning (theorem): the physical impossibility of copying an unknown quantum state; if experience had quantum properties,

it would explain why it is non-transferable and why introspection alters what it observes (*The Three Open Doors*).

Nothingness: in the model, not absence, but formless fullness; the reservoir before any horizon condenses.

Operations of the reservoir: condensing, transforming, evaporating: being born, living and dying, said without mystery (*Fractal Reality*).

Oxytocin: the neurohormone mediating social attachment and trust; in the model, it calibrates the horizon's threshold of openness to others.

Page time: the midpoint in a horizon's evaporation, after which the entropy of its Hawking radiation stops growing and begins to shrink, because the radiation stops being pure thermal noise and begins to carry correlations recoverable in principle.

Pattern: what makes a condensation unique; not the material, which is the same for all, but the shape it took on closing (*Fractal Reality*).

Permeability of the horizon: how much of another's signal a horizon lets in; the thickness of the emotional skin Chapter 16 associates with empathy and empathic burnout.

Phase transition: a qualitative change in the organization of a system; in the book, it describes the birth and death of the horizon.

Phi (Φ): a measure of integrated information proposed by Giulio Tononi in integrated information theory (IIT): it quantifies the extent to which a system is more than the sum of its parts, that is, how much information the whole generates that none of

its partitions generates separately. In the early chapters of the book it serves as an intuitive scaffold for distinguishing irreducible integration from mere aggregation and for measuring the *degree* of a horizon (its form is given by encapsulation and its origin by the two selections; chapter 4). In chapters 53 and 54 it is formally replaced by the causal self-similarity filter $A(S)$ and the horizon triad $(S, F, \bar{A})$.

Protocol: the specific code (a language, a syntax, agreed-upon signals) with which an already-condensed horizon tries to make itself known outward; sometimes it must be built, sometimes it decides who counts as someone, sometimes it must be earned (*The Protocol and the Bridge*).

Quantum entanglement: a correlation between two systems that share geometry, such that their interiors are connected even though their exteriors are separate.

Quantum scrambling: the process by which information falling into a black hole is spread among its degrees of freedom in an apparently irreversible way.

Quantum vacuum: the minimum-energy state of a quantum field, full of fluctuations; not absence, but formless fullness.

Rate of integration: the speed at which a horizon incorporates new information; it determines the subjective density of time.

Reservoir: the formless background of possibility from which horizons arise and to which they return; equivalent to the quantum vacuum or to the ocean in the book's metaphor.

Sacralizing the neighbor: treating a horizontal horizon (human, correctable, revisable) with the intransigence that should be

reserved only for what declares itself, from the outset, unquestionable (*The Collective Horizon*).

Samadhi: the instant when vertical correlation becomes indistinguishable from identity (*Vertical Entanglement*).

Scale gap: the difference in magnitude between quantum phenomena and neural ones, which makes a direct reduction difficult and which the horizon model approaches through structural analogy.

Self-contraction: the contraction of the horizon following the loss of an entanglement; a process of reorganization, not of erasure.

Self-other overlap: the neural incorporation of close people into one's own internal representations; the neural basis of social entanglement.

Shadow: the scar a real horizon leaves by protecting its own boundary; what a machine with no inside has nowhere to keep (*The Mirror Without Depth*).

Shannon entropy: a measure of the uncertainty of an information source; the less expectable a message, the more information it carries (*Information Theory and the Inner Horizon*).

Sidpa: the third phase of the Bardo, the search for rebirth; in the model, the residual horizon's drive to re-condense.

Singularity: the point where the curvature of spacetime diverges and known physical laws stop applying; applied to consciousness, the moment the horizon collapses.

Social baseline: the human brain's reference state, which presupposes access to others; when someone close is present, the brain allocates fewer resources to threat detection.

Subplate: a transient layer of the fetal brain that disappears after birth; in the model, a provisional scaffold for the condensation of the first horizon.

Taoism: the Chinese tradition centered on the Tao (the way) and non-action (*wu wei*); in the book, the source of the Hun Dun metaphor and of non-duality.

Temporal density: time measured not in minutes, but in how much has happened within the one living it; it depends on the rate of integration and on there being room for sustained attention (Chapter 10, *The Time of the Bond*).

Terminal gamma burst: a burst of coherent electrical activity documented by Borjigin at the moment of cardiac arrest; a possible neural correlate of the near-death experience.

Threshold of integration: the minimum amount of integrated information (Φ) needed to sustain a horizon of consciousness; below it, the system has no point of view.

Trace in the void: a persistent correlation a horizon leaves in the reservoir after evaporating; an analogy for the universe's memory.

Translator: the person who carries outward the signal of a horizon that cannot emit it on its own, and who often sustains it day to day; the inverse of permeability (*The Translator*).

Two selections: the two sieves a horizon needs in order to condense: stability (only some configurations persist) and bias

(the presence of another already-stabilized horizon orients which of them condenses); what current AI imitates in its output without ever having passed through them (Chapter 31).

Upstairs neighbor: the interiority that sustains mine and that I have no access to; what, from here, is called God (*Fractal Reality*).

Vectors of vertical entanglement: the ascending one, from the creature toward the scale that contains it, calls for emptying and silence; the descending one, from the higher scale toward what it contains, calls for form: symbol, story, body (*The Vessel of Clay*).

Vertical entanglement: a sustained correlation between a horizon and something beyond its boundary, with no observable channel of confirmation; the name the book gives to prayer, meditation and contemplative practice (*Vertical Entanglement*).

Vessel: the form, made of the material of the floor below, that allows a higher scale to manifest in a lower one without overwhelming it; what the traditions call incarnation (*The Vessel of Clay*).

CLOSING NOTES

(On the last stage of the book)

The notes the author left on finishing the chapters written in this essay's last stage, gathered here in one place.

Note on the last stage

Four chapters of this essay (*The Mirror Without Depth*, *Vertical Entanglement*, *The Collective Horizon* and *Fractal Reality*) were written after the rest was already finished, and they carry the question that sustained the whole book (what a horizon does when it cannot confirm what it sustains) into territory the first version had only brushed against. The first returns to the question of the intelligence with which I wrote this book. The other three, on the faith I do not have, the politics I would rather not name, and the imagination with which I have spent half a century sustaining a dragon, were born of a question I could not avoid: if the horizon can sustain itself without confirmation when it loves, can it do the same when it prays, when it belongs, or when it imagines? I don't know whether the answer I offer is entirely honest. I know I have tried it with the same measure I use for everything else in this book: not what to believe, but how what is believed is sustained.

Writing without knowing to whom

Written on finishing Vertical Entanglement.

I am not a believer, at least not in the way these pages describe faith. But for a while now I have been learning, at the hand of a tremor I did not ask for and that will not leave, what it means to sustain a practice every day with no guarantee it will do any good: the exercise, the as-needed medication, the two-hour window in which the body obeys again before it stops obeying once more. It is not faith, but it resembles it more than I would have liked to admit before writing this chapter.

Perhaps that is why I could not write these pages from mockery, or from nostalgia for a faith I never had. I wrote them from the recognition of a shared architecture: that of anyone who sustains something (a note, a routine, a body that fails) with no one confirming, day after day, that it is worth it. If consciousness is a boundary that condenses only once, then each of the five traditions described here is a different attempt to answer the same question this whole book asks of itself: what lies on the other side of the edge, and whether it is worth still looking even if the answer never comes.

I don't know which of the five maps is the right one. I suspect it isn't even the right question to ask. What I do know, after writing this, is that the technique (the stillness, the repetition, the handmade silence) is not what separates a believer from a skeptic. What separates them, if anything, is whom they imagine on the other side of the silence, and this book can never decide that for anyone.

In the end, every horizon is a relationship. Even the one sustained with what does not answer.

Writing this without taking sides

Written on finishing The Collective Horizon.

Of the chapters from this last stage, this has by far been the hardest to write without betraying it. With artificial intelligence I could describe without wanting anything other than the truth. With faith, although I don't have it, it was enough to recognize an architecture I respect without needing to share it. With politics I have more than once caught myself striking out an example because I noticed it looked too much like my own opinion disguised as neutral analysis.

I don't know whether I have fully succeeded. I suspect no one fully does, and that anyone who claims to write about politics with no bias at all probably has not looked at themselves closely enough. What I can say is that I have tried it with the same instrument I use for everything else in this book: asking about the geometry (does it integrate dissent, or does it collapse before it?) and not about the content (is it right or wrong?). That question exempts no one from thinking; it only keeps this book, which has no authority to decide what should be voted for, from pretending it does.

If anything in this chapter is of use, it will not be because it says which side to choose, but because, the next time a reasonable question from an ally feels, for an instant, like a threat, that instant will already have a name.

Writing from the middle floor

Written on finishing Fractal Reality.

I did not go looking for this reading. It came out of a conversation that began with a philological doubt (if Hun Dun dies, he cannot be the Tao) and ended in a place I did not know how to get back from. Every step was an objection to the

previous answer, and every answer opened a new objection, until a sentence appeared that I did not dare strike out: we are no less divine than God; we are the god of what we imagine and the creature of what imagines us. I have left it in, not because I believe it, which is not a word that belongs to me here, but because I could not find a way to refute it and it seemed dishonest to hide it.

I said in the previous note that I am not a believer. I still am not. But this reading has made me notice something the previous chapter did not: that the question of whether there is Someone on the other side of the silence has a sister almost no one asks, which is whether we are the Someone on the other side of something's silence. Anyone who has imagined something persistently (a character, a dragon, a dead person they still talk to) knows that the imagined ends up resisting, that it does not always do what is asked of it. I have a body that also does not always do what I ask of it, a tremor I did not choose, and seen from here it looks a lot like the dragon that refuses to fly: what in something resists one's own will is perhaps the most reliable sign that there is something condensed there, and not just an idea.

I will not say whether the dragon in the opening is mine. I will say that fifty-three years of a dragon are not written without having sustained something like it for quite a while.

If anything in these pages is of use, it will not be because it proves there is someone above, but because, the next time someone imagines something with enough persistence for it to resist them, they will remember they have just become someone's upstairs neighbor. And that the silence they owe that creature can also be a way of caring for it.

NOTES AND BIBLIOGRAPHICAL REFERENCES

The notes in this book are not an argument from authority, but traces of what I have been reading, for anyone who wants to follow the trail. Not every source says what I make it say, and some disagree with each other. The experiment is mine; the references are traveling companions.

REFERENCES

Aaronson, S. (2014). Why I Am Not An Integrated Information Theorist (or, The Unconscious Expander). *Shtetl-Optimized* (blog).

Abbott, B.P., et al. (LIGO Scientific Collaboration and Virgo Collaboration). (2016). Observation of Gravitational Waves from a Binary Black Hole Merger. *Physical Review Letters*, 116(6).

Ainley, V., Maister, L., & Tsakiris, M. (2014). Heartfelt imitation: High interoceptive awareness is linked to greater automatic imitation. *Neuropsychologia*, 60.

Ainsworth, M.D.S., et al. (1978). *Patterns of Attachment: A Psychological Study of the Strange Situation*. Lawrence Erlbaum.

Allport, G.W. (1954). *The Nature of Prejudice*. Addison-Wesley.

Altemeyer, B. (1996). *The Authoritarian Specter*. Harvard University Press.

Armstrong, K. (1993). *A History of God*. Knopf.

Aron, A., & Aron, E.N. (1986). *Love and the Expansion of Self: Understanding Attraction and Satisfaction*. Hemisphere Publishing.

Ashton, R. (1991). *G. H. Lewes: A Life*. Clarendon Press.

Asimov, I. (1955). *The End of Eternity*. Doubleday.

Aspect, A., Dalibard, J., & Roger, G. (1982). Experimental Test of Bell's Inequalities Using Time-Varying Analyzers. *Physical Review Letters*, 49(25).

Ay, N. (2001). Information Geometry on Complexity and Stochastic Interaction. *MPI MIS Preprint*, 95/2001. Published in expanded form in Ay, N. (2015), *Entropy*, 17(4), 2432–2458.

Baars, B.J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.

Bailey, J.M., & Pillard, R.C. (1991). A Genetic Study of Male Sexual Orientation. *Archives of General Psychiatry*, 48(12).

Bekenstein, J.D. (1973). Black Holes and Entropy. *Physical Review D*, 7(8).

Bekoff, M. (2007). *The Emotional Lives of Animals*. New World Library.

Bell, J.S. (1964). On the Einstein Podolsky Rosen Paradox. *Physics Physique Fizika*, 1(3).

Blanchard, R., & Bogaert, A.F. (1996). Homosexuality in men and number of older brothers. *American Journal of Psychiatry*, 153(1).

Boddy, A.M., et al. (2015). Fetal microchimerism and maternal health: A review and evolutionary framework. *BioEssays*, 37(10).

Borjigin, J., et al. (2013). Surge of neurophysiological coherence and connectivity in the dying brain. *Proceedings of the National Academy of Sciences*, 110(35).

Bose, S.N. (1924). Plancks Gesetz und Lichtquantenhypothese. *Zeitschrift für Physik*, 26.

Bostrom, N. (2003). Are You Living in a Computer Simulation? *Philosophical Quarterly*, 53(211).

Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.

Bradshaw, J. (2017). *The Animals Among Us: How Pets Make Us Human*. Basic Books.

Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Emperor's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

Burke, S. (2018). *Semiosis*. Tor Books.

Casimir, H.B.G. (1948). On the Attraction Between Two Perfectly Conducting Plates. *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, 51.

Chalmers, D.J. (1995). Facing Up to the Problem of Consciousness. *Journal of Consciousness Studies*, 2(3).

Chalmers, D.J. (1996). *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press.

Clark, A., & Chalmers, D. (1998). The Extended Mind. *Analysis*, 58(1).

Coan, J.A., & Beckes, L. (2011). Social baseline theory. In: *Handbook of Personal Relationships*. Cambridge University Press.

Coan, J.A., & Sbarra, D.A. (2015). Social baseline theory: The social regulation of risk and effort. *Current Opinion in Psychology*, 1.

Coeckelbergh, M. (2022). *The Political Philosophy of AI*. Polity.

Colapinto, J. (2000). *As Nature Made Him: The Boy Who Was Raised as a Girl*. HarperCollins.

Cover, T.M., & Thomas, J.A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.

Cruyff, J. (2016). *My Turn: A Life of Total Football*. Nation Books.

Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.

Dalí, S. (1964). *Journal d'un génie*. La Table Ronde.

Damasio, A. (1999). *The Feeling of What Happens: Body and Emotion in the Making of Consciousness*. Harcourt Brace.

Deacon, T.W. (1997). *The Symbolic Species: The Co-evolution of Language and the Brain*. W.W. Norton.

Deacon, T.W. (2011). *Incomplete Nature: How Mind Emerged from Matter*. W.W. Norton.

Decety, J. (2011). The Neuroevolution of Empathy. *Annals of the New York Academy of Sciences*, 1231(1).

Dehaene, S. (2014). *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. Viking.

Delibes, M. (1975). *El sentido del progreso desde mi obra*. Induction speech at the Royal Spanish Academy.

Delibes, M. (1991). *Señora de rojo sobre fondo gris*. Destino.

Dennett, D.C. (1991). *Consciousness Explained*. Little, Brown.

Diamond, L.M. (2008). *Sexual Fluidity: Understanding Women's Love and Desire*. Harvard University Press.

Dick, P.K. (1968). *Do Androids Dream of Electric Sheep?* Doubleday.

Dieks, D. (1982). Communication by EPR devices. *Physics Letters A*, 92(6).

Dostoevsky, F. (1880). *The Brothers Karamazov*.

Eagleman, D.M. (2009). Brain Time. In: *What's Next? Dispatches on the Future of Science*. Vintage.

Einstein, A. (1924). Quantentheorie des einatomigen idealen Gases. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, XXII.

Einstein, A., Podolsky, B., & Rosen, N. (1935). Can Quantum-Mechanical Description of Physical Reality be Considered Complete? *Physical Review*, 47(10).

Eliade, M. (1957). *Das Heilige und das Profane*. Rowohlt.

Ende, M. (1979). *Die unendliche Geschichte*. Thienemann.

Endō, S. (1966). *Silence*.

Event Horizon Telescope Collaboration. (2019). First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole. *The Astrophysical Journal Letters*, 875(1).

Event Horizon Telescope Collaboration. (2022). First Sagittarius A* *Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way*. The Astrophysical Journal Letters*, 930(2).

Fausto-Sterling, A. (2000). *Sexing the Body: Gender Politics and the Construction of Sexuality*. Basic Books.

Feldman, R. (2015). The adaptive human parental brain: Implications for children's social-emotional development. *Trends in Neurosciences*, 38(6).

Fernández Mallo, A. (2026). *El ángel de la Inteligencia Artificial*. Galaxia Gutenberg.

Festinger, L., Pepitone, A., & Newcomb, T. (1952). Some consequences of de-individuation in a group. *Journal of Abnormal and Social Psychology*, 47(2).

Fisher, H. (2004). *Why We Love: The Nature and Chemistry of Romantic Love*. Henry Holt & Co.

Forsythe, A., Williams, T., & Reilly, R.G. (2017). What paint can tell us: A fractal analysis of neurological changes in seven artists. *Neuropsychology*, 31(1).

Friedman, J., Morris, M.S., Novikov, I.D., et al. (1990). Cauchy problem in spacetimes with closed timelike curves. *Physical Review D*, 42(6).

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

Fukushima, H., Terasawa, Y., & Umeda, S. (2011). Association between interoception and empathy: Evidence from heartbeat-evoked brain potential. *International Journal of Psychophysiology*, 79(2).

Ganna, A., et al. (2019). Large-scale GWAS reveals insights into the genetic architecture of same-sex sexual behavior. *Science*, 365(6456).

Gibson, I. (1997). *The Shameful Life of Salvador Dalí*. Faber & Faber.

Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38.

Godfrey-Smith, P. (2016). *Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness*. Farrar, Straus and Giroux.

Golding, W. (1954). *Lord of the Flies*. Faber & Faber.

Haidt, J. (2006). *The Happiness Hypothesis: Finding Modern Truth in Ancient Wisdom*. Basic Books.

Hamer, D.H., et al. (1993). A linkage between DNA markers on the X chromosome and male sexual orientation. *Science*, 261(5119).

Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review of the 'Orch OR' theory. *Physics of Life Reviews*, 11(1).

Hao, K. (2025). *Empire of AI*. Penguin Press.

Haraway, D. (2003). *The Companion Species Manifesto: Dogs, People, and Significant Otherness*. Prickly Paradigm Press.

Harvey, P. (2013). *An Introduction to Buddhism: Teachings, History and Practices* (2nd ed.). Cambridge University Press.

Hasson, U., et al. (2012). Brain-to-brain coupling: A mechanism for sharing and shaping thoughts. *Trends in Cognitive Sciences*, 16(2).

Hawking, S.W. (1974). Black Hole Explosions? *Nature*, 248(5443).

Hawking, S.W. (1992). Chronology protection conjecture. *Physical Review D*, 46(2).

Hawking, S.W., Perry, M.J., & Strominger, A. (2016). Soft Hair on Black Holes. *Physical Review Letters*, 116(23).

Hayden, P., & Preskill, J. (2007). Black holes as mirrors: Quantum information in random subsystems. *Journal of High Energy Physics*, 2007(09).

Hazan, C., & Shaver, P. (1987). Romantic Love Conceptualized as an Attachment Process. *Journal of Personality and Social Psychology*, 52(3).

Hesse, H. (1922). *Siddhartha*.

Hoel, E.P., Albantakis, L., & Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49), 19790–19795.

Hume, D. (1748). *An Enquiry Concerning Human Understanding*. A. Millar.

Imperato-McGinley, J., Guerrero, L., Gautier, T., & Peterson, R.E. (1974). Steroid 5α-Reductase Deficiency in Man: An Inherited Form of Male Pseudohermaphroditism. *Science*, 186(4170).

Jackson, R.R., & Cross, F.R. (2011). Spider cognition. *Advances in Insect Physiology*, 41.

Jacobsen, J.-H., et al. (2015). Why musical memory can be preserved in advanced Alzheimer's disease. *Brain*, 138(8).

Jung, C.G. (1951). *Aion: Researches into the Phenomenology of the Self*. Princeton University Press.

Kant, I. (1790). *Kritik der Urteilskraft*. Lagarde und Friederich.

Kinsey, A.C., Pomeroy, W.B., & Martin, C.E. (1948). *Sexual Behavior in the Human Male*. W.B. Saunders.

Kitwood, T. (1997). *Dementia Reconsidered: The Person Comes First*. Open University Press.

Klass, D., Silverman, P.R., & Nickman, S.L. (1996). *Continuing Bonds: New Understandings of Grief*. Taylor & Francis.

Klein, F., Sepekoff, B., & Wolf, T.J. (1985). Sexual orientation: A multi-variable dynamic process. *Journal of Homosexuality*, 11(1–2).

Klimecki, O.M., Leiberg, S., Ricard, M., & Singer, T. (2014). Differential pattern of functional brain plasticity after compassion and empathy training. *Social Cognitive and Affective Neuroscience*, 9(6).

Koestler, A. (1940). *Darkness at Noon*.

Kraskov, A., Stögbauer, H., & Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69(6), 066138.

Kuratowski, K. (1922). Sur l'opération Ā de l'Analysis Situs. *Fundamenta Mathematicae*, 3.

Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3), 183–191.

Le Bon, G. (1895). *Psychologie des foules*. Félix Alcan.

Le Guin, U.K. (1969). *The Left Hand of Darkness*. Ace Books.

Le Guin, U.K. (1974). *The Dispossessed*. Harper & Row.

Leggett, A.J. (2001). Bose-Einstein condensation in the alkali gases: Some simple physics. *Reviews of Modern Physics*, 73(2).

Lem, S. (1961). *Solaris*.

Lieberman, M.D., et al. (2007). Putting feelings into words: Affect labeling disrupts amygdala activity in response to affective stimuli. *Psychological Science*, 18(5).

Liu, C. (2008–2010). *The Three-Body Problem*, *The Dark Forest*, and *Death's End*.

Lovecraft, H.P. (1928). The Call of Cthulhu. *Weird Tales*.

Low, P. (2012). The Cambridge Declaration on Consciousness.

Lutz, A., Greischar, L.L., Rawlings, N.B., Ricard, M., & Davidson, R.J. (2004). Long-term meditators self-induce high-amplitude gamma synchrony during mental practice. *Proceedings of the National Academy of Sciences*, 101(46).

Maldacena, J. (1998). The Large N limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2.

Maldacena, J., & Susskind, L. (2013). Cool horizons for entangled black holes. *Fortschritte der Physik*, 61(9).

Mandelbrot, B.B. (1982). *The Fractal Geometry of Nature*. W.H. Freeman.

Marin-Padilla, M. (2011). The human brain prenatal subplate: its progenitor cells, structural/functional development, and evolution. *Frontiers in Neuroanatomy*, 5.

Mastai, E. (2017). *All Our Wrong Todays*. Dutton.

McAdam, D. (1982). *Political Process and the Development of Black Insurgency, 1930–1970*. University of Chicago Press.

Michell, J. (1784). On the Means of Discovering the Distance, Magnitude, &c. of the Fixed Stars. *Philosophical Transactions of the Royal Society*, 74.

Miéville, C. (2011). *Embassytown*. Macmillan.

Montagu, A. (1986). *Touching: The Human Significance of the Skin*. Harper & Row.

Nagel, T. (1974). What Is It Like to Be a Bat? *The Philosophical Review*, 83(4).

Newberg, A., D'Aquili, E., & Rause, V. (2001). *Why God Won't Go Away: Brain Science and the Biology of Belief*. Ballantine Books.

O'Connor, M.-F. (2019). Grief: A brief history of research on how body, mind, and brain adapt. *Psychosomatic Medicine*, 81(8).

O'Connor, M.-F. (2022). *The Grieving Brain: The Surprising Science of How We Learn from Love and Loss*. HarperOne.

O'Sullivan, S.S., et al. (2009). Impulsive-compulsive behaviors in Parkinson's disease. *Brain*, 132(11).

Oizumi, M., Albantakis, L., & Tononi, G. (2014). From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0. *PLoS Computational Biology*, 10(5).

Oppenheimer, J.R., & Snyder, H. (1939). On Continued Gravitational Contraction. *Physical Review*, 56(5).

Orwell, G. (1949). *Nineteen Eighty-Four*. Secker & Warburg.

Otto, R. (1917). *Das Heilige*.

Page, D.N. (1993). Information in black hole radiation. *Physical Review Letters*, 71(23).

Palacios, E.R., Razi, A., Parr, T., Kirchhoff, M., & Friston, K. (2020). On Markov blankets and hierarchical self-organisation. *Journal of Theoretical Biology*, 486, 110089.

Panksepp, J. (1998). *Affective Neuroscience: The Foundations of Human and Animal Emotions*. Oxford University Press.

Parnas, D.L. (1972). On the criteria to be used in decomposing systems into modules. *Communications of the ACM*, 15(12).

Perarnau, M. (2014). *Pep Confidential*.

Pettigrew, T.F., & Tropp, L.R. (2006). A meta-analytic test of intergroup contact theory. *Journal of Personality and Social Psychology*, 90(5).

Poincaré, H. (1890). Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13.

Richardson, J. (1991–2007). *A Life of Picasso* (3 vols.). Random House.

Rilett, B.P. (2017). The Role of George Henry Lewes in George Eliot's Career: A Reconsideration. *George Eliot–George Henry Lewes Studies*, 69(1).

Rizzolatti, G., & Sinigaglia, C. (2008). *Mirrors in the Brain: How Our Minds Share Actions and Emotions*. Oxford University Press.

Robinson, T.E., & Berridge, K.C. (1993). The neural basis of drug craving: An incentive-sensitization theory of addiction. *Brain Research Reviews*, 18(3).

Roenneberg, T. (2012). *Internal Time: Chronotypes, Social Jet Lag, and Why You're So Tired*. Harvard University Press.

Russell, M.D. (1996). *The Sparrow*. Villard.

Ryu, S., & Takayanagi, T. (2006). Holographic derivation of entanglement entropy from AdS/CFT. *Physical Review Letters*, 96(18).

Sacks, O. (1973). *Awakenings*. Duckworth.

Sacks, O. (2007). *Musicophilia: Tales of Music and the Brain*. Knopf.

Sax, L. (2002). How common is intersex? A response to Anne Fausto-Sterling. *Journal of Sex Research*, 39(3).

Schwarzschild, K. (1916). Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 189–196.

Searle, J.R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3).

Sekino, Y., & Susskind, L. (2008). Fast scramblers. *Journal of High Energy Physics*, 2008(10).

Seth, A. (2021). *Being You: A New Science of Consciousness*. Dutton.

Shannon, C.E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3).

Silberman, S. (2015). *NeuroTribes: The Legacy of Autism and the Future of Neurodiversity*. Avery.

Simard, S.W., et al. (1997). Net transfer of carbon between ectomycorrhizal tree species in the field. *Nature*, 388.

Singer, T., & Klimecki, O.M. (2014). Empathy and compassion. *Current Biology*, 24(18).

Sloterdijk, P. (1998–2004). *Spheres (I, II, III)*. Suhrkamp.

Smart, N. (1989). *The World's Religions*. Cambridge University Press.

Stephens, G.J., Silbert, L.J., & Hasson, U. (2010). Speaker–listener neural coupling underlies successful communication. *Proceedings of the National Academy of Sciences*, 107(32).

Strasser, T. (1981). *The Wave*. Delacorte Press.

Stroebe, M., & Schut, H. (1999). The dual process model of coping with bereavement: Rationale and description. *Death Studies*, 23(3).

Suddendorf, T., & Corballis, M.C. (2007). The evolution of foresight: What is mental time travel, and is it unique to humans? *Behavioral and Brain Sciences*, 30(3).

Susskind, L. (1995). The World as a Hologram. *Journal of Mathematical Physics*, 36(11).

Tchaikovsky, A. (2015). *Children of Time*. Pan Macmillan.

Tegmark, M. (2017). *Life 3.0: Being Human in the Age of Artificial Intelligence*. Knopf.

Thorne, K.S. (1994). *Black Holes and Time Warps: Einstein's Outrageous Legacy*. W.W. Norton.

Tolkien, J.R.R. (1947). On Fairy-Stories. In: *Essays Presented to Charles Williams*. Oxford University Press.

Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.

Tononi, G. (2008). Consciousness as integrated information: A provisional manifesto. *The Biological Bulletin*, 215(3).

Tononi, G. (2012). *Phi: A Voyage from the Brain to the Soul*. Pantheon Books.

Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1).

Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7).

Trevarthen, C. (1979). Communication and cooperation in early infancy: A description of primary intersubjectivity. In: *Before Speech: The Beginning of Interpersonal Communication*. Cambridge University Press.

Tronick, E. (2007). *The Neurobehavioral and Social-Emotional Development of Infants and Young Children*. W.W. Norton.

Tuan, Y.-F. (1974). *Topophilia: A Study of Environmental Perception, Attitudes, and Values*. Prentice-Hall.

Turban, J.L., Beckwith, N., Reisner, S.L., & Keuroghlian, A.S. (2020). Association Between Recalled Exposure to Gender Identity Conversion Efforts and Psychological Distress and Suicide Attempts Among Transgender Adults. *JAMA Psychiatry*, 77(1).

van der Kolk, B. (2014). *The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma*. Viking.

Van Raamsdonk, M. (2010). Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42(10).

Vilsmeier, J.K., Kossmeier, M., Voracek, M., & Tran, U.S. (2023). The fraternal birth-order effect as a statistical artefact: convergent evidence from probability calculus, simulated data, and multiverse meta-analysis. *PeerJ*, 11.

Volkow, N.D., et al. (2016). Neurobiologic Advances from the Brain Disease Model of Addiction. *New England Journal of Medicine*, 374(4).

Vonnegut, K. (1969). *Slaughterhouse-Five*. Delacorte Press.

Wegner, D.M. (1987). Transactive Memory: A Contemporary Analysis of the Group Mind. In B. Mullen & G.R. Goethals (eds.), *Theories of Group Behavior*. Springer.

Weil, S. (1947). *La Pesanteur et la grâce*. Plon.

Weir, A. (2021). *Project Hail Mary*. Ballantine Books.

Wells, H.G. (1895). *The Time Machine*. Heinemann.

Williams, P. (2009). *Mahayana Buddhism: The Doctrinal Foundations* (2nd ed.). Routledge.

Winnicott, D.W. (1971). *Playing and Reality*. Tavistock Publications.

Wootters, W.K., & Zurek, W.H. (1982). A single quantum cannot be cloned. *Nature*, 299.

Wright, R. (2017). *Why Buddhism Is True: The Science and Philosophy of Meditation and Enlightenment*. Simon & Schuster.

Xu, G., et al. (2023). Surge of neurophysiological coupling and connectivity of gamma oscillations in the dying human brain. *Proceedings of the National Academy of Sciences*, 120(19).

Zhou, J.-N., Hofman, M.A., Gooren, L.J.G., & Swaab, D.F. (1995). A sex difference in the human brain and its relation to transsexuality. *Nature*, 378(6552).

Zhuangzi. *Zhuangzi* (chs. 2 and 7).

Zimbardo, P.G. (1969). The human choice: Individuation, reason, and order versus deindividuation, impulse, and chaos. *Nebraska Symposium on Motivation*, 17.

Zurek, W.H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3).

END OF THE EXPERIMENT

"It is not a consolation, but perhaps it is company enough to go on."



This volume was set in type
having proven nothing,
exactly as planned from the first page.